

Curriculum Guide

Every skill from counting to college mathematics, explained in plain English for parents, grandparents and anyone helping a learner.

I	II	III	IV	V	VI	VII
Count	Operate	Relate	Solve	Prove	Change	Space

How to read this guide

Mathera is a free app that teaches math from first counting all the way to second-year college mathematics. It is laid out like a tree that grows from the roots up. This guide lists every skill in that tree, in order, with a plain-English description of each one.

Eras, branches and skills

Eras. The math is divided into seven eras, from Era I (Count) to Era VII (Space). Each era is roughly a stage of schooling.

Branches. Each era is split into branches, each one a single topic such as *Fractions* or *Quadratics*.

Skills. Each branch holds a handful of skills: one clear thing a learner can learn to do, like *Add and subtract tens*.

Each skill below has its name, its code (for example I . 4 . 21: Era I, branch 4, skill 21), one sentence on what it is, and a line for each level.

Four levels for every skill

Every skill is practiced in four levels, from a gentle first try to real mastery:

1	The simplest version of the idea, often with small numbers or a picture to help.
2	The same idea with a little more to keep track of.
3	The idea in a less familiar form, or with a step added.
4	The full skill, often applied to a real situation or explained in the learner's own words.

A level is **planted** once the learner gets three questions in a row right, and planting a level also plants every easier level. When level 4 is planted, the skill is **proven**.

Brave and Legend: the optional challenge levels

For learners who finish quickly, some skills have two extra levels: about half the skills in Era IV, and about a quarter in Eras I-III and V, chosen where a challenge really helps. They open only once the skill is proven. They are always optional and never hold anyone back.

★ Brave	The skill with a twist: working backwards, combining it with an earlier skill, or finding a missing value that makes something work.
★★ Legend	Puzzle-style problems in the spirit of math competitions, which need a flash of insight rather than longer arithmetic.

Eras VI and VII arrive in Mathera 2.0; their skills are listed here so you can see where the road leads, but they have no practice yet.

Keeping skills alive

Skills in the tree need occasional watering. The app brings back each proven skill for a quick review, at first after a day, then after longer and longer gaps. A skill left too long starts to wilt, and the app gently steers the learner back to it. This spaced review is what turns something learned once into something kept for good.

A note for helpers

You do not need to know the math to help. Use this guide to see what a skill is about, ask the learner to show you a question, and celebrate each level planted. Later editions may include versions written for different kinds

of helper, such as parents, tutors and classroom teachers.

The seven eras at a glance

The school stages are a rough guide only. Learners of any age can start wherever the math feels new.

Era I Count	Era I is where numbers begin. Learners count real things, compare which group has more, learn what the digits in a number mean, and add and take away within 100. Alongside the numbers come early patterns and logic, flat and solid shapes, telling time, money, and simple measuring. About ages 4–8 · kindergarten to grade 2	7 branches 108 skills
Era II Operate	Era II builds the four operations into strong tools. Learners work with numbers into the millions, master multiplication and division, meet factors and multiples, and learn to handle fractions and decimals with confidence. The era closes with first expressions (using letters for numbers), measurement, area and volume, and reading data. About ages 8–11 · grades 3 to 5	10 branches 135 skills
Era III Relate	Era III is about how quantities relate to each other. Negative numbers, ratios, rates and percentages come first, then powers and roots, then algebra: writing expressions and solving equations. It ends with first graphs of straight lines, basic non-proof geometry of angles, circles and solids, and basic statistics and chance. About ages 11–14 · grades 6 to 8	9 branches 146 skills
Era IV Solve	Era IV is algebra in full. Learners solve every common kind of equation, study functions and their graphs (lines, parabolas, polynomials, exponentials and logarithms), meet complex numbers, and work with sequences, counting, probability and data. About half of its skills also have the two optional challenge levels, Brave and Legend. About ages 14–17 · high-school algebra through precalculus	15 branches 199 skills
Era V Prove	Era V is the era of reasons. This is where proof begins: learners learn how a mathematical argument works, then use short proofs throughout geometry: lines, triangles, polygons, similar shapes and circles. Trigonometry (the math of angles and waves) follows, then coordinate geometry, conic sections, transformations, solids and classic ruler-and-compass constructions. About ages 14–17 · high-school geometry and trigonometry	15 branches 173 skills
Era VI Change	Era VI is the mathematics of change. It covers calculus from limits to derivatives and integrals, then differential equations, curves described in new ways, and infinite series. It closes with statistical inference (drawing conclusions from data) and a short unit on mechanics, the math of forces and motion. About ages 16–19 · final years of high school and first-year college · coming in Mathera 2.0	12 branches 174 skills
Era VII Space	Era VII takes math into many dimensions. Linear algebra studies vectors and matrices, the language behind computer graphics, data science and engineering. Multivariable calculus extends calculus to surfaces and 3-D space. Differential equations model how real systems change over time. About ages 18–20 · second-year college mathematics · coming in Mathera 2.0	19 branches 197 skills

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Era I • Count

About ages 4–8 • kindergarten to grade 2

Era I is where numbers begin. Learners count real things, compare which group has more, learn what the digits in a number mean, and add and take away within 100. Alongside the numbers come early patterns and logic, flat and solid shapes, telling time, money, and simple measuring.

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Counting

Counting is where all math begins. This branch builds counting from 1 to 120, recognizing small amounts at a glance, and counting in steps of 2, 5 and 10.

I.1.01

Count objects one by one

Counting a group of things by pointing to each one once and saying one number for each.

- 1 Counts up to 3 things, touching each one.
 - 2 Counts up to 5 things.
 - 3 Counts up to 10 things in a line.
 - 4 Counts up to 10 things scattered around, without skipping or counting any twice.
-

I.1.02

Number words in order

Knowing the counting words (one, two, three...) and saying them in the right order.

- 1 Says the number words from one to three in order.
 - 2 Says the number words up to five and knows which comes next.
 - 3 Says the number words up to ten and knows which comes next.
 - 4 Starts counting from any number, such as seven, and carries on.
-

I.1.03

Count to 20

Counting past ten up to twenty, and noticing that the teen numbers are ten plus a few more.

- 1 Fills in missing numbers when counting from 10 to 15.
 - 2 Fills in missing numbers when counting from 15 to 20.
 - 3 Sees that a teen number like 12 is ten and 2 more.
 - 4 Counts up to 20 objects, using a full row of ten as a head start.
-

I.1.04

Count to 50

Counting up to fifty, including the tricky moment when one group of ten ends and the next begins.

- 1 Knows the names of the tens: ten, twenty, thirty, forty, fifty.
 - 2 Fills in missing numbers when counting within one group of ten, like 14 to 17.
 - 3 Counts across into the next ten, such as 19 to 20 or 29 to 30.
 - 4 Fills in missing numbers anywhere from 1 to 50.
-

I.1.05

Count to 120

Counting from fifty all the way up to one hundred twenty, including going past one hundred.

- 1 Fills in missing numbers when counting from 50 to 100.
 - 2 Knows that 100 comes after 99.
 - 3 Reads and counts numbers from 100 to 120, such as one hundred sixteen.
 - 4 Continues counting from any number up to 120.
-

I.1.06

Read and write numerals to 20

Recognizing written numbers (numerals) from 0 to 20 and matching them to their names and amounts.

- 1 Reads the numbers 0 to 10 and matches each to its word.
 - 2 Reads the numbers 11 to 20, such as finding 12 for twelve.
 - 3 Picks out a number that is written the right way round, not flipped.
 - 4 Matches a written number to a picture showing that many.
-

I.1.07

How many in all

Understanding that the last number said when counting tells how many there are altogether.

- 1 Knows the last number counted tells how many in all.
 - 2 Answers how many from someone else's count, without counting again.
 - 3 Counts out an exact amount, such as finding the box with exactly 4.
 - 4 Knows the total stays the same whichever end you start counting from.
-

I.1.08

Zero

Understanding zero as the number for nothing at all.

- 1 Knows zero means none, such as an empty plate.
 - 2 Picks out the picture that shows zero things.
 - 3 Writes 0 when there is nothing to count.
 - 4 Knows that 0 comes after 1 when counting backward.
-

I.1.09

See amounts to 5 at a glance

Seeing how many are in a small group, up to five, at a quick glance without counting one by one.

- 1 Says how many at a glance for 1 to 3 dots.
 - 2 Says how many at a glance for 4 or 5 dots.
 - 3 Recognizes the dot patterns on dice.
 - 4 Recognizes how many fingers are held up without counting.
-

I.1.10

See amounts to 10 at a glance

Seeing amounts up to ten at a glance, often by spotting a group of five and some more.

- 1 Says how many counters are in a row of five boxes (a five-frame).
- 2 Says how many counters are in two rows of five boxes (a ten-frame).
- 3 Sees two groups, like 5 and 1, and knows the total quickly.
- 4 Says how many dots at a glance, however they are arranged.

I.1.11

Count on

Adding a few more by starting from a number and counting up from there, instead of counting everything.

- 1 Starts at a number and counts on 1 more, staying within 10.
 - 2 Starts at a number and counts on 2 more, staying within 10.
 - 3 Starts from a given number and counts on the extra dots shown.
 - 4 Chooses to start from the bigger number, which makes counting on quicker.
 - ★ Works backwards: finds how many were counted on, or which number it started from.
 - ★★ Follows a pattern of jumps by 2 and decides which number it will land on.
-

I.1.12

Count back

Counting backward, one number at a time, which later helps with taking away.

- 1 Counts back from 5, such as jumping back along a number line.
 - 2 Counts back from 10.
 - 3 Counts back from 20.
 - 4 Counts back past 10 into single digits, such as 14 down to 9.
-

I.1.13

One more, one less

Knowing what number is just one more or one less than a given number.

- 1 Finds one more than a number up to 10.
 - 2 Finds one less than a number up to 10.
 - 3 Finds one more or one less for numbers up to 20.
 - 4 The same with numbers up to 100.
 - ★ Follows a chain of steps, like 2 more then 1 less, to find where it ends.
 - ★★ Solves a puzzle: two neighbouring numbers add up to a total; finds one of them.
-

I.1.14

Skip count by 10s

Counting in jumps of ten: 10, 20, 30, and so on.

- 1 Counts by tens from 10 to 50.
 - 2 Counts by tens up to 100.
 - 3 Counts by tens from any number, such as 8, 18, 28, 38.
 - 4 Counts bars of ten cubes by tens to find the total.
 - ★ Works backwards, counting down by tens to fill in missing numbers.
 - ★★ Works out how many numbers are said when counting by tens, such as from 8 to 88.
-

I.1.15

Skip count by 5s

Counting in jumps of five: 5, 10, 15, 20, and so on.

- 1 Counts by fives from 5 to 25.
- 2 Counts by fives up to 50.
- 3 Counts by fives up to 100.
- 4 Counts hands of five fingers, or 5-cent coins, by fives.

Skip count by 2s

Counting in jumps of two: 2, 4, 6, 8, and so on.

- ① Counts by twos up to 10.
- ② Counts by twos up to 20.
- ③ Counts objects in pairs to find how many.
- ④ Counts by twos starting from any even number, such as 30.
- ★ Spots the one wrong number in a count by twos.
- ★★ Counts how many even numbers lie between two numbers, such as 13 and 27.

Comparing & ordering

Once learners can count, they learn to compare: which group has more, which number is bigger, and how to put numbers in order. This is the start of seeing numbers as having sizes and places.

I.2.01

More or fewer by matching

Telling which of two groups has more or fewer by pairing the items up one to one.

- 1 Pairs up two small rows and spots which one has some left over.
- 2 Pairs up two rows to decide which has more.
- 3 Pairs up two rows to decide which has fewer.
- 4 Says how many more one row has, by counting the leftovers.

I.2.02

Equal groups

Recognizing when two groups have the same number of things, even if they look different.

- 1 Checks whether two rows have the same number by giving each item a partner.
- 2 Works out how many to add to make two groups the same.
- 3 Finds groups with the same number even when arranged differently.
- 4 Counts to check, instead of assuming a longer row has more.

I.2.03

Number line 0–10

Using a number line from 0 to 10, where numbers sit in order along a line like steps.

- 1 Works out which number belongs at a marked spot on the line.
- 2 Finds where a given number sits on the line.
- 3 Makes jumps of 1 forward or back along the line.
- 4 Fills in several missing numbers on the line.

I.2.04

Compare numbers to 10

Deciding which of two numbers up to 10 is bigger or smaller.

- 1 Picks the bigger number, with pictures to help.
- 2 Picks the smaller number, with pictures to help.
- 3 Uses the number line: the number further right is bigger.
- 4 Compares two numbers with no pictures or line to help.

I.2.05

Number line 0–20

Using a number line from 0 to 20, including the teen numbers.

- 1 Places teen numbers, like 12, on the line.
- 2 Counts how many numbers lie between two numbers.
- 3 Finds the number a few steps before or after a given number.
- 4 Makes jumps forward and back along the line and says where they land.
- ★ Finds the number exactly halfway between two numbers on the line.
- ★★ Works out how many equal jumps of 2 it takes to get from one number to another.

I.2.06

Order numbers

Putting a set of numbers in order from smallest to biggest, or biggest to smallest.

- 1 Orders three numbers up to 10.
 - 2 Orders three numbers up to 20.
 - 3 Picks out the greatest and the least in a small list.
 - 4 Orders numbers up to 100, in either direction.
-

I.2.07

Ordinal numbers

Using position words like first, second and third to say where something is in a line.

- 1 Uses first, second and third to name places in a line.
 - 2 Uses position words all the way up to tenth.
 - 3 Works out how many are in front of or behind someone in a line.
 - 4 Understands last and next-to-last.
 - ★ Works out how many people are in front of someone, given their place in line.
 - ★★ Solves a puzzle: someone is seventh from the front and seventh from the back; how many in line?
-

I.2.08

Compare to 20

Comparing numbers up to 20 by thinking about tens and ones, or using a number line.

- 1 Compares teen numbers by looking at the ones, since both have one ten.
 - 2 Uses a number line to decide which number is smaller or bigger.
 - 3 Decides whether statements like "7 is more than 15" are true or false.
 - 4 Solves short stories asking who has more or fewer.
-

I.2.09

Compare to 100

Comparing numbers up to 100 by looking at the tens first, then the ones.

- 1 Compares two-digit numbers by their tens, such as 78 and 87.
 - 2 Compares numbers with the same tens by looking at the ones.
 - 3 Decides which ten a number is closer to, such as 88 being nearer 90.
 - 4 Solves short stories comparing numbers up to 100.
-

I.2.10

Symbols =, > and <

Using the signs = (the same as), > (greater than) and < (less than) to compare numbers.

- 1 Learns that = means "the same as."
 - 2 Chooses > or < between two small numbers; the open side faces the bigger one.
 - 3 Writes comparisons with these signs for numbers up to 20.
 - 4 The same with numbers up to 100.
-

I.2.11

Estimate a quantity

Making a sensible guess of how many there are without counting, then checking.

- 1 Glances at a group and says whether it is more or less than 10.
- 2 Picks a sensible estimate, like about 5 or about 20.
- 3 Counts to check someone's guess and sees how close it was.
- 4 Uses a group of 10 as a guide to improve a poor guess.

How many more, how many fewer

Working out the difference between two amounts: how many more one has, or how many fewer.

- 1 Lines up two cube trains up to 10 and counts the extra cubes.
- 2 The same with up to 20 cubes.
- 3 Counts the jumps between two numbers on a number line.
- 4 Solves short stories like "Mia has 11, Zoe has 3; how many fewer?"
- ★ Works backwards: knows one child has 2 more and how many they have, finds the other's amount.
- ★★ Solves a puzzle: after one child gives 5 to even things up, finds how big the gap was.

Place value

Place value is the idea that a digit's worth depends on where it sits: the 4 in 40 means four tens, not four. This branch shows how tens and hundreds are built from ones, which makes bigger numbers and later arithmetic make sense.

I.3.01 Ten ones make a ten

Understanding that ten single things can be grouped together and thought of as one ten.

- 1 Works out how many more sticks are needed to make a bundle of 10.
- 2 Works out how many more counters fill a box of ten spaces (a ten-frame).
- 3 Recognizes a group of 10 as "one ten."
- 4 Knows how many ones make a given number of tens, such as 2 tens being 20 ones.

I.3.02 Teen numbers are a ten and some ones

Seeing that each teen number is one ten plus some extra ones, like 16 being ten and six.

- 1 Says how many tens and ones make 11 to 15, using blocks (a rod is ten, a small cube is one).
- 2 The same for 16 to 19.
- 3 Picks the blocks that show a given teen number.
- 4 Reads the teen number shown by a set of blocks.

I.3.03 Tens and ones to 50

Breaking numbers up to 50 into tens and ones.

- 1 Knows the names for groups of tens: ten, twenty, thirty and so on.
- 2 Finds a total by counting the tens first, then counting on the ones.
- 3 Says how many tens and ones are needed to build a number.
- 4 Reads the number shown by a set of tens and ones blocks.

I.3.04 Tens and ones to 99

Understanding two-digit numbers up to 99: the left digit counts tens and the right digit counts ones.

- 1 Says whether a digit stands for tens or ones.
- 2 Gives what a digit is worth, such as the 8 in 28 being worth 8.
- 3 Builds a number from tens and ones, such as 3 tens and 1 one making 31.
- 4 Tells apart numbers with swapped digits, like 28 and 82.

I.3.05 Bundle and unbundle

Swapping between ones and tens: ten ones can become a ten, and a ten can be broken back into ones.

- 1 Bundles 10 loose ones into a ten and says the new count of tens and ones.
- 2 Breaks a ten into 10 ones and says the new count.
- 3 Recognizes a number shown two ways, like 27 as 1 ten and 17 ones.
- 4 Works out a missing number of tens in such an unusual arrangement.
- ★ Spots the one arrangement of tens and ones that does not make the given number.
- ★★ Finds every different way to show a number with tens and ones blocks.

I.3.06

Hundreds

Understanding that ten tens make a hundred, and building numbers in the hundreds.

- 1 Works out how many more tens are needed to make 100.
 - 2 Counts in hundreds up to 900.
 - 3 Says how many hundreds, tens and ones make a three-digit number.
 - 4 Reads the three-digit number shown by a set of blocks.
-

I.3.07

Three-digit place value

Knowing what each digit in a three-digit number stands for: hundreds, tens or ones.

- 1 Finds the hundreds digit, which is the one on the left.
 - 2 Finds the tens digit (middle) and the ones digit (right).
 - 3 Says what a digit is worth, such as the 8 in 819 being worth 800.
 - 4 Understands a zero digit means none of that place, as in 208.
 - ★ Builds a number from hundreds, tens and ones given in a mixed-up order.
 - ★★ Arranges three digit cards to make the biggest number, sometimes one that must be even.
-

I.3.08

Expanded form

Writing a number as the sum of what each digit is worth, like 47 as $40 + 7$ (expanded form).

- 1 Writes a two-digit number as tens plus ones, like $28 = 20 + 8$.
 - 2 Writes a three-digit number as hundreds plus tens plus ones.
 - 3 Turns a split-up number back into an ordinary one, like $200 + 80 + 8 = 288$.
 - 4 Handles numbers with a zero in them, like 280 or 305.
-

I.3.09

Number names

Reading and writing numbers as words, such as thirty-two or two hundred eighty-five.

- 1 Matches number words to numbers up to 100.
 - 2 Writes numbers up to 20 as words.
 - 3 Writes numbers up to 100 as words.
 - 4 Reads and writes numbers up to 1,000 as words.
-

I.3.10

10 more, 10 less

Finding the number that is 10 more or 10 less, and later 100 more or less, than a given number.

- 1 Finds 10 more or less by moving down or up a row on a chart of 1 to 100.
 - 2 Finds 10 more or less mentally by changing the tens digit.
 - 3 Finds 100 more or less by changing the hundreds digit.
 - 4 Finds 10 more or less when it crosses a hundred, like 108 down to 98.
-

I.3.11

The hundred chart

Getting to know the hundred chart, a grid of the numbers 1 to 100 arranged in rows of ten.

- 1 Finds a given number's spot on the chart.
- 2 Knows that numbers go up by 1 along a row and by 10 down a column.
- 3 Fills in missing numbers on the chart.
- 4 Spots patterns, such as which numbers share a row.

Compare using place value

Comparing numbers by looking at the biggest place first, such as the hundreds or tens.

- ① Compares two-digit numbers by their tens.
- ② Knows a three-digit number is bigger than any two-digit number.
- ③ Compares numbers with the same first digit by looking at the next digit.
- ④ Puts a list of numbers in order from least to greatest.
- ★ Chooses the missing digit that makes a comparison true, such as 'blank 8 is more than 59'.
- ★★ Works out how many different digits could fill the gap and keep a comparison true.

Addition & subtraction

This branch introduces adding and taking away, first with objects and stories, then with quick mental strategies. By the end, learners know their facts up to 20 by heart and can add and subtract two-digit numbers.

I.4.01

Put together

Adding by putting two groups together and finding how many there are altogether.

- 1 Joins two small groups and finds the total, up to 5.
- 2 The same with totals up to 10.
- 3 Counts every item in both groups once to find the total.
- 4 Matches a picture to a number sentence using the plus sign, like $3 + 1 = 4$.

I.4.02

Take away

Subtracting by taking some away from a group and finding how many are left.

- 1 Takes some away from a group of up to 5 and counts what remains.
- 2 The same with groups of up to 10.
- 3 Solves simple giving-away situations by counting what is left.
- 4 Matches a picture to a number sentence using the minus sign, like $4 - 3 = 1$.

I.4.03

Addition stories

Solving short stories where things are added, like more fish swimming in.

- 1 Solves an adding story with objects or pictures to count.
- 2 Picks the picture that shows what happens in the story.
- 3 Picks the number sentence, like $2 + 2 = 4$, that matches the story.
- 4 Solves adding stories with no pictures at all.

I.4.04

Subtraction stories

Solving short stories where things are taken away, like giving away stickers.

- 1 Solves a taking-away story with objects or pictures to count.
- 2 Picks the picture that shows what happens in the story.
- 3 Picks the number sentence, like $4 - 3 = 1$, that matches the story.
- 4 Solves taking-away stories with no pictures at all.

I.4.05

Part-part-whole

Seeing a number as a whole made of two parts, like 4 being 3 and 1.

- 1 Picks two parts that together make a given number.
- 2 Puts two parts together to find the whole.
- 3 Knows the whole and one part, and finds the hidden part.
- 4 Uses a diagram of a whole joined to its two parts to find a missing number.

I.4.06

Number bonds to 5

Knowing all the pairs of numbers that add up to 3, 4 and 5 (called number bonds).

- 1 Picks pairs that make 3, like 2 and 1.
 - 2 Picks pairs that make 4.
 - 3 Picks pairs that make 5.
 - 4 Lists every pair that makes 5, from 0 and 5 up to 5 and 0.
-

I.4.07

Number bonds to 10

Knowing the pairs of numbers that add up to 6, 7, 8, 9 and 10, especially the pairs that make 10.

- 1 Picks pairs that make 6.
 - 2 Finds the missing partner to make 7 or 8, like 5 and what makes 7.
 - 3 Finds the missing partner to make 9 or 10.
 - 4 Quickly gives the partner that makes 10 for any number, like 2 and 8.
 - ★ Finds the missing number when three numbers must add up to 10.
 - ★★ Counts every way to share a few apples between two baskets, following a rule.
-

I.4.08

Doubles

Adding a number to itself, like $6 + 6$, which are facts that are easy to remember.

- 1 Adds doubles up to $5 + 5$, often using matching dice.
 - 2 Adds doubles up to $10 + 10$.
 - 3 Recognizes doubles and knows them by heart.
 - 4 Solves short stories that use a double, like 3 pages and 3 more.
 - ★ Works backwards: given the result of doubling (and maybe taking away), finds the start number.
 - ★★ Solves a story where two people start equal, one gets extra, and the total is known.
-

I.4.09

Near doubles

Using a known double to solve a sum that is one away from it, like $4 + 3$ being one less than $4 + 4$.

- 1 Solves sums like $3 + 2$ as a double plus one.
 - 2 Solves sums like $2 + 3$ as a double minus one, using $3 + 3$.
 - 3 Picks which double helps with a given sum.
 - 4 Solves a mix of doubles and near doubles.
-

I.4.10

Make ten

Adding by first filling up to 10, then adding what is left, like $8 + 5$ as $8 + 2 + 3$.

- 1 Chooses how to split a number so the first part makes ten.
- 2 Adds 9 and a number by moving 1 over to make 10.
- 3 Adds 8 and a number by moving 2 over to make 10.
- 4 Uses the same trick with 7 and other numbers.

I.4.11

Count on to add

Adding a small number by starting at the other number and counting up.

- 1 Adds 0, 1 or 2, knowing that adding 0 changes nothing.
 - 2 Adds 3 by counting on three steps.
 - 3 Starts from the bigger number, since that means fewer steps.
 - 4 Counts on to add, with totals up to 20.
-

I.4.12

Add in any order

Understanding that numbers can be added in any order, and using that to add three numbers easily.

- 1 Knows $3 + 5$ gives the same as $5 + 3$, and spots when a swap is wrong.
 - 2 Adds three small numbers together.
 - 3 Looks for two numbers that make ten and adds them first.
 - 4 Looks for a double to add first, like $4 + 7 + 4$.
-

I.4.13

Subtract by counting back

Subtracting a small number by starting at the bigger number and counting backward.

- 1 Takes away 0, 1 or 2, knowing that taking away 0 changes nothing.
 - 2 Takes away 3 by counting back three steps.
 - 3 Counts back to subtract, starting from numbers up to 20.
 - 4 Recognizes when counting back is the easiest method: taking away a small number.
-

I.4.14

Subtract by counting up

Subtracting by counting up from the smaller number to the bigger one, like finding $12 - 9$ by counting 10, 11, 12.

- 1 Turns a take-away into an adding question, like $3 + ? = 4$.
 - 2 Counts up to find a difference, staying within 10.
 - 3 Counts up in two hops, first to 10 and then beyond, like 6 to 10 to 14.
 - 4 Decides whether counting back or counting up is quicker.
-

I.4.15

Subtract through ten

Subtracting from teen numbers by stepping down to 10 first, then taking away the rest.

- 1 Takes a number away from 10, using the pairs that make 10.
- 2 Takes away just enough to land on 10, like $14 - 4$.
- 3 Subtracts in two steps through 10, like $14 - 6$ as $14 - 4 - 2$.
- 4 Chooses the easiest method for a given subtraction.

I.4.16

Fact families

Seeing how three numbers, like 8, 1 and 9, make a family of two additions and two subtractions.

- 1 Picks the other addition in the family, like $1 + 8 = 9$ for $8 + 1 = 9$.
 - 2 Picks the subtractions in the same family, like $9 - 1 = 8$.
 - 3 Writes all four facts in a family.
 - 4 Uses one fact in a family to answer another.
 - ★ Spots the one fact that does not belong to a given family.
 - ★★ Given two members of a family, finds the smallest or biggest the third could be.
-

I.4.17

Missing addends

Finding the missing number in an addition, like $3 + ? = 5$.

- 1 Finds how many more are needed to reach a small number, with pictures.
 - 2 Finds the missing number in additions up to 10.
 - 3 The same with totals up to 20.
 - 4 Finds a missing number that comes first, like $? + 6 = 9$.
-

I.4.18

The equal sign as a balance

Understanding that the equal sign means both sides have the same value, like a balanced scale.

- 1 Decides whether statements like $3 = 5$ are true or false.
 - 2 Finds the number that balances a sum, like $? = 3 + 1$.
 - 3 Balances a sum on each side, like $4 + 1 = 3 + ?$.
 - 4 Decides whether number sentences such as $6 - 2 = 3$ are true or false.
 - ★ Balances a sum on each side with bigger numbers, like $6 + 3 = 7 + ?$.
 - ★★ Solves balance-scale picture puzzles to find how many cubes weigh the same as one shape.
-

I.4.19

Addition facts within 20

Knowing the addition facts with totals up to 20 quickly and accurately.

- 1 Adds pairs of numbers with totals up to 10.
 - 2 Adds pairs that go past 10, like $3 + 9$.
 - 3 Answers a mix of both kinds.
 - 4 Recalls these facts from memory, quickly.
-

I.4.20

Subtraction facts within 20

Knowing the subtraction facts starting from numbers up to 20 quickly and accurately.

- 1 Subtracts within 10.
- 2 Subtracts from teen numbers without going below 10, like $12 - 2$.
- 3 Subtracts from teen numbers going below 10, like $11 - 8$.
- 4 Recalls these facts from memory, quickly.

I.4.21

Add and subtract tens

Adding and subtracting whole tens, like $30 + 40$ or $45 - 30$, by thinking in tens.

- 1 Adds ten to some tens, like $10 + 40$.
 - 2 Adds two numbers of tens, like $20 + 70$.
 - 3 Adds tens to a two-digit number, like $23 + 10$, changing only the tens digit.
 - 4 Takes tens away from a two-digit number, like $45 - 30$.
-

I.4.22

Two-digit addition with models

Adding two-digit numbers, using blocks or a number line to see the tens and ones.

- 1 Adds tens to tens and ones to ones when the ones don't reach ten.
 - 2 Adds when the ones make a new ten, bundling them into the tens.
 - 3 Adds by making jumps of ten and then ones along a number line.
 - 4 Solves story problems that need two-digit addition.
 - ★ Finds a missing digit in a two-digit addition, like $19 + 6 \text{ blank} = 84$.
 - ★★ Arranges four digit cards into two numbers to make the smallest possible total.
-

I.4.23

Two-digit subtraction with models

Subtracting two-digit numbers, using blocks or a number line to see the tens and ones.

- 1 Takes away tens from tens and ones from ones when there are enough ones.
- 2 Breaks a ten into ones when there are not enough ones to take away.
- 3 Subtracts by jumping back in tens and then ones along a number line.
- 4 Solves story problems that need two-digit subtraction.
- ★ Works backwards: knows what was taken away and what is left, finds the starting amount.
- ★★ Arranges four digit cards into two numbers to make the smallest possible difference.

Patterns & logic

This branch trains careful thinking: spotting and continuing patterns, sorting things by what they have in common, and reasoning from clues. These habits of noticing and explaining sit underneath all later math.

I.5.01

Copy a pattern

Copying a repeating pattern of colors or shapes exactly.

- 1 Copies a pattern of two colors taking turns, like purple, blue, purple, blue.
- 2 Copies a pattern of two shapes taking turns.
- 3 Copies a pattern where one item comes once and the next twice, like red, blue, blue.
- 4 Copies a pattern of three different items, like red, yellow, blue.

I.5.02

Extend a repeating pattern

Working out what comes next in a repeating pattern.

- 1 Says what comes next when two items take turns.
- 2 Says what comes next in a pattern like square, square, hexagon.
- 3 Says what comes next in a pattern of three different items.
- 4 Picks out the part of a pattern that keeps repeating.
- ★ Counts how many of one item appear in the first few places of a pattern.
- ★★ Works out which item comes far ahead, such as the 17th, without drawing them all.

I.5.03

Patterns and rules

Understanding that a pattern follows a rule, and using that rule to check and compare patterns.

- 1 Tells a real pattern from a random row, using two kinds of item.
- 2 The same using three kinds of item.
- 3 Finds the one spot where a pattern has a mistake.
- 4 Matches two patterns that follow the same rule with different items.

I.5.04

Growing patterns

Understanding patterns that grow each step, like towers that get one block taller each time.

- 1 Says how tall the next tower is when each grows by 1.
- 2 Says how tall the next tower is when each grows by 2.
- 3 Predicts a tower several steps ahead, not just the next one.
- 4 Describes the rule, such as "add 1 each time."

I.5.05

Number patterns

Finding the rule in a list of numbers that goes up by the same amount each time.

- 1 Continues a list that goes up by 1.
 - 2 Continues a list that goes up by 2.
 - 3 Continues a list that goes up by 5 or by 10.
 - 4 Finds a missing number in the middle of the list.
 - ★ Finds the rule and fills in two missing numbers, in lists going up or down.
 - ★★ Spots a hidden rule, like adding 2 then 3 in turn, or adding one more each time.
-

I.5.06

Sort by one attribute

Sorting things into groups by one feature, such as color, shape or size.

- 1 Puts a new item in the right group when things are sorted by color.
 - 2 The same when things are sorted by shape.
 - 3 The same when things are sorted by size.
 - 4 Looks at sorted groups and says what the sorting rule was.
-

I.5.07

Sort by two attributes

Sorting things by two features at once, such as items that are both purple and square.

- 1 Finds items that match two features, like purple squares.
 - 2 Places a shape in two overlapping circles, such as "red" and "stars."
 - 3 Spots the one item that doesn't belong with the others, and why.
 - 4 Works out a two-part sorting rule from examples.
-

I.5.08

Odd and even

Telling odd numbers from even numbers: even numbers split into pairs with none left over.

- 1 Groups objects into pairs and counts the pairs.
 - 2 Knows that one left over after pairing means the number is odd.
 - 3 Lists the even numbers up to 20 by counting in twos.
 - 4 Says whether any number is odd or even by looking at its last digit.
-

I.5.09

Same and different

Looking closely to see what is the same and what is different between things.

- 1 Spots and counts the differences between two pictures.
 - 2 Finds the item that exactly matches in color, shape and size.
 - 3 Says which feature is different between two shapes.
 - 4 Chooses the sentence that correctly describes what two shapes share.
-

I.5.10

True or false

Deciding whether a statement is true or false, and saying why.

- 1 Checks statements about a picture, like "there are 3 purple shapes."
- 2 Checks statements about numbers, like $8 + 6 = 15$.
- 3 Corrects a false statement by finding the right answer.
- 4 Chooses the reason that shows why a statement is false.

I.5.11

Always, sometimes, never

Deciding whether a statement is always true, sometimes true or never true, and backing it up with examples.

- 1 Judges statements about shapes, like "a rectangle has 5 sides."
 - 2 Judges statements about numbers.
 - 3 Finds an example that fits a description.
 - 4 Finds one example that breaks a claim (a counterexample).
-

I.5.12

If-then

Following "if this, then that" rules and working out what must happen.

- 1 Follows everyday if-then rules, like "if the light is red, cars stop."
 - 2 Uses if-then rules about numbers to answer yes or no.
 - 3 Follows a two-way rule, such as "if less than 13, add 5; if not, take 3."
 - 4 Tests a rule and finds the example that breaks it.
-

I.5.13

Logic grid puzzles

Solving small logic puzzles, using a grid to track clues about who has what.

- 1 Solves a two-person, two-item puzzle from one clue.
 - 2 Solves a three-person, three-item puzzle.
 - 3 Combines several clues, needing all of them to reach the answer.
 - 4 Chooses which conclusion must be true from the clues.
-

I.5.14

Order by clues

Putting people or things in order using clues like "taller than" or "first in line."

- 1 Uses a taller-or-shorter clue to compare two people.
 - 2 Uses first-and-last clues to find who is in the middle.
 - 3 Combines two clues to order three people.
 - 4 Orders people using number clues, like "4 more than" or "2 fewer than."
 - ★ Uses a 'not' clue, like 'Ann was not first', to order three runners.
 - ★★ Orders four runners from several clues, including 'right after'.
-

I.5.15

Guess my number

Finding a mystery number from clues, and learning which clues are useful.

- 1 Finds the number from bigger-than and smaller-than clues.
- 2 Uses an odd-or-even clue to narrow it down.
- 3 Combines two clues, like "between 70 and 80" and "ends in 8."
- 4 Chooses a set of clues that points to just one number.
- ★ Finds the number using three clues together, such as range, odd or even, and digits.
- ★★ Solves a riddle about a two-digit number's digits, such as their sum and difference.

Shape & space

This branch is about shapes and space: naming flat and solid shapes, noticing their features, fitting shapes together, and describing where things are. It also brings the first taste of fractions through halves and quarters.

I.6.01

Name flat shapes

Naming common flat shapes such as circles, squares, rectangles and triangles.

- 1 Recognizes a circle and tells it apart from similar shapes like an oval.
- 2 Tells a square (four equal sides) from a rectangle.
- 3 Recognizes a triangle.
- 4 Names less common shapes: hexagon (6 sides), rhombus (a slanted square) and trapezoid.

I.6.02

Shape features

Describing flat shapes by their features: how many sides and corners, and whether the sides are straight or curved.

- 1 Counts the sides of a shape.
- 2 Counts the corners, where two sides meet.
- 3 Says whether a shape's sides are straight, curved or both.
- 4 Counts both sides and corners, noticing the numbers match.
- ★ Adds up the corners of several triangles and squares.
- ★★ Solves a puzzle: from the number of shapes and total sides, finds how many are squares.

I.6.03

Shapes in any position

Recognizing a shape whichever way it is turned or stretched, by focusing on what makes it that shape.

- 1 Decides whether a tilted four-sided shape really is a square.
- 2 Recognizes tall, thin or turned triangles as triangles.
- 3 Rejects look-alikes with a gap or a curved side.
- 4 Explains what makes a shape a triangle, whatever its color, size or direction.

I.6.04

Name solid shapes

Naming common solid (3D) shapes such as cubes, spheres, cones and cylinders.

- 1 Recognizes a cube, shaped like a dice.
- 2 Recognizes a sphere, round all over like a ball.
- 3 Recognizes a cone and a cylinder (a can shape).
- 4 Recognizes a box shape, called a rectangular prism.

I.6.05

Solid shape features

Describing solid shapes: how they move and what flat faces they have.

- 1 Says whether a solid can roll, stack or slide.
 - 2 Counts the flat faces (surfaces) on a solid, including hidden ones.
 - 3 Names the flat shape of a face, like the square face of a cube.
 - 4 Matches everyday objects to solid shapes, like an ice cube to a cube.
-

I.6.06

Put shapes together

Combining smaller shapes to make bigger ones, like two triangles making a square.

- 1 Says what shape two triangles make when put together.
 - 2 Works out how many small blocks cover a bigger shape.
 - 3 Chooses blocks that fill an outline exactly, with no gaps.
 - 4 Finds more than one set of blocks that makes the same shape.
-

I.6.07

Take shapes apart

Breaking a shape into smaller shapes, and seeing the smaller shapes hidden inside a picture.

- 1 Says what shapes you get when a square is cut along a line.
 - 2 Finds which smaller shapes make up a bigger one.
 - 3 Counts the small triangles inside a picture.
 - 4 Works out which shape the pieces make when pushed back together.
 - ★ Counts every rectangle in a picture, big and small, including ones made of smaller pieces.
 - ★★ Counts every triangle, big and small, in a triangle crossed by lines.
-

I.6.08

Solids from blocks

Building and describing structures made from cubes and other solid blocks.

- 1 Counts the blocks in a simple tower.
 - 2 Picks the building that matches a model exactly.
 - 3 Works out which blocks were used to build a pictured tower.
 - 4 Counts all the cubes in a larger building, column by column.
-

I.6.09

Position words

Using position words like above, below, between, behind and left to describe where things are.

- 1 Uses above and below.
- 2 Uses beside and between.
- 3 Uses in front of and behind.
- 4 Uses left and right, as seen from where the learner is looking.

I.6.10

Directions on a grid

Following and giving directions to move a character around a grid of squares.

- 1 Counts steps forward or back to reach a target.
- 2 Turns left or right from the character's own point of view.
- 3 Finds the shortest route to a target, going around obstacles.
- 4 Chooses the directions, like "4 right, then 1 up," that reach the target.
- ★ Works out the moves that take the robot back to where it started.
- ★★ Follows a long list of moves and works out how far right of the start the robot ends.

I.6.11

Halves

Understanding halves: two equal parts of a shape or a group.

- 1 Decides whether two parts are equal in size.
- 2 Picks out the shape that is cut into halves.
- 3 Finds half of a small group by sharing it into two equal groups.
- 4 Spots shapes cut into two parts that are not halves because they are unequal.

I.6.12

Quarters

Understanding quarters: four equal parts, each one half of a half.

- 1 Names the shaded part of a shape as a half or a quarter.
- 2 Picks out the shape cut into four equal parts.
- 3 Finds a quarter of a small group by sharing it into four equal groups.
- 4 Knows a half of a shape is bigger than a quarter of the same shape.
- ★ Works backwards from a quarter to find a half or the whole amount.
- ★★ Solves stories with a half of a half, working back to the starting amount.

I.6.13

Symmetry

Understanding symmetry: a shape is symmetrical if it can be folded so the two halves match exactly.

- 1 Decides whether two sides match when folded along a line.
- 2 Picks the fold line (line of symmetry) that splits a shape into matching halves.
- 3 Completes a picture by drawing its mirror-image half.
- 4 Finds symmetry in everyday things like letters and numbers.

I.6.14

Shape riddles

Solving riddles that describe a shape by its features, like "I have 3 corners."

- 1 Guesses a shape from one clue about its features.
- 2 Uses two clues together to pin down the shape.
- 3 Chooses which riddle describes a given shape.
- 4 Checks a set of shapes against a riddle and counts how many fit.

Measure & data

This branch connects math to everyday life: comparing lengths, weights and amounts, telling time, reading a calendar, counting money, and reading simple charts. It shows learners that numbers are useful for describing the world.

I.7.01

Compare lengths

Comparing how long things are, to say which is longer or shorter.

- 1 Says which of two things is longer or shorter when their ends line up.
- 2 Compares lengths whose ends don't line up, by counting grid spaces.
- 3 Puts three things in order from shortest to longest.
- 4 Compares two things by checking each against a third, like a piece of string.

I.7.02

Measure with units

Measuring length by laying small objects, like paper clips, end to end and counting them.

- 1 Knows the objects must touch with no gaps or overlaps.
- 2 Measures something by counting paper clips laid end to end.
- 3 Counts longer rows of units carefully, counting on from 10.
- 4 Understands that bigger units give a smaller count for the same length.

I.7.03

Measure with a ruler

Using a ruler to measure length in centimeters or inches.

- 1 Reads a length from a ruler when the object starts at zero.
- 2 Finds the length when the object doesn't start at zero.
- 3 Gives a length to the nearest whole centimeter.
- 4 Works out where to stop to draw a line of a given length.

I.7.04

Length stories

Solving everyday problems about length, and making sensible guesses of how long things are.

- 1 Adds two lengths together, like a ribbon made longer.
- 2 Works out how much longer one thing is than another.
- 3 Picks a sensible estimate for a real object, like a fork being about 18 cm.
- 4 Measures to check a guess and says how far off it was.
- ★ Works backwards: knows the whole ribbon and one piece, finds the other piece.
- ★★ Works out every length two sticks can measure, end to end or by their difference.

I.7.05

Compare weight

Comparing how heavy things are, using a balance scale or small cubes as weights.

- 1 Says which of two familiar things is heavier or lighter.
- 2 Reads a balance scale: the heavier side goes down.
- 3 Compares weights measured in cubes.
- 4 Puts three things in order from heaviest to lightest.

I.7.06

Compare capacity

Comparing how much different containers hold (their capacity), using cups of water.

- 1 Says which of two containers holds more.
 - 2 Counts how many cups it took to fill a container.
 - 3 Compares two containers by how many cups each holds.
 - 4 Puts three containers in order from holding least to most.
-

I.7.07

Time to the hour

Telling time to the hour on a clock with hands and on a digital clock.

- 1 Knows the short hand is the hour hand.
 - 2 Reads o'clock times from a clock with hands.
 - 3 Matches a digital time like 10:00 to a clock with hands.
 - 4 Picks the clock that shows a given o'clock time.
-

I.7.08

Time to the half hour

Telling time to the half hour, such as half past 2 or 2:30.

- 1 Reads half past times when the long hand points to 6.
 - 2 Works out where the long hand should point from where the short hand is.
 - 3 Matches half past times to digital times, like 2:30.
 - 4 Picks the clock that shows a given half past time.
-

I.7.09

Time to five minutes

Telling time to five minutes, and knowing morning (a.m.) from afternoon or evening (p.m.).

- 1 Counts by fives around the clock to find the minutes past the hour.
 - 2 Reads any time to five minutes, like 2:45.
 - 3 Reads quarter past and quarter to times.
 - 4 Decides whether a time is a.m. (midnight to noon) or p.m.
 - ★ Works out how many minutes pass between a start time and an end time.
 - ★★ Finds the real time from a clock that runs a few minutes fast or slow.
-

I.7.10

Days, weeks and the calendar

Knowing the days of the week and months of the year, and reading a calendar.

- 1 Knows the order of the days and counts days ahead.
- 2 Knows the months and facts like how many days are in February.
- 3 Finds information on a calendar page.
- 4 Works out dates a few days before or after a given date.
- ★ Counts on or back several days, as in '5 days ago it was Sunday; what day is today?'
- ★★ Uses the fact that days repeat every 7 to find the weekday of a later date.

I.7.11

Coins

Recognizing US coins and counting up their value in cents.

- 1 Names the penny, nickel, dime and quarter.
- 2 Knows each coin's value, noticing that size doesn't match value.
- 3 Counts the total of several coins of the same kind.
- 4 Counts a mix of coins, starting with the biggest.

I.7.12

Make amounts and change

Choosing coins to pay an amount and working out the change.

- 1 Picks coins that add up to a given amount.
- 2 Checks whether two different sets of coins make the same amount.
- 3 Works out change from 10 cents by counting up.
- 4 Works out change from a dollar, which is 100 cents.
- ★ Finds the fewest coins that make an amount.
- ★★ Counts every different way to make an amount using 1, 5 and 10 cent coins.

I.7.13

Tally charts

Recording and reading counts with tally marks, where every fifth mark crosses a group of four.

- 1 Counts a small set of tally marks.
- 2 Reads and makes tallies grouped in fives.
- 3 Turns a set of votes into a tally chart.
- 4 Answers questions using a tally chart, such as the total number of votes.

I.7.14

Picture graphs

Reading and making picture graphs, where each small picture stands for one vote or item.

- 1 Reads how many chose something by counting pictures in a row.
- 2 Picks the picture graph that matches a tally chart.
- 3 Compares rows to find the most or fewest chosen.
- 4 Works out how many more or fewer one row has than another.

I.7.15

Bar graphs

Reading and making bar graphs, where the height of each bar shows how many.

- 1 Reads the value of a bar from the scale beside it.
- 2 Picks the bar graph that matches a tally chart.
- 3 Compares bars to find how many more one has than another.
- 4 Adds bars together to find a total.

I.7.16

Ask a question with data

Asking a question, collecting answers, and showing them on a chart to learn something.

- 1 Chooses a good survey question, one that gets different answers from different people.
- 2 Sorts a list of answers into a tally chart.
- 3 Picks which graph answers a given question.
- 4 Chooses a true statement about what a graph shows.



Era II • Operate

About ages 8–11 • grades 3 to 5

Era II builds the four operations into strong tools. Learners work with numbers into the millions, master multiplication and division, meet factors and multiples, and learn to handle fractions and decimals with confidence. The era closes with first expressions (using letters for numbers), measurement, area and volume, and reading data.

II.1	Place value to millions	12	II.6	Fractions II	14
II.2	Multiplication	15	II.7	Decimals	14
II.3	Division	15	II.8	Expressions	9
II.4	Factors & multiples	12	II.9	Measurement & geometry	20
II.5	Fractions I	15	II.10	Data	9

Place value to millions

This branch grows whole numbers up to the millions and teaches the written methods for adding and subtracting them. Knowing what each digit is worth makes rounding, estimating and column arithmetic make sense rather than feel like magic.

II.1.01

Thousands

Understanding numbers in the thousands, where each digit's position tells you whether it counts thousands, hundreds, tens or ones.

- 1 Sees that ten hundreds make one thousand.
- 2 Reads four-digit numbers shown on a place value chart.
- 3 Writes a four-digit number from its parts, such as 9 thousands, 8 hundreds and 6 ones.
- 4 Breaks a number into the value of each digit, as in $9,806 = 9,000 + 800 + 6$.

II.1.02

Numbers to a million

Reading, writing and understanding numbers all the way up to a million.

- 1 Tells what a digit is worth in the ten thousands place, such as the 8 in 83,256.
- 2 Does the same in the hundred thousands place.
- 3 Uses the comma to read big numbers, seeing 832,563 as 832 thousand and 563.
- 4 Matches a large number to how it is written in words.

II.1.03

Place value patterns

Seeing that each place is worth ten times the place to its right, and what that means for multiplying or dividing by 10.

- 1 Explains that a digit in one place is worth ten times the same digit one place to the right.
- 2 Multiplies by 10 by moving each digit one place to the left.
- 3 Divides by 10 by moving each digit one place to the right.
- 4 Understands zeros that hold an empty place in the middle of a number.

II.1.04

Compare and order large numbers

Deciding which of two or more large numbers is bigger, and putting them in order.

- 1 Compares four-digit numbers by looking at digits from the left.
- 2 Compares six-digit numbers the same way.
- 3 Puts a list of numbers in order from greatest to least.
- 4 Chooses the right sign: greater than, less than, or equals.

II.1.05

Round to tens and hundreds

Rounding a number to the nearest ten or hundred to get a simpler number that is close to it.

- 1 Rounds a number to the nearest ten.
- 2 Rounds a number to the nearest hundred.
- 3 Uses the rule that a number exactly halfway, like 795, rounds up.
- 4 Rounds by seeing which end is closer on a number line.

II.1.06

Round large numbers

Rounding big numbers to the nearest thousand, ten thousand or any place, and knowing when a rounded number is good enough.

- 1 Rounds a number to the nearest thousand.
 - 2 Rounds a number to the nearest ten thousand.
 - 3 Rounds to whatever place is asked, from tens to hundred thousands.
 - 4 Decides when a rounded number is good enough and when an exact one is needed.
 - ★ Works backwards: finds the smallest and largest whole numbers that round to a given number.
 - ★★ Arranges five digit cards into the largest number that rounds to a given thousand.
-

II.1.07

Estimate sums and differences

Estimating the answer to an addition or subtraction by rounding first, to get a quick sense of the answer.

- 1 Rounds each number, then adds to estimate a sum.
 - 2 Rounds each number, then subtracts to estimate a difference.
 - 3 Estimates by keeping only the first digit of each number and zeroing the rest.
 - 4 Uses an estimate to judge whether someone's answer is reasonable.
-

II.1.08

Column addition, no regrouping

Adding large numbers by writing them in columns, one on top of the other, when no carrying is needed.

- 1 Adds two three-digit numbers in columns.
 - 2 Picks the layout where ones sit under ones, tens under tens, and so on.
 - 3 Adds four-digit numbers in columns.
 - 4 Adds numbers with five or more digits.
-

II.1.09

Column addition with regrouping

Column addition when a column adds to 10 or more, so a digit is carried into the next column.

- 1 Carries from the ones column into the tens.
 - 2 Carries from the tens column into the hundreds.
 - 3 Carries in two different columns in one sum.
 - 4 Adds three numbers at once in columns, with carrying.
 - ★ Finds missing digits in a column addition, using what must have been carried.
 - ★★ Solves a puzzle: counts the numbers that, added to themselves with digits swapped, give a total.
-

II.1.10

Column subtraction, no regrouping

Subtracting large numbers in columns when each top digit is big enough, so no borrowing is needed.

- 1 Subtracts one three-digit number from another in columns.
- 2 Picks the layout where the places are lined up correctly.
- 3 Subtracts four-digit numbers in columns.
- 4 Checks a subtraction by adding the answer back on.

II.1.11

Column subtraction with regrouping

Column subtraction when a top digit is too small, so the learner borrows from the next column (called regrouping).

- 1 Borrows a ten to subtract in the ones column.
- 2 Borrows a hundred to subtract in the tens column.
- 3 Borrows across zeros, as in 300 minus 14.
- 4 Checks a subtraction with regrouping by adding back.
- ★ Finds missing digits in a column subtraction, using what must have been borrowed.
- ★★ Solves a puzzle: counts three-digit numbers that, minus their digits reversed, give a set answer.

II.1.12

Addition and subtraction problems

Using addition and subtraction to solve everyday word problems with large numbers.

- 1 Solves a word problem that takes one addition or subtraction.
- 2 Solves a problem that takes two steps, such as adding then subtracting.
- 3 Chooses which calculation matches a word problem.
- 4 Picks the best estimate to check an answer to a word problem.

Multiplication

This branch builds multiplication from equal groups, through the times tables, to multiplying large numbers on paper. Multiplication is a fast way to add equal groups, and it underpins division, fractions and area.

II.2.01

Equal groups

Seeing multiplication as a number of equal groups, such as 3 groups of 2.

- 1 Counts groups of 2 and finds how many there are in all.
- 2 Finds the total for groups of 5 and groups of 10.
- 3 Writes equal groups as a repeated addition, like $8 + 8$.
- 4 Rewrites a repeated addition as a multiplication, like 2 times 8.

II.2.02

Arrays

Using arrays (objects set out in neat rows and columns) to picture multiplication.

- 1 Counts the rows and columns in an array.
- 2 Writes the multiplication that an array shows.
- 3 Sees that turning an array on its side gives the same total, so 8 times 3 equals 3 times 8.
- 4 Plans an array to show a given multiplication fact.

II.2.03

Multiply by 0, 1, 2, 5 and 10

Learning the easiest times tables first: multiplying by 0, 1, 2, 5 and 10.

- 1 Knows that times 0 gives 0 and times 1 leaves a number unchanged.
- 2 Knows the 2 times table.
- 3 Knows the 5 times table.
- 4 Knows the 10 times table.

II.2.04

Facts for 3, 4 and 6

Learning the 3, 4 and 6 times tables.

- 1 Knows the 3 times table.
- 2 Finds 4 times a number by doubling it, then doubling again.
- 3 Knows the 6 times table.
- 4 Answers a mix of 3, 4 and 6 times table facts.

II.2.05

Facts for 7, 8 and 9

Learning the harder 7, 8 and 9 times tables.

- 1 Uses patterns in the 9 times table to find facts.
- 2 Knows the 8 times table.
- 3 Knows the 7 times table.
- 4 Answers a mix of 7, 8 and 9 times table facts.

II.2.06

All facts to 10×10

Bringing all the times tables up to 10 times 10 together until they are quick and reliable.

- 1 Knows the square facts, where a number is multiplied by itself, like 7 times 7.
 - 2 Works out a new fact from one already known, such as doubling 4 times 8 to get 8 times 8.
 - 3 Answers a mix of facts from across all the tables.
 - 4 Recalls any fact from memory, including finding a missing number.
-

II.2.07

The area model

Picturing multiplication as the area of a rectangle, which later explains how to multiply bigger numbers.

- 1 Counts the unit squares that cover a rectangle.
 - 2 Finds the area by multiplying length by width.
 - 3 Splits a rectangle into two parts and adds their areas.
 - 4 Splits a multiplication the same way, as in 3 times 18 = 3 times 10 plus 3 times 8.
-

II.2.08

Properties of multiplication

Learning the rules multiplication always follows, which make calculations easier.

- 1 Knows the order doesn't matter: 4 times 7 equals 7 times 4.
 - 2 Knows grouping doesn't matter when multiplying three numbers.
 - 3 Splits a number to multiply in parts, as in 3 times 18 = 3 times 10 plus 3 times 8.
 - 4 Uses these rules to make a calculation easy, like 2 times 16 times 5.
 - ★ Regroups a product like 25 times 83 times 4 to make it easy (25 times 4 is 100).
 - ★★ Spots a shared number in two products and combines them, as in 7 times 37 plus 3 times 37.
-

II.2.09

Multiply by multiples of 10

Multiplying by 10, 100, 1,000 and numbers like 40 or 300 using known facts and zeros.

- 1 Multiplies a number by 10.
 - 2 Multiplies a one-digit number by a multiple of ten, like 3 times 80.
 - 3 Multiplies two multiples of ten, like 30 times 80.
 - 4 Multiplies by 100 and by 1,000.
-

II.2.10

Two-digit by one-digit

Multiplying a two-digit number by a one-digit number, such as 28 times 8.

- 1 Uses a split rectangle (area model) to multiply tens and ones separately.
 - 2 Writes the tens and ones products separately, then adds them.
 - 3 Uses the standard written method with carrying.
 - 4 Estimates first by rounding, then finds the exact answer.
-

II.2.11

Three- and four-digit by one-digit

Multiplying a three- or four-digit number by a one-digit number.

- 1 Multiplies each place separately, then adds the parts.
- 2 Handles the carrying from one column to the next.
- 3 Uses the standard written method from start to finish.
- 4 Estimates to pick out the correct answer.

II.2.12

Two-digit by two-digit

Multiplying two two-digit numbers, such as 28 times 86.

- 1 Uses a rectangle split into four parts to find the product.
 - 2 Writes the four partial products, then adds them.
 - 3 Finishes the standard written method with two rows.
 - 4 Estimates by rounding both numbers, then finds the exact answer.
 - ★ Finds a missing digit in a two-digit multiplication.
 - ★★ Arranges four digit cards into two numbers that make the largest possible product.
-

II.2.13

Multi-digit multiplication

Multiplying bigger numbers, such as a three-digit number by a two-digit number.

- 1 Multiplies a three-digit number by a two-digit number.
 - 2 Does the same when zeros appear in the numbers.
 - 3 Works through the written method row by row with confidence.
 - 4 Picks out which answer is reasonable by estimating.
-

II.2.14

Multiplication problems

Using multiplication to solve everyday word problems.

- 1 Solves problems about equal groups, like vases holding the same number of flowers.
 - 2 Solves problems about rows and columns, like tiles on a wall.
 - 3 Solves problems saying one thing is some number of times as much as another.
 - 4 Solves problems that take more than one step.
 - ★ Works backwards: undoes a chain of steps to find the number someone started with.
 - ★★ Solves classic puzzles like 'chickens and cows: 26 heads, 78 legs; how many chickens?'
-

II.2.15

Multiplication patterns

Spotting and explaining patterns in multiplication.

- 1 Recognizes which numbers belong to a given times table.
- 2 Tells whether a product will be even or odd without multiplying.
- 3 Makes a product easier by halving one number and doubling the other.
- 4 Explains why a pattern happens, such as why doubling one number doubles the answer.
- ★ Finds the last digit of a big product using only the last digits.
- ★★ Finds the last digit of a number multiplied by itself many times, using a repeating cycle.

Division

This branch teaches division as sharing and grouping, then builds up to long division. Division undoes multiplication, and knowing how to handle leftovers makes it useful in real life.

II.3.01

Sharing equally

Dividing by sharing things out equally, so each group gets the same amount.

- 1 Shares objects equally between 2 groups.
 - 2 Shares objects equally among 3, 4 or 5 groups.
 - 3 Works out how many each gets in a sharing story, without a picture.
 - 4 Writes a sharing situation as a division, like 10 divided by 2.
-

II.3.02

Making groups

Dividing by making groups of a set size and counting how many groups there are.

- 1 Makes groups of a given size from a set of objects.
 - 2 Works out how many groups a story makes, like packs of 3 cards.
 - 3 Writes a grouping situation as a division.
 - 4 Tells whether a story is about sharing out or making groups.
-

II.3.03

Division as a missing factor

Seeing division as a multiplication with a missing number: 12 divided by 3 asks what times 3 makes 12.

- 1 Answers a division by thinking of the matching multiplication.
 - 2 Knows the fact family: the four related facts made from three numbers, like 2, 9 and 18.
 - 3 Uses known times table facts to answer divisions.
 - 4 Finds a missing number wherever it sits in a multiplication or division.
-

II.3.04

Division facts

Learning the division facts that go with the times tables.

- 1 Divides by 1, 2, 5 and 10.
 - 2 Divides by 3 and 4.
 - 3 Divides by 6, 7, 8 and 9.
 - 4 Recalls any division fact from memory.
-

II.3.05

Dividing with 0 and 1

Learning the special cases of dividing with 0 and 1, including why nothing can be divided by zero.

- 1 Knows that dividing by 1 leaves a number unchanged.
- 2 Knows that any number divided by itself is 1.
- 3 Knows that 0 divided by any number is 0.
- 4 Explains why dividing by 0 has no answer.

II.3.06

Remainders

Dealing with remainders, the amount left over when a number doesn't divide exactly.

- 1 Finds how many are left over after making equal groups.
 - 2 Writes a division with a remainder, like 26 divided by 3 is 8 R 2.
 - 3 Knows the remainder must always be smaller than the number being divided by.
 - 4 Works backwards from an answer and remainder to find the starting number.
 - ★ Finds the largest or smallest number that leaves a given remainder.
 - ★★ Finds a mystery number from two remainder clues, testing numbers in a range.
-

II.3.07

Interpreting remainders

Deciding what a remainder means in a real situation, since the right answer depends on the story.

- 1 Rounds up when everyone needs a place, like tables or buses.
 - 2 Drops the remainder when only full groups count, like complete teams.
 - 3 Shares the remainder as a fraction, like cutting ribbon into equal pieces.
 - 4 Chooses which of these makes sense for a given problem.
 - ★ Solves problems that need a step or two before dividing, like adding teachers to students.
 - ★★ Uses the remainder after dividing by 7 to find the weekday many days from now.
-

II.3.08

Divide multiples of 10

Dividing numbers that end in zeros, like 2,400 divided by 3, using known facts.

- 1 Divides a tens number by a single digit, like 240 divided by 3.
 - 2 Divides a hundreds or thousands number by a single digit.
 - 3 Divides by a multiple of ten, like 240 divided by 30.
 - 4 Uses one fact to complete a pattern, such as 24, 240 and 2,400 divided by 3.
-

II.3.09

Two-digit by one-digit division

Dividing a two-digit number by a one-digit number, such as 84 divided by 4.

- 1 Shares place value blocks (tens and ones) into equal groups.
 - 2 Takes away easy chunks, like 40 groups of 2, and adds up the chunks.
 - 3 Follows the steps of long division: divide, subtract, bring down.
 - 4 Does long division when there is a remainder.
-

II.3.10

Long division by one digit

Using long division to divide bigger numbers by a one-digit number.

- 1 Divides a three-digit number by a single digit.
- 2 The same with four-digit numbers.
- 3 Handles answers with a zero in the middle.
- 4 Checks a division by multiplying the answer back.
- ★ Finds missing digits in a long division, checking by multiplying back.
- ★★ Counts how many numbers in a range divide exactly by a given number.

II.3.11

Division by two digits

Dividing by a two-digit number, such as 1,377 divided by 17.

- 1 Picks a sensible estimate for the answer.
 - 2 Takes away large easy chunks of the divisor and adds them up.
 - 3 Uses long division with a two-digit divisor.
 - 4 Adjusts a first guess that turns out too small or too big.
-

II.3.12

Divisibility rules

Using quick tests to tell whether a number divides exactly by 2, 3, 4, 5, 6, 9 or 10.

- 1 Tests for 2, 5 and 10 by looking at the last digit.
 - 2 Tests for 3 and 9 by adding up the digits.
 - 3 Tests for 4 and 6.
 - 4 Applies the right test to decide about a given number.
 - ★ Finds the smallest missing digit that makes a number pass a divisibility test.
 - ★★ Counts numbers in a range that divide by one number but not by another.
-

II.3.13

Division problems

Using division to solve everyday word problems.

- 1 Solves problems about sharing equally.
 - 2 Solves problems about making groups of a set size.
 - 3 Solves problems with leftovers, giving full groups and the amount left.
 - 4 Solves problems that take more than one step.
-

II.3.14

Mixed operations

Solving word problems that could need any of the four operations: adding, subtracting, multiplying or dividing.

- 1 Chooses which operation a problem needs.
 - 2 Solves problems that take two steps with different operations.
 - 3 Picks out which answer is reasonable.
 - 4 Writes an equation, using a letter for the unknown amount, that fits a story.
-

II.3.15

Estimating quotients

Estimating the answer to a division to get a quick idea of its size.

- 1 Swaps in nearby numbers that divide easily, like 540 divided by 6.
- 2 Rounds both numbers to the nearest ten before dividing.
- 3 Tells which two numbers the answer lies between.
- 4 Uses an estimate to spot a wrong answer.

Factors & multiples

This branch looks at how whole numbers are built from other numbers: factors (numbers that divide in exactly) and multiples (the numbers in a times table). These ideas are the tools for simplifying and adding fractions later on.

II.4.01

Factor pairs

Finding factor pairs: two whole numbers that multiply to make a given number, like 6 and 7 for 42.

- 1 Finds a factor pair of a small number, such as 2 and 4 for 8.
- 2 Reads a factor pair from the rows and columns of an array.
- 3 Finds all the factor pairs of numbers up to 50, spotting any that are missing.
- 4 The same for numbers up to 100.

II.4.02

Multiples

Understanding multiples: the numbers you land on when counting in steps of a given number, like 3, 6, 9, 12.

- 1 Continues a list of multiples by counting in steps.
- 2 Decides whether a number is a multiple of another.
- 3 Predicts which squares on a hundred chart will be shaded for a given times table.
- 4 Sees that if 5 times 7 is 35, then 35 is a multiple of 5 and 5 is a factor of 35.

II.4.03

Primes and composites

Sorting numbers into primes, which have exactly two factors (1 and themselves), and composites, which have more.

- 1 Uses a list of factors to decide whether a number is prime or composite.
- 2 Recognizes the prime numbers up to 20.
- 3 Recognizes the prime numbers up to 100.
- 4 Explains why 1 is neither prime nor composite: it has only one factor.

II.4.04

The sieve of Eratosthenes

Finding all the primes up to 100 with an ancient method: crossing out the multiples of each prime on a number chart.

- 1 Crosses out the multiples of 2, keeping 2 itself.
- 2 Then crosses out the multiples of 3 and of 5.
- 3 Then crosses out the multiples of 7.
- 4 Understands that the numbers left over are exactly the primes.

II.4.05

Prime factorization

Breaking a number down into the primes that multiply to make it, like $12 = 2 \text{ times } 2 \text{ times } 3$.

- 1 Reads the prime factors from a factor tree, a diagram that splits a number into branches.
 - 2 Divides by primes, smallest first, until reaching 1.
 - 3 Writes a repeated prime with a small raised number (a power), so 3 times 3 times 3 is 3 cubed.
 - 4 Knows each number has only one prime breakdown, and spots one that isn't finished.
 - ★ Finds the smallest number to multiply by to get a square number, using the prime breakdown.
 - ★★ Counts how many factors a number has, straight from its prime breakdown.
-

II.4.06

Common factors

Finding the factors two numbers share, and the largest of these, called the greatest common factor (GCF).

- 1 Lists all the factors of each number.
 - 2 Picks out the factors the two lists have in common.
 - 3 Finds the greatest common factor of two numbers.
 - 4 Finds the greatest common factor using the primes the two numbers share.
-

II.4.07

Using the GCF

Putting the greatest common factor to work in fractions, packing problems and calculations.

- 1 Simplifies a fraction by dividing top and bottom by their greatest common factor.
 - 2 Splits two kinds of items into identical packs with none left over.
 - 3 Rewrites a sum by pulling out the common factor, as in $12 + 15 = 3 \text{ times } (4 + 5)$.
 - 4 Solves word problems, like cutting two ribbons into the longest equal pieces.
-

II.4.08

Common multiples

Finding the multiples two numbers share, and the smallest of these, called the least common multiple (LCM).

- 1 Continues the list of multiples for each number.
 - 2 Picks out the multiples the two lists have in common.
 - 3 Finds the least common multiple of two numbers.
 - 4 Finds the least common multiple using prime breakdowns.
-

II.4.09

Using the LCM

Putting the least common multiple to work in fractions and in problems about things lining up.

- 1 Finds the smallest shared bottom number for two fractions.
- 2 Works out when two repeating events next happen together, like two flashing lights.
- 3 Solves word problems, like the smallest class that splits evenly into teams of 7 or 9.
- 4 Uses the fact that the GCF times the LCM equals the two numbers multiplied together.
- ★ Finds the least common multiple of three numbers, as in three lights flashing together.
- ★★ Counts numbers in a range that are multiples of one number or another, without counting twice.

II.4.10

Square numbers

Understanding square numbers, made by multiplying a number by itself, and why they are called squares.

- 1 Sees a square number as dots arranged in a square.
- 2 Knows the square numbers up to 12 times 12.
- 3 Explains why square numbers have an odd number of factors: one factor pairs with itself.
- 4 Spots patterns, like the gaps between squares going up by odd numbers.

II.4.11

Even and odd rules

Knowing whether a sum or product will be even or odd without working it out, and why.

- 1 Predicts whether a sum is even or odd.
- 2 Predicts whether a product is even or odd.
- 3 Decides if a long sum or product is even or odd, for example by counting the odd numbers.
- 4 Solves puzzles, like whether six odd numbers can add up to 35.

II.4.12

Number puzzles with factors

Solving puzzles and games that use factors and multiples.

- 1 Solves mystery-number riddles from clues about factors.
- 2 Adds up a number's factors to test for perfect numbers, like $6 = 1 + 2 + 3$.
- 3 Plays a factor game, working out which factors a partner scores.
- 4 Explains a winning strategy, like why picking a large prime is a good first move.
- ★ Finds a mystery number from three or more clues, including how many factors it has.
- ★★ Finds two numbers from their sum and product, by checking factor pairs.

Fractions I

This branch introduces fractions: numbers that describe equal parts of a whole, like one half or three quarters. It focuses on what fractions mean, how to picture them and how to compare them, before any calculating.

II.5.01

Equal parts

Understanding that fractions start with cutting a whole into equal parts, and naming those parts.

- 1 Names equal parts as halves, thirds or fourths.
- 2 Names equal parts as sixths or eighths.
- 3 Tells whether a shape is cut into equal parts or not.
- 4 Names what fraction of a whole a shaded part is.

II.5.02

Unit fractions

Understanding unit fractions: one single part of a whole, like one half, one third or one tenth.

- 1 Recognizes one half, one third and one fourth in a picture.
- 2 Recognizes unit fractions with more parts, like one sixth.
- 3 Sees that cutting a whole into more parts makes each part smaller.
- 4 Compares unit fractions, knowing one third is bigger than one eleventh.

II.5.03

Fractions of a whole

Using a fraction to describe several equal parts of one whole, like three quarters of a shape.

- 1 Names the fraction of a shape that is shaded.
- 2 Knows the bottom number (denominator) counts the equal parts and the top (numerator) counts the ones chosen.
- 3 Writes a fraction from words, like two thirds.
- 4 Picks the picture that shows a given fraction.

II.5.04

Fractions of a set

Finding a fraction of a group of things, like two thirds of 18 cookies.

- 1 Finds one part of a group, such as one half of 10.
- 2 Finds several parts of a group, such as two thirds of 18.
- 3 Says what fraction of a group of dots is a certain color.
- 4 Solves word problems about a fraction of a group.
- ★ Works backwards: knows a fraction of the total, finds the whole amount.
- ★★ Works back through two giveaways, each a fraction of what's left, to find the start.

II.5.05

Fractions on a number line

Placing fractions on a number line, seeing each fraction as a point with its own place among the numbers.

- 1 Names the size of each equal step between 0 and 1.
- 2 Names a unit fraction marked on the line.
- 3 Finds where other fractions sit, like three quarters.
- 4 Places fractions greater than 1 on a longer number line.

II.5.06

Whole numbers as fractions

Seeing that whole numbers can be written as fractions, like two halves making one.

- 1 Knows how many equal parts make one whole, as in three thirds make 1.
- 2 Writes a whole number as a fraction over 1, as in 3 is three over one.
- 3 Writes a whole number as a fraction with any bottom number, as in 5 is ten halves.
- 4 Spots which fractions on a number line land on whole numbers.

II.5.07

Equivalent fractions with models

Using pictures to see that different fractions can be the same amount, like one half and five tenths.

- 1 Uses fraction strips of the same length to find equal fractions.
- 2 Uses two number lines to find fractions at the same spot.
- 3 Cuts a shaded square into smaller parts and counts to find an equal fraction.
- 4 Picks out which fractions are equal to one half and which are not.

II.5.08

Equivalent fractions by rule

Finding equal fractions with a rule: multiplying or dividing the top and bottom by the same number.

- 1 Multiplies top and bottom by the same number to make an equal fraction.
- 2 Divides top and bottom by the same number to make an equal fraction.
- 3 Writes a fraction in simplest form, with the smallest possible numbers.
- 4 Tests whether two fractions are equal.
- ★ Finds an equal fraction from a second clue, such as how much bigger the bottom is than the top.
- ★★ Finds what number to add to the top and bottom of a fraction to make it equal another.

II.5.09

Compare, same denominator

Comparing fractions with the same bottom number, where the parts are the same size.

- 1 Sees that more parts of the same size is more.
- 2 Uses the greater than, less than and equals signs to compare.
- 3 Compares by seeing which fraction is further along a number line.
- 4 Answers story problems, like who used more ribbon.

II.5.10

Compare, same numerator

Comparing fractions with the same top number, where the counts match but the part sizes differ.

- 1 Sees that one big part beats one small part.
- 2 Uses the comparison signs, such as two tenths is less than two eighths.
- 3 Explains why one fifth is more than one seventh.
- 4 Answers story problems, like who ate more pizza.

II.5.11

Compare any fractions

Comparing any two fractions, even when their tops and bottoms are both different.

- 1 Decides whether a fraction is less than, equal to or more than one half.
- 2 Rewrites both fractions with the same bottom number, then compares.
- 3 Uses a quick cross-multiplying check to compare.
- 4 Puts a list of fractions in order from least to greatest.
- ★ Finds the smallest whole number top that makes a fraction bigger than another.
- ★★ Counts how many fractions with a given bottom number fit between two fractions.

II.5.12

Mixed numbers

Understanding mixed numbers: a whole number plus a fraction, like one and three quarters.

- 1 Reads a mixed number aloud correctly.
- 2 Names a mixed number marked on a number line.
- 3 Picks the picture that shows a given mixed number.
- 4 Puts a whole number and a fraction together as a mixed number.

II.5.13

Improper fractions

Understanding improper fractions, where the top is bigger than the bottom, like 13 thirds, and switching them to mixed numbers.

- 1 Recognizes which fractions are greater than 1.
- 2 Turns an improper fraction into a mixed number.
- 3 Turns a mixed number into an improper fraction.
- 4 Compares an improper fraction with a mixed number.
- ★ Finds a missing bottom or top number so an improper fraction matches a mixed number.
- ★★ Counts every fraction in simplest form, greater than 1, whose top and bottom add to a set total.

II.5.14

Fractions as division

Seeing a fraction as a division: three quarters is the same as 3 divided by 4.

- 1 Writes a fraction as a division, and a division as a fraction.
- 2 Works out each person's share when pizzas are shared equally.
- 3 Gives a division answer as a mixed number, as in 9 divided by 2 is four and a half.
- 4 Solves sharing word problems with mixed-number answers.

Fraction sense

Building a feel for how big a fraction is without exact calculation.

- 1 Understands that two thirds of a big cake is more than two thirds of a small one.
- 2 Tells whether a fraction is close to 0, one half or 1.
- 3 Estimates a sum of fractions by rounding each to a nearby easy value.
- 4 Says which of two fractions is greater and explains why.

Fractions II

This branch teaches calculating with fractions: adding, subtracting, multiplying and simple dividing. The key idea is that fractions can only be added or subtracted when the parts are the same size.

II.6.01 Add, like denominators

Adding fractions that have the same bottom number (denominator), so the parts are the same size.

- 1 Adds fractions using a picture of a fraction bar.
- 2 Adds them with numbers alone, adding the tops and keeping the bottom.
- 3 Adds fractions whose total is more than 1, writing it as a mixed number.
- 4 Solves story problems, like how much pizza two people ate.

II.6.02 Subtract, like denominators

Subtracting fractions that have the same bottom number.

- 1 Subtracts fractions by crossing out parts in a picture.
- 2 Subtracts them with numbers alone.
- 3 Subtracts a fraction from one whole.
- 4 Solves story problems, like how much cake is left.

II.6.03 Break fractions apart

Splitting a fraction into smaller fractions that add up to it, as a tool for adding.

- 1 Writes three quarters as one quarter plus one quarter plus one quarter.
- 2 Finds several different ways to split the same fraction.
- 3 Splits an improper fraction into wholes plus a fraction.
- 4 Splits one fraction to make a whole first, which makes adding easier.
- ★ Finds the missing fraction with 1 on top that completes a sum, like one third plus one over what.
- ★★ Splits a fraction with 1 on top into two different such fractions, finding the largest bottom possible.

II.6.04 Mixed numbers, like denominators

Adding and subtracting mixed numbers (a whole number plus a fraction) with the same bottom number.

- 1 Adds the wholes and the fraction parts separately.
- 2 Swaps one whole for its fraction pieces, so five and a fifth becomes four and six fifths.
- 3 Subtracts mixed numbers when a whole must be broken up first.
- 4 Solves story problems, like total distance walked over two days.

II.6.05 Common denominators

Rewriting two fractions so they have the same bottom number (a common denominator), which is needed before adding them.

- 1 Finds the smallest common bottom number using least common multiples.
- 2 Uses the two bottom numbers multiplied together as a common bottom number.
- 3 Rewrites a fraction with a new, larger bottom number.
- 4 Chooses the smallest common bottom number from several options.

II.6.06

Add, unlike denominators

Adding fractions with different bottom numbers by first making the parts the same size.

- 1 Adds using fraction strips cut into matching pieces.
 - 2 Rewrites both fractions with a common bottom number, then adds.
 - 3 Gives the answer in simplest form.
 - 4 Adds three fractions with different bottom numbers.
 - ★ Finds the missing top number in a sum of fractions with different bottoms.
 - ★★ Adds a long chain of fractions using a trick where almost every piece cancels out.
-

II.6.07

Subtract, unlike denominators

Subtracting fractions with different bottom numbers.

- 1 Subtracts using fraction strips cut into matching pieces.
 - 2 Rewrites both fractions with a common bottom number, then subtracts.
 - 3 Gives the answer in simplest form.
 - 4 Solves story problems, like how much pizza is left.
-

II.6.08

Mixed numbers, unlike denominators

Adding and subtracting mixed numbers whose fraction parts have different bottom numbers.

- 1 Adds two such mixed numbers.
 - 2 Subtracts two such mixed numbers.
 - 3 Handles fraction parts that add to more than a whole, or need a whole broken up.
 - 4 Estimates the answer first by rounding to whole numbers.
-

II.6.09

Fraction $\times$ whole number

Multiplying a fraction by a whole number, like 3 lots of two fifths.

- 1 Writes it as a repeated addition of the fraction.
 - 2 Multiplies the top by the whole number, keeping the bottom.
 - 3 Shows it as equal jumps along a number line.
 - 4 Solves word problems, like the total length of several equal pieces of string.
-

II.6.10

Multiply fractions

Multiplying one fraction by another, which finds a fraction of a fraction.

- 1 Uses an overlapping-rectangle picture to find the answer.
- 2 Multiplies the tops together and the bottoms together.
- 3 Simplifies before multiplying to keep the numbers small.
- 4 Multiplies mixed numbers.
- ★ Finds the missing top number that makes a fraction product come out right.
- ★★ Multiplies a long chain of fractions where each top cancels the bottom next to it.

II.6.11

Multiplying as scaling

Understanding multiplication as resizing: multiplying by a fraction less than 1 makes a number smaller, and more than 1 makes it bigger.

- 1 Predicts that multiplying by a fraction less than 1 shrinks a number.
 - 2 Predicts that multiplying by a fraction more than 1 grows a number.
 - 3 Compares two such products without working them out.
 - 4 Judges and explains statements about whether a product grows or shrinks.
-

II.6.12

Unit fraction $\div$ whole number

Dividing a unit fraction (one part, like one half) by a whole number, like sharing half a liter among 5 cups.

- 1 Splits a shaded part of a bar into equal pieces and names each piece.
 - 2 Divides with numbers alone, as in one third divided by 5 is one fifteenth.
 - 3 Solves sharing story problems.
 - 4 Checks an answer by multiplying back.
-

II.6.13

Whole number $\div$ unit fraction

Dividing a whole number by a unit fraction, which asks how many of those parts fit into it.

- 1 Counts the fraction pieces in whole bars to find the answer.
 - 2 Works out how many sevenths fit into 3, and similar.
 - 3 Solves story problems, like how many children can have a fifth of a pizza each.
 - 4 Checks an answer by multiplying back.
-

II.6.14

Fraction problems

Solving word problems that involve calculating with fractions.

- 1 Chooses which operation a fraction problem needs.
- 2 Solves problems that take more than one step.
- 3 Solves problems that involve mixed numbers, like scaling up a recipe.
- 4 Judges whether an answer is reasonable by estimating.
- ★ Works backwards: from how much a fraction change took, finds the whole amount.
- ★★ Solves a puzzle: from a bottle's full and part-full weights, finds the empty bottle's weight.

Decimals

This branch introduces decimals, the numbers with a point like 2.75, which are another way to write tenths and hundredths. They connect fractions to money and measurement, and the learner learns to calculate with them.

II.7.01

Tenths

Understanding tenths written as decimals, where 0.1 means one tenth of a whole.

- 1 Sees that 0.1 is one tenth, so 0.2 is two tenths.
- 2 Names the decimal shown by a bar cut into ten parts.
- 3 Names decimals marked on a number line.
- 4 Matches a decimal like 2.8 to its words, two and eight tenths.

II.7.02

Hundredths

Understanding hundredths written as decimals, where 0.01 means one hundredth of a whole.

- 1 Writes a number of hundredths as a decimal, as in 81 hundredths is 0.81.
- 2 Names the decimal shown on a grid of 100 squares.
- 3 Matches a decimal like 1.83 to its words.
- 4 Knows that 0.8 (eight tenths) is more than 0.06 (six hundredths).

II.7.03

Decimal place value

Knowing what each digit after the decimal point is worth: tenths, then hundredths, then thousandths.

- 1 Tells the value of a digit in the tenths place.
- 2 Tells the value of a digit in the hundredths place.
- 3 Tells the value of a digit in the thousandths place.
- 4 Breaks a decimal into the value of each digit, as in $2.786 = 2 + 0.7 + 0.08 + 0.006$.

II.7.04

Fractions and decimals

Switching between fractions and decimals, since they are two ways of writing the same amount.

- 1 Writes tenths as fractions and decimals, as in 0.8 is eight tenths.
- 2 Writes hundredths as decimals, including fractions that can be scaled to hundredths.
- 3 Knows the common ones: one half is 0.5, one quarter 0.25, three quarters 0.75.
- 4 Writes a decimal as a fraction in simplest form.

II.7.05

Compare decimals

Deciding which of two or more decimals is bigger, and putting them in order.

- 1 Compares decimals with the same number of digits after the point.
- 2 Compares decimals of different lengths, knowing 0.5 is more than 0.45.
- 3 Names decimals on a finely marked number line.
- 4 Puts a list of decimals in order from least to greatest.
- ★ Counts which digits can fill a gap and keep a decimal comparison true.
- ★★ Counts every decimal with up to two places that lies between two given numbers.

II.7.06

Round decimals

Rounding decimals to a simpler nearby number, such as the nearest whole number or tenth.

- 1 Rounds a decimal to the nearest whole number.
 - 2 Rounds a decimal to the nearest tenth.
 - 3 Rounds a decimal to the nearest hundredth.
 - 4 Rounds to estimate an answer, as in 9.8 times 6 is about 60.
-

II.7.07

Add decimals

Adding decimals, keeping the decimal points lined up so tenths add to tenths.

- 1 Adds decimals with tenths.
 - 2 Adds decimals with hundredths in columns.
 - 3 Lines up the points correctly when adding a whole number to a decimal.
 - 4 Adds decimals with different numbers of digits after the point.
 - ★ Finds missing digits in a decimal addition, using what was carried.
 - ★★ Places four digits in two decimals to make a sum as close as possible to a target.
-

II.7.08

Subtract decimals

Subtracting decimals, keeping the decimal points lined up.

- 1 Subtracts decimals with tenths.
 - 2 Subtracts decimals with hundredths in columns, borrowing when needed.
 - 3 Lines up the points correctly when the numbers have different lengths.
 - 4 Subtracts a decimal from a whole number by filling in zeros, as in 17 minus 13.85.
-

II.7.09

Powers of ten and the point

Multiplying and dividing decimals by 10, 100 and 1,000, and understanding what really happens.

- 1 Multiplies a decimal by 10.
 - 2 Multiplies a decimal by 100 and by 1,000.
 - 3 Divides a decimal by 10.
 - 4 Explains that the digits move to new places while the point stays still.
-

II.7.10

Decimal $\times$ whole number

Multiplying a decimal by a whole number, like 8.3 times 8.

- 1 Uses pictures of bars to multiply, such as 2 lots of 0.8.
- 2 Uses the written method to multiply.
- 3 Multiplies as if there were no point, then puts the point back in the right place.
- 4 Estimates the answer by rounding first.

II.7.11

Multiply decimals

Multiplying one decimal by another, like 1.8 times 3.6.

- 1 Uses a split rectangle (area model) to multiply the parts.
 - 2 Counts the digits after the points to place the point in the answer.
 - 3 Uses the written method to multiply decimals.
 - 4 Estimates the answer by rounding first.
 - ★ Works backwards: finds the missing number in a decimal multiplication.
 - ★★ Uses one known product, like 15 times 23, to work out related decimal products quickly.
-

II.7.12

Decimal ÷ whole number

Dividing a decimal by a whole number, including money amounts.

- 1 Shares a decimal amount on a hundred grid into equal groups.
 - 2 Uses long division, keeping the point in line.
 - 3 Adds zeros after the point to keep dividing until nothing is left.
 - 4 Shares a sum of money equally.
-

II.7.13

Divide by a decimal

Dividing by a decimal, such as how many 0.8 m pieces fit in 16.8 m.

- 1 Multiplies both numbers by 10 or 100 to make the divisor a whole number.
 - 2 Uses the written method to divide by a decimal.
 - 3 Estimates the answer by rounding first.
 - 4 Solves word problems, like cutting a ribbon into equal pieces.
-

II.7.14

Money

Using decimals to work with money in dollars and cents.

- 1 Writes cents as part of a dollar, as in 83 cents is \$0.83.
- 2 Adds and subtracts money, such as working out change.
- 3 Multiplies a price by a number of items.
- 4 Works out what is left of a budget after several purchases.
- ★ Solves shopping problems that need several steps to find one item's price.
- ★★ Solves a puzzle: two items' total and difference in price are known; finds each price.

Expressions

This branch moves from single calculations to expressions: short strings of numbers and operations, like $32 + 8$ times 6. Learning the order to work them out, and spotting rules and missing numbers, prepares the ground for algebra.

II.8.01

Order of operations

Following the agreed order of operations, so everyone gets the same answer to a calculation with several steps.

- 1 Works left to right when there is only adding and subtracting.
- 2 Does multiplying and dividing before adding and subtracting.
- 3 Works out anything in brackets first.
- 4 Handles calculations that mix all of these rules.
- ★ Chooses where the brackets go so a calculation gives a target answer.
- ★★ Places plus, minus and times signs between numbers to make the largest possible value.

II.8.02

Write numerical expressions

Turning a description in words into a calculation written with numbers and symbols, and back again.

- 1 Picks the calculation that matches a description in words.
- 2 Writes a calculation using brackets to show what comes first, then finds its value.
- 3 Compares calculations without working them out, as in 6 times a sum is six times as big.
- 4 Picks the words that correctly describe a written calculation.

II.8.03

Evaluate expressions

Working out the value of a calculation with several steps.

- 1 Works out a calculation with two operations.
- 2 Works out a calculation with three operations.
- 3 Works out a calculation that includes brackets.
- 4 Finds the first wrong step in someone else's working.

II.8.04

Number patterns and rules

Finding the rule behind a number pattern and using it to continue the pattern.

- 1 Finds the rule of a pattern, such as add 10.
- 2 Uses the rule to find the next numbers.
- 3 Follows two patterns at once and compares them step by step.
- 4 Pairs up numbers from two patterns and predicts which pairs will appear.
- ★ Works out which position in a pattern a given number appears at.
- ★★ Adds all the numbers from 1 up to a big number by pairing the ends.

II.8.05

Input-output tables

Using input-output tables, where a rule turns each number going in into a number coming out.

- 1 Uses the rule to find a missing output.
 - 2 Finds the rule that fits every column of the table.
 - 3 Works backwards from an output to find the input.
 - 4 Works with two-step rules, like times 2 then add 8, including backwards.
-

II.8.06

Missing numbers

Finding the missing number that makes a calculation true.

- 1 Finds the missing number in an addition.
 - 2 Finds the missing number in a multiplication.
 - 3 Finds a missing number anywhere in a calculation, including subtraction.
 - 4 Finds the number in a box inside a longer calculation with brackets.
-

II.8.07

Equivalent expressions

Recognizing that two calculations written differently can have the same value.

- 1 Sees that 3 times 8 plus 3 times 3 is the same as 3 times 11.
 - 2 Regroups numbers when multiplying, as in 2 times 5 times 16 is 10 times 16.
 - 3 Decides whether two calculations are equal.
 - 4 Explains why moving brackets changes a subtraction but not an addition.
-

II.8.08

Problems to expressions

Writing a word problem as a single calculation, then solving it.

- 1 Picks the calculation for a one-step problem.
 - 2 Picks the calculation for a two-step problem.
 - 3 Picks a calculation that needs brackets, like adults and children buying tickets at one price.
 - 4 Writes one calculation for a problem and works it out.
-

II.8.09

Balance puzzles

Solving balance puzzles, where both sides of a scale weigh the same, as a first step toward equations.

- 1 Finds the weight of one box on a level pan balance.
- 2 Finds the weight of one box when several equal boxes are on one side.
- 3 Uses two balances together to find two different unknown weights.
- 4 Writes a balance as an equation, using a box for the unknown weight.
- ★ Swaps one shape for what it equals to solve two linked balance clues.
- ★★ Finds three unknown weights from three pair clues, by adding all the clues together.

Measurement & geometry

This branch applies arithmetic to the physical world: measuring length, weight, volume and time, and working with shapes and angles. It covers perimeter, area and volume, and how to describe and sort shapes.

II.9.01

Metric length

Measuring length in metric units (millimeters, centimeters, meters, kilometers) and converting between them.

- 1 Converts between millimeters, centimeters and meters.
- 2 Converts kilometers to meters.
- 3 Converts meters with decimals into centimeters, as in 8.8 m is 880 cm.
- 4 Chooses the best unit for measuring something.

II.9.02

Imperial length

Measuring length in US units (inches, feet, yards, miles) and converting between them.

- 1 Converts between inches, feet and yards.
- 2 Converts miles to feet.
- 3 Converts to a mix of units, as in 81 inches is 6 feet 9 inches.
- 4 Chooses the best unit for measuring something.

II.9.03

Mass

Measuring how heavy things are, in grams and kilograms or ounces and pounds.

- 1 Converts kilograms to grams.
- 2 Converts pounds to ounces.
- 3 Converts decimal amounts, as in 8.6 kg is 8,600 g.
- 4 Estimates the weight of everyday objects, like a cat.

II.9.04

Capacity

Measuring how much a container holds, in milliliters and liters or cups, pints, quarts and gallons.

- 1 Converts liters to milliliters.
- 2 Converts between cups, pints, quarts and gallons.
- 3 Converts decimal amounts, as in 8.8 liters is 8,800 milliliters.
- 4 Estimates how much everyday containers hold, like a glass of milk.

II.9.05

Time

Working with time: how long things last, converting between units, and reading timetables.

- 1 Works out how long something lasts from its start and end times.
- 2 Converts between hours and minutes.
- 3 Converts between other units of time, such as days and hours.
- 4 Reads a bus timetable to answer questions.

II.9.06

Perimeter

Finding the perimeter of a shape: the total distance around its edge.

- 1 Counts unit lengths around a shape on a grid.
 - 2 Adds up the lengths of all the sides.
 - 3 Uses a shortcut for rectangles: twice the length plus twice the width.
 - 4 Finds a missing side when the perimeter is known.
 - ★ Works backwards from a rectangle's perimeter and a clue about its sides to find its area.
 - ★★ Finds the perimeter of one strip after a square is cut into equal strips.
-

II.9.07

Area by counting

Finding area, the amount of flat space a shape covers, by counting squares.

- 1 Counts whole squares to find the area of a shape on a grid.
 - 2 Counts half squares too, pairing them up to make wholes.
 - 3 Compares the areas of two shapes.
 - 4 Sees that shapes that look different can have the same area.
-

II.9.08

Area of rectangles

Finding the area of rectangles by multiplying length by width, and using this for other shapes.

- 1 Multiplies length by width to find a rectangle's area.
 - 2 Finds a missing side when the area is known.
 - 3 Finds the area of an L-shape or similar by splitting it into rectangles.
 - 4 Solves word problems, like the area of a garden.
 - ★ Works backwards from the area and one side to find the missing side, then the perimeter.
 - ★★ Finds the area of a path running around a garden, by taking the garden from the whole.
-

II.9.09

Perimeter vs area

Understanding that perimeter (distance around) and area (space inside) are different, and knowing which one a job needs.

- 1 Sees that rectangles with the same perimeter can have different areas.
 - 2 Sees that rectangles with the same area can have different perimeters.
 - 3 Decides whether a real task needs perimeter or area.
 - 4 Finds the biggest pen that can be enclosed with a given length of fence.
 - ★ Finds a rectangle's sides when both its area and perimeter are given.
 - ★★ Counts every whole-number rectangle with a given perimeter whose area is above a target.
-

II.9.10

Volume by counting cubes

Finding volume, the amount of space a solid takes up, by counting unit cubes.

- 1 Counts the unit cubes in a solid.
- 2 Counts the cubes in one layer, then multiplies by the number of layers.
- 3 Compares which of two boxes is made of more cubes.
- 4 Picks out a box built from an exact number of cubes.

II.9.11

Volume of boxes

Finding the volume of a box by multiplying length, width and height.

- 1 Multiplies length times width times height.
 - 2 Uses the area of the base times the height, including finding a missing height.
 - 3 Finds the volume of a solid made of two boxes by adding them.
 - 4 Solves word problems, like the volume of a fish tank.
 - ★ Works out how many small cubes fit exactly inside a box.
 - ★★ Solves the painted-cube puzzle: how many small cubes have paint on exactly one or two faces.
-

II.9.12

Points, lines and rays

Learning the basic words of geometry: points, lines, segments and rays, and how lines can meet.

- 1 Tells apart a point, a line (endless both ways) and a segment (with two ends).
 - 2 Recognizes a ray: a line with one end that goes on forever the other way.
 - 3 Recognizes parallel lines, which never meet.
 - 4 Recognizes perpendicular lines, which meet at a square corner.
-

II.9.13

Angles

Understanding angles as an amount of turn, measured in degrees, and naming the different kinds.

- 1 Knows a full turn is 360 degrees.
 - 2 Sorts angles as right (a square corner), acute (smaller) or obtuse (bigger).
 - 3 Knows a straight angle is a half turn, and two make a full turn.
 - 4 Names an angle using three letters, with the corner in the middle.
-

II.9.14

Measure angles

Using a protractor, a half-circle measuring tool, to measure and draw angles.

- 1 Chooses which of the protractor's two scales to read.
 - 2 Measures an angle with a protractor.
 - 3 Draws an angle of a given size with a protractor.
 - 4 Estimates the size of an angle before measuring.
-

II.9.15

Add angles

Adding angles together and finding missing angles.

- 1 Adds two angles side by side to find the whole angle.
- 2 Finds a missing part of a known angle, such as a right angle.
- 3 Uses the fact that angles all the way around a point add to 360 degrees.
- 4 Solves word problems, like how far a door has swung open.
- ★ Splits a straight or right angle into two parts when one is several times the other.
- ★★ Finds the angle between a clock's hands at a given time, allowing for the hour hand's drift.

II.9.16

Classify triangles

Sorting triangles by their sides and by their angles.

- 1 Names triangles by their sides: all equal, two equal, or none equal.
 - 2 Names triangles by their angles: all acute, one right angle, or one obtuse angle.
 - 3 Names a triangle by both its sides and its angles.
 - 4 Decides whether a given kind of triangle can exist.
-

II.9.17

Classify quadrilaterals

Sorting four-sided shapes (quadrilaterals) and seeing how their families fit together.

- 1 Names squares, rectangles and rhombuses (four equal sides).
 - 2 Names parallelograms (two pairs of parallel sides) and trapezoids (exactly one pair).
 - 3 Places shapes in a family tree, seeing a square as a special rectangle.
 - 4 Judges statements like every square is a rhombus as true or false.
-

II.9.18

Lines of symmetry

Finding lines of symmetry: fold lines where one half of a shape matches the other exactly.

- 1 Picks which line is a line of symmetry.
 - 2 Counts how many lines of symmetry a shape has.
 - 3 Completes a shape on a grid so it has a given line of symmetry.
 - 4 Recognizes shapes that have no line of symmetry.
-

II.9.19

The first quadrant

Locating points on a grid using two numbers (coordinates): how far across, then how far up.

- 1 Knows the two axes and the starting corner, and reads points on an axis.
 - 2 Finds the point with given coordinates.
 - 3 Reads the coordinates of a point.
 - 4 Spots a pattern in a set of points and predicts the next one.
-

II.9.20

Measurement problems

Solving word problems that bring together measurement, perimeter, area and volume.

- 1 Solves problems that need converting units, like meters and centimeters.
 - 2 Solves problems about perimeter and area, like fencing a garden.
 - 3 Solves problems about volume, like how many liters fill a tank.
 - 4 Solves problems with several steps, like the cost of fencing with a gate.
-

Data

This branch is about data: collecting information, showing it in charts and tables, and reading what it says. It also teaches how to summarize data with a typical value and how to spot a graph that misleads.

II.10.01 Line plots

Using line plots, where each value is marked with an X above a number line.

- 1 Reads a line plot to answer simple questions.
 - 2 Makes a line plot from a list of values.
 - 3 Reads line plots whose values are fractions.
 - 4 Answers questions that need calculating, like the total of all the values.
-

II.10.02 Bar graphs with scales

Reading and making bar graphs whose scale counts in steps like 2, 5 or 10.

- 1 Reads a bar whose top falls between numbered lines on the scale.
 - 2 Picks the bar graph that matches a table.
 - 3 Compares bars to find how many more.
 - 4 Chooses a scale big enough for the data.
-

II.10.03 Pictographs with keys

Reading and making pictographs, where each picture stands for a set number of things, shown in a key.

- 1 Reads a pictograph where each picture stands for 2.
 - 2 Reads half pictures, which stand for half the key's amount.
 - 3 Builds a row of pictures to show a given number.
 - 4 Compares and combines rows to answer questions.
-

II.10.04 Tables and frequency

Reading and making frequency tables, which count how often each answer appears.

- 1 Reads a table to find the most or least common answer.
- 2 Counts a list of answers to fill in a table.
- 3 Uses totals to find missing entries.
- 4 Compares entries in a table to find how many more.
- ★ Fills in a two-way table, such as girls and boys who walk or not, to find a missing count.
- ★★ Solves problems about overlapping groups, like kids who like apples, bananas, both or neither.

II.10.05

The mean as a fair share

Understanding the mean (the average) as what each would get if everything were shared out equally.

- 1 Evens out towers of cubes so they are all the same height.
 - 2 Finds the mean by adding the values and dividing by how many there are.
 - 3 Works out how the mean changes when a value is added.
 - 4 Solves word problems about sharing totals fairly.
 - ★ Works backwards: knows the mean and all but one score, finds the missing score.
 - ★★ Works out which number was removed, from the mean before and after.
-

II.10.06

A typical value

Describing a set of data with one typical value: the most common value (mode), the middle value (median) or the mean.

- 1 Finds the mode, the value that appears most often.
 - 2 Finds the median by putting values in order and taking the middle one.
 - 3 Compares the mean and median of the same data.
 - 4 Decides which typical value best describes data with an unusual value in it.
 - ★ Finds the missing value that makes the mean and the median of a list equal.
 - ★★ Builds five numbers to fit a given mean, median and mode, making the biggest as small or large as possible.
-

II.10.07

Line graphs

Reading and making line graphs, which show how something changes over time.

- 1 Reads a value at a given time from a line graph.
 - 2 Checks plotted points and spots one placed wrongly.
 - 3 Describes the overall trend: rising, falling or staying level.
 - 4 Compares two lines on the same graph.
-

II.10.08

Collect and organize data

Running a simple survey: asking a question, recording answers, displaying them and drawing a conclusion.

- 1 Chooses a clear survey question with sensible answer choices.
 - 2 Counts tally marks, which are grouped in fives.
 - 3 Chooses the best kind of chart for the data.
 - 4 Picks the conclusion that the data actually support.
-

II.10.09

Read graphs critically

Reading graphs with a critical eye, noticing when a graph is misleading or can't answer a question.

- 1 Checks where the scale starts before reading a bar.
- 2 Explains how a scale that doesn't start at zero makes small differences look big.
- 3 Recognizes questions a graph cannot answer.
- 4 Picks the fairer, clearer version of a survey question.



Era III • Relate

About ages 11–14 • grades 6 to 8

Era III is about how quantities relate to each other. Negative numbers, ratios, rates and percentages come first, then powers and roots, then algebra: writing expressions and solving equations. It ends with first graphs of straight lines, basic non-proof geometry of angles, circles and solids, and basic statistics and chance.

III.1	Integers & rationals	15	III.6	Equations & inequalities	12
III.2	Ratios & rates	15	III.7	Lines & first functions	12
III.3	Percent	17	III.8	Basic Non-Proof Geometry	23
III.4	Exponents & roots	17	III.9	Statistics & probability	20
III.5	Expressions	15			

Integers & rationals

This branch introduces negative numbers (numbers below zero) and fractions and decimals that can be negative too. They are the everyday language of temperatures, debts, and depths, and every later topic in algebra depends on them.

III.1.01 Negative numbers in context

Understanding numbers below zero through real situations like cold weather, depths, and money owed.

- 1 Reads temperatures below zero and works out the new temperature after a drop.
- 2 Writes heights below sea level as negative numbers.
- 3 Writes a debt as a negative amount of money.
- 4 Finds the opposite of a number, the same distance from zero on the other side.

III.1.02 Integers on a number line

Placing positive and negative whole numbers (integers) on a number line and using it to compare them.

- 1 Names the whole number marked at a point on a number line, including below zero.
- 2 Finds a pair of opposites on a number line from the gap between them.
- 3 Compares two negative numbers, knowing that further left means smaller.
- 4 Puts a mixed list of positive and negative numbers in order from least to greatest.

III.1.03 Absolute value

Absolute value is a number's distance from zero, ignoring whether it is positive or negative.

- 1 Says how far a number is from zero on a number line.
- 2 Reads the absolute value symbol, two straight bars around a number, and works inside it first.
- 3 Compares the sizes of two numbers using their absolute values.
- 4 Uses absolute value in real settings, like which of two things is farther from sea level.
- ★ Finds both numbers a set distance from a point, or counts whole numbers within that distance.
- ★★ Finds the point that makes a total of distances to three numbers as small as possible.

III.1.04 Add integers

Adding positive and negative numbers, and understanding why the answer comes out the way it does.

- 1 Adds two numbers with the same sign by adding their sizes.
- 2 Adds a negative and a positive number by moving along a number line.
- 3 Uses plus and minus counters that cancel in pairs to add numbers with different signs.
- 4 Uses the rules directly to add numbers with different signs, without a picture.
- ★ Finds the missing number in a subtraction with negative answers, like 18 minus what makes negative 1.
- ★★ Adds a long list with alternating plus and minus signs by pairing terms up.

III.1.05

Subtract integers

Subtracting positive and negative numbers, including taking away a negative.

- 1 Rewrites a subtraction as adding the opposite, so taking away minus 15 means adding 15.
- 2 Subtracts by moving left along a number line.
- 3 Finds the distance between two points on a number line, even across zero.
- 4 Works through a string of additions and subtractions with mixed signs.

III.1.06

Multiply integers

Multiplying positive and negative numbers and knowing whether the answer is positive or negative.

- 1 Continues a multiplication pattern past zero to see why a positive times a negative is negative.
- 2 Uses the sign rules: same signs give a positive, different signs give a negative.
- 3 Multiplies three or more numbers, counting the negatives to decide the sign.
- 4 Uses multiplication with negatives in real settings, like a steady temperature fall.

III.1.07

Divide integers

Dividing positive and negative numbers, using the same sign rules as multiplication.

- 1 Divides using the sign rules, such as a negative divided by a negative is positive.
- 2 Uses a known multiplication fact to answer the matching division.
- 3 Reads a fraction bar as division and works out fractions that include negatives.
- 4 Works through mixed division and multiplication, left to right.

III.1.08

Order of operations with integers

Doing calculations in the right order (brackets, then multiplying and dividing, then adding and subtracting) with negative numbers.

- 1 Works out calculations with two operations, doing multiplying or dividing first.
- 2 Works out the part in brackets first, then the rest.
- 3 Handles longer calculations that mix brackets, operations, and negative numbers.
- 4 Spots the mistake in someone else's worked calculation and explains what went wrong.
- ★ Chooses where brackets go, with negative numbers, to reach a target answer.
- ★★ Chooses plus, minus or times signs between negative and positive numbers to make the smallest value.

III.1.09

Rational numbers

A rational number is any number that can be written as a fraction of two whole numbers, including negatives.

- 1 Shows that a whole number like minus 13 can be written as a fraction.
- 2 Places negative fractions correctly on a number line.
- 3 Rewrites a negative decimal as a fraction in simplest form.
- 4 Simplifies fractions with negative signs, such as a negative over a negative.

III.1.10

Add and subtract rationals

Adding and subtracting fractions and decimals that may be negative.

- 1 Adds and subtracts negative fractions and simplifies the answer.
 - 2 Adds and subtracts negative decimals.
 - 3 Combines a fraction and a decimal in one calculation.
 - 4 Solves real problems, like a temperature warming up from below zero.
-

III.1.11

Multiply and divide rationals

Multiplying and dividing fractions and decimals that may be negative.

- 1 Multiplies fractions where one is negative and simplifies the answer.
 - 2 Multiplies negative decimals.
 - 3 Finds a reciprocal (the number that multiplies to make 1), such as minus one third for minus 3.
 - 4 Finds a fraction of a negative fraction, such as four fifths of minus three fifths.
-

III.1.12

Terminating and repeating decimals

Turning fractions into decimals and noticing whether the decimal stops or repeats forever.

- 1 Divides the top of a fraction by the bottom to get a decimal.
 - 2 Matches decimals that end, like 0.8, with their fractions.
 - 3 Writes a repeating decimal, like 0.181818..., using a bar over the repeating part.
 - 4 Predicts from the bottom of a fraction whether its decimal will end or repeat.
-

III.1.13

Repeating decimals to fractions

Turning a repeating decimal back into an exact fraction.

- 1 Turns a single repeating digit, like 0.222..., into a fraction.
 - 2 Turns a repeating pair of digits, like 0.141414..., into a fraction.
 - 3 Handles decimals where the repeat starts later, like 0.1888....
 - 4 Understands that a repeating 9 equals the next number up, so 0.999... is exactly 1.
 - ★ Turns a decimal where the repeat starts one place later, like 0.7555..., into a fraction.
 - ★★ Adds two decimals with repeating pairs of digits and gives the answer as a simple fraction.
-

III.1.14

Compare and order rationals

Comparing and ordering fractions and decimals, both positive and negative.

- 1 Compares a fraction with a decimal to see which is larger.
 - 2 Compares negative fractions and decimals, including ones that are equal.
 - 3 Finds where a negative fraction sits on a number line.
 - 4 Orders a mixed list of positive and negative fractions and decimals.
-

III.1.15

All four quadrants

Using a full coordinate grid, with negative numbers on both axes, to locate points.

- 1 Names which of the four quarters of the grid a point lies in.
- 2 Finds a point on the grid from its pair of coordinates, including negatives.
- 3 Reflects a point across the axes, like a mirror, and gives where it lands.
- 4 Finds the distance between two points that sit on the same row or column.

Ratios & rates

This branch is about ratios (comparing two amounts, like 2 cups of flour for every 3 eggs) and rates (like miles per hour or price per item). They are the math behind recipes, maps, shopping deals, and speed.

III.2.01 Ratio language

Using a ratio to compare two amounts, such as the number of girls to boys in a group.

- 1 Writes a ratio by counting two kinds of objects in a picture.
 - 2 Writes a part-to-part ratio, like girls to boys, in simplest form.
 - 3 Writes one part as a fraction of the whole group.
 - 4 Recognizes the three ways to write a ratio: 9 to 7, 9 : 7, and as a fraction.
-

III.2.02 Equivalent ratios

Finding ratios that describe the same relationship, like 2 : 3 and 4 : 6.

- 1 Makes an equal ratio by multiplying both parts by the same number.
 - 2 Fills a gap in a table of equal ratios.
 - 3 Uses a bar split into parts (a tape diagram) to share a total in a ratio.
 - 4 Uses two matched number lines side by side to find a missing value.
-

III.2.03 Ratio tables

Using a ratio table, a table of matching amounts, to scale a relationship up or down.

- 1 Completes the next entry in a ratio table.
 - 2 Finds two missing values anywhere in a ratio table.
 - 3 Scales down to one unit, then up to any amount.
 - 4 Decides whether two tables show the same ratio.
-

III.2.04 Unit rates

Finding a unit rate, the amount for just one of something, like the price of one item.

- 1 Works out how many per one, such as pencils per pack.
 - 2 Works out the price of a single item.
 - 3 Works out speed as distance per hour.
 - 4 Compares two deals by finding the price per item in each.
-

III.2.05 Rates with fractions

Finding a unit rate when the amounts are fractions, like walking a third of a mile in two thirds of an hour.

- 1 Finds a speed per hour when both distance and time are fractions.
- 2 Finds how much of a job gets done per hour from fractional amounts.
- 3 Uses fractional rates in everyday settings, like a tap filling a jug.
- 4 Compares two fractional rates to see which is faster.

III.2.06

Ratios on a graph

Showing a ratio as points on a graph and seeing that they form a straight line through zero.

- 1 Spots which plotted point does not fit a given ratio.
 - 2 Picks the straight line through zero that shows a ratio.
 - 3 Reads missing values off a ratio's line.
 - 4 Compares two lines on one graph, such as which walker is faster.
-

III.2.07

Proportions

A proportion is a statement that two ratios are equal, used to find a missing amount.

- 1 Picks a ratio that is equal to a given one.
 - 2 Finds a missing number by scaling one ratio up to match the other.
 - 3 Finds a missing number using cross multiplication when scaling is awkward.
 - 4 Sets up and solves a proportion from a word problem, like scaling a recipe.
 - ★ Solves proportions where the unknown sits inside a sum, like x plus 5, over 4, equals 7 over 4.
 - ★★ Finds the number to add to both parts of a ratio to turn it into a new ratio.
-

III.2.08

Proportional relationships

Deciding whether two quantities are proportional, meaning one is always the same multiple of the other.

- 1 Checks a table to see if the ratio stays the same in every row.
 - 2 Checks whether a graph is a straight line through zero.
 - 3 Checks whether an equation has the form y equals a number times x .
 - 4 Picks out the situation that is not proportional from a list.
-

III.2.09

The constant of proportionality

The constant of proportionality is the fixed multiplier linking two proportional amounts, like the price per kilogram.

- 1 Finds the multiplier from a table.
 - 2 Finds the multiplier from a graph.
 - 3 Writes the equation, y equals the multiplier times x , from one pair of values.
 - 4 Explains what the multiplier means in a real setting, such as cost per kilogram.
-

III.2.10

Scale drawings

Using scale drawings and maps, where every length is shrunk or enlarged by the same amount.

- 1 Finds the scale factor, how many times bigger one drawing is than the other.
- 2 Uses a drawing's scale to find a real-life length.
- 3 Redraws a length at a new scale.
- 4 Uses a map scale like 1 : 10,000 to convert between map and real distances.

III.2.11

Unit conversion with ratios

Converting between units, such as centimeters and meters, by multiplying by the right conversion number.

- 1 Converts between two units in one step, like centimeters to meters.
- 2 Converts in several steps, like kilometers to centimeters.
- 3 Converts rates, such as kilometers per hour to meters per second.
- 4 Converts area units, keeping careful track of squared units.

III.2.12

Sharing in a ratio

Dividing an amount between people or ingredients in a given ratio.

- 1 Shares an amount of money in a given ratio.
- 2 Works out the total when only the difference between shares is known.
- 3 Works out how much to add to change one ratio into another.
- 4 Finds how much of each ingredient is in a mixture.
- ★ Works out the shares when only the gap between two shares is known.
- ★★ Works out starting amounts when a gift from one person to another changes their ratio.

III.2.13

Speed, distance, time

Using the link between speed, distance, and time to solve travel problems.

- 1 Finds the distance traveled from the speed and the time.
- 2 Finds the time a trip takes from the distance and the speed.
- 3 Finds the average speed over a trip with two different speeds.
- 4 Works out when two travelers heading toward each other will meet.
- ★ Finds the true average speed for a round trip, which is not just halfway between the two speeds.
- ★★ Solves classic puzzles: when one traveler catches another, or how far a bird flies between two trains.

III.2.14

Comparing ratios

Comparing two ratios to see which is bigger, such as which drink mix is stronger.

- 1 Decides which of two mixes has more juice for its water.
- 2 Rewrites both ratios with a shared amount so they are easy to compare.
- 3 Compares ratios by writing them as fractions.
- 4 Picks the reasoning that correctly shows which mix is stronger.

III.2.15

Ratio puzzles

Solving puzzle-like ratio problems from everyday life.

- 1 Scales a recipe for a different number of people.
- 2 Finds how much of each color is in a paint mixture.
- 3 Works out that more workers take less time, and how much less.
- 4 Explains why an adding shortcut gives the wrong answer in a ratio problem.
- ★ Links two ratios that share a middle item, such as red to blue and blue to green.
- ★★ Combines two groups' averages, working out the class average or what share each group is.

Percent

This branch is all about percent, which simply means "out of 100." Percents appear in sales, taxes, tips, interest, and the news, so this branch focuses on real money and real data.

III.3.01 Percent means per hundred

Understanding that a percent is a number out of 100, so 15% means 15 out of every 100.

- 1 Reads a percent from a grid of 100 squares.
- 2 Writes a percent as a fraction.
- 3 Writes a percent as a decimal.
- 4 Uses familiar percents like 25%, 50%, and 75% to find part of a group.

III.3.02 Fractions, decimals, percents

Switching between fractions, decimals, and percents, which are three ways to write the same amount.

- 1 Turns a fraction into a percent, like three quarters into 75%.
- 2 Turns a percent into a decimal, including percents like 62.5%.
- 3 Turns a decimal into a percent, including ones over 100%.
- 4 Handles fractions like one third, whose decimals repeat forever.

III.3.03 Percent of a number

Finding a percent of a number, such as 15% of 480.

- 1 Finds 10% and 50% of a number.
- 2 Finds 25% and 75% of a number.
- 3 Finds any percent of a number.
- 4 Builds harder percents in the head from easy ones, like 60% from 10%.

III.3.04 Find the whole

Working backwards to find the whole amount when only a percent of it is known.

- 1 Finds the whole when told what a simple percent of it is.
- 2 Uses a table to step from a known percent down to 10%, then up to 100%.
- 3 Picks the equation that matches a find-the-whole problem.
- 4 Solves word problems, like finding class size from how many walk to school.
- ★ Works backwards through two percent spends in a row to find the starting amount.
- ★★ Finds a tank's full size from how much it takes to go from one percent full to another.

III.3.05 Find the percent

Working out what percent one number is of another.

- 1 Finds what percent one number is of another, like 32 of 160.
- 2 Uses a table to rescale a part out of a whole to a part out of 100.
- 3 Picks the right calculation to find a percent.
- 4 Solves word problems, like the percent of games a team won.

III.3.06

Over 100% and under 1%

Understanding percents bigger than 100% and smaller than 1%.

- 1 Works with percents over 100%, like 120 being 120% of 100.
 - 2 Finds tiny percents, like 0.2%, of a number.
 - 3 Knows that more than 100% of a number is bigger than the number itself.
 - 4 Uses large percents in real settings, like a town growing to 200% of its size.
-

III.3.07

Percent increase

Working with amounts that go up by a percent, like a price rise.

- 1 Finds the new amount after a percent increase.
 - 2 Increases an amount in one multiplication, like times 1.09 for a 9% rise.
 - 3 Finds the percent increase between an old and a new amount.
 - 4 Solves real problems, like the percent increase in rent.
-

III.3.08

Percent decrease

Working with amounts that go down by a percent, like a price drop.

- 1 Finds the new amount after a percent decrease.
 - 2 Decreases an amount in one multiplication, like times 0.87 for a 13% drop.
 - 3 Finds the percent decrease between an old and a new amount.
 - 4 Solves real problems, like the percent drop in a phone's price.
-

III.3.09

Discounts and sale prices

Working out discounts and sale prices when shopping.

- 1 Finds how much money a percent discount takes off.
 - 2 Finds the sale price after a discount.
 - 3 Works out the total effect of one discount taken after another.
 - 4 Compares a percent-off deal with a dollars-off deal.
 - ★ Handles one discount taken after another, finding the single discount that matches.
 - ★★ Works out an original price from a clue linking the sale price to half the price.
-

III.3.10

Tax and tips

Adding sales tax and tips to a bill.

- 1 Adds sales tax to a price to find the total.
 - 2 Works out a tip as a percent of a meal's cost.
 - 3 Finds the full cost of an item including tax.
 - 4 Estimates a tip in the head by rounding.
-

III.3.11

Markup and profit

Understanding how shops set prices by adding a markup, and how profit is measured as a percent.

- 1 Finds the markup, the amount a shop adds to what it paid.
- 2 Finds the selling price after a markup.
- 3 Finds profit as a percent of what the item cost.
- 4 Solves several-step problems, like a markup followed by a sale.

III.3.12

Simple interest

Simple interest is money earned (or owed) each year as a fixed percent of the starting amount.

- 1 Finds the interest from the amount, the yearly rate, and the number of years.
 - 2 Works backwards to find the yearly interest rate.
 - 3 Works backwards to find how many years it took.
 - 4 Finds the total amount at the end, including periods shorter than a year.
 - ★ Works backwards to find the years, the rate or the starting amount from the interest.
 - ★★ Works out how money was split between two accounts earning different rates.
-

III.3.13

Percent error

Percent error measures how far off a guess or measurement is, compared with the true value.

- 1 Finds the error, the gap between a guess and the true amount.
 - 2 Writes that error as a percent of the true value.
 - 3 Finds the percent error of a real measurement.
 - 4 Uses percent error to decide which of two measurements is more accurate.
-

III.3.14

Percent change traps

Avoiding common mistakes with percent changes that trip people up.

- 1 Tells a rise in percentage points apart from a percent increase.
 - 2 Sees why going down 10% and then up 10% does not return to the start.
 - 3 Finds the overall change after two percent changes in a row.
 - 4 Explains why a claim about percent changes in the news is misleading.
 - ★ Finds the overall change after a percent rise and a percent fall, which don't simply cancel.
 - ★★ Works out the percent rise needed to undo a fall, which is much bigger than the fall.
-

III.3.15

Percent in data

Reading and using percents in charts and survey results.

- 1 Turns survey counts from a bar chart into a percent.
 - 2 Finds the missing slice of a pie chart, knowing the whole is 100%.
 - 3 Uses a pie chart's percent to find how many people chose an option.
 - 4 Compares groups of different sizes fairly by using percents.
-

III.3.16

Estimating with percents

Estimating percents quickly in the head to get a sensible rough answer.

- 1 Rounds the number first to estimate 10% of it.
- 2 Uses a nearby easy percent, like 25%, to estimate.
- 3 Picks the reasonable answer from a list by estimating.
- 4 Rounds both the percent and the number to estimate in the head.

Multi-step percent problems

Solving percent problems that take several steps.

- ① Applies two percent changes, one after the other.
- ② Works backwards to find the original price before a percent rise.
- ③ Applies a discount and then sales tax to find a final price.
- ④ Picks the single calculation that gives a final price and explains why.
- ★ Undoes a shop's markup and then the sales tax, in reverse order, to find the shop's cost.
- ★★ Works out how much to add to a mix to change its percent, using the part that stays fixed.

Exponents & roots

This branch covers exponents, a short way to write repeated multiplication, and roots, which undo them. These tools let us handle very large and very small numbers, and they lead to the discovery of numbers that cannot be written as fractions.

III.4.01 Whole-number exponents

An exponent (a small raised number) says how many times to multiply a number by itself, so 3 to the 5th means $3 \times 3 \times 3 \times 3 \times 3$.

- 1 Writes out what a number with an exponent means as repeated multiplication.
- 2 Names the base (the number multiplied) and the exponent (how many times).
- 3 Works out the value of a number with an exponent, including negative bases.
- 4 Finds the missing base when a number is written as a power.
- ★ Adds several copies of the same power and writes the result as one power.
- ★★ Finds the last digit of a huge power, using the repeating cycle of last digits.

III.4.02 Powers of ten

Using powers of ten, like $10 \times 10 \times 10 = 1000$, to understand place value and multiply quickly.

- 1 Works out powers of ten, like 10 to the 2nd is 100.
- 2 Says which power of ten a digit is worth in a large number.
- 3 Multiplies a number by a power of ten by shifting the decimal point.
- 4 Gets a first look at negative powers of ten, which make small decimals like 0.001.

III.4.03 Exponents in the order of operations

Knowing when to work out exponents in a longer calculation: after brackets, before multiplying and adding.

- 1 Works out the exponent first, then adds or subtracts.
- 2 Works out the brackets first, then applies the exponent.
- 3 Tells apart minus 3 squared (the answer is negative) from minus 3 in brackets, squared (positive).
- 4 Works through calculations that mix exponents with all four operations.

III.4.04 Multiplying powers

A shortcut for multiplying powers of the same number: add the exponents.

- 1 Multiplies powers of the same letter by adding the exponents.
- 2 Uses the rule with numbers and checks the value.
- 3 Multiplies expressions with numbers and several letters.
- 4 Multiplies three or more powers, including a letter on its own.
- ★ Rewrites powers of 4, 8, 16 or 32 as powers of 2, then multiplies by adding exponents.
- ★★ Counts the digits of a huge product by pairing 2s with 5s to make tens.

III.4.05

Dividing powers

A shortcut for dividing powers of the same number: subtract the exponents.

- 1 Divides powers of the same number by subtracting the exponents.
- 2 Knows that any number to the power zero equals 1.
- 3 Divides expressions with numbers and several letters.
- 4 Combines multiplying and dividing powers in one expression.

III.4.06

Power of a power

Rules for raising a power to another power, and for raising products and fractions to a power.

- 1 Raises a power to a power by multiplying the exponents.
- 2 Raises a product, like 2 times n, to a power by raising each part.
- 3 Raises a fraction to a power by raising top and bottom.
- 4 Uses all the exponent rules together to simplify one expression.
- ★ Decides which of two powers is larger by rewriting them with the same exponent.
- ★★ Orders three huge powers by rewriting them all as powers of the same number.

III.4.07

Negative exponents

A negative exponent means one divided by the power, so 2 to the minus 1 is one half.

- 1 Matches a negative exponent with the fraction it means.
- 2 Rewrites expressions with negative exponents as fractions.
- 3 Simplifies divisions that leave a negative exponent, writing the answer with a positive one.
- 4 Uses the exponent rules with negative exponents.

III.4.08

Scientific notation

Scientific notation writes very large or very small numbers as a small number times a power of ten.

- 1 Recognizes which numbers are written in scientific notation.
- 2 Writes a very large number, like 8 billion, in scientific notation.
- 3 Writes a very small decimal in scientific notation.
- 4 Turns scientific notation back into an ordinary number.

III.4.09

Calculating in scientific notation

Doing arithmetic with numbers written in scientific notation.

- 1 Multiplies two numbers in scientific notation.
- 2 Divides two numbers in scientific notation.
- 3 Adds and subtracts numbers in scientific notation with different powers of ten.
- 4 Solves real problems with huge numbers, like how far light travels.

III.4.10

Comparing huge and tiny numbers

Comparing very large and very small numbers written with powers of ten.

- 1 Picks the bigger of two numbers in scientific notation.
- 2 Works out how many times bigger one power of ten is than another.
- 3 Puts several tiny or huge numbers in order.
- 4 Finds the nearest power of ten to a number, its rough size or order of magnitude.

III.4.11

Square roots

A square root undoes squaring: the square root of 81 is 9, because $9 \times 9 = 81$.

- 1 Finds the side of a square from the number of small squares inside it.
 - 2 Uses the square root sign to work out square roots.
 - 3 Sees that taking the square root undoes squaring.
 - 4 Solves x squared equals a number, finding both the positive and negative answers.
 - ★ Counts the whole numbers that lie between two square roots.
 - ★★ Finds the square root of a big product without a calculator, by spotting squares hidden inside it.
-

III.4.12

Cube roots

A cube root undoes cubing: the cube root of 27 is 3, because $3 \times 3 \times 3 = 27$.

- 1 Recognizes perfect cubes, numbers like 8 and 27.
 - 2 Finds the edge of a cube from its volume.
 - 3 Finds cube roots of negative numbers.
 - 4 Solves x cubed equals a number.
-

III.4.13

Estimate roots

Estimating square roots that are not whole numbers, like the square root of 23.

- 1 Says which two whole numbers a square root lies between.
 - 2 Estimates a square root to one decimal place.
 - 3 Marks a square root at about the right place on a number line.
 - 4 Squares nearby decimals to decide which is the better estimate.
-

III.4.14

Irrational numbers

Irrational numbers, like pi and the square root of 2, cannot be written as fractions; their decimals go on forever without repeating.

- 1 Picks the number that cannot be written as a fraction.
 - 2 Knows true facts about famous irrational numbers like pi.
 - 3 Tells apart decimals that repeat (fractions) from those that never repeat.
 - 4 Decides whether a number like 3 times pi is rational or irrational.
-

III.4.15

The real number system

Sorting numbers into families: counting numbers, integers (whole numbers and their negatives), fractions, and irrational numbers.

- 1 Names the smallest family a number belongs to.
- 2 Counts which numbers in a list are irrational.
- 3 Places a number in the right spot on a diagram of nested families.
- 4 Lists every family a number belongs to.

Compare irrationals

Comparing irrational numbers, like square roots, with each other and with ordinary decimals.

- 1 Compares a square root with a decimal by estimating.
- 2 Matches square roots and decimals to points on a number line.
- 3 Puts a mixed list of square roots and decimals in order.
- 4 Decides which whole number a square root is closer to.

Growth with exponents

Exponential growth is when an amount keeps multiplying, such as doubling every day, instead of adding the same amount.

- 1 Works out how big an amount gets after doubling several times.
- 2 Works out powers of 2.
- 3 Continues a table where each value is multiplied by the same number.
- 4 Solves real growth problems, like bacteria tripling every hour.
- ★ Runs growth forwards or backwards: when an amount passes a target, or how many there were at first.
- ★★ Thinks backwards from full: if a patch triples daily, finds the day it covered one ninth of a pond.

Expressions

This branch starts algebra: using letters to stand for numbers and writing expressions (math phrases like $3n + 5$) to describe patterns and situations. Learners practice tidying expressions and see why they work, which prepares them for solving equations next.

III.5.01 Variables

A variable is a letter, like n , that stands for a number we don't know yet or that can change.

- 1 Works out which number a letter stands for in a simple statement like $n + 8 = 17$.
- 2 Puts a given number in place of the letter and works out the result.
- 3 Writes a phrase like "a number plus 10" using a letter.
- 4 Uses letters in formulas, like the perimeter of a square is 4 times its side.

III.5.02 Expressions from words

Turning word phrases into math expressions, such as "10 times a number" into $10n$.

- 1 Writes sums and differences, like "a number minus 17."
- 2 Writes products and divisions, like "10 times a number."
- 3 Handles tricky wording where the order flips, like "17 subtracted from a number."
- 4 Writes phrases with two steps, like "a number plus 10, then divided by 8."

III.5.03 Evaluate expressions

Evaluating an expression: putting in numbers for the letters and working out the value.

- 1 Works out an expression with one letter.
- 2 Works out an expression with two letters, including negative values.
- 3 Works out expressions with squares, taking care with negative numbers.
- 4 Uses real formulas, like the area of a triangle, with given measurements.
- ★ Finds an expression's value without knowing the letters, by spotting it as a multiple of a known one.
- ★★ Finds x squared plus y squared from just the sum (or difference) and the product of x and y .

III.5.04 Parts of an expression

Naming the parts of an expression: terms (the pieces added together), coefficients, constants, and factors.

- 1 Counts the terms, the separate pieces joined by plus or minus.
- 2 Finds a coefficient, the number multiplying a letter.
- 3 Finds the constant, the plain number with no letter.
- 4 Breaks one term into the things multiplied together to make it.

III.5.05

Like terms

Like terms have exactly the same letters, such as $6a$ and $8a$, and can be added together into one.

- 1 Picks out terms that are alike.
 - 2 Adds like terms to simplify an expression.
 - 3 Combines like terms when some are being subtracted.
 - 4 Combines terms with several letters, noticing that xy and yx are the same.
-

III.5.06

The distributive property

The distributive property: a number outside brackets multiplies everything inside, so $3(x + 2)$ becomes $3x + 6$.

- 1 Multiplies out brackets with a positive number in front.
 - 2 Multiplies out brackets with a negative number in front.
 - 3 Multiplies out brackets with a subtraction inside, keeping signs right.
 - 4 Multiplies out brackets with a decimal or fraction in front.
-

III.5.07

Factor out a GCF

Factoring out the greatest common factor: pulling the biggest shared number or letter outside brackets, the reverse of multiplying out.

- 1 Pulls the largest shared number out of an expression like $21p + 15$.
 - 2 Pulls out shared letters as well as numbers.
 - 3 Pulls out a negative number, adjusting the signs inside.
 - 4 Checks a factored answer by multiplying it back out.
-

III.5.08

Equivalent expressions

Equivalent expressions look different but always give the same value, whatever number the letter is.

- 1 Tests two expressions by putting the same number into both.
 - 2 Simplifies an expression to find which one it equals.
 - 3 Picks out the pair of expressions that are truly equal.
 - 4 Explains why agreeing for one number is not enough to be equivalent.
 - ★ Finds the missing number that makes two expressions equal for every value of x .
 - ★★ Finds two hidden numbers by matching the x parts and the plain numbers on both sides.
-

III.5.09

Simplify expressions

Simplifying expressions by multiplying out brackets and then combining like terms.

- 1 Multiplies out one bracket, then combines like terms.
- 2 Simplifies expressions with two brackets.
- 3 Simplifies expressions with decimals.
- 4 Finds the first line where a worked simplification goes wrong.

III.5.10

Expressions in geometry

Writing expressions for the perimeter and area of shapes whose sides are given with letters.

- 1 Writes and simplifies the perimeter of a rectangle.
 - 2 Writes the area of a rectangle without brackets.
 - 3 Writes the perimeter of an L-shape, working out any missing sides.
 - 4 Writes the perimeter of shapes joined together, leaving out the shared edge.
-

III.5.11

Expressions in context

Writing and reading expressions that describe real situations, like the cost of tickets.

- 1 Writes an expression for a total cost, such as tickets plus a fee.
 - 2 Writes an expression for ages, including ages in the future.
 - 3 Explains what each number in a real formula stands for.
 - 4 Reads meaning from a rewritten form, like 1.04p meaning price plus 4% tax.
-

III.5.12

Expressions with powers

Working with letters raised to powers, like x squared, and not mixing them up with doubling.

- 1 Tells apart x squared (x times x) from $2x$ (x plus x).
 - 2 Works out expressions with powers, including negative values.
 - 3 Combines like terms that have the same power.
 - 4 Multiplies terms with powers, multiplying the numbers and adding the exponents.
-

III.5.13

Patterns to expressions

Writing a rule with a letter that gives any term in a number or shape pattern.

- 1 Writes a rule for a pattern given in a table.
 - 2 Writes a rule for a growing shape pattern, like matchstick shapes.
 - 3 Writes the rule for the n th term (any position) of a sequence, including ones that go down.
 - 4 Checks whether someone's rule really fits the pattern.
 - ★ Works out which position in a pattern a given number appears at, including patterns that go down.
 - ★★ Uses a pattern's rule to find the longest toothpick row that can be built from a set number.
-

III.5.14

Algebra tiles

Using algebra tiles, small shapes standing for x squared, x , and 1, to see how expressions work.

- 1 Writes the expression that a set of tiles shows.
- 2 Combines tiles, removing pairs of positive and negative that cancel.
- 3 Fills a rectangle with tiles to see how multiplying out brackets works.
- 4 Arranges tiles into a rectangle to find the two sides, which is factoring.

Why number tricks work

Using algebra to explain "think of a number" tricks that always give the same answer.

- 1 Follows a number trick step by step with a starting number.
- 2 Follows the trick with a letter instead of a number.
- 3 Finishes a trick so everyone ends up back at their starting number.
- 4 Picks the algebra reason a trick always works, rather than trusting a few tested numbers.
- ★ Undoes a four-step number trick in reverse to find the starting number.
- ★★ Uses algebra to count two-digit numbers that, minus their digits reversed, give a set answer.

Equations & inequalities

This branch teaches how to solve equations (statements that two things are equal, with an unknown number to find) and inequalities (statements that one thing is bigger or smaller). It is the core skill of algebra, used to answer real questions like how many tickets you can afford.

III.6.01 What an equation is

An equation is like a balanced scale: both sides are equal, and a solution is a number that keeps it balanced.

- 1 Uses a balance picture to see that both sides must change the same way.
- 2 Explains what it means for a number to be a solution.
- 3 Tests whether a given number makes an equation true.
- 4 Decides whether a number statement like $8 + 8 = 12 + 3$ is true or false.

III.6.02 One-step equations: + and –

Solving equations that need just one step of adding or subtracting, such as $x + 21 = 44$.

- 1 Solves one-step equations with whole numbers.
- 2 Solves one-step equations that include negative numbers.
- 3 The same with fractions.
- 4 The same with decimals.

III.6.03 One-step equations: $\times$ and $\div$

Solving equations that need just one step of multiplying or dividing, such as $10x = 130$.

- 1 Solves one-step equations with whole numbers.
- 2 Solves one-step equations with negative numbers.
- 3 Solves equations with a fraction in front of x by multiplying by the flipped fraction.
- 4 The same with decimals.

III.6.04 Two-step equations

Solving equations that need two steps, such as $8x + 12 = 92$: undo the adding, then the multiplying.

- 1 Solves two-step equations with whole numbers.
- 2 Solves two-step equations with negative numbers.
- 3 Solves two-step equations that involve a fraction.
- 4 Checks a proposed answer by putting it back into the equation.

III.6.05 Multi-step equations

Solving equations that need several steps, like multiplying out brackets or combining terms first.

- 1 Multiplies out brackets first, then solves.
- 2 Combines like terms first, then solves.
- 3 Solves equations with fractions of x .
- 4 Checks a proposed answer by putting it back into the equation.

III.6.06

Variables on both sides

Solving equations where the unknown appears on both sides, and spotting ones with no answer or endless answers.

- 1 Gathers the x terms on one side and solves.
 - 2 Multiplies out brackets on both sides, then solves.
 - 3 Recognizes an equation that has no solution at all.
 - 4 Recognizes an equation that is true for every number.
 - ★ Given one solution of an equation, finds the missing number in the equation itself.
 - ★★ Counts the whole numbers k that give an equation a positive whole-number solution.
-

III.6.07

Equations from word problems

Turning a word problem into an equation, solving it, and making sense of the answer.

- 1 Decides exactly what the letter should stand for.
 - 2 Writes the equation that matches the story.
 - 3 Writes and solves the equation.
 - 4 Uses the solution to answer the actual question asked.
 - ★ Solves problems about numbers in a row, like three even numbers in a row adding to 132.
 - ★★ Writes and solves equations for classic puzzles, like chickens and cows from heads and legs.
-

III.6.08

Inequalities

An inequality says one side is less than or greater than the other, so it has many solutions, not just one.

- 1 Reads the less-than and less-than-or-equal signs and tests a number.
 - 2 Understands that an inequality has endlessly many solutions.
 - 3 Picks the number line picture that shows an inequality.
 - 4 Writes an inequality from words, like "your age must be below 11."
-

III.6.09

One-step inequalities

Solving inequalities in one step, and learning the one new rule: multiplying or dividing by a negative flips the sign.

- 1 Solves by adding or subtracting.
 - 2 Solves by multiplying or dividing by a positive number.
 - 3 Flips the inequality sign when dividing by a negative number.
 - 4 Solves and then shows the answer on a number line.
-

III.6.10

Two-step inequalities

Solving inequalities that need two steps, such as $8x + 2 < 58$.

- 1 Solves two-step inequalities.
- 2 Flips the sign when a step divides by a negative number.
- 3 Solves and shows the answer on a number line.
- 4 Solves real problems, like the most tickets a budget allows.
- ★ Solves an inequality that needs a sign flip, then finds the smallest or largest whole number that fits.
- ★★ Counts the whole numbers that fit a double inequality, with the number trapped between two limits.

Rearrange formulas

Rearranging a formula to make a different letter the subject, such as finding time from distance and speed.

- ① Rearranges the distance formula to find the time.
- ② Rearranges the rectangle area formula to find the width.
- ③ Rearranges the perimeter formula, which needs two steps.
- ④ Rearranges the Fahrenheit formula to find Celsius.

Choosing a strategy

Choosing a good method for solving an equation and checking the answer.

- ① Solves a "think of a number" puzzle by undoing each step in reverse.
- ② Picks a correct next step that keeps both sides balanced.
- ③ Checks an answer by working out both sides of the original equation.
- ④ Picks the quickest first step for solving an equation.

Lines & first functions

This branch connects rules, tables, and graphs, focusing on straight lines and the idea of a function (a rule that gives exactly one output for each input). It shows how math describes steady change, like a plant growing the same amount each week.

III.7.01 Input-output rules

Using a rule that turns an input number into an output number, like "multiply by 9, then add 10."

- 1 Applies a rule to an input, or undoes it to find the input from the output.
- 2 Fills in a table of inputs and outputs from a rule.
- 3 Picks the rule that fits every pair in a table.
- 4 Writes a table's rule as an equation.
- ★ Runs a number backwards through two machines in a row to find what went in.
- ★★ Follows a machine that feeds its output back in, spotting the loop to predict a far-off pass.

III.7.02 Graph from a table

Turning a table of number pairs into points on a graph, and reading pairs back off a graph.

- 1 Finds the point on a grid that matches a pair from a table.
- 2 Reads a missing value off a graph.
- 3 Decides whether plotted points line up in a straight line.
- 4 Extends a straight-line pattern on a graph to predict a new value.

III.7.03 Rate of change

Rate of change is how much one amount changes for each step in another, like centimeters per minute.

- 1 Finds the rate of change between two points: the change in y divided by the change in x.
- 2 Finds the rate of change from a table.
- 3 Finds the rate of change from a graph.
- 4 Explains what a rate means in real terms, using its units.

III.7.04 Slope

Slope measures how steep a line is: how far it goes up for each step across.

- 1 Finds the slope on a grid by counting the rise and the run.
- 2 Works out the slope from two points' coordinates.
- 3 Sorts slopes into uphill, downhill, flat, and straight up and down.
- 4 Compares slopes to decide which line is steepest.
- ★ Finds a missing coordinate of a point, given the slope and one other point.
- ★★ Counts the points with whole-number coordinates that lie on a line segment.

III.7.05

Slope and similar triangles

Using small right triangles drawn on a line to check that a straight line has the same slope everywhere.

- 1 Reads the rise and run from a triangle drawn on a line.
- 2 Works out rise divided by run for two slope triangles and checks they match.
- 3 Sees that big and small triangles on one line give the same slope.
- 4 Finds the slope from different pairs of points and checks whether three points lie on one line.

III.7.06

$y = mx + b$

Every straight line can be written as $y = mx + b$, where m is the slope and b is where it crosses the vertical axis.

- 1 Reads the slope and the starting value from an equation.
- 2 Matches an equation with its line on a graph.
- 3 Writes the equation of a line shown on a graph.
- 4 Writes the equation of a line from just two points.
- ★ Finds where a parallel line crosses the vertical axis, given one point it passes through.
- ★★ Finds the area of the triangle a line makes with the two axes.

III.7.07

Proportional vs non-proportional

Telling proportional relationships (straight lines through zero) apart from other straight lines.

- 1 Checks whether a line on a graph passes through zero.
- 2 Checks whether y divided by x stays the same in a table.
- 3 Checks an equation for an added starting amount that breaks proportionality.
- 4 Decides from a story, like a taxi fare with a starting fee, whether it is proportional.

III.7.08

Linear vs nonlinear

Telling linear relationships (straight-line graphs, steady change) from nonlinear ones (curves).

- 1 Checks a table for the same increase at each step.
- 2 Decides whether a graph is straight or curved.
- 3 Spots equations that are not linear, such as ones with x squared.
- 4 Picks the correct answer and the correct reason for it.

III.7.09

Functions

A function is a rule that gives exactly one output for each input.

- 1 Checks a list of pairs to see if any input has two outputs.
- 2 Reads an arrow diagram to decide if it shows a function.
- 3 Uses the vertical line test: if an up-down line hits a graph twice, it is not a function.
- 4 Decides whether a real-life pairing, like person to height, is a function.

III.7.10

Compare functions

Comparing two functions shown in different ways, such as one as a table and one as a graph.

- 1 Compares two equations to see which changes faster.
 - 2 Compares starting values when one is an equation and one is a table.
 - 3 Compares functions shown as a graph, a table, and an equation.
 - 4 Finds where two functions are equal, after which the faster-growing one stays ahead.
-

III.7.11

Describe graphs qualitatively

Reading the shape of a graph to tell the story it shows, without needing exact numbers.

- 1 Finds the parts of a graph that go up and the parts that go down.
 - 2 Knows that a flat part of a graph means nothing is changing.
 - 3 Picks the graph that matches a story, like a bike ride away from home.
 - 4 Picks the story that matches a graph.
-

III.7.12

Linear models

Using a straight-line equation as a model of a real situation, then using it to predict.

- 1 Writes an equation for a situation, like a plant's height after some weeks.
- 2 Explains what the slope means in the story, such as growth per week.
- 3 Explains what the starting value means in the story.
- 4 Uses the model to predict and judges whether the prediction is sensible.

Basic Non-Proof Geometry

This branch covers the hands-on side of geometry: angles, area, volume and circles, plus the Pythagorean theorem, a famous rule for right triangles. It is about measuring and working things out, from carpeting a room to filling a can; proving why the rules are true comes later, in Era V.

III.8.01 Angle relationships

Using the ways angles fit together, like two angles on a straight line always adding up to 180 degrees.

- 1 Checks whether two angles are complementary, meaning they add up to 90 degrees.
- 2 Finds angles that are supplementary, meaning they add up to 180 degrees.
- 3 Finds all four angles where two straight lines cross, since opposite angles are equal.
- 4 Writes and solves an equation to find a missing angle.

III.8.02 Parallel lines & a transversal

Finding the angles formed when one line crosses two parallel lines (lines that never meet).

- 1 Finds the matching angle in the same position at the other crossing.
- 2 Finds angle pairs on opposite sides of the crossing line, which are equal.
- 3 Finds angle pairs on the same side between the parallel lines, which add to 180 degrees.
- 4 Finds all eight angles from just one.

III.8.03 Triangle angle sum

Using the fact that the three angles inside any triangle always add up to 180 degrees.

- 1 Checks whether three angles could belong to a triangle.
- 2 Finds a missing angle in a triangle.
- 3 Uses an angle outside the triangle, which equals the two far inside angles added together.
- 4 Finds a missing angle in two steps, such as working out a second angle first.
- ★ Finds a triangle's angles when they are given as a ratio, such as 9 to 7 to 4.
- ★★ Finds the angle where two corner-splitting lines (angle bisectors) meet inside a triangle.

III.8.04 Area of parallelograms

Finding the area (the space inside) of a parallelogram, a slanted rectangle.

- 1 Cuts a triangle off one end and slides it over to make a rectangle.
- 2 Finds the area by multiplying the base by the height.
- 3 Picks the true straight-up height and ignores the slanted side.
- 4 Works backwards from the area to find a missing height or base.

III.8.05 Area of triangles

Finding the area of a triangle, which is half of a matching parallelogram.

- 1 Sees that a triangle is half of a parallelogram.
- 2 Finds the area as half of base times height.
- 3 Finds the area of a wide-angled triangle whose height falls outside it.
- 4 Works backwards from the area to find a missing base or height.

III.8.06

Area of trapezoids

Finding the area of a trapezoid, a four-sided shape with one pair of parallel sides.

- 1 Splits a trapezoid into a rectangle and triangles and adds their areas.
- 2 Averages the two parallel sides, then multiplies by the height.
- 3 Uses the trapezoid formula, ignoring measurements it does not need.
- 4 Works backwards from the area to find a missing side.

III.8.07

Composite area

Finding the area of irregular shapes by breaking them into simpler pieces.

- 1 Splits a shape into two rectangles and finds each area.
- 2 Finds the total area of a shape made of several parts.
- 3 Subtracts a cut-out hole from a larger area.
- 4 Finds the area of a shape drawn on a coordinate grid.
- ★ Finds both the area and the perimeter after a notch or corner is cut from a rectangle.
- ★★ Finds the area of a tilted square drawn inside a square, by cutting off four corner triangles.

III.8.08

Circle parts

Naming the parts of a circle and meeting pi, the special number (about 3.14) that links a circle's width to its distance around.

- 1 Names the center, radius (center to edge), and diameter (all the way across).
- 2 Uses the fact that the diameter is twice the radius.
- 3 Names a chord (a straight line across) and an arc (a curved piece of the edge).
- 4 Divides the distance around a circle by its size to see pi appear.

III.8.09

Circumference

Finding the circumference, the distance around a circle.

- 1 Estimates the distance around from the width, using pi.
- 2 Finds the exact distance around from the radius, leaving the answer in terms of pi.
- 3 Works backwards from the distance around to find the width.
- 4 Turns an exact answer in terms of pi into a decimal using 3.14.

III.8.10

Area of a circle

Finding the area inside a circle: pi times the radius times the radius.

- 1 Finds the exact area from the radius.
- 2 Finds the area from the diameter, halving it first.
- 3 Finds the area of half and quarter circles.
- 4 Works backwards from the area to find the size of the circle.
- ★ Finds the area of a ring between two circles with the same center.
- ★★ Finds the area of a circle drawn around a square, using the square's diagonal.

III.8.11

Surface area of prisms

Finding surface area: the total area of all the flat faces of a 3D shape, like wrapping paper for a box.

- 1 Picks a net (the flat pattern that folds into a shape) that makes a cube.
- 2 Finds the area of each different face of a box.
- 3 Adds up all the faces to find a box's total surface area.
- 4 Finds the surface area of a pyramid from its flat pattern.

III.8.12

Volume of prisms

Finding volume, the space inside a 3D shape, for boxes and other prisms (shapes with the same cross-section all the way through).

- 1 Counts the small cubes that fill a box.
- 2 Multiplies the area of the base by the height.
- 3 Finds the volume when edge lengths are fractions.
- 4 Works backwards from the volume to find a missing length.
- ★ Counts how many whole cubes really fit in a box, since leftover gaps can't hold one.
- ★★ Finds a box's volume from just the areas of its three different faces.

III.8.13

Volume of cylinders

Finding the volume of a cylinder, like a can: the circle's area times the height.

- 1 Multiplies a given base area by the height.
- 2 Finds the circle's area first, then multiplies by the height.
- 3 Starts from the width instead of the radius and gives an exact answer.
- 4 Solves real problems, like how much a can holds.

III.8.14

Volume of cones & spheres

Finding the volume of cones and spheres (balls), and how they compare with cylinders.

- 1 Sees that three cones of water fill a cylinder of the same size.
- 2 Finds a cone's volume as one third of the matching cylinder.
- 3 Finds the volume of a sphere using its formula.
- 4 Compares a cone, sphere, and cylinder of the same width and height.

III.8.15

Pythagorean theorem

The Pythagorean theorem: in a right triangle, the two short sides squared add up to the longest side squared.

- 1 Names the hypotenuse, the longest side opposite the square corner.
- 2 Finds the longest side from the two shorter ones.
- 3 Finds a shorter side from the other two.
- 4 Solves real problems, like how high a leaning ladder reaches.
- ★ Finds the right triangles hidden inside a shape, like a diamond, to work out a side or an area.
- ★★ Works out how far the top of a ladder slides down when its foot slides out.

III.8.16

Converse of Pythagoras

Using the Pythagorean rule in reverse to test whether a triangle has a square corner.

- 1 Tests whether three side lengths make a right triangle.
 - 2 Recognizes Pythagorean triples, whole-number sets like 3, 4, 5 that make right triangles.
 - 3 Decides if a triangle's widest angle is smaller than, equal to, or larger than a square corner.
 - 4 Tests for a right angle when sides are decimals or the triangle is drawn on a grid.
-

III.8.17

Distance on the plane

Using the Pythagorean theorem to find the straight-line distance between two points on a grid.

- 1 Counts squares to find the across and up legs of a right triangle.
 - 2 Works out those leg lengths from the points' coordinates.
 - 3 Finds the straight distance between two points.
 - 4 Finds the perimeter of a shape drawn on a grid.
-

III.8.18

Pythagoras in 3D

Using the Pythagorean theorem to find slanted lengths in 3D shapes like boxes and cones.

- 1 Finds the diagonal across one face of a box.
 - 2 Finds the diagonal from one corner of a box to the far opposite corner.
 - 3 Finds the slanted side length of a cone.
 - 4 Solves real problems, like the longest stick that fits in a box.
-

III.8.19

Translations

A translation slides a shape without turning or flipping it.

- 1 Slides a shape on a grid and says where a corner lands.
 - 2 Uses a coordinate rule, like add 4 to each number, to slide a point.
 - 3 Knows a slide changes position only, not size, shape, or direction.
 - 4 Decides whether one shape is a slide of another.
-

III.8.20

Reflections

A reflection flips a shape over a line, like a mirror image.

- 1 Flips a shape over the horizontal axis.
 - 2 Flips a shape over the vertical axis.
 - 3 Flips a shape over the diagonal line where x and y are equal.
 - 4 Finds the mirror line between a shape and its reflection.
-

III.8.21

Rotations

A rotation turns a shape around a fixed point, like the hands of a clock.

- 1 Turns a shape a quarter turn about the center of the grid.
- 2 Turns a shape a half turn.
- 3 Turns a shape three quarters of a turn, paying attention to direction.
- 4 Picks the coordinate rule that describes a turn.

Dilations

A dilation enlarges or shrinks a shape by a scale factor, keeping its shape the same.

- 1 Finds new lengths when a shape is enlarged.
- 2 Finds new lengths when a shape is shrunk.
- 3 Finds the scale factor from a point and its new position.
- 4 Knows angles stay the same while area grows by the scale factor times itself.
- ★ Works back from the change in area to the scale factor, remembering area grows by its square.
- ★★ Finds the fixed center point of a dilation from two points and where they end up.

Congruence & similarity informally

Congruent shapes are exact copies; similar shapes are the same shape but possibly a different size.

- 1 Knows slides, flips, and turns keep size the same, and only a dilation changes it.
- 2 Decides whether two rectangles are similar by checking their proportions.
- 3 Follows a sequence of moves to find where a point ends up.
- 4 Decides whether two triangles are similar by checking two of their angles.

Statistics & probability

This branch covers statistics (collecting, summarizing, and picturing data) and probability (how likely something is to happen). These skills help people read surveys, charts, and news claims with a sensible, questioning eye.

III.9.01 Statistical questions

A statistical question is one whose answer varies, so you need to collect data to answer it.

- 1 Tells a statistical question apart from one with a single answer.
- 2 Expects many different answers to a statistical question.
- 3 Picks the best-worded statistical question on a topic.
- 4 Decides what data to collect to answer a question.

III.9.02 Mean

The mean (average) is what each value would be if everything were shared out equally.

- 1 Evens out stacks of cubes to see the mean as a fair share.
- 2 Finds the mean by adding the values and dividing by how many there are.
- 3 Works out the score needed next to reach a target average.
- 4 Sees how one unusual value, an outlier, pulls the mean up or down.
- ★ Works backwards: knows the mean and all but one value, finds the missing value.
- ★★ Works out which number was removed, from the mean before and after.

III.9.03 Median & mode

The median is the middle value when data are in order; the mode is the most common value.

- 1 Orders a list and finds the middle value.
- 2 Finds the median of an even-sized list by averaging the two middle values.
- 3 Finds the mode, and knows a list may have none.
- 4 Decides whether the mean or the median better describes a typical value.
- ★ Finds the missing value that makes the mean and the median of a list equal.
- ★★ Finds the largest number a list could contain, given its mean, median and mode.

III.9.04 Range

The range is the gap between the largest and smallest values, a simple measure of spread.

- 1 Finds the range, including data with negative numbers.
- 2 Finds the range from a dot plot.
- 3 Sees how one extreme value can stretch the range.
- 4 Compares the ranges of two sets of data.

III.9.05

Quartiles & IQR

Quartiles split ordered data into four equal parts; the interquartile range (IQR) is the spread of the middle half.

- 1 Splits ordered data into a lower half and an upper half.
 - 2 Finds the lower and upper quartiles, the middles of each half.
 - 3 Finds the interquartile range by subtracting the lower quartile from the upper.
 - 4 Uses a standard rule based on the IQR to decide which values are outliers.
-

III.9.06

Box plots

A box plot is a chart that shows the spread of data using five key numbers.

- 1 Finds the five key numbers: smallest, lower quartile, median, upper quartile, largest.
 - 2 Picks the box plot that matches five given numbers.
 - 3 Reads values like the median from a box plot.
 - 4 Compares two box plots to decide which statements are true.
-

III.9.07

Dot plots

A dot plot shows each data value as a dot above a number line.

- 1 Picks the dot plot that matches a list of data.
 - 2 Reads the median and range from a dot plot.
 - 3 Describes the shape: where the peak is and which value stands apart.
 - 4 Answers questions from a dot plot, like how many had 4 or more.
-

III.9.08

Histograms

A histogram is a bar chart of data grouped into ranges, like ages 10 to 19, 20 to 29.

- 1 Chooses sensible equal-width groups that cover all the data.
 - 2 Counts how many values fall in each group to set each bar's height.
 - 3 Reads how many values are in a group from the chart.
 - 4 Describes the shape as balanced or lopsided to one side.
-

III.9.09

Mean absolute deviation

Mean absolute deviation measures how spread out data are: the average distance of the values from the mean.

- 1 Finds how far each value is from the mean.
 - 2 Averages those distances to find the mean absolute deviation.
 - 3 Explains what a larger or smaller mean absolute deviation says about consistency.
 - 4 Compares the spread of two sets of data.
-

III.9.10

Sampling

Sampling means surveying a smaller group to learn about a whole population.

- 1 Tells the sample (who is surveyed) from the population (everyone of interest).
- 2 Picks a method that gives a fair, random sample.
- 3 Spots a biased sample, one that leans toward certain answers.
- 4 Knows larger random samples give more trustworthy estimates.

III.9.11

Inferences from samples

Using results from a sample to make estimates about the whole population.

- 1 Finds the fraction of a sample with some feature.
 - 2 Scales a sample result up to estimate a whole population.
 - 3 Combines several samples for a better estimate.
 - 4 Recognizes that samples vary, and draws a conclusion that allows for this.
-

III.9.12

Compare populations

Comparing two groups using their typical values and how spread out they are.

- 1 Compares the medians of two groups.
 - 2 Compares the spread of two groups using box plots.
 - 3 Measures the gap between two means in units of their typical spread.
 - 4 Writes a fair conclusion about which group scored higher.
-

III.9.13

Probability basics

Probability measures how likely something is, from 0 (impossible) to 1 (certain).

- 1 Knows a probability must lie between 0 and 1.
 - 2 Describes an event as impossible, unlikely, likely, or certain.
 - 3 Finds a probability as the number of ways it can happen over the total number of outcomes.
 - 4 Finds the chance something does not happen by subtracting from 1.
 - ★ Works backwards from a probability to find how many marbles of the other color are in a bag.
 - ★★ Works out how many marbles to add to a bag to change the chance of red to a target.
-

III.9.14

Experimental probability

Experimental probability is estimated from what actually happens when you repeat something many times.

- 1 Records the results of repeated trials in a tally.
 - 2 Finds the fraction of trials with a given result.
 - 3 Compares actual results with what theory predicts.
 - 4 Knows that more trials usually give results closer to the true chance.
-

III.9.15

Sample spaces

A sample space is the full list of possible outcomes, such as every result of tossing two coins.

- 1 Picks the complete list of outcomes.
- 2 Uses a table to list all outcomes of two events together.
- 3 Uses a tree diagram, a branching picture, to count outcomes.
- 4 Multiplies the number of choices at each stage to count all possibilities.
- ★ Counts codes where no digit can be used twice, multiplying one fewer choice each time.
- ★★ Counts every three-digit number whose digits add up to a set total.

III.9.16

Compound events

Finding the probability of two things happening, such as rolling a 6 and drawing a vowel card.

- 1 Multiplies probabilities for two events that don't affect each other.
 - 2 Finds a probability by counting in a table of all outcomes.
 - 3 Tells apart the chance of "this and that" from "this or that."
 - 4 Solves word problems, like picking two marbles with replacement.
 - ★ Finds the chance of picking two marbles in a row without putting the first one back.
 - ★★ Finds a chance with two dice, like getting a total that is a multiple of 4, by careful counting.
-

III.9.17

Simulations

A simulation uses coins, dice, or random numbers to act out a situation and estimate a probability.

- 1 Picks a random device that matches a given chance.
 - 2 Plans how random digits will stand for outcomes in each trial.
 - 3 Runs a simulation from a list of random digits and counts results.
 - 4 Estimates a probability from the simulation results.
-

III.9.18

Scatter plots

A scatter plot shows pairs of data as dots, to see whether two things are related.

- 1 Finds the dot that shows a given pair of values.
 - 2 Describes whether the two go up together, go opposite ways, or show no link.
 - 3 Spots clusters and dots that stand apart from the rest.
 - 4 Decides whether the pattern follows a straight line or a curve.
-

III.9.19

Trend lines

A trend line is a straight line drawn through a scatter plot to show the overall pattern and make predictions.

- 1 Picks the line that best fits the dots, by eye.
 - 2 Writes the equation of a trend line.
 - 3 Uses a trend line's equation to make a prediction.
 - 4 Explains what the slope and starting value of a trend line mean.
-

III.9.20

Two-way tables

A two-way table counts people or things sorted two ways at once, such as grade and whether they play an instrument.

- 1 Reads counts and totals from a two-way table.
- 2 Fills in missing counts so rows and columns add up.
- 3 Turns counts into percents within one row or column.
- 4 Decides whether the two ways of sorting are linked.

IV

Era IV • Solve

About ages 14–17 • high-school algebra through precalculus

Era IV is algebra in full. Learners solve every common kind of equation, study functions and their graphs (lines, parabolas, polynomials, exponentials and logarithms), meet complex numbers, and work with sequences, counting, probability and data. About half of its skills also have the two optional challenge levels, Brave and Legend.

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Equations & inequalities

This branch sharpens equation solving for trickier cases: fractions, formulas full of letters, distances (absolute value), and inequalities that describe a whole range of answers. These are the everyday tools of all later algebra.

IV.1.01

Fractions & decimals in equations

Solving equations that contain fractions or decimals by first multiplying through to get rid of them.

- 1 Multiplies every term by a common bottom number so the fractions disappear.
- 2 Multiplies by 10, 100 or 1000 so the decimals become whole numbers.
- 3 Solves the equation once it has been cleaned up.
- 4 Checks the answer by putting it back into the original equation.
- ★ Finds the extra letters that make an equation true for every x , or for no x at all.
- ★★ Solves an equation with a long chain of fractions that collapses neatly once the hidden pattern is seen.

IV.1.02

Literal equations

Rearranging a formula, such as area equals length times width, to get a different letter on its own.

- 1 Rearranges a simple formula to get one letter by itself.
- 2 Treats the other letters just like ordinary numbers while rearranging.
- 3 Rearranges real science formulas, such as turning Fahrenheit into Celsius.
- 4 Notes which values are not allowed, because dividing by zero is impossible.
- ★ Solves for a letter that appears in two places, gathering it together first.
- ★★ Finds the ratio of two letters, like x divided by y , without knowing either letter.

IV.1.03

Absolute value equations

Solving equations with absolute value (the distance of a number from zero, ignoring its sign).

- 1 Reads an absolute value as a distance on the number line.
- 2 Splits the equation into two cases, one positive and one negative, and solves both.
- 3 Gets the absolute value part alone on one side before splitting.
- 4 Recognizes when there is no solution, since a distance can never be negative.
- ★ Solves equations with an absolute value on both sides.
- ★★ Finds every solution when distances are added together or absolute values sit inside each other.

IV.1.04

Compound inequalities

Solving pairs of inequalities joined by "and" or "or", where the answer is a range of numbers rather than one number.

- 1 Treats "and" as keeping only the numbers that satisfy both conditions.
- 2 Treats "or" as keeping every number that satisfies at least one condition.
- 3 Solves inequalities squeezed between two limits, like 12 is at most $3x + 6$, which is at most 24.
- 4 Shows the answer on a number line and writes it as a range in brackets.

Absolute value inequalities

Solving inequalities with absolute value, such as "x is within 8 of the number minus 7", which describe ranges of distance.

- ① Turns "distance less than" into one range sitting between two limits.
- ② Turns "distance greater than" into two separate ranges going outward.
- ③ Solves these inequalities and picks the matching number-line picture.
- ④ Solves tolerance problems, such as how far a bolt may be from its target length.

Functions

A function is a rule that turns each input into exactly one output, like a machine. This branch teaches how to write, read, graph and combine functions, which is the language most of higher math is written in.

IV.2.01

Function notation

Using the shorthand $f(x)$, read "f of x", to mean the output of rule f when the input is x .

- 1 Explains in words what a statement like $f(7) = 8$ means.
- 2 Tells apart finding an output, $f(3)$, from finding the input that gives 3.
- 3 Writes real situations, like the cost of 10 rides, in this shorthand.
- 4 Writes a function rule from a description in words.
- ★ Works backwards from a rule written in a disguised form to find the plain rule.
- ★★ Untangles two linked equations about the same function to find one of its outputs.

IV.2.02

Evaluate functions

Evaluating a function: putting a given input into the rule and working out the output.

- 1 Puts whole numbers into a rule and works out the result.
- 2 Puts in negative numbers and fractions.
- 3 Puts in a whole expression, like $a + 1$, and simplifies what comes out.
- 4 Reads outputs off a graph or a table instead of a formula.
- ★ Uses one known output to find a missing number in the rule, then evaluates.
- ★★ Builds outputs up one step at a time from a rule that refers to the previous output.

IV.2.03

Domain & range

Finding a function's domain (all inputs allowed) and range (all outputs it can produce).

- 1 Lists the inputs and outputs from a set of number pairs.
- 2 Reads the domain and range off a graph.
- 3 Finds inputs to exclude from a formula, such as those causing division by zero or square roots of negatives.
- 4 Writes the domain and range as ranges in bracket form.
- ★ Finds the domain when two restrictions apply at once.
- ★★ Finds the range of a trickier rule, where a clever idea is needed.

IV.2.04

Reading function graphs

Reading information from the graph of a function, such as outputs, solutions and where two graphs meet.

- 1 Reads an output from a graph at a given input.
 - 2 Finds every input on a graph that gives a chosen output.
 - 3 Uses two graphs to find where one lies above the other.
 - 4 Reads the points where two graphs cross.
 - ★ Solves equations where the input has been doubled or shifted, still using the graph.
 - ★★ Applies a graphed function twice in a row and finds every input that lands on a given value.
-

IV.2.05

Functions in context

Understanding what a function means in a real situation, such as cost depending on number of rides.

- 1 Names what the input and the output stand for.
 - 2 Explains a statement like $C(10) = 50$ in plain words.
 - 3 Decides which inputs make sense in the situation, such as no negative time.
 - 4 Gives the units of inputs, outputs and rates, such as dollars per ride.
-

IV.2.06

Average rate of change

Finding the average rate of change: how much the output changes for each one-unit change in input, over a stretch.

- 1 Computes change in output divided by change in input using a formula.
 - 2 Finds the average rate of change from a table of values.
 - 3 Finds it from a graph.
 - 4 Explains what the rate means in context, such as liters drained per minute.
-

IV.2.07

Increasing & decreasing

Identifying where a graph goes up (increasing), goes down (decreasing), or stays flat as you read left to right.

- 1 Spots whether a graph rises or falls over a given stretch.
 - 2 Writes the rising and falling stretches as ranges of x .
 - 3 Finds stretches where the graph stays flat.
 - 4 Links rising to a positive rate of change and falling to a negative one.
-

IV.2.08

Intercepts & zeros

Finding where a graph crosses the axes: the y -intercept, and the zeros (inputs that give an output of zero).

- 1 Finds where the graph crosses the vertical axis by putting in zero.
- 2 Finds the zeros of a simple straight-line rule.
- 3 Reads the zeros and the crossing point off a graph.
- 4 Finds the zeros of a curved rule using algebra.

IV.2.09

Maxima & minima

Finding the highest and lowest points of a graph, either nearby peaks and valleys or the overall top and bottom.

- 1 Tells whether a point is a local peak, a local valley, or neither.
 - 2 Finds the overall highest and lowest values of a whole graph.
 - 3 Reads a minimum or maximum value and where it happens from a graph.
 - 4 Explains what a maximum means in context, such as the best price for profit.
-

IV.2.10

Piecewise functions

Working with piecewise functions, which use different rules for different ranges of input, like tax brackets.

- 1 Chooses the right rule for a given input and evaluates it.
 - 2 Matches each piece to its part of the graph.
 - 3 Reads filled and open dots to tell which piece owns a boundary point.
 - 4 Writes each piece's rule from a graph.
-

IV.2.11

Step functions

Working with step functions, whose graphs look like stairs, such as parking fees charged per started hour.

- 1 Rounds numbers down or up to a whole number using the floor and ceiling rules.
 - 2 Matches a step rule to its staircase graph.
 - 3 Solves postage and parking problems charged per started unit.
 - 4 Works out the value exactly at a jump and just after it.
 - ★ Works backwards from a step output to find all the inputs that give it.
 - ★★ Solves puzzle equations that mix a number with its rounded-down value.
-

IV.2.12

Absolute value function

Working with the absolute value function, whose graph is a V shape.

- 1 Matches V-shaped graphs to their equations.
 - 2 Finds the tip of the V and the steepness of each side.
 - 3 Rewrites the V as two straight-line pieces.
 - 4 Solves equations by seeing where the V meets a flat line.
-

IV.2.13

Combining functions

Combining two functions by adding, subtracting, multiplying or dividing their outputs.

- 1 Adds two functions' outputs at a given input.
- 2 Subtracts one function from another and simplifies.
- 3 Multiplies or divides two functions and simplifies.
- 4 Finds which inputs are allowed after combining, avoiding division by zero.
- ★ Works backwards from a sum and a difference to recover the original functions.
- ★★ Splits a function into a mirror-symmetric part and a flip-symmetric part.

Composition

Composition: feeding the output of one function straight into another, like two machines in a row.

- 1 Applies one function, then the other, to a number.
- 2 Writes the combined rule as a single expression.
- 3 Shows that doing the two functions in the other order usually gives a different result.
- 4 Breaks a complicated rule into an inside function and an outside function.
- ★ Finds a missing number so that both orders give the same result.
- ★★ Applies a function hundreds of times by spotting that the results repeat in a cycle.

Inverse functions

Finding the inverse of a function: the rule that undoes it, taking each output back to its input.

- 1 Describes in words the steps that undo a function, in reverse order.
- 2 Finds the inverse rule by swapping x and y and solving.
- 3 Sees the inverse's graph as a mirror image across the diagonal line $y = x$.
- 4 Checks whether two functions undo each other.

Linear functions

Linear functions have straight-line graphs and describe steady change, like a fixed hourly wage. This branch covers the different ways to write a line's equation, how lines relate to each other, and how to fit a line to real data.

IV.3.01

Point-slope form

Writing a line's equation from one point on it and its slope (steepness), called point-slope form.

- 1 Reads the slope and the point straight from an equation in this form.
- 2 Writes the equation from a given point and slope.
- 3 Finds the slope from two points, then writes the equation.
- 4 Matches an equation in this form to its graph.
- ★ Works backwards to find a missing slope from a point the line passes through.
- ★★ Finds the single point that a whole family of changing lines all pass through.

IV.3.02

Standard form

Writing a line as a number of x plus a number of y equals a total, called standard form, handy for budgets.

- 1 Tidies an equation into standard form with whole numbers.
- 2 Finds where the line crosses each axis.
- 3 Picks the right graph using the two crossing points.
- 4 Solves budget problems, like mixing adult and child tickets for a fixed total.
- ★ Works backwards from the two crossing points to the equation.
- ★★ Counts every whole-number solution, like all ways to hit a total exactly.

IV.3.03

Converting forms

Switching a line's equation between its different forms, each of which shows different information.

- 1 Rearranges standard form into the form showing slope and starting value.
- 2 Rearranges point-slope form into slope and starting value form.
- 3 Rearranges slope and starting value form into standard form with whole numbers.
- 4 Chooses which form suits a given job best.

IV.3.04

Horizontal & vertical lines

Working with perfectly flat (horizontal) and straight-up (vertical) lines, which have special equations.

- 1 Writes the equation of a flat line through a point.
- 2 Writes the equation of a vertical line through a point.
- 3 Knows a flat line has slope zero and a vertical line has no slope at all.
- 4 Explains why a vertical line is not a function but a flat line is.

IV.3.05

Parallel lines

Parallel lines never meet because they have exactly the same slope.

- 1 Gives the slope of any line parallel to a given one.
- 2 Tests whether two equations give parallel lines.
- 3 Writes the equation of a parallel line through a given point.
- 4 Compares slopes to check whether two sides of a shape on a grid are parallel.
- ★ Finds the missing number that makes two lines parallel.
- ★★ Finds when two lines are parallel but not the very same line.

IV.3.06

Perpendicular lines

Perpendicular lines meet at a right angle; their slopes are flipped and have opposite signs, like 3 and minus one third.

- 1 Gives the slope of a line at right angles to a given one.
- 2 Tells whether two lines are parallel, perpendicular or neither.
- 3 Writes a perpendicular line through a given point.
- 4 Finds the line that cuts a segment in half at a right angle.
- ★ Finds the missing number that makes two lines perpendicular.
- ★★ Finds the point on a line closest to a given point, or reflects a point in a line.

IV.3.07

Intercepts of lines

Finding where a line crosses the x-axis and y-axis, and what those points mean.

- 1 Finds where a line crosses the horizontal axis by setting y to zero.
- 2 Finds where a line crosses the vertical axis by setting x to zero.
- 3 Explains the crossing points in context, such as a full tank at time zero.
- 4 Uses the two crossing points to pick the graph.
- ★ Finds the area of the triangle a line makes with the two axes.
- ★★ Finds every line through a given point that makes a triangle of a given area.

IV.3.08

Two-variable inequalities

Graphing inequalities in x and y, where the answer is a whole shaded region on one side of a line.

- 1 Decides whether the boundary line is solid (included) or dashed (not included).
- 2 Tests a point to decide which side to shade.
- 3 Checks whether a given point is a solution.
- 4 Writes the inequality that a shaded graph shows.

IV.3.09

Linear regression & residuals

Fitting a best straight line to real data and measuring how well it fits, using leftover errors called residuals.

- 1 Uses a calculator to find the line of best fit for data.
- 2 Matches a scatter plot to a number (r) that measures how strongly the points line up.
- 3 Computes a residual: actual value minus the value the line predicts.
- 4 Reads a plot of residuals to judge whether a straight line suits the data.

Systems

A system is two or more equations that must be true at the same time, like two facts about the same ticket sale. This branch teaches several ways to find the numbers that satisfy every equation at once, and how to use them to solve real problems.

IV.4.01

What a solution is

Understanding that a solution of a system is a pair of numbers that makes every equation true at once.

- 1 Checks whether a pair of numbers works in both equations.
- 2 Explains that the solution is where the two lines cross on a graph.
- 3 Picks the pair that fits both facts in a word problem, such as ticket counts and total cost.
- 4 Estimates the solution by looking at where two graphed lines cross.

IV.4.02

Solve by graphing

Solving a system by drawing both lines and reading off where they cross.

- 1 Picks the graph that shows both lines correctly.
- 2 Reads the crossing point from the graph.
- 3 Checks the crossing point in both equations to be sure it is exact.
- 4 Sees that graphs only give estimates, and finds the exact answer with algebra.

IV.4.03

Substitution

Solving a system by substitution: getting one letter alone, then putting that expression into the other equation.

- 1 Rearranges one equation to get a letter by itself.
- 2 Replaces that letter in the other equation, leaving just one unknown.
- 3 Solves for one letter, then puts it back to find the other.
- 4 Checks a proposed answer in both equations.
- ★ Clears fractions first, then substitutes.
- ★★ Solves equations with x and y in the bottom of fractions by renaming one over x and one over y .

IV.4.04

Elimination

Solving a system by elimination: adding or subtracting the equations so that one letter cancels out.

- 1 Lines up the equations so matching letters sit in the same column.
- 2 Adds or subtracts the equations to cancel one letter.
- 3 Solves for one letter, then puts it back to find the other.
- 4 Checks the answer in both original equations.
- ★ Tidies messy equations into line before eliminating.
- ★★ Adds and subtracts whole equations to reveal a much simpler pair.

IV.4.05

Elimination with multiplying

Elimination when the letters do not cancel right away, so one or both equations are first multiplied by a number.

- 1 Multiplies one equation so a letter will cancel.
- 2 Multiplies both equations by different numbers so a letter will cancel.
- 3 Cancels a letter and solves the system.
- 4 Multiplies every term, including signs, correctly, avoiding common slips.

IV.4.06

Choosing a method

Choosing the quickest method, graphing, substitution or elimination, for a given system.

- 1 Recognizes when a rough graph is good enough.
- 2 Recognizes when substitution is easiest, because a letter is already nearly alone.
- 3 Recognizes when elimination is easiest, because a letter cancels straight away.
- 4 Gives a reason for the chosen method.
- ★ Finds a combination of the unknowns directly, without solving for each one.
- ★★ Factors a squared equation so the system turns into a simple straight-line one.

IV.4.07

No or infinite solutions

Recognizing systems with no solution (parallel lines) or infinitely many solutions (the same line twice).

- 1 Sees that parallel lines never cross, so there is no solution.
- 2 Sees that two equations for the same line share every point.
- 3 Spots these cases while solving, when the letters all cancel out.
- 4 Spots these cases by comparing slopes and starting values.
- ★ Finds the missing numbers that make a system have no solution.
- ★★ Finds when a system with an unknown throughout has no solution.

IV.4.08

Systems word problems

Turning a word problem into two equations and solving them.

- 1 Chooses two letters to stand for the two unknown amounts.
- 2 Writes two equations from the facts given.
- 3 Solves the system.
- 4 Answers the actual question asked, in words and units.

IV.4.09

Mixture problems

Solving mixture problems, such as combining a weak and a strong salt solution to get a chosen strength.

- 1 Writes an equation for the total amount of liquid.
- 2 Writes an equation for the amount of pure ingredient.
- 3 Solves the two equations together.
- 4 Checks whether an answer makes sense, for example that the mix lies between the two strengths.

IV.4.10

Rate & distance systems

Solving speed and distance problems with two unknowns, such as a boat's speed and a river's current.

- 1 Adds the current's speed going downstream and subtracts it going upstream.
- 2 Finds a plane's own speed and the wind speed from its speeds with and against the wind.
- 3 Sets up equations using distance equals speed times time.
- 4 Solves the system and explains the answer in context.

IV.4.11

Systems of inequalities

Graphing several inequalities together and finding the region where all of them are true.

- 1 Graphs one inequality as a shaded side of a line.
- 2 Finds the region where the shaded areas overlap.
- 3 Tests whether a point lies in every region.
- 4 Writes the inequalities from a graph.
- ★ Finds the values of an unknown that put a point inside the overlap.
- ★★ Counts every whole-number grid point inside the overlap.

IV.4.12

Linear programming basics

Linear programming: finding the best choice, like greatest profit, when there are several limits to respect.

- 1 Writes the limits, such as at least 34 items, as inequalities.
- 2 Identifies which points lie in the allowed region.
- 3 Finds the corner points of the allowed region.
- 4 Tests the corners to find the greatest or smallest value.
- ★ Finds the smallest cost when the region stretches out forever.
- ★★ Works out which goals would make a chosen corner the best one.

IV.4.13

Three-variable systems

Solving three equations with three unknowns, x , y and z , all at once.

- 1 Combines equations to remove one letter, leaving two equations in two unknowns.
- 2 Solves that smaller pair.
- 3 Puts known values back in to find the third unknown.
- 4 Checks a solution in all three equations.
- ★ Solves systems where each equation is missing one of the letters.
- ★★ Adds all three equations to find the total $x + y + z$ in one step.

IV.4.14

Linear-quadratic systems

Solving a system of a straight line and a parabola (the U-shaped graph of a squared equation).

- 1 Picks the graph showing both the line and the parabola.
- 2 Substitutes the line into the parabola to get one squared equation.
- 3 Decides whether the line crosses the curve 0, 1 or 2 times.
- 4 Solves for the crossing points and checks them.
- ★ Finds the value that makes the line just touch the curve.
- ★★ Finds the midpoint of the two crossing points without finding the points themselves.

Matrices for systems

Using a matrix (a grid of numbers) to record a system and solve it by tidying rows step by step.

- ① Writes a system's numbers in a grid called an augmented matrix.
- ② Changes a row by adding a multiple of another row.
- ③ Uses row changes to get zeros below the diagonal, a staircase shape.
- ④ Reads the solution from the finished staircase grid.

Exponents & radicals

This branch works with roots (like square roots) and powers, including fractional powers. Learners simplify them, calculate with them, and solve equations that contain them.

IV.5.01

Simplify radicals

Simplifying a root, like the square root of 192, by pulling whole squares out from under the root sign.

- 1 Pulls perfect-square factors out of a square root, like the square root of 192 becoming 8 root 3.
- 2 Does the same when letters appear under the root.
- 3 Simplifies cube roots by pulling out perfect cubes.
- 4 Simplifies fourth roots and higher.
- ★ Factors numbers first to simplify a root of a product without multiplying it out.
- ★★ Simplifies a root that has another root nested inside it.

IV.5.02

Add & subtract radicals

Adding and subtracting roots, which works only when the roots match, like adding apples to apples.

- 1 Adds and subtracts matching roots, like 8 root 2 plus 6 root 2.
- 2 Simplifies roots first so they match, then combines them.
- 3 Does the same with letters under the roots.
- 4 Recognizes roots that cannot be combined, even after simplifying.

IV.5.03

Multiply radicals

Multiplying expressions that contain roots.

- 1 Multiplies two square roots by multiplying the numbers inside.
- 2 Multiplies a root across a bracket.
- 3 Multiplies two brackets containing roots, term by term.
- 4 Multiplies matching pairs like 8 minus 3 root 6 and 8 plus 3 root 6, where the roots vanish.

IV.5.04

Rationalize denominators

Rewriting a fraction so there is no root in the bottom, called rationalizing the denominator.

- 1 Removes a single square root from the bottom of a fraction.
- 2 Removes a cube root from the bottom.
- 3 Removes a root from a bottom like 5 minus 2 root 2 by using its matching pair.
- 4 Simplifies the result fully.
- ★ Handles bottoms with two roots multiplied by numbers, or two fractions at once.
- ★★ Handles bottoms with cube roots or three terms.

IV.5.05

Rational exponents

Understanding fractional powers, such as a number to the power one half, which means its square root.

- 1 Evaluates a power of one half or one third as a square or cube root.
 - 2 Evaluates powers like four thirds: take the root, then raise to the power.
 - 3 Converts between fractional powers and root signs, both ways.
 - 4 Simplifies using the usual power rules, like adding powers when multiplying.
 - ★ Solves equations where the unknown is raised to a fractional power.
 - ★★ Finds one power from another given power, or decides which of several powers is largest.
-

IV.5.06

Radical equations

Solving equations with the unknown under a square root sign.

- 1 Gets the root alone on one side.
 - 2 Squares both sides to remove the root.
 - 3 Solves the resulting equation.
 - 4 Checks answers, rejecting false ones that squaring can introduce.
 - ★ Solves equations with two roots, squaring more than once.
 - ★★ Solves an equation that is really a squared equation in the square root of x .
-

IV.5.07

Same-base exponent equations

Solving equations where the unknown is in the power, by writing both sides as powers of the same number.

- 1 Rewrites numbers like 32 as a power of 2.
- 2 Sets the powers equal once the bases match.
- 3 Solves harder ones, including fractions and negative powers.
- 4 Checks whether a value really solves the equation.

Polynomials

A polynomial is an expression built from powers of x added together, like $3x$ squared plus $2x$ minus 5 . This branch covers multiplying polynomials out and, in reverse, factoring them (breaking them into pieces that multiply together), a key skill for solving equations later.

IV.6.01

Polynomial vocabulary

Learning the words used to describe polynomials, such as term, coefficient (the number in front) and degree (the highest power).

- 1 Picks out the terms, the numbers in front of each, and the plain number on its own.
- 2 Finds the degree, the highest power of x .
- 3 Writes a polynomial in standard order, highest power first.
- 4 Names a polynomial by its degree and number of terms, such as "quadratic trinomial".

IV.6.02

Add & subtract polynomials

Adding and subtracting polynomials by combining matching terms, like x squared with x squared.

- 1 Adds two polynomials by combining matching terms.
- 2 Subtracts, remembering the minus sign changes every term in the second bracket.
- 3 Lines up matching terms in columns to add or subtract.
- 4 Writes the perimeter of a shape whose sides are expressions.

IV.6.03

Monomial $\times$ polynomial

Multiplying a single term, like $5x$ cubed, across a bracket of several terms.

- 1 Multiplies a number across a bracket.
- 2 Multiplies a term with letters and powers across a bracket, adding the powers.
- 3 Handles a negative term in front, changing every sign inside.
- 4 Writes the area of a rectangle whose sides are expressions.
- ★ Finds the missing term that was multiplied across a bracket.
- ★★ Uses a known equation to evaluate a bigger expression without finding x .

IV.6.04

Multiply binomials

Multiplying two brackets with two terms each, like $(x + 2)$ times $(x + 8)$.

- 1 Uses a box diagram to multiply two brackets.
- 2 Multiplies every term in one bracket by every term in the other.
- 3 Does the same when there are numbers in front of the x .
- 4 Does the same with negative signs.

IV.6.05

Special products

Recognizing special products, patterns like $(a + b)$ squared, that can be multiplied out quickly.

- 1 Squares a sum, like $(x + 8)$ squared, using the pattern.
 - 2 Squares a difference, like $(x \text{ minus } 8)$ squared.
 - 3 Multiplies a sum by a difference, like $(x \text{ minus } 8)(x + 8)$, where the middle terms cancel.
 - 4 Uses the patterns for mental math, such as working out 93 squared.
 - ★ Treats a group of terms as a single block to use a pattern.
 - ★★ Chains patterns together, such as going from x plus one over x to its fourth powers.
-

IV.6.06

Multiply polynomials

Multiplying longer polynomials, where each bracket may have three or more terms.

- 1 Multiplies a two-term bracket by a three-term bracket.
 - 2 Organizes the multiplying in a box diagram.
 - 3 Multiplies three brackets together.
 - 4 Checks a claimed answer by trying a number for x on both sides.
 - ★ Finds one chosen term of the answer without multiplying everything out.
 - ★★ Finds hidden numbers in the brackets from terms of the answer.
-

IV.6.07

GCF factoring

Factoring out the greatest common factor: the biggest piece every term shares.

- 1 Finds the largest factor shared by several terms.
 - 2 Takes it out in front of a bracket.
 - 3 Takes out a negative factor so the bracket starts with a positive term.
 - 4 Takes out a shared bracket, like $(x + 6)$, as the common factor.
-

IV.6.08

Factor by grouping

Factoring by grouping: splitting four terms into pairs, factoring each pair, then pulling out the bracket they share.

- 1 Splits four terms into two sensible pairs.
 - 2 Factors each pair separately.
 - 3 Pulls out the bracket the two pairs share.
 - 4 Regroups the terms differently when the first grouping gets stuck.
 - ★ Groups three terms against one instead of two pairs.
 - ★★ Uses grouping to crack an equation and find all its solutions.
-

IV.6.09

Factor $x^2 + bx + c$

Factoring expressions like x squared minus x minus 42 into two brackets, by finding two numbers that multiply and add correctly.

- 1 Finds two numbers with a given product and a given sum.
- 2 Works out whether those numbers are positive or negative.
- 3 Writes the two brackets.
- 4 Checks the factoring by multiplying the brackets back out.

IV.6.10

Factor $ax^2 + bx + c$

Factoring squared expressions that have a number other than 1 in front of the x squared.

- 1 Multiplies the first and last numbers, then finds two numbers with that product and the right sum.
- 2 Splits the middle term into two parts using those numbers.
- 3 Factors the four terms by grouping.
- 4 Uses guess and check as another route, testing the middle term.

IV.6.11

Difference of squares

Factoring a difference of two squares, such as x squared minus 49, into (x minus 7)(x + 7).

- 1 Recognizes when an expression is one square minus another.
- 2 Factors it into a minus bracket and a plus bracket.
- 3 Does the same with numbers in front, like 64x squared minus 49.
- 4 Keeps factoring when a piece is again a difference of squares.
- ★ Factors when one of the squares is a whole bracket.
- ★★ Uses the pattern to count whole-number solutions of a puzzle.

IV.6.12

Perfect square trinomials

Factoring perfect square trinomials, three-term expressions that are really one bracket squared.

- 1 Recognizes the pattern of a perfect square.
- 2 Factors it into a bracket squared.
- 3 Does the same with numbers in front of the x.
- 4 Chooses the last number to make an expression a perfect square.

IV.6.13

Sum & difference of cubes

Factoring a sum or difference of two cubes, such as x cubed plus 216, using a set pattern.

- 1 Recognizes numbers and terms that are perfect cubes.
- 2 Factors a sum of two cubes using the pattern.
- 3 Factors a difference of two cubes.
- 4 Uses a memory trick for getting the signs right.

IV.6.14

Factor completely

Factoring completely: using every tool in turn until no piece can be broken down further.

- 1 Takes out the common factor first.
- 2 Chooses which pattern fits what is left.
- 3 Keeps going until every piece is fully factored.
- 4 Recognizes expressions that cannot be factored with whole numbers.
- ★ Works through three rounds of factoring.
- ★★ Adds and subtracts a clever term to turn an expression into a difference of squares.

Polynomial division

Dividing one polynomial by another, much like long division with numbers.

- 1 Divides each term by a single term.
- 2 Uses long division to divide by a bracket like $x + 5$.
- 3 Uses synthetic division, a quick shorthand for dividing by x plus or minus a number.
- 4 Writes the answer as quotient and remainder.
- ★ Divides by harder brackets, or finds a missing number to make it divide evenly.
- ★★ Finds missing numbers so one polynomial divides another exactly, without full division.

Polynomial identities

Working with identities: equations that are true for every value of x , and using them as tools.

- 1 Checks whether an equation is true for every x by expanding both sides.
- 2 Checks a famous identity with real numbers.
- 3 Uses that identity to generate whole-number right-triangle sides, like 5, 12, 13.
- 4 Uses identities for mental math, such as 82 squared minus 68 squared.
- ★ Rewrites a polynomial in powers of a bracket like $x + 3$.
- ★★ Uses an identity to do a big calculation with almost no arithmetic.

Quadratics

Quadratics are equations with an x squared in them; their graphs are U-shaped curves called parabolas. This branch covers graphing them, solving them in several ways, and using them for real problems like the path of a thrown ball.

IV.7.01

Parabola features

Identifying the key features of a parabola: its tip (vertex), its mirror line, which way it opens, and where it crosses the axes.

- 1 Finds the vertex, the lowest or highest point of the U.
- 2 Finds the vertical mirror line through the vertex.
- 3 Tells whether the U opens upward or downward.
- 4 Finds where the parabola crosses the x -axis and the y -axis.

IV.7.02

Graph vertex form

Graphing a quadratic written in vertex form, which shows the tip of the U directly.

- 1 Reads the vertex straight from the equation.
- 2 Explains how the number in front makes the U narrower, wider or upside down.
- 3 Uses the mirror line to find matching points on each side.
- 4 Picks the correct sketch of the parabola.
- ★ Finds the vertex when there is a number multiplying x inside the bracket.
- ★★ Finds the lowest vertex among a whole family of changing parabolas.

IV.7.03

Graph standard form

Graphing a quadratic written in standard form, x squared terms first, then x , then a plain number.

- 1 Uses a short formula to find the x -position of the vertex.
- 2 Finds the full vertex point.
- 3 Reads where the curve crosses the y -axis from the plain number.
- 4 Picks the correct sketch.

IV.7.04

Graph factored form

Graphing a quadratic written in factored form, as brackets multiplied together, which shows the x -axis crossings directly.

- 1 Reads where the curve crosses the x -axis from the brackets.
- 2 Finds the vertex halfway between those crossings.
- 3 Tells from the number in front whether the U opens up or down.
- 4 Picks the correct sketch.

IV.7.05

Convert quadratic forms

Switching a quadratic between vertex, standard and factored forms.

- 1 Multiplies out vertex form into standard form.
- 2 Multiplies out factored form into standard form.
- 3 Factors standard form into brackets.
- 4 Knows which form shows the vertex, the crossings or the y-intercept best.

IV.7.06

Solve by square roots

Solving quadratics by getting the squared part alone and taking the square root of both sides.

- 1 Gets x squared alone on one side.
- 2 Takes the square root, remembering both the positive and negative answers.
- 3 Solves when a whole bracket is squared.
- 4 Recognizes when there is no real solution, since no square is negative.

IV.7.07

Solve by factoring

Solving a squared equation (a quadratic) by splitting it into two simpler pieces that multiply to zero.

- 1 Rearranges the equation so one side is zero.
- 2 Factors the quadratic into two brackets.
- 3 Solves by setting each bracket to zero, since a product is zero only if a piece is.
- 4 Checks each answer in the original equation.
- ★ Expands and tidies a messier equation before factoring.
- ★★ Spots a quadratic in disguise, such as an equation in x to the fourth power, and finds all solutions.

IV.7.08

Completing the square

Completing the square: adding the right number to make a perfect squared bracket, which lets any quadratic be solved.

- 1 Finds the number that completes the square: half the x number, squared.
- 2 Adds that number to both sides to keep the equation balanced.
- 3 Does the same when there is a number in front of the x squared.
- 4 Solves a quadratic fully by completing the square.
- ★ Handles odd middle numbers, where fractions appear.
- ★★ Completes two squares at once, in x and in y, to find a smallest value.

IV.7.09

Vertex form by completing

Using completing the square to rewrite a quadratic in vertex form and find its tip.

- 1 Groups the x terms and takes out any number in front.
- 2 Completes the square inside the bracket.
- 3 Adjusts the plain number so the expression stays the same.
- 4 Reads off the vertex.
- ★ Finds a vertex that has fractions in it.
- ★★ Pairs up four brackets cleverly to find the smallest value of a product.

IV.7.10

Quadratic formula

Using the quadratic formula, a recipe that solves any quadratic from its three numbers a , b and c .

- 1 Identifies the three numbers a , b and c in the equation.
 - 2 Puts them into the formula carefully, starting with the part under the root.
 - 3 Solves and simplifies the square root in the answer.
 - 4 Rearranges the equation into the usual form first, then uses the formula.
 - ★ Works backwards from the answers to find the original equation.
 - ★★ Uses the formula when the equation itself contains square roots.
-

IV.7.11

Discriminant

Using the discriminant, the part under the root in the formula, to tell how many solutions a quadratic has without solving.

- 1 Calculates $b^2 - 4ac$.
 - 2 Uses its sign to say whether there are two, one or no real solutions.
 - 3 Tells whether the solutions are whole or fraction numbers, or messy roots.
 - 4 Links the discriminant to how many times the graph crosses the x -axis.
 - ★ Finds the missing number that gives exactly one solution.
 - ★★ Finds when an equation with an unknown throughout has a repeated solution.
-

IV.7.12

Choosing a method

Choosing the quickest way to solve a given quadratic: square roots, factoring or the formula.

- 1 Uses square roots when there is no plain x term.
 - 2 Checks whether an expression factors neatly.
 - 3 Uses the formula, which always works, when nothing else is quick.
 - 4 Gives a reason for the chosen method.
-

IV.7.13

Projectile motion

Using quadratics to model something thrown or launched, whose height rises then falls.

- 1 Uses the height formula to find the height at a given time.
 - 2 Finds the highest point the object reaches.
 - 3 Finds when the object lands, at height zero.
 - 4 Finds the two times when it passes a given height, going up and coming down.
-

IV.7.14

Area & optimization

Using the vertex of a parabola to find the biggest area or best value in real problems.

- 1 Writes an area as a quadratic expression.
- 2 Finds the vertex to locate the best value.
- 3 Finds the largest area or revenue possible, such as for a fenced pen.
- 4 Explains what the vertex means in the situation.

IV.7.15

Quadratic inequalities

Solving quadratic inequalities, finding the range of x where a squared expression is above or below zero.

- 1 Finds where the expression equals zero.
- 2 Tests the sign in each region between those points.
- 3 Reads the answer from the graph of the parabola.
- 4 Writes the answer as ranges in bracket form.
- ★ Rearranges the inequality first so one side is zero.
- ★★ Finds when an expression is negative for at least one x .

IV.7.16

Write a quadratic from features

Writing the equation of a parabola from information about it, such as its tip or crossing points.

- 1 Writes the equation from the vertex and one other point.
- 2 Writes it from the two x -axis crossings and another point.
- 3 Writes it from any three points by solving a system.
- 4 Matches an equation to a graph as a check.
- ★ Writes it from the mirror line and two points.
- ★★ Finds every parabola that just touches the x -axis and passes through two points.

IV.7.17

Quadratic regression

Fitting a quadratic to data, and recognizing quadratic patterns in tables of numbers.

- 1 Tells whether a table grows steadily or follows a curved pattern.
- 2 Finds the differences of the differences, which stay constant for a quadratic.
- 3 Uses a calculator to fit a quadratic to data.
- 4 Judges whether a prediction far beyond the data is sensible.

Complex numbers

Complex numbers add a new number, i , whose square is minus 1, so that every squared equation has solutions. A complex number has an ordinary part and an i part, like $3 + 2i$, and can be pictured as a point on a flat map; this branch covers calculating with them and what they mean geometrically.

IV.8.01

The number i

Meeting the imaginary number i , defined so that i times i equals minus 1.

- 1 Uses the rule that i squared is minus 1 to simplify.
- 2 Explains why i was invented: to solve equations like x squared equals minus 1.
- 3 Sees that powers of i repeat in a cycle of four: i , minus 1, minus i , 1.
- 4 Simplifies a large power of i using that cycle.
- ★ Adds a long list of powers of i , using the fact that each group of four cancels.
- ★★ Multiplies a long chain of powers of i , or adds them with weights.

IV.8.02

Square roots of negatives

Taking square roots of negative numbers by writing them with i , such as the root of minus 100 equals $10i$.

- 1 Writes the square root of a negative number using i .
- 2 Simplifies such roots fully.
- 3 Avoids the trap of combining two negative roots into one before using i .
- 4 Multiplies roots of negatives correctly, converting to i first.

IV.8.03

Standard form $a + bi$

Writing complex numbers in the standard way: an ordinary (real) part plus an imaginary part, like $6 + 6i$.

- 1 Names the real part and the imaginary part of a complex number.
- 2 Uses the fact that two complex numbers are equal only if both parts match.
- 3 Rewrites a complex expression in standard form.
- 4 Classifies numbers as real, purely imaginary, or neither.

IV.8.04

Add & subtract complex

Adding and subtracting complex numbers by combining the real parts and the imaginary parts separately.

- 1 Adds the real parts together.
- 2 Adds both the real and imaginary parts.
- 3 Subtracts, applying the minus sign to both parts.
- 4 Pictures addition as joining arrows on the complex plane.
- ★ Solves a simple equation for an unknown complex number.
- ★★ Finds every possible fourth corner of a parallelogram from three corners.

IV.8.05

Multiply complex

Multiplying complex numbers, using i squared equals minus 1 to tidy the answer.

- 1 Multiplies a complex number by an ordinary number.
 - 2 Multiplies a complex number by i .
 - 3 Multiplies two complex numbers like two brackets, then replaces i squared by minus 1.
 - 4 Squares a complex number.
 - ★ Finds a value that makes a product purely real or purely imaginary.
 - ★★ Finds the square roots of a complex number by matching the two parts.
-

IV.8.06

Conjugates

Using the conjugate of a complex number: the same number with the sign of its i part flipped.

- 1 Writes the conjugate of a complex number.
 - 2 Multiplies a number by its conjugate, getting an ordinary real number.
 - 3 Sees the conjugate as a mirror image across the horizontal axis.
 - 4 Knows that non-real solutions of real equations come in conjugate pairs.
-

IV.8.07

Divide complex

Dividing complex numbers by multiplying top and bottom by the conjugate, so the bottom becomes an ordinary number.

- 1 Multiplies the top of the fraction by the conjugate of the bottom.
 - 2 Simplifies the bottom into an ordinary number.
 - 3 Gives the final answer in standard form.
 - 4 Finds one divided by a complex number.
-

IV.8.08

Complex plane

Picturing complex numbers as points on a flat map, with real numbers across and imaginary numbers up.

- 1 Knows the horizontal axis holds real numbers and the vertical axis holds imaginary ones.
 - 2 Plots a complex number as a point.
 - 3 Reads the complex number at a plotted point.
 - 4 Sees adding a complex number as a move across and up.
-

IV.8.09

Modulus

Finding the modulus of a complex number: its distance from zero on the complex plane.

- 1 Uses the right-triangle rule to find the modulus.
- 2 Sees the modulus as the distance from zero.
- 3 Finds the distance between two complex numbers.
- 4 Finds the point halfway between two complex numbers.
- ★ Finds a missing part from a given distance.
- ★★ Finds the nearest and farthest a point on a circle can be from another point.

IV.8.10

Complex roots of quadratics

Solving quadratics whose answers are complex numbers, when the graph never crosses the x-axis.

- 1 Recognizes a negative value under the root means non-real answers.
- 2 Uses the quadratic formula to find complex answers.
- 3 Sees that the answers come as a conjugate pair.
- 4 Checks a complex answer by putting it back in.

IV.8.11

Fundamental theorem of algebra

The fundamental theorem of algebra: a polynomial with highest power n has exactly n roots, once complex numbers are allowed.

- 1 States how many roots a polynomial has from its highest power.
- 2 Counts repeated roots the right number of times.
- 3 Uses the rule that non-real roots come in conjugate pairs.
- 4 Builds a polynomial from its given roots.

IV.8.12

Polar form

Writing a complex number in polar form, by its distance from zero and its angle, like giving a direction and a distance.

- 1 Finds the angle a complex number makes with the positive real axis.
- 2 Recognizes a number written using its distance and angle.
- 3 Converts a standard complex number into distance-and-angle form.
- 4 Converts distance-and-angle form back to standard form.

IV.8.13

Multiply & divide in polar

Multiplying and dividing complex numbers in polar form, where distances multiply and angles add.

- 1 Multiplies the distances from zero.
- 2 Adds the angles when multiplying.
- 3 Divides the distances and subtracts the angles when dividing.
- 4 Sees multiplying as rotating and stretching.
- ★ Finds the missing number from a product.
- ★★ Adds up many angles to find where a long chain of turns ends up.

IV.8.14

De Moivre's theorem

De Moivre's theorem: raising a complex number to a power multiplies its angle, making big powers easy.

- 1 Raises a number in polar form to a power by multiplying the angle.
- 2 Finds the distance and angle of a large power.
- 3 Converts the answer back into standard form.
- 4 Uses the theorem to find the sine or cosine of double or triple an angle.
- ★ Finds the smallest power that makes a complex number purely real or imaginary.
- ★★ Works out a huge power of z plus one over z using the angle pattern.

IV.8.15

Roots of complex numbers

Finding all the roots of a complex number, such as its three cube roots, which are evenly spread around a circle.

- 1 Knows there are exactly n different n th roots.
- 2 Uses the formula that spaces the roots by equal angles.
- 3 Finds the roots of 1, which lie evenly around a circle.
- 4 Sees the roots as the corners of a regular polygon.
- ★ Finds all the roots from just one of them.
- ★★ Finds the product or sum of all the roots at once.

IV.8.16

Complex loci

Describing shapes on the complex plane, such as circles and lines, using distance and angle conditions on z .

- 1 Recognizes a fixed distance from a point as a circle, and finds its center and radius.
- 2 Recognizes equal distances from two points as a straight line between them.
- 3 Recognizes a fixed angle from a point as a half-line.
- 4 Shades regions such as the inside of a circle.
- ★ Takes out a number first to find the true center and radius.
- ★★ Finds a circle hidden in a condition where one distance is twice another.

Function toolkit

This branch builds a toolkit of basic graph shapes and the moves that change them: sliding, flipping and stretching. Knowing these lets a learner sketch and understand a new function quickly instead of plotting point by point.

IV.9.01

Parent functions

Getting to know the parent functions: the basic graph shapes, like the straight line and the U-shaped parabola, that others are built from.

- 1 Recognizes the straight line, parabola and cubic (S-shaped) graphs.
- 2 Recognizes the square root, V-shaped absolute value and reciprocal (one over x) graphs.
- 3 Recognizes exponential (rapid growth) and logarithm (slow growth) graphs.
- 4 Matches each graph to its equation.
- ★ Recognizes a basic function written in an unfamiliar way.
- ★★ Counts how many times two basic graphs cross by picturing them.

IV.9.02

Vertical shifts

Sliding a graph up or down by adding or subtracting a number to the whole function.

- 1 Sees that adding a number moves every point of the graph up by that amount.
- 2 Tells whether a graph has moved up or down.
- 3 Picks the correct shifted graph.
- 4 Writes the equation of a shifted graph.

IV.9.03

Horizontal shifts

Sliding a graph left or right by changing the input, which moves it the opposite way to the sign.

- 1 Finds the new center of a graph shifted sideways.
- 2 Explains why subtracting inside the bracket moves the graph right.
- 3 Picks the correct shifted graph.
- 4 Writes the equation of a graph shifted sideways.
- ★ Spots a hidden sideways shift in an expanded equation.
- ★★ Shifts a graph until it is symmetric about the vertical axis.

IV.9.04

Reflections

Flipping a graph in the x-axis (upside down) or the y-axis (left to right).

- 1 Flips a point upside down by changing the sign of its height.
- 2 Flips a point left to right by changing the sign of its x.
- 3 Picks the correct flipped graph.
- 4 Identifies which flip an equation shows.

IV.9.05

Vertical stretch & compress

Stretching or squashing a graph vertically by multiplying the whole function by a number.

- 1 Multiplies each point's height by the stretch number.
- 2 Tells a stretch (number above 1) from a squash (number between 0 and 1).
- 3 Tracks how key points like zeros and turning points move.
- 4 Writes the equation of a stretched graph.
- ★ Chooses a stretch so the graph reaches a target value.
- ★★ Finds the points that a vertical stretch leaves in place.

IV.9.06

Horizontal stretch & compress

Stretching or squashing a graph sideways by multiplying the input by a number.

- 1 Divides each point's x by the number, bringing points closer to the axis.
- 2 Sees that multiplying the input by 2 squashes the graph to half width.
- 3 Tracks how zeros and turning points move sideways.
- 4 Writes the equation of a sideways-stretched graph.

IV.9.07

Combined transformations

Combining several moves, like a shift, a flip and a stretch, and doing them in the right order.

- 1 Applies a stretch and a shift to a point, in the correct order.
- 2 Tracks key points through several moves.
- 3 Picks the graph of a fully transformed function.
- 4 Describes a set of moves in words.

IV.9.08

Even & odd functions

Recognizing even functions (mirror-symmetric left to right) and odd functions (unchanged by a half-turn about the center).

- 1 Uses the fact that an even function gives the same output for x and $-x$.
- 2 Uses the fact that an odd function gives opposite outputs for x and $-x$.
- 3 Tests a formula to see whether it is even or odd.
- 4 Recognizes functions that are neither.
- ★ Finds a value that makes a function even or odd.
- ★★ Separates off the odd part of a function to find an unknown output.

IV.9.09

Symmetry of graphs

Using symmetry to find more points on a graph and complete a sketch.

- 1 Finds a mirror point for a graph symmetric about the vertical axis.
- 2 Finds the matching point for a graph symmetric through the center point.
- 3 Finds other mirror lines, such as through the tip of a V.
- 4 Uses symmetry to complete half a graph.
- ★ Finds a mirror point through a center other than the origin.
- ★★ Sees that two mirror lines force the graph to repeat.

IV.9.10

End behavior

Describing end behavior: what a graph does far to the left and far to the right.

- 1 Reads arrow notation describing where a graph heads.
 - 2 Describes the ends of a polynomial from its highest power.
 - 3 Reads the end behavior from a graph.
 - 4 Works out the end behavior from an equation in brackets.
-

IV.9.11

Continuity informally

Understanding continuity informally: whether a graph can be drawn without lifting the pencil.

- 1 Tells whether a graph can be drawn in one stroke.
 - 2 Measures the size of a jump between two pieces.
 - 3 Finds a hole, a single missing point.
 - 4 Finds where a graph breaks at a hole or at an asymptote (a line the graph approaches).
-

IV.9.12

Asymptotes informally

Understanding asymptotes informally: lines a graph gets closer and closer to without reaching.

- 1 Finds a vertical asymptote, where the graph shoots up or down.
 - 2 Finds a horizontal asymptote, the height the graph levels off at.
 - 3 Reads both kinds from a graph.
 - 4 Explains an asymptote in context, such as the lowest average cost per item.
-

IV.9.13

One-to-one & inverses on graphs

Using graphs to see whether a function can be undone (is one-to-one) and to picture its inverse.

- 1 Uses the rule that a flat line may cross a one-to-one graph only once.
 - 2 Picks the inverse's graph as the mirror image across the diagonal line.
 - 3 Swaps a point's coordinates to get a point on the inverse.
 - 4 Swaps the domain and range for the inverse.
 - ★ Finds an inverse value by spotting the input that gives it.
 - ★★ Finds when a function is its own inverse.
-

IV.9.14

Restrict a domain

Restricting the domain: using only part of a graph so the function can be undone.

- 1 Explains why part of a graph must be cut away to allow an inverse.
- 2 Chooses half of a parabola that passes the one-crossing test.
- 3 Finds the inverse on that half.
- 4 Checks that a proposed inverse really undoes the function.

IV.9.15

Composition on graphs & tables

Composing functions, feeding one output into another, using tables and graphs instead of formulas.

- 1 Uses a table to apply one function, then another.
 - 2 Does the same by reading two graphs.
 - 3 Finds which inputs are allowed for the combined function.
 - 4 Uses composition in real situations, such as cost after hours of baking.
 - ★ Works backwards through a composition to find the input.
 - ★★ Applies a function many times by finding the repeating cycle.
-

IV.9.16

Choosing a model

Choosing a model for data: straight-line, quadratic or exponential, by looking at patterns in a table.

- 1 Recognizes equal differences as a straight-line pattern.
 - 2 Recognizes equal second differences as a quadratic pattern.
 - 3 Recognizes equal ratios, repeated multiplying, as an exponential pattern.
 - 4 Chooses a model from a description of a real situation.
-

IV.9.17

Growth races

Comparing growth: seeing that repeated multiplying (exponential) eventually beats any steady or power-based growth.

- 1 Compares a straight-line and an exponential function at one point.
 - 2 Finds when exponential growth finally overtakes a steady one.
 - 3 Compares the two using a table.
 - 4 Compares the two using a graph.
 - ★ Finds the last crossing, after which the exponential stays ahead for good.
 - ★★ Compares two enormous powers without calculating them.
-

IV.9.18

Modulus graphs

Graphing functions with absolute value (distance from zero) applied, which folds parts of a graph over.

- 1 Flips the parts of a graph below the axis to above it.
- 2 Replaces the left half of a graph with a mirror of the right half.
- 3 Solves an equation using the folded graph.
- 4 Combines folding with shifts and other moves.
- ★ Counts solutions for different values of an unknown using the graph.
- ★★ Solves equations with absolute values inside absolute values.

Polynomial functions

This branch studies polynomial functions (expressions built from powers of x , like x cubed minus $4x$) as graphs. Learners find where the graphs cross zero, what shape they take, and how to find every solution of a polynomial equation.

IV.10.01 Degree & leading coefficient

Finding the degree (highest power of x) and the leading coefficient (the number in front of that power) of a polynomial.

- 1 Finds the highest power of x in a polynomial.
- 2 Finds the number in front of that highest power.
- 3 Finds both from a polynomial written as brackets multiplied together.
- 4 Uses the degree to state the most zeros the graph can have.
- ★ Finds both when brackets are raised to powers.
- ★★ Finds both when the biggest terms cancel out on subtraction.

IV.10.02 End behavior of polynomials

Predicting which way a polynomial graph heads at its far ends, from its highest power and the sign in front.

- 1 Knows that an even highest power sends both ends the same way.
- 2 Knows that an odd highest power sends the ends opposite ways.
- 3 Uses the sign of the leading number to decide up or down.
- 4 Matches graphs with the same end behavior.

IV.10.03 Zeros & multiplicity

Finding the zeros of a polynomial and their multiplicity: how many times each zero is repeated.

- 1 Reads the zeros from brackets multiplied together.
- 2 Counts how many times a zero repeats from the power on its bracket.
- 3 Checks the total degree by adding up the repeats.
- 4 Matches a graph to a possible equation using its zeros.

IV.10.04 Behavior at zeros

Predicting how a polynomial graph behaves at each zero: crossing straight through, bouncing back, or flattening as it crosses.

- 1 Knows a graph crosses the axis at a zero that appears an odd number of times.
- 2 Knows a graph touches and bounces back at a zero that appears an even number of times.
- 3 Knows a graph flattens as it crosses at a zero that appears three times.
- 4 Predicts the overall shape and picks the matching graph.
- ★ Factors first to find how the graph behaves at each zero.
- ★★ Works out the zeros of a new graph built by putting x squared into an old one.

IV.10.05

Sketch from factored form

Sketching a polynomial graph from its factored form, using zeros, the y-intercept and the ends.

- 1 Marks where the graph meets the x-axis.
 - 2 Finds where the graph crosses the y-axis.
 - 3 Works out which way each end of the graph heads.
 - 4 Joins the pieces into a correct sketch.
-

IV.10.06

Turning points

Finding turning points: where a graph changes from going up to going down, or the other way.

- 1 States the most turning points possible: one less than the degree.
 - 2 Uses a graphing tool to locate turning points.
 - 3 Finds the local highest and lowest values.
 - 4 Explains a turning point in context, such as the top of a roller-coaster hill.
 - ★ Finds turning points without technology by treating x^2 as one quantity.
 - ★★ Finds when a graph has exactly one turning point instead of three.
-

IV.10.07

Remainder theorem

The remainder theorem: the remainder after dividing by $x - a$ equals the value of the polynomial at a .

- 1 Finds a remainder by putting a number into the polynomial.
 - 2 Uses synthetic division, a quick shorthand, to divide and find the remainder.
 - 3 Uses synthetic division as a quick way to evaluate.
 - 4 Finds a missing number from a given remainder.
-

IV.10.08

Factor theorem

The factor theorem: if a polynomial gives zero at a , then $x - a$ is one of its factors.

- 1 Uses a zero output to name a factor.
 - 2 Tests whether a bracket is a factor by evaluating.
 - 3 Divides out a known factor to find what is left.
 - 4 Factors the rest completely.
-

IV.10.09

Rational root theorem

The rational root theorem: a short list of fractions that are the only possible neat roots of a polynomial.

- 1 Lists the candidate fraction roots from the first and last numbers.
- 2 Tests the candidates to find every root that works.
- 3 Finds a first root of a harder cubic.
- 4 Divides out that root to leave a simpler quadratic.
- ★ Clears fractions in the equation first, then finds the roots.
- ★★ Finds all whole-number values of an unknown that allow a neat root.

IV.10.10

Find all zeros

Finding every zero of a polynomial, including irrational and complex ones, by combining earlier tools.

- 1 Finds the neat roots first.
 - 2 Divides one out and solves the quadratic that is left.
 - 3 Finds irrational and complex zeros as well.
 - 4 Writes the polynomial fully factored.
 - ★ Uses a given root's matching partner (its conjugate) to find the rest.
 - ★★ Solves a symmetric fourth-power equation with a clever substitution.
-

IV.10.11

Write a polynomial from zeros

Building a polynomial when its zeros are given, working backwards from answers to equation.

- 1 Turns each zero into a bracket and multiplies them.
 - 2 Includes repeated zeros the right number of times.
 - 3 Adds the matching partner of an irrational or complex zero.
 - 4 Scales the polynomial so it passes through a given point.
 - ★ Reads zeros and a point from a graph to write its equation.
 - ★★ Finds a polynomial from three of its values using a clever subtraction.
-

IV.10.12

Polynomial inequalities

Solving polynomial inequalities: finding where a polynomial is above or below zero.

- 1 Finds the zeros, where the sign may change.
 - 2 Tests the sign in each region between zeros.
 - 3 Reads the answer from a graph.
 - 4 Writes the answer as ranges in bracket form.
-

IV.10.13

Polynomial models

Using polynomials to model real situations, such as the volume of a box folded from a sheet of card.

- 1 Works out a box's volume for a given corner cut.
 - 2 Uses a calculator to fit a cubic to data.
 - 3 Uses a graphing tool to find the cut giving the biggest box.
 - 4 Decides which cut sizes are physically possible.
-

IV.10.14

Roots & coefficients

Connecting a polynomial's roots to its numbers: for example, the roots of a quadratic add to a known value.

- 1 Finds the sum and product of a quadratic's roots without solving.
- 2 Builds a quadratic from the sum and product of its roots.
- 3 Uses the matching rules for the three roots of a cubic.
- 4 Writes a new equation whose roots are changed versions of old ones.
- ★ Finds symmetric expressions, like the sum of the roots cubed, without solving.
- ★★ Finds a missing number when one root is a multiple of the other.

Rational & radical functions

Rational functions are fractions with polynomials on top and bottom. This branch covers calculating with such algebraic fractions, solving equations that contain them, and graphing them, including the lines their graphs approach but never touch.

IV.11.01 Simplify rational expressions

Simplifying algebraic fractions (rational expressions) by factoring and cancelling, just like reducing $\frac{6}{8}$ to $\frac{3}{4}$.

- 1 Factors the top and bottom completely.
- 2 Cancels factors shared by the top and bottom.
- 3 Lists the x-values that are not allowed because they make the bottom zero.
- 4 Recognizes that a minus b over b minus a equals minus 1.

IV.11.02 Multiply & divide rational expressions

Multiplying and dividing algebraic fractions.

- 1 Factors every part before multiplying.
- 2 Multiplies and cancels shared factors.
- 3 Divides by flipping the second fraction and multiplying.
- 4 Lists all x-values that must be excluded, including from the flipped fraction.
- ★ Works backwards to find a missing fraction in a product.
- ★★ Multiplies a long chain of fractions in which almost everything cancels.

IV.11.03 Add & subtract rational expressions

Adding and subtracting algebraic fractions by giving them a common bottom.

- 1 Adds fractions that already share a bottom.
- 2 Finds the lowest common bottom.
- 3 Rewrites each fraction over the common bottom and combines.
- 4 Simplifies the result fully.
- ★ Works backwards: splits one fraction into two simpler ones.
- ★★ Adds a long patterned list of fractions in which the middle terms cancel.

IV.11.04 Complex fractions

Simplifying complex fractions: fractions that contain smaller fractions inside them.

- 1 Combines the top and the bottom each into one fraction.
- 2 Multiplies top and bottom by a common bottom to clear the small fractions.
- 3 Simplifies the result fully.
- 4 Lists all excluded values, including those from the inner fractions.
- ★ Simplifies a fraction nested three levels deep.
- ★★ Solves an equation built from a fraction that repeats inside itself.

IV.11.05

Rational equations

Solving equations that contain algebraic fractions.

- 1 Finds the common bottom to multiply through by.
 - 2 Clears the fractions and solves.
 - 3 Uses cross-multiplying when one fraction equals another.
 - 4 Checks answers, rejecting any that make a bottom zero.
-

IV.11.06

Extraneous solutions

Spotting extraneous solutions: false answers that appear during solving but do not work in the original equation.

- 1 Explains how squaring both sides can create a false answer.
 - 2 Spots a false answer that makes a fraction's bottom zero.
 - 3 Spots false answers in square-root equations.
 - 4 States the final set of true solutions, or that there are none.
 - ★ Rearranges first, then solves and checks.
 - ★★ Squares twice to solve an equation with two roots, then checks.
-

IV.11.07

Vertical asymptotes

Finding vertical asymptotes: upright lines where a graph shoots up or down because the bottom of the fraction is zero.

- 1 Finds x-values that make the bottom zero.
 - 2 Cancels shared factors first, since those give holes rather than asymptotes.
 - 3 Describes whether the graph shoots up or down near the asymptote.
 - 4 Writes the equations of the asymptotes.
-

IV.11.08

Horizontal asymptotes

Finding horizontal asymptotes: the height a graph levels off at far to the left and right.

- 1 Compares the highest powers on top and bottom.
 - 2 Divides the leading numbers when the highest powers match.
 - 3 Recognizes when the graph levels off at zero.
 - 4 Checks whether a graph crosses its horizontal asymptote.
-

IV.11.09

Holes

Finding holes: single missing points in a graph, caused by a factor that cancels from top and bottom.

- 1 Finds where a hole happens from a cancelled factor.
- 2 Finds both coordinates of the hole.
- 3 Picks the graph showing the hole.
- 4 Tells a hole from an asymptote.

IV.11.10

Graph rational functions

Sketching the graph of a rational function by putting together its key features.

- 1 Finds where the graph crosses each axis.
 - 2 Finds the asymptotes and any holes.
 - 3 Tests whether the graph is above or below the axis in each region.
 - 4 Picks or draws the correct sketch.
 - ★ Builds a function to match given asymptotes and a crossing point.
 - ★★ Solves an inequality with a fraction, using the graph's features.
-

IV.11.11

Slant asymptotes

Finding slant asymptotes: sloping lines a graph approaches when the top's power is one more than the bottom's.

- 1 Recognizes when a slanted asymptote exists.
 - 2 Finds it by dividing the top by the bottom.
 - 3 Reads it from a graph.
 - 4 Tells whether the graph sits above or below it at the ends.
 - ★ Finds where the graph crosses its slanted asymptote.
 - ★★ Finds values of an unknown in top and bottom that move the asymptote through a point.
-

IV.11.12

Inverse variation

Inverse variation: when one quantity goes up, the other goes down so their product stays the same, like speed and travel time.

- 1 Recognizes the equation for inverse variation.
 - 2 Finds the fixed constant from one pair of values.
 - 3 Handles quantities that depend on two others at once.
 - 4 Solves word problems, such as brightness fading with distance.
-

IV.11.13

Root functions

Working with square root and cube root functions and their graphs.

- 1 Knows the basic shapes of the square root and cube root graphs.
 - 2 Shifts, flips and stretches these graphs.
 - 3 Finds the allowed inputs and possible outputs.
 - 4 Uses a square root to undo half of a parabola.
-

IV.11.14

Work & rate problems

Solving work problems, such as how long two people take to finish a job together.

- 1 Treats a rate as the fraction of the job done per hour.
- 2 Adds two workers' rates to get their combined rate.
- 3 Sets up the equation for working together.
- 4 Solves it and answers in context.
- ★ Handles one worker starting alone before the other joins.
- ★★ Solves when one worker's time is a set amount longer than the other's.

Exponential & logarithmic

Exponential functions describe things that grow or shrink by the same percent each step, like savings with interest or a cooling cup of tea. Logarithms are the reverse tool, answering "what power do I need?", and this branch uses both to solve real problems about growth, decay and scales like decibels.

IV.12.01 Exponential growth

Understanding exponential growth: a quantity that is multiplied by the same number each step, like bacteria doubling every hour.

- 1 Works out a growing amount by multiplying step after step.
- 2 Uses a growth formula, starting amount times growth factor to the power x .
- 3 Writes the formula from a starting amount and growth factor.
- 4 Finds the starting amount and growth factor from a table.
- ★ Finds the formula from two points on the curve.
- ★★ Finds a missing value from two others without working out the full formula.

IV.12.02 Exponential decay

Understanding exponential decay: a quantity that shrinks by the same percent each step, like a car losing value.

- 1 Tells growth from decay by whether the multiplier is above or below 1.
- 2 Converts a decay multiplier, like 0.96, into a percent loss, like 4 percent.
- 3 Finds the decay multiplier and percent loss from a table.
- 4 Solves real problems, such as a car's value after several years.

IV.12.03 Graph exponentials

Graphing exponential functions and recognizing their shape: a curve that levels off on one side and shoots up or down on the other.

- 1 Finds key points on an exponential graph.
- 2 Describes the flat line the graph approaches on one side.
- 3 Gives the allowed inputs and possible outputs.
- 4 Tells a growth curve from a decay curve.
- ★ Finds the growth factor from a point on the graph.
- ★★ Counts solutions of an equation by picturing two graphs, without solving.

IV.12.04 Transform exponentials

Shifting, flipping and stretching exponential graphs.

- 1 Describes shifts up, down, left and right.
- 2 Recognizes flips over the x -axis.
- 3 Tracks how a point moves under a stretch and flip.
- 4 Finds the new flat line the graph approaches and the new range.
- ★ Finds three unknown numbers in the formula from three points.
- ★★ Shows that a stretch of an exponential graph is really a sideways shift.

IV.12.05

Percent growth models

Writing growth and decay models from percent changes, like prices rising 3 percent a year.

- 1 Turns a percent increase into a multiplier, such as 3 percent becoming 1.03.
- 2 Turns a percent decrease into a multiplier, such as 3 percent becoming 0.97.
- 3 Writes the full model from a starting amount and a percent rate.
- 4 Converts a weekly rate into the equivalent yearly rate.

IV.12.06

Compound interest

Calculating compound interest, where interest is added to savings regularly and then earns interest itself.

- 1 Uses the compound interest formula to find a final amount.
- 2 Works out the rate per period and the number of periods, such as quarterly.
- 3 Compares two savings offers to see which pays more.
- 4 Finds how long money takes to reach a target, using logarithms.

IV.12.07

The number e

Meeting the number e, about 2.718, which arises when interest is compounded continuously and appears throughout science.

- 1 Sees e emerge by compounding more and more often.
- 2 Uses the continuous growth formula for money.
- 3 Evaluates points on the graph of e to the power x.
- 4 Uses continuous growth for populations and similar settings.
- ★ Finds exact rates and times, such as a doubling time.
- ★★ Finds a limit value that turns out to be a power of e.

IV.12.08

What a logarithm is

Understanding a logarithm as the answer to "what power?": log base 3 of 243 asks what power of 3 gives 243.

- 1 Finds a logarithm by asking which power gives the number.
- 2 Uses the link between a logarithm statement and a power statement.
- 3 Uses common logs (base 10) and natural logs (base e).
- 4 Knows the log of 1 is 0 and the log of the base is 1, and that logs and powers undo each other.

IV.12.09

Convert exponential and log

Switching between power form, such as 2 to the 4 equals 16, and logarithm form, log base 2 of 16 equals 4.

- 1 Rewrites a power statement as a logarithm.
- 2 Rewrites a logarithm as a power statement.
- 3 Does the same with e and the natural log.
- 4 Solves simple log equations by switching forms.

IV.12.10

Evaluate logs

Working out the value of logarithms, by hand when possible and by calculator otherwise.

- 1 Evaluates logs of exact powers by hand.
 - 2 Evaluates logs whose answers are fractions or negative numbers.
 - 3 Estimates a log between two whole numbers.
 - 4 Uses a calculator for other logs.
 - ★ Evaluates logs involving roots and bases that are themselves powers.
 - ★★ Multiplies a long chain of logs that collapses to one simple value.
-

IV.12.11

Log properties

Using the log rules, which turn multiplying into adding, dividing into subtracting, and powers into multiplying.

- 1 Uses the rule that the log of a product is a sum of logs.
 - 2 Uses the rule that the log of a quotient is a difference of logs.
 - 3 Uses the rule that the log of a power brings the power to the front.
 - 4 Expands one log into several, and combines several into one.
-

IV.12.12

Change of base

Changing the base of a logarithm, which lets a calculator work out logs in any base.

- 1 Recognizes the change-of-base formula.
 - 2 Evaluates a log in any base with a calculator.
 - 3 Explains why the formula works using power form.
 - 4 Compares the sizes of logs in different bases.
-

IV.12.13

Graph logs

Graphing logarithmic functions, which rise quickly at first and then ever more slowly.

- 1 Sees the log graph as the mirror image of an exponential graph.
 - 2 Finds the upright line the graph approaches but never reaches.
 - 3 Tracks how points move under shifts and stretches.
 - 4 Finds the allowed inputs, which must keep the inside positive.
 - ★ Finds the base and shift from features of the graph.
 - ★★ Solves an equation mixing a log and a straight line by spotting the single crossing.
-

IV.12.14

Solve exponential equations

Solving exponential equations, where the unknown is in the power.

- 1 Rewrites both sides with the same base when possible.
- 2 Takes logs of both sides to bring the unknown down.
- 3 Gets the power alone before taking logs.
- 4 Gives both an exact answer and a decimal.
- ★ Pulls out a shared power before solving.
- ★★ Solves an equation that is really a quadratic in 3 to the power x.

IV.12.15

Solve log equations

Solving equations that contain logarithms.

- 1 Rewrites a log equation in power form to solve it.
 - 2 Combines logs into one first, then solves.
 - 3 Sets the insides equal when a log equals a log in the same base.
 - 4 Checks answers, rejecting any that make a log's inside zero or negative.
 - ★ Solves when logs have different bases.
 - ★★ Solves an equation that is a quadratic in the log itself.
-

IV.12.16

Half-life & doubling time

Using half-life (time to halve) and doubling time to model decay and growth, such as carbon dating.

- 1 Finds how much remains after a time, given the half-life.
 - 2 Finds how many there are after a time, given the doubling time.
 - 3 Finds the time taken to reach a given amount.
 - 4 Estimates the age of an old object by carbon dating.
-

IV.12.17

Log scales & regression

Using log scales, like decibels, pH and earthquake magnitudes, and fitting exponential models to data.

- 1 Uses the formulas for pH and decibels.
- 2 Compares earthquake sizes, where each step up means ten times the shaking.
- 3 Uses a calculator to fit an exponential model to data.
- 4 Takes logs of data to turn a curve into a straight line.

Sequences & series

A sequence is a list of numbers following a pattern, and a series is what you get by adding them up. This branch covers the main patterns, quick ways to add long lists, and real uses such as savings plans and loans.

IV.13.01 Sequence notation

Using the notation for sequences, where a_n means the n th number in the list.

- 1 Picks out a numbered term from a list.
- 2 Uses a formula to find a given term.
- 3 Tells a sequence that ends from one that goes on forever.
- 4 Writes a formula for the n th term from the first few terms.
- ★ Works backwards to find which term has a given value.
- ★★ Finds a term far down a list with a clever insight instead of listing.

IV.13.02 Arithmetic sequences

Working with arithmetic sequences, which go up or down by the same amount each step, like 3, 7, 11, 15.

- 1 Finds the fixed step, called the common difference.
- 2 Writes a formula for any term, where the step acts like a slope.
- 3 Finds a term far down the list.
- 4 Finds which term has a given value.

IV.13.03 Geometric sequences

Working with geometric sequences, where each term is multiplied by the same number, like 2, 6, 18, 54.

- 1 Finds the fixed multiplier, called the common ratio.
- 2 Writes a formula for any term.
- 3 Finds a term far down the list, including with a negative multiplier.
- 4 Finds which term has a given value, using logarithms if needed.

IV.13.04 Recursive vs explicit

Comparing recursive rules (each term from the one before) with explicit formulas (any term directly from its position).

- 1 Explains a step-by-step rule in words.
- 2 Generates the next few terms from the rule.
- 3 Turns a step-by-step rule into a direct formula.
- 4 Decides which kind of rule suits a given pattern.
- ★ Works backwards through the rule to find the first term.
- ★★ Finds a far term by spotting that the sequence repeats in a cycle.

IV.13.05

Arithmetic series

Adding up an arithmetic sequence quickly, using the trick of pairing first with last.

- 1 Adds 1 up to a large number by pairing terms from each end.
 - 2 Uses the sum formula for any arithmetic list.
 - 3 Finds how many terms or what first term gives a certain total.
 - 4 Solves stacking and seating problems, like seats in a theater.
 - ★ Finds a term and the step from a formula for the running total.
 - ★★ Finds one total from two others that are linked.
-

IV.13.06

Finite geometric series

Adding up a geometric sequence with a set number of terms, using a formula.

- 1 Identifies the first term, multiplier and number of terms for the formula.
 - 2 Adds up the terms by multiplying the sum and subtracting, so the middle terms cancel.
 - 3 Applies the formula, including with a negative multiplier.
 - 4 Finds how many terms give a certain total.
-

IV.13.07

Sigma notation

Using sigma notation, the Greek letter sigma used as shorthand for "add up all of these".

- 1 Reads the shorthand: how many terms, and what the first one is.
 - 2 Writes out and adds the terms.
 - 3 Writes a long sum using the shorthand.
 - 4 Uses rules for sums to evaluate quickly.
 - ★ Handles sums that start at a later term.
 - ★★ Adds a sum in which nearly every term cancels with its neighbor.
-

IV.13.08

Infinite geometric series

Adding up an infinite geometric series, which can have a finite total when the terms shrink fast enough.

- 1 Tells whether an endless sum settles to a total, by checking the multiplier.
 - 2 Uses the formula to find the total.
 - 3 Writes a repeating decimal, like 0.777 and so on, as a fraction.
 - 4 Handles sums starting later, and problems like a bouncing ball.
 - ★ Finds when a sum with an unknown settles, or works backwards from the total.
 - ★★ Finds the first term and multiplier from two different totals.
-

IV.13.09

Convergence informally

Seeing informally whether an endless sum settles to a total (converges) or keeps growing (diverges).

- 1 Works out running totals of the first few terms.
- 2 Decides whether an endless sum settles or grows without limit.
- 3 Sees the surprise that one plus a half plus a third and so on grows forever.
- 4 Uses a table and graph of running totals to see where they head.

IV.13.10

Special recursive sequences

Exploring famous sequences, such as the Fibonacci numbers (each the sum of the two before) and triangular numbers.

- 1 Continues the Fibonacci sequence.
- 2 Finds the golden ratio that the Fibonacci ratios approach.
- 3 Finds triangular and square numbers.
- 4 Spots the pattern and continues a sequence.

IV.13.11

Sums of powers

Using ready-made shortcut formulas to add up whole numbers, their squares or their cubes, without adding every term.

- 1 Adds up 1, 2, 3 and so on to a given number with a shortcut formula.
- 2 Adds up square numbers (a number times itself) with their own formula.
- 3 Adds up cubes (a number times itself three times) with their formula.
- 4 Splits a mixed sum into whole numbers, squares and plain numbers, then combines the formulas.

IV.13.12

Pascal's triangle

Exploring Pascal's triangle, a triangle of numbers where each is the sum of the two above it.

- 1 Builds the next row of the triangle.
- 2 Spots patterns in the rows, such as how many entries each has.
- 3 Links the entries to counting choices, such as choosing 4 of 6 questions.
- 4 Knows each row adds up to a power of 2.
- ★ Uses the hockey-stick pattern to add a diagonal quickly.
- ★★ Finds hidden patterns, such as how many odd numbers a row has.

IV.13.13

Binomial theorem

Using the binomial theorem to multiply out a bracket raised to a power, like $(x - 2)$ to the 5th, without doing it step by step.

- 1 Uses Pascal's triangle to find the numbers in an expansion.
- 2 Multiplies out a bracket raised to a power.
- 3 Finds one particular term without writing them all.
- 4 Handles brackets with negatives and numbers in front.
- ★ Finds the term with no x in it.
- ★★ Uses shortcuts, like finding the total of all the numbers at once.

IV.13.14

Series in money

Using series to handle money over time, such as regular savings, loan payments and pensions.

- 1 Works out a savings balance after regular yearly deposits.
- 2 Tracks a loan as interest is added and payments are made.
- 3 Uses the formula for regular deposits over many years.
- 4 Compares a one-time investment with a regular savings plan.

Counting & probability

This branch teaches clever ways to count possibilities without listing them all, and uses that counting to work out chances (probabilities). These ideas lie behind games, lotteries, insurance and medical test results.

IV.14.01 Factorials & counting

Counting possibilities by multiplying choices, and using factorials: 5 factorial means 5 times 4 times 3 times 2 times 1.

- 1 Multiplies the number of choices at each stage, such as for a lock code.
- 2 Works out a factorial, like 8 factorial.
- 3 Counts the ways to arrange a set of different objects in a row.
- 4 Simplifies a factorial divided by another by cancelling.
- ★ Solves an equation involving factorials for the unknown number.
- ★★ Finds how many zeros end a huge factorial, without calculating it.

IV.14.02 Permutations

Counting permutations: the number of ways to arrange items when order matters, like gold, silver and bronze.

- 1 Counts ways to fill first, second and third places from a group.
- 2 Uses the permutation formula.
- 3 Counts arrangements of letters in a word with repeated letters.
- 4 Counts seatings around a round table, where turning the table changes nothing.
- ★ Adds a restriction, such as a number not starting with zero.
- ★★ Counts arrangements where no two matching letters sit side by side.

IV.14.03 Combinations

Counting combinations: the number of ways to choose items when order does not matter, like pizza toppings.

- 1 Counts ways to choose a group of items from a larger set.
- 2 Uses the combination formula.
- 3 Decides whether order matters, so whether to arrange or just choose.
- 4 Links combinations to the entries of Pascal's triangle.
- ★ Works backwards from a number of combinations to find the group size.
- ★★ Counts something cleverly using combinations, such as a polygon's diagonals.

IV.14.04

Counting with conditions

Counting when there are rules, such as a person who must be included or two who refuse to serve together.

- 1 Counts groups that must include or leave out a certain person.
- 2 Counts "at least one" by subtracting the groups with none from the total.
- 3 Counts orders where two people must stand together.
- 4 Counts committees with rules about how many of each kind.

IV.14.05

Probability by counting

Finding probability by counting: favorable outcomes divided by all equally likely outcomes.

- 1 Finds the chance of an outcome when all outcomes are equally likely.
- 2 Finds chances with cards and dice.
- 3 Uses combinations to count outcomes, such as drawing two marbles at once.
- 4 Works out lottery chances.
- ★ Works backwards from a probability to find how many items there are.
- ★★ Counts outcomes cleverly, such as pairs of numbers with an even sum.

IV.14.06

Addition rule

Using the addition rule to find the chance that one thing or another happens, without counting the overlap twice.

- 1 Finds the chance of one event or another by counting outcomes.
- 2 Adds chances when the two events cannot both happen.
- 3 Subtracts the overlap when both can happen at once.
- 4 Uses Venn diagrams (overlapping circles) to find chances.

IV.14.07

Complements & "at least one"

Using the complement, the chance something does not happen, especially to find the chance of "at least one".

- 1 Finds the chance something does not happen by subtracting from 1.
- 2 Finds the chance of at least one by subtracting the chance of none.
- 3 Works out chances over repeated tries, such as several seeds sprouting.
- 4 Solves the birthday problem: the chance two people share a day.
- ★ Works backwards to find the single-try chance.
- ★★ Uses the complement cleverly, such as for no two heads in a row.

IV.14.08

Independent events

Working with independent events: events where one happening does not change the chance of the other.

- 1 Multiplies chances to find the chance both independent events happen.
- 2 Tests whether two events are independent.
- 3 Finds chances when an item is put back before the next draw.
- 4 Tells independent events from events that cannot happen together.
- ★ Finds a missing chance from other information.
- ★★ Finds who is more likely to win a turn-taking game that could go on forever.

IV.14.09

Conditional probability

Finding conditional probability: the chance of one thing given that another thing is already known to have happened.

- 1 Explains in words what "the chance of A given B" means.
- 2 Reads a conditional chance from a two-way table.
- 3 Uses the formula: chance of both, divided by chance of the given event.
- 4 Finds chances when items are not put back.

IV.14.10

Tree diagrams

Using tree diagrams, branching pictures that track each possible outcome step by step with its chance.

- 1 Multiplies along the branches to find the chance of one path.
- 2 Adds the paths that lead to the same result.
- 3 Uses branches whose chances depend on what happened first.
- 4 Reads a tree backwards to find the chance of the first event given the second.

IV.14.11

Bayes' theorem

Using Bayes' theorem to update a chance after new evidence, such as what a positive medical test really means.

- 1 Uses the formula to reverse a conditional chance.
- 2 Works out the chance someone is ill given a positive test.
- 3 Sees the trap: with a rare disease, most positive tests can be false alarms.
- 4 Uses a tree or a table of counts to do the same calculation.

IV.14.12

Expected value

Finding expected value: the average result per try if something were repeated many times.

- 1 Works out a weighted average of outcomes using their chances.
- 2 Works out the average gain in a game to see if it is fair.
- 3 Applies expected value to insurance and lotteries.
- 4 Chooses between options by comparing expected values.
- ★ Finds missing chances from a given expected value.
- ★★ Uses shortcuts, such as adding the expected values of many small parts.

Data & distributions

This branch covers how to measure the spread of data and how to model chance with standard patterns, like the bell curve. These are the basic tools behind surveys, test scores and scientific studies.

IV.15.01 Standard deviation

Finding the standard deviation: a measure of how spread out data are around the average.

- 1 Finds how far each value is from the average.
- 2 Finds the variance, the average of those distances squared.
- 3 Finds the standard deviation, the square root of the variance.
- 4 Uses the slightly different formula for a sample rather than a whole population.

IV.15.02 Summary statistics

Calculating summary statistics, such as the average and spread, from tables and totals rather than raw lists.

- 1 Finds the mode and other summaries from a frequency table.
- 2 Estimates the average from data grouped into ranges.
- 3 Finds the average and spread from given totals.
- 4 Works out how the average and spread change when every value is scaled or shifted.
- ★ Combines the summaries of two groups into one.
- ★★ Corrects the average and spread after one value turns out to be wrong.

IV.15.03 z-scores

Using z-scores, which say how many standard deviations a value is above or below average.

- 1 Converts a value into a z-score.
- 2 Compares performance across different groups using z-scores.
- 3 Identifies unusual values, more than 2 standard deviations from average.
- 4 Converts a z-score back into the original value.
- ★ Finds the average and spread from two values and their z-scores.
- ★★ Works out how a z-score changes when every score is transformed.

IV.15.04 Random variables

Working with random variables: a number whose value depends on chance, described by a table of values and probabilities.

- 1 Reads chances from a probability table.
- 2 Uses the fact that all the probabilities must add up to 1.
- 3 Finds the expected value, the long-run average.
- 4 Finds the variance, a measure of spread.

IV.15.05

The binomial setting

Recognizing a binomial setting: a fixed number of independent tries, each with the same chance of success.

- 1 Knows the four conditions a binomial setting must meet.
- 2 Identifies the number of tries and the chance of success.
- 3 Decides whether a real situation is binomial.
- 4 Identifies which condition fails when it is not.

IV.15.06

Binomial probabilities

Calculating binomial probabilities: the chance of getting a certain number of successes in a set of tries.

- 1 Uses the formula for exactly a given number of successes.
- 2 Adds chances to find the chance of up to a given number.
- 3 Finds chances of "at least" and "at most" a number.
- 4 Uses a calculator for bigger problems.
- ★ Finds how many tries are needed to reach a target chance.
- ★★ Finds the success chance that makes two counts equally likely.

IV.15.07

Binomial mean & spread

Finding the average and spread of a binomial count, using short formulas.

- 1 Finds the expected count: number of tries times chance of success.
- 2 Finds the standard deviation with its formula.
- 3 Uses them to judge whether a result is unusual.
- 4 Describes the shape of the distribution, balanced or lopsided.
- ★ Works backwards from the average and variance to the tries and chance.
- ★★ Finds the tries and chance from the average of the count squared.

IV.15.08

The normal distribution

Understanding the normal distribution, the familiar bell curve that describes many measurements like heights.

- 1 Recognizes the bell curve and finds its center.
- 2 Uses the rule that about 68, 95 and 99.7 percent lie within 1, 2 and 3 standard deviations.
- 3 Finds the average and standard deviation from the rule.
- 4 Judges whether data are roughly bell-shaped.
- ★ Runs the rule backwards to find the average and spread from percentages.
- ★★ Uses the rule to find a chance given that a value is above average.

IV.15.09

Normal probabilities

Finding probabilities for bell-curve data using z-scores or a calculator.

- 1 Uses z-scores to find the chance of being above or below a value.
- 2 Finds the chance of falling between two values.
- 3 Works backwards from a chance to the value that gives it.
- 4 Uses a calculator for normal chances.

Scatter plots & correlation

Reading scatter plots and correlation: how strongly two measurements move together in a straight-line pattern.

- 1 Describes the direction, shape and strength of a pattern in a scatter plot.
- 2 Matches a scatter plot to its correlation number r , between minus 1 and 1.
- 3 Explains why two things moving together does not prove one causes the other.
- 4 Sees how one far-off point can strengthen or weaken the pattern.



Era V • Prove

About ages 14–17 • high-school geometry and trigonometry

Era V is the era of reasons. This is where proof begins: learners learn how a mathematical argument works, then use short proofs throughout geometry: lines, triangles, polygons, similar shapes and circles. Trigonometry (the math of angles and waves) follows, then coordinate geometry, conic sections, transformations, solids and classic ruler-and-compass constructions.

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Logic & proof

This branch teaches how mathematicians reason: making careful statements, testing them, and building a proof (a step-by-step argument that leaves no room for doubt). These habits underpin all the geometry that follows.

V.1.01

Undefined terms & definitions

Learning the basic words of geometry and what makes a good definition. A few words, like point and line, are left undefined; everything else is built from them.

- 1 Names which basic idea (point, line or plane) an everyday object models, like a pencil tip.
- 2 Names segments and rays from points on a line, such as the ray pointing the opposite way.
- 3 Picks the precise definition, spotting missing details such as 'greater than 1' for prime numbers.
- 4 Checks whether a definition still holds when turned around, and spots ones that fail.

V.1.02

Conditional statements

Working with 'if ... then ...' statements, such as 'If it rains, then the ground gets wet.' These are the building blocks of every proof.

- 1 Picks out the 'if' part (hypothesis) and the 'then' part (conclusion) of a statement.
- 2 Rewrites statements like 'The game is canceled if it rains' in if-then form.
- 3 Decides whether an if-then statement is true, including ones whose 'if' part is false.
- 4 Turns everyday claims, like 'Every student who scores 90 or more gets an A,' into if-then form.

V.1.03

Related conditionals

Flipping and negating an if-then statement to make three related versions, and seeing which ones mean the same thing as the original.

- 1 Picks the converse: the statement with the 'if' and 'then' parts swapped.
- 2 Picks the inverse: the statement with 'not' added to both parts.
- 3 Picks the contrapositive: swapped and with 'not' added to both parts.
- 4 Picks which version always agrees with the original, knowing a statement and its contrapositive always match.

V.1.04

Biconditionals

Working with 'if and only if' statements, which say two things always go together: each one guarantees the other.

- 1 Decides whether an 'if and only if' claim is true by checking it works both ways.
- 2 Finds the second if-then statement hidden inside an 'if and only if' claim.
- 3 Tests each direction of a claim and says which direction, if any, fails.
- 4 Picks a true 'if and only if' statement that could be a definition, or rewrites a definition that way.

V.1.05

Truth tables

Using truth tables: charts that list every true/false combination to show exactly when a combined statement is true.

- 1 Works out 'not', 'and' and 'or' for real statements and fills in table entries.
- 2 Decides when an if-then statement is true, including the surprising rows where the 'if' part is false.
- 3 Builds a full truth table column by column and counts the rows where it comes out true.
- 4 Uses a table to check whether two statements mean exactly the same thing.
- ★ Finds the true-or-false value of each letter so that three linked statements all hold at once.
- ★★ Solves knights-and-knaves puzzles, where some islanders always tell the truth and others always lie.

V.1.06

Inductive vs deductive

Telling apart two kinds of reasoning: spotting a pattern and guessing a rule (inductive), versus reasoning from known facts to a certain conclusion (deductive).

- 1 Spots a pattern in a list of numbers and predicts the next one.
- 2 Finds where a pattern that looked reliable breaks, such as a formula that stops giving prime numbers.
- 3 Reaches a certain answer from a known fact, such as a triangle's angles adding to 180 degrees.
- 4 Classifies an argument as pattern-spotting (inductive) or step-by-step reasoning (deductive).

V.1.07

Counterexamples

Using a counterexample: a single case that shows a general claim is false.

- 1 Decides whether a given number really breaks a claim.
- 2 Finds a number that breaks a false claim.
- 3 Picks the change that turns a false claim into a true one.
- 4 Tells which claims a single example can prove ('there is...') and which it never can ('all...').

V.1.08

Detachment & syllogism

Two basic rules for drawing valid conclusions from if-then statements, and spotting arguments that only look valid.

- 1 Draws the conclusion from a rule plus a fact: late book, so there is a fine.
- 2 Links two if-then statements into one new statement.
- 3 Chains three or more if-then statements, or puts them in order so each leads to the next.
- 4 Decides whether an argument is valid, spotting tempting but faulty reasoning.
- ★ Runs the chain backwards: when the final result is false, works out what else must be false.
- ★★ Untangles a jumbled list of rules about made-up creatures, rewording some so they link into one chain.

V.1.09

Properties of equality

The basic rules that let us work with equal quantities, such as 'anything equals itself' and 'you may add the same amount to both sides.'

- 1 Names which basic rule a statement shows: equals itself, works both ways, or chains along.
- 2 Names the rule behind a step such as adding the same number to both sides.
- 3 Recognizes substitution: swapping in a value known to be equal.
- 4 Uses the same rules for congruence (lengths or angles that match exactly) and spots correctly written statements.

V.1.10

Algebraic proofs

Giving a reason for every step of solving an equation. It connects ordinary algebra to the idea of proof.

- 1 Names the rule that justifies one step of solving an equation.
- 2 Fills in a missing reason in a table of steps and reasons.
- 3 Finds the one wrong step in a proof, including steps that multiply out brackets.
- 4 Supplies a missing step of a proof from the reason written beside it.

V.1.11

Two-column proofs

Writing a two-column proof: a table with statements on the left and the reason for each on the right, leading from what is given to what must be shown.

- 1 Picks out what is given and what must be proved from a stated theorem.
- 2 Plans a proof by working backwards: which line must come just before the last one?
- 3 Fills in a missing reason in a partly written proof about angles.
- 4 Puts the steps of a complete proof in the right order.

V.1.12

Paragraph & flow proofs

Other ways to write a proof: as a flowchart of boxes and arrows, or as an ordinary paragraph.

- 1 Fills in a missing box in a flow proof, a diagram of statements joined by arrows.
- 2 Works out which earlier statements feed into a step when a table proof becomes a flow proof.
- 3 Fills a gap in a paragraph proof or puts its sentences in order.
- 4 Spots a sentence with no valid reason, and tells the proof formats apart.

V.1.13

Indirect proof

Proving something by showing that its opposite leads to an impossibility. This is called indirect proof or proof by contradiction.

- 1 Starts an indirect proof by assuming the opposite, such as 'x is not 8.'
- 2 Follows that assumption to an impossibility, like a triangle with more than 180 degrees.
- 3 Concludes from the impossibility that the original claim must be true.
- 4 Puts in order the steps proving the square root of 2 is not a fraction, or finds a faulty step.

V.1.14

Proof by induction

Proof by induction proves a claim for every whole number, like a row of dominoes: show the first falls, and that each one knocks over the next.

- 1 Names the starting case and what the 'each one leads to the next' step assumes.
- 2 Puts the steps of an induction proof of a sum formula in order, or names the term added.
- 3 Fills in the key rewriting step when proving a number always divides an expression.
- 4 Finds where an 'always bigger' claim starts to hold, which is the starting case for the proof.
- ★ Finds the faulty step in a fake induction proof, such as a starting case that is wrong.
- ★★ Works out exactly which numbers an unusual domino chain reaches, such as only doubling from 1.

Sets & Venn diagrams

Working with sets (collections of things) and Venn diagrams, the overlapping circles that show what groups share.

- 1 Reads set notation and decides whether a number belongs to a set.
- 2 Lists what is in either set, in both sets, or outside a set.
- 3 Uses counts in overlapping circles to answer questions like 'how many do neither?'
- 4 Lists the members of a set described by a rule, such as 'whole numbers from 2 to 8.'

Quantifiers

Understanding claims that begin 'for all' or 'there exists', and how to tell when they are true or how to disprove them.

- 1 Tells whether a sentence is about every case or about at least one case.
- 2 Decides whether such a statement is true for a given list of numbers.
- 3 Writes the opposite of such a statement correctly, swapping 'all' and 'some'.
- 4 Finds the one member of a list that breaks a 'for all' claim.
- ★ Writes the opposite of statements with two such phrases, like 'some x is smaller than every y.'
- ★★ Sees how swapping the order of 'for all' and 'there exists' changes whether a statement is true.

Lines & angles

This branch uses the new proof skills on the simplest shapes: segments, angles and lines. The learner proves facts about angles that were only measured or noticed in earlier eras.

V.2.01

Segments & midpoints

Working with line segments and their midpoints (the point exactly halfway), using basic rules about how lengths add up.

- 1 Finds a whole length from its two parts when a point sits between the ends.
 - 2 Uses a midpoint to find a whole length from half of it.
 - 3 Solves for x when a midpoint splits a segment into two equal expressions, giving reasons for each step.
 - 4 Finds the midpoint of a segment from its endpoints' coordinates on a grid.
-

V.2.02

Angles & bisectors

Working with angles and angle bisectors (lines that split an angle into two equal halves).

- 1 Finds a whole angle from its two parts.
 - 2 Uses a bisector to find a whole angle from half, or puts the steps of a short proof in order.
 - 3 Solves for x when a bisector splits an angle into two equal expressions.
 - 4 Uses the fact that angles along a straight line add to 180 degrees.
 - ★ Finds an angle when a bisector and one extra ray split a larger angle.
 - ★★ Finds the angle between two bisectors in a fan of four rays.
-

V.2.03

Angle pair theorems

Proving facts about pairs of angles, such as why the opposite angles made by two crossing lines are always equal.

- 1 Finds the angles made where two lines cross, using that opposite angles are equal.
 - 2 Uses that two angles adding to 180 degrees with the same angle are equal, then solves for x .
 - 3 Uses the same idea with angles that add to 90 degrees.
 - 4 Puts the steps of a short proof about angle pairs in order.
-

V.2.04

Parallel line theorems

Proving angle facts when a line crosses two parallel lines (lines that never meet), such as which angles are equal.

- 1 Spots which angles match in position when a line crosses two parallel lines.
- 2 Finds x using equal 'Z-shaped' angles, or finds the mistaken step in a proof of that fact.
- 3 Finds x using angles on the same side that add to 180 degrees, or finds a proof's mistake.
- 4 Solves for x using any of these angle pairs.
- ★ Adds a helpful parallel line through an outside point to find an angle.
- ★★ Finds the angle at a corner of a zigzag path between two parallel lines.

V.2.05

Proving lines parallel

Working the other way: using angle measurements to prove that two lines must be parallel.

- ① Finds the value that makes two lines parallel, or decides whether they are.
 - ② Picks which angle fact alone would prove two lines parallel.
 - ③ Puts the steps of a short proof that two lines are parallel in order.
 - ④ Uses the fact that two lines parallel to the same line are parallel to each other.
-

V.2.06

Perpendicular lines

Proving facts about perpendicular lines, which meet at a right angle (a square corner).

- ① Uses right angles at perpendicular lines to find a nearby angle.
- ② Uses that a line perpendicular to one of two parallel lines is perpendicular to both.
- ③ Finds the shortest distance from a point to a straight line on a grid.
- ④ Finds the mistaken step, or the missing reason, in a proof about perpendicular lines.

Triangles

This branch is about triangles, the most important shape in geometry. The learner proves why triangle facts are true and learns the shortcuts for showing two triangles are congruent (exact copies of each other).

V.3.01

Classify triangles

Sorting triangles by their side lengths and by their angles.

- 1 Decides whether a triangle with given side lengths is scalene, isosceles or equilateral.
- 2 Classifies a triangle as acute, right or obtuse from two marked angles.
- 3 Classifies a triangle by its sides and its angles at the same time.
- 4 Does the same for a triangle drawn on a grid, working out side lengths and angles from coordinates.

V.3.02

Triangle sum proof

Proving that the three angles of any triangle always add up to 180 degrees.

- 1 Explains why drawing a line parallel to one side helps prove the angles add to 180 degrees.
- 2 Uses the equal 'Z-shaped' angles beside that parallel line to find a missing angle.
- 3 Puts the steps of the proof in order, or finds the one mistaken step.
- 4 Uses the 180-degree fact to find missing angles, including in right triangles.
- ★ Finds an angle on a bent path between two parallel lines by drawing an extra parallel line.
- ★★ Finds the angle at the fifth point of a star, using that a star's five point angles add to 180 degrees.

V.3.03

Exterior angle theorem

Proving that an outside angle of a triangle equals the two far inside angles added together.

- 1 Picks out the two far inside angles that match an outside angle, and uses them to find x.
- 2 Puts the steps of the proof in order, or finds the one mistaken step.
- 3 Solves for x or finds the size of an outside angle from expressions.
- 4 Decides whether an inside angle could have a given size, knowing it must be smaller than the outside angle.

V.3.04

Congruent figures

Understanding congruent figures, which are exact copies of each other, and how to match their parts correctly.

- 1 Names the matching angle or side in two congruent triangles from the way they are labelled.
- 2 Picks the correct congruence statement from the marks on a diagram.
- 3 Uses the order of the letters to find a matching angle, and spots reorderings that break the match.
- 4 Puts in order the steps proving that if two angles match, the third angles match too.

V.3.05

SSS

The side-side-side rule: if all three sides of one triangle match the three sides of another, the triangles are congruent.

- 1 Uses three matching sides to set up and solve for an unknown length.
 - 2 Finds the third pair of matching sides, often a side the two triangles share.
 - 3 Decides whether two triangles are congruent by the side-side-side rule.
 - 4 Completes a short congruence proof: names a missing reason or finds the one mistaken step.
-

V.3.06

SAS

The side-angle-side rule: two matching sides and the matching angle between them guarantee congruent triangles.

- 1 Names the angle that sits between two given sides.
 - 2 Picks the pair of sides still needed to use the side-angle-side rule.
 - 3 Counts how many different triangles fit two sides and an angle not between them, showing that shortcut fails.
 - 4 Completes a short congruence proof: names a missing reason or finds the one mistaken step.
-

V.3.07

ASA

The angle-side-angle rule: two matching angles and the matching side between them guarantee congruent triangles.

- 1 Names the side that sits between two given angles.
 - 2 Picks the pair of angles still needed to use the angle-side-angle rule.
 - 3 Chooses which congruence rule the marked parts prove.
 - 4 Completes a short congruence proof: names a missing reason or finds the one mistaken step.
-

V.3.08

AAS

The angle-angle-side rule: two matching angles and a matching side not between them also guarantee congruent triangles.

- 1 Picks the extra matching parts that would let the angle-angle-side rule work.
 - 2 Shows why angle-angle-side works by turning it into angle-side-angle: orders steps or finds the mistake.
 - 3 Chooses which congruence rule the marked parts prove.
 - 4 Completes a short congruence proof: names a missing reason or finds the one mistaken step.
-

V.3.09

HL

The hypotenuse-leg rule, a shortcut for right triangles: matching longest sides and one other matching side mean congruent triangles.

- 1 Decides whether the hypotenuse-leg rule can be used, remembering it works for right triangles only.
- 2 Identifies the hypotenuse and picks the extra matching side needed.
- 3 Chooses which congruence rule the marked parts prove.
- 4 Completes a short proof for right triangles: names a missing reason or finds the one mistaken step.

V.3.10

CPCTC proofs

Proving two parts are equal by first proving the triangles containing them are congruent, since matching parts of congruent triangles are equal.

- 1 Says which triangles must be shown congruent first before claiming two parts are equal.
- 2 Names a pair of parts that must match once triangles are congruent, and uses it to solve for x .
- 3 Completes a proof with overlapping triangles that share a side, finding a reason or a mistake.
- 4 Completes a two-stage proof: one pair of congruent triangles leads to a second pair.

V.3.11

Isosceles triangles

Proving facts about isosceles triangles, which have two equal sides, and equilateral triangles, which have three.

- 1 Completes the proof that the angles opposite the equal sides match: names a reason or finds a mistake.
- 2 Uses two equal angles to conclude two sides are equal and find a length.
- 3 Uses the fact that an equilateral triangle has three 60-degree angles to find x .
- 4 Solves for x in an isosceles triangle using its equal sides or equal angles.
- ★ Chains two isosceles triangles, one inside the other, to find a hidden angle.
- ★★ Finds the angles when a triangle is cut into isosceles pieces with several equal lengths, competition style.

V.3.12

Triangle inequality

The triangle inequality: any two sides of a triangle together are longer than the third, so not every three lengths make a triangle.

- 1 Picks a possible third side, knowing any two sides together must be longer than the third.
- 2 Decides whether three given lengths can form a triangle.
- 3 Finds the range of possible lengths for the third side, or counts the whole-number options.
- 4 Orders sides or angles, knowing the biggest angle is opposite the longest side.
- ★ Finds the perimeter of an isosceles triangle when only one arrangement of its sides is possible.
- ★★ Counts every triangle with whole-number sides and a given perimeter.

V.3.13

Midsegment theorem

The midsegment of a triangle joins the midpoints of two sides; it is always parallel to the third side and half as long.

- 1 Names the side that a segment joining two midpoints runs parallel to.
- 2 Finds the length of that segment, which is half the third side.
- 3 Solves for x or for a missing side using the half-as-long fact.
- 4 Explains what a coordinate calculation of slopes and lengths has proved about the midsegment.

Points of concurrency

Special points where three lines drawn inside a triangle all meet at one spot, such as the balance point (centroid).

- 1 Shows that the three perpendicular bisectors meet at one point: orders proof steps or finds a mistake.
- 2 Shows that the three angle bisectors meet at one point: names a reason or finds a mistake.
- 3 Uses the balance point (centroid), which splits each median two-to-one, to find lengths.
- 4 Locates where the three altitudes (heights) meet: inside, on or outside the triangle.
- ★ Finds areas of pieces cut off by the medians.
- ★★ Finds exact distances between the special points of a right triangle.

Polygons

This branch moves from triangles to polygons (flat shapes with straight sides), especially four-sided ones. The learner proves their properties and sorts them into a family tree.

V.4.01

Interior angle sums

Finding the total of the inside angles of any polygon by splitting it into triangles.

- 1 Counts the triangles formed by drawing lines from one corner of a polygon.
- 2 Finds the total of the inside angles of a polygon with any number of sides.
- 3 Finds each inside angle of a regular polygon (one with all sides and angles equal).
- 4 Works backwards from one inside angle to find how many sides a regular polygon has.
- ★ Finds the gap angle where two different regular polygons share a side.
- ★★ Solves puzzles where the angles grow by the same amount each time, finding the largest angle or the number of sides.

V.4.02

Exterior angle sum

Understanding why the outside angles of any polygon always add up to 360 degrees, one full turn.

- 1 Puts the steps of a short proof that outside angles add to 360 degrees in order, or finds a wrong reason.
- 2 Orders the steps of the 'walk around the shape and make one full turn' explanation.
- 3 Finds each outside angle of a regular polygon.
- 4 Works backwards from one outside angle to find the number of sides.

V.4.03

Parallelogram properties

Proving the properties of a parallelogram, a four-sided shape whose opposite sides are parallel.

- 1 Uses equal opposite sides of a parallelogram to find a length or solve for x.
- 2 Uses equal opposite angles to find an angle or solve for x.
- 3 Uses the fact that the diagonals cut each other in half.
- 4 Puts the steps of a short proof of a parallelogram property in order.
- ★ Uses an angle bisector that creates an isosceles triangle inside a parallelogram.
- ★★ Finds a parallelogram's area from two triangles drawn to a point inside it.

V.4.04

Proving parallelograms

Working the other way: using given facts to prove that a four-sided shape must be a parallelogram.

- 1 Finds the value that makes both pairs of opposite sides equal, or decides if given sides prove it.
- 2 Puts in order a short proof that one pair of sides both parallel and equal is enough.
- 3 Completes a proof that diagonals cutting each other in half make a parallelogram: orders steps or finds a wrong reason.
- 4 Finds the missing fourth corner of a parallelogram on a grid.

V.4.05

Rectangles, rhombi, squares

The extra properties of special parallelograms: rectangles, rhombuses (all sides equal) and squares.

- 1 Proves that a rectangle's diagonals are equal: orders the steps or finds a wrong reason.
- 2 Proves that a rhombus's diagonals cross at right angles: orders the steps or finds a wrong reason.
- 3 Decides which shapes are sure to have equal or perpendicular diagonals.
- 4 Solves for x using angle and diagonal facts of rectangles, rhombuses and squares.

V.4.06

Trapezoids & kites

The properties of trapezoids (one pair of parallel sides) and kites (two pairs of equal neighboring sides).

- 1 Uses the equal base angles of an isosceles trapezoid to find angles or pick true facts.
- 2 Uses the midsegment of a trapezoid, which equals the average of the two parallel sides.
- 3 Puts in order a short proof about the equal angles of a kite.
- 4 Solves for x in kites, using their equal angles and diagonals that cross at right angles.

V.4.07

Quadrilateral family tree

Seeing how four-sided shapes are related, like a family tree: every square is a rectangle, but not every rectangle is a square.

- 1 Decides true or false for statements like 'every square is a parallelogram'.
- 2 Fills in 'always', 'sometimes' or 'never' for facts about shapes.
- 3 Picks the single extra fact that turns a parallelogram into a rectangle or rhombus.
- 4 Gives the most specific name for a shape from a list of its facts.

V.4.08

Regular polygon area

Finding the area of a regular polygon using its apothem, the distance from the center straight to the middle of a side.

- 1 Picks out the apothem and the radius in a drawing of a regular polygon.
- 2 Uses area equals half the apothem times the perimeter, forwards or backwards.
- 3 Uses trigonometry (side-and-angle ratios) to find the apothem.
- 4 Finds the area of a polygon with many sides, seeing it come close to a circle's area.
- ★ Finds the exact area of a regular hexagon, using square roots.
- ★★ Finds shaded areas, such as a star or triangle inside a regular hexagon.

V.4.09

Coordinate proofs

Proving facts about shapes by placing them on a grid and using coordinates to measure slopes, lengths and midpoints.

- 1 Chooses smart coordinates for a shape, such as the fourth corner of a square.
- 2 Finds slopes from letter coordinates to show sides are parallel.
- 3 Fills in a missing length in a proof that all four sides are equal.
- 4 Puts in order the steps of a coordinate proof that one segment cuts another in half.

Similarity

This branch is about similar shapes: same shape, different size, like a photo and its enlargement. Similarity lets us measure things we cannot reach, such as the height of a tree.

V.5.01

Proportions in figures

Using proportions (two equal fractions) to find unknown lengths in shapes.

- 1 Picks the proportion that matches sides of two similar triangles to find x .
 - 2 Solves a proportion by cross multiplying.
 - 3 Splits a total in a given ratio, such as triangle angles in the ratio 7 to 4 to 1.
 - 4 Uses the golden ratio, about 1.618, to find the long side of a golden rectangle.
-

V.5.02

Similar polygons

Understanding similar polygons: shapes with the same angles whose sides are all scaled by the same amount.

- 1 Uses matching angles of similar shapes to find a missing angle.
 - 2 Uses side ratios of similar shapes to find a missing length.
 - 3 Picks the correct similarity statement, matching corners in the right order.
 - 4 Decides whether all shapes of one kind, such as all parallelograms, must be similar.
-

V.5.03

Scale factor

The scale factor is the number every length is multiplied by to go from one similar shape to the other.

- 1 Finds the scale factor from one similar triangle to the other.
 - 2 Uses a scale factor to find a missing side.
 - 3 Tells from the scale factor whether a copy is an enlargement or a reduction.
 - 4 Uses map scales, like 1 to 25,000, to find real distances.
-

V.5.04

AA similarity

The angle-angle rule: if two angles of one triangle match two of another, the triangles are similar.

- 1 Decides whether two triangles are similar by checking two pairs of angles.
 - 2 Uses similar triangles to find a missing angle.
 - 3 Finds a length when a line parallel to the base cuts off a smaller triangle.
 - 4 Puts the steps of a short proof that two triangles are similar in order.
 - ★ Spots a hidden pair of similar triangles that share an angle, then finds a length.
 - ★★ Finds where two wires between poles cross, using two hidden pairs of similar triangles.
-

V.5.05

SSS similarity

The side-side-side similarity rule: if all three pairs of sides are in the same ratio, the triangles are similar.

- 1 Picks which three side ratios to compare to test for similarity.
- 2 Decides whether two triangles are similar by comparing all three side ratios.
- 3 Finds the scale factor or a missing side of similar triangles.
- 4 Puts the steps of a short proof in order, or names a missing reason.

V.5.06

SAS similarity

The side-angle-side similarity rule: two sides in the same ratio with an equal angle between them mean similar triangles.

- 1 Picks the missing fact (a ratio or the angle between the sides) needed for this rule.
 - 2 Decides whether the marked parts prove two triangles similar.
 - 3 Solves for x in a triangle with a smaller similar triangle inside it.
 - 4 Completes a short proof: names a missing reason or finds the one mistaken step.
-

V.5.07

Triangle proportionality

A line drawn parallel to one side of a triangle cuts the other two sides into pieces with equal ratios.

- 1 Puts in order the proof that a parallel line splits two sides in equal ratios.
 - 2 Decides from the side lengths whether a segment is parallel to a side.
 - 3 Uses three parallel lines crossing two lines to find a length.
 - 4 Solves for x when a line parallel to one side cuts a triangle.
-

V.5.08

Angle bisector theorem

The angle bisector theorem: a line splitting a triangle's angle in half divides the opposite side in the same ratio as the two nearby sides.

- 1 Decides whether an angle bisector hits the middle of the opposite side, which usually it does not.
 - 2 Finds a missing piece of the opposite side using the ratio of the two nearby sides.
 - 3 Names a missing reason in the proof of the theorem.
 - 4 Finds where an angle bisector meets the opposite side, on a coordinate grid.
-

V.5.09

Similar right triangles

Drawing the height from the right angle of a right triangle creates three similar triangles, which leads to a proof of the Pythagorean theorem.

- 1 Finds matching angles when the height is drawn from the right angle to the longest side.
 - 2 Picks the correct statement naming the three similar triangles.
 - 3 Finds an exact length using the geometric mean (the square root of a product).
 - 4 Proves the Pythagorean theorem this way: puts steps in order or finds the one mistaken step.
 - ★ Combines these length facts with the Pythagorean theorem to find an exact length.
 - ★★ Finds the longest side when only the height and the difference of the two pieces are given.
-

V.5.10

Perimeter & area ratios

When shapes are scaled, perimeters scale by the scale factor but areas scale by the factor multiplied by itself.

- 1 Uses the scale factor to find the perimeter of a similar shape.
- 2 Uses the squared scale factor to find the area of a similar shape.
- 3 Works back from two areas to the scale factor or a matching side.
- 4 Solves everyday problems, such as how much more pizza is in a bigger pizza.
- ★ Finds areas when a line parallel to the base cuts a triangle into a small triangle and a strip.
- ★★ Finds a hidden triangle's area inside a trapezoid from the areas of two others.

V.5.11

Volume ratios

When solids are scaled, volumes grow by the scale factor cubed, so doubling every length makes eight times the volume.

- ① Uses the scale factor cubed to find the volume of a similar solid.
 - ② Uses the scale factor squared to find surface area.
 - ③ Works back from two volumes to the scale factor, a length or the surface-area ratio.
 - ④ Explains why a giant would be too heavy for its bones: weight grows faster than strength.
-

V.5.12

Indirect measurement

Using similar triangles to measure things that are too tall or far away to measure directly.

- ① Finds the height of a tall object by comparing its shadow with a person's shadow.
- ② Finds a height using a mirror on the ground and similar triangles.
- ③ Finds the width of a river using similar triangles marked out on one bank.
- ④ Picks the sensible answer and rules out ones that cannot be right.

Right-triangle trig

This branch introduces trigonometry (the math of triangle measurements) for right triangles. Knowing one angle and one side, the learner can work out the rest, which is how builders, pilots and surveyors find heights and distances.

V.6.01

Naming the sides

Naming the three sides of a right triangle relative to one chosen angle.

- 1 Names the hypotenuse, the longest side, across from the right angle.
- 2 Names the side opposite a given angle.
- 3 Names the side adjacent to (next to) a given angle, other than the hypotenuse.
- 4 Sees that the opposite and adjacent sides swap when the other angle is chosen.

V.6.02

Trig ratios from similarity

Discovering that right triangles with the same angle always have the same side ratios. These ratios are called sine, cosine and tangent.

- 1 Proves that the ratios depend only on the angle, not the triangle's size: orders steps or names a reason.
- 2 Compares measured side ratios to spot which triangle has a different angle.
- 3 Writes sine, cosine or tangent as a fraction of two sides.
- 4 Writes all three ratios exactly for an angle in a right triangle.

V.6.03

Find a side with sine

Using sine (the opposite side divided by the hypotenuse) to find a missing side.

- 1 Decides whether sine is the right ratio to find a side.
- 2 Picks the correct sine equation for the triangle.
- 3 Finds a side by multiplying when the unknown is on top of the fraction.
- 4 Finds a side by dividing when the unknown is on the bottom, including word problems like ramps.

V.6.04

Find a side with cosine

Using cosine (the adjacent side divided by the hypotenuse) to find a missing side.

- 1 Decides whether cosine is the right ratio to find a side.
- 2 Picks the correct cosine equation for the triangle.
- 3 Finds a side by multiplying when the unknown is on top of the fraction.
- 4 Finds a side by dividing when the unknown is on the bottom, such as a ladder's length.

V.6.05

Find a side with tangent

Using tangent (the opposite side divided by the adjacent side) to find a missing side.

- 1 Decides whether tangent is the right ratio to find a side.
- 2 Picks the correct tangent equation for the triangle.
- 3 Finds a side by multiplying when the unknown is on top of the fraction.
- 4 Finds a side by dividing when the unknown is on the bottom, such as a shadow's length.

V.6.06

Find an angle

Finding an unknown angle from two known sides by running the trig ratios backwards.

- 1 Explains what 'inverse sine' means: the angle whose sine is a given number.
- 2 Uses the inverse keys to find an angle from a ratio.
- 3 Chooses the right ratio from two known sides and finds the angle.
- 4 Finds an angle in a real situation, like a leaning ladder, rounded to the nearest degree.

V.6.07

Solve a right triangle

Solving a right triangle means finding every missing side and angle.

- 1 Finds the missing angle of a right triangle.
- 2 Finds every missing side and angle given two sides.
- 3 Finds every missing side and angle given one side and one angle.
- 4 Checks a solution with the Pythagorean theorem and the angle total.
- ★ Splits a triangle into two right triangles with a height line to find a side.
- ★★ Finds exact answers without a calculator using the similar triangles hidden in a right triangle.

V.6.08

45–45–90 triangles

The half-square triangle, with angles 45, 45 and 90 degrees, has sides in a fixed pattern, so no calculator is needed.

- 1 Proves the side pattern of this triangle: orders the steps or finds the wrong step.
- 2 Finds the longest side exactly from a shorter side.
- 3 Finds a shorter side exactly from the longest side.
- 4 Finds the diagonal of a square exactly.

V.6.09

30–60–90 triangles

The half-equilateral triangle, with angles 30, 60 and 90 degrees, has sides in a fixed pattern, so no calculator is needed.

- 1 Proves the side pattern by halving an equilateral triangle, finding the one wrong step.
- 2 Finds the other two sides exactly from the shortest side.
- 3 Finds the shorter sides exactly from the longest or middle side.
- 4 Finds the exact height of an equilateral triangle.
- ★ Splits a triangle with a height line into a 45-degree and a 60-degree special triangle to find a side exactly.
- ★★ Finds an equilateral triangle's side from a point's distances to its three sides.

V.6.10

Elevation & depression

Solving real problems with angles of elevation (looking up) and depression (looking down), such as finding a building's height.

- 1 Identifies an angle of elevation or depression in a picture.
- 2 Finds a height from the angle looking up and the distance along the ground.
- 3 Finds a distance from the angle looking down, such as from a cliff to a boat.
- 4 Finds a height from two angles measured at two spots along the ground.
- ★ Solves two-angle problems exactly using the special 30 and 60 degree triangles.
- ★★ Finds a tower's height from sightings in two different directions, a three-dimensional puzzle.

V.6.11

Cofunctions

The sine of an angle equals the cosine of the other sharp angle in the same right triangle; that is where the 'co' in cosine comes from.

- 1 Finds the angle whose cosine equals the sine of a given angle.
 - 2 Proves why this works using a right triangle: names a reason or orders the steps.
 - 3 Solves for x when a sine and a cosine of two expressions are equal.
 - 4 Picks or finds the matching cosine for a given sine, and the reverse.
-

V.6.12

Pythagorean identity

For any angle, sine squared plus cosine squared always equals 1. It comes straight from the Pythagorean theorem.

- 1 Proves the identity from the Pythagorean theorem: orders the steps or finds the wrong one.
 - 2 Finds the exact cosine of an angle from its sine.
 - 3 Finds the exact tangent of an angle from its sine.
 - 4 Checks whether a given sine and cosine could belong to the same angle.
 - ★ Works backwards from the tangent to find the sine and cosine.
 - ★★ Uses clever algebra on sine and cosine together, competition style.
-

V.6.13

Reciprocal ratios

Three more ratios, cosecant, secant and cotangent, which are simply sine, cosine and tangent turned upside down.

- 1 Recognizes cosecant as the hypotenuse divided by the opposite side.
 - 2 Recognizes secant as the hypotenuse divided by the adjacent side.
 - 3 Recognizes cotangent as the adjacent side divided by the opposite side.
 - 4 Finds all three exactly from a right triangle's sides.
-

V.6.14

Calculator trig

Using a calculator correctly for trigonometry, avoiding the common traps.

- 1 Tells from an answer whether the calculator is set to degrees or radians.
- 2 Uses the inverse keys correctly to find an angle.
- 3 Rounds only at the end of a two-step calculation to avoid creeping errors.
- 4 Gives exact answers with square roots when asked, instead of decimals.

General triangles

This branch extends trigonometry to triangles without a right angle. Two new rules, the law of sines and the law of cosines, let the learner solve any triangle, which is used in navigation and land surveying.

V.7.01

Law of sines

The law of sines: in any triangle, each side divided by the sine of its opposite angle gives the same number.

- 1 Completes the law of sines and pairs each side with its opposite angle.
- 2 Finds a missing side using the law of sines.
- 3 Finds a missing angle using the law of sines.
- 4 Shows where the law comes from by drawing a height line: finds the height or orders the steps.

V.7.02

Ambiguous case

The 'ambiguous case': when two sides and an angle not between them are known, there may be no triangle, one, or two.

- 1 Recognizes the tricky setup (two sides and an angle not between them) and what to compare.
- 2 Decides whether zero, one or two triangles fit the given facts.
- 3 Finds both possible angles when two triangles fit.
- 4 Pictures a side that can swing to two places and finds a length or angle in each triangle.
- ★ Finds the exact range of side lengths that gives exactly two triangles.
- ★★ Finds the sum or product of the two possible lengths at once, without solving each triangle.

V.7.03

Law of cosines

The law of cosines finds sides and angles in any triangle; it is the Pythagorean theorem with a correction for the angle.

- 1 Puts numbers into the law of cosines, or names a reason in its coordinate proof.
- 2 Finds the third side from two sides and the angle between them.
- 3 Finds an angle from all three sides.
- 4 Shows that with a right angle the law becomes the Pythagorean theorem.
- ★ Finds a side next to the known angle, which means solving a squared equation.
- ★★ Uses the law twice in a parallelogram to find the exact length of the second diagonal.

V.7.04

Choosing the law

Deciding which law to use based on what information is given.

- 1 Chooses the law of cosines when given three sides, or two sides and the angle between.
- 2 Chooses the law of sines when given two angles and a side, and uses it.
- 3 Checks which of two possible angles from the law of sines really gives a triangle.
- 4 Puts the steps of a full solution plan in order, or picks the best first step.

V.7.05

Solve any triangle

Finding every missing side and angle of any triangle, step by step.

- 1 Finds the first missing part with the right law.
- 2 Finds the third angle from the other two.
- 3 Finds the last missing side.
- 4 Checks whether three side lengths can make a triangle at all.

V.7.06

Area with sine

Finding a triangle's area from two sides and the angle between them, without knowing its height.

- 1 Finds a triangle's area from two sides and the angle between them.
- 2 Puts the steps of the proof of this area rule in order.
- 3 Works backwards from the area to find a missing side or angle.
- 4 Finds a parallelogram's area the same way, or a side from its area.
- ★ Compares the areas of two triangles that share a corner angle.
- ★★ Finds the area of a large triangle built by stretching each side of a smaller one.

V.7.07

Heron's formula

Heron's formula finds a triangle's area from its three side lengths alone.

- 1 Finds the semi-perimeter, which is half the distance around the triangle.
- 2 Picks the correct form of Heron's formula.
- 3 Finds a triangle's area from its three sides using Heron's formula.
- 4 Uses the area from both methods to find the exact sine of an angle.

V.7.08

Bearings & navigation

Using bearings (compass directions given as angles) and triangle laws to solve navigation problems.

- 1 Converts between a compass bearing (measured clockwise from north) and an ordinary angle.
- 2 Pictures a two-leg route and finds the angle inside the triangle at the turn.
- 3 Finds the straight-line distance back to the start after a two-leg trip.
- 4 Finds the bearing of the end point from the start, or the heading home.

V.7.09

Surveying problems

Using triangle laws to measure land and distances that cannot be walked or reached directly.

- 1 Finds the distance to a tree across a river from two points on this bank.
- 2 Finds a height or distance using angles looking up from two spots.
- 3 Finds the area of a triangular plot from two sides and the angle between them.
- 4 Picks the reasonable answer for a surveying problem, ruling out impossible ones.

Circles

This branch studies circles in depth: the lines, angles and arcs (curved pieces) connected to them, and why they always relate in fixed ways. It ends by describing circles with equations on a grid.

V.8.01

Circle vocabulary

Learning the names for lines and parts connected to a circle.

- 1 Decides true or false about chords, radii and diameters.
- 2 Tells minor arcs, major arcs and semicircles (half circles) apart.
- 3 Describes two circles as sharing a center or as the same size.
- 4 Picks out a radius, chord, diameter or tangent in a diagram.

V.8.02

Central angles & arcs

Measuring arcs of a circle using central angles, the angles whose corner is at the center.

- 1 Uses the fact that an arc and its center angle have the same measure.
- 2 Adds or subtracts neighboring arcs and angles.
- 3 Uses the fact that equal chords cut off equal arcs.
- 4 Solves for x when the angles around the center add to 360 degrees.

V.8.03

Arc length

Finding the length of an arc, the curved distance along part of a circle.

- 1 Finds an arc's length as a fraction of the whole distance around.
- 2 Finds an arc's exact length in terms of π from the radius and angle.
- 3 Works backwards from an arc length to the angle or radius.
- 4 Solves everyday problems, like how far a clock hand's tip travels.

V.8.04

Sector area

Finding the area of a sector, a pizza-slice shaped piece of a circle.

- 1 Finds a sector's area as a fraction of the whole circle, or the reverse.
- 2 Finds a sector's exact area in terms of π .
- 3 Finds the area between a chord and its arc by subtracting a triangle.
- 4 Solves everyday problems, like the area a sprinkler waters.
- ★ Works backwards from a sector's area to find the radius.
- ★★ Finds the exact area of the lens where two quarter circles in a square overlap.

V.8.05

Radians

Radians are a natural way to measure angles by comparing arc length to radius. A full turn is 2 times π radians.

- 1 Explains that one radian is the angle made by an arc as long as the radius.
- 2 Converts between degrees and radians.
- 3 Finds arc length as radius times angle, with the angle in radians.
- 4 Finds a sector's area using the angle in radians.

V.8.06

Inscribed angle theorem

An angle with its corner on the circle is always half the arc it cuts off. This is the inscribed angle theorem.

- 1 Finds an angle with its corner on the circle as half its arc, or the reverse.
 - 2 Identifies which of the three proof cases applies, based on where the center lies.
 - 3 Uses the fact that angles cutting off the same arc are equal.
 - 4 Solves for x using the half-the-arc rule.
 - ★ Adds radii to make an isosceles triangle and finds a hidden angle.
 - ★★ Finds angles between corners of a regular polygon drawn inside a circle.
-

V.8.07

Inscribed angle corollaries

Useful results that follow from the inscribed angle theorem, such as an angle in a half circle always being a right angle.

- 1 Uses the fact that an angle standing on a diameter is a right angle.
 - 2 Uses the fact that opposite angles of a four-sided shape inside a circle add to 180 degrees.
 - 3 Finds unknown angles in such shapes by solving for x .
 - 4 Finds the one wrong step in a proof about these facts.
-

V.8.08

Tangent & radius

A tangent line (one that just touches a circle) is always at right angles to the radius at that point.

- 1 Uses the right angle between a tangent and the radius to find an angle.
 - 2 Uses the Pythagorean theorem to find exact lengths with a tangent.
 - 3 Tests whether a line touches the circle by checking for a right angle.
 - 4 Completes a short proof: names a missing reason or orders the steps.
-

V.8.09

Two tangents from a point

The two tangent lines drawn to a circle from the same outside point are always equal in length.

- 1 Uses equal tangent lengths from one point to find a length.
 - 2 Finds the perimeter of a triangle whose sides all touch a circle inside it.
 - 3 Solves for x using equal tangent lengths.
 - 4 Puts the steps of the proof that the two tangents are equal in order.
 - ★ Finds the radius of the circle that fits inside a right triangle.
 - ★★ Finds the perimeter of a triangle cut off by a third tangent, or works backwards from it.
-

V.8.10

Chord properties

Facts about chords (lines joining two points on a circle) and how they relate to the center.

- 1 Finds the one wrong step in a proof that a line from the center at right angles cuts a chord in half.
- 2 Uses the fact that equal chords are equally far from the center.
- 3 Finds a chord's length from the radius and its distance to the center.
- 4 Orders the steps for finding a circle's center, and explains why they work.

V.8.11

Angles from chords

When two chords cross inside a circle, the angle between them depends on the two arcs they cut off.

- 1 Finds the angle where two chords cross inside a circle.
- 2 Uses the rule that this angle is half the sum of the two arcs.
- 3 Works backwards from the angle to find a missing arc.
- 4 Completes the proof of the rule: names a missing reason or orders the steps.

V.8.12

Angles from secants & tangents

When lines meet outside a circle, or a tangent meets a chord, the angle depends on the arcs in a similar way.

- 1 Finds the angle where two lines cutting the circle meet outside it.
- 2 Uses the half-the-difference rule with a tangent and a cutting line.
- 3 Finds the angle between a tangent and a chord, which is half its arc.
- 4 Solves for x using the half-the-difference rule.
- ★ Uses the full 360 degrees of the circle to find the angle between two tangents.
- ★★ Finds angles when the circle is split into arcs in a given ratio.

V.8.13

Intersecting chord lengths

When two chords cross, the two pieces of one chord multiply to the same answer as the two pieces of the other.

- 1 Finds a missing piece where two chords cross, using equal products of the pieces.
- 2 Picks the correct equation for the pieces.
- 3 Solves for x , including pieces written as expressions.
- 4 Completes the proof using similar triangles: names a missing reason or finds the wrong step.

V.8.14

Secant & tangent lengths

Similar length rules for lines that cut through or touch a circle from an outside point.

- 1 Finds a missing length for two cutting lines (secants) from an outside point.
- 2 Finds a missing length for a tangent and a secant from the same point.
- 3 Solves for x in these setups.
- 4 Completes the proof of the rule: names a missing reason or finds the wrong step.
- ★ Finds the radius when the cutting line passes through the center.
- ★★ Uses the 'power of a point' idea across two circles to find an exact tangent length.

V.8.15

Equation of a circle

Describing a circle with an equation on a grid, based on its center and radius.

- 1 Reads the center and radius from a circle's equation.
- 2 Builds the equation from the Pythagorean theorem, and tests whether a point is inside or outside.
- 3 Writes the equation of a circle from the ends of a diameter.
- 4 Finds tangent lines to a circle on a grid.

V.8.16

General form of a circle

Recognizing a circle's equation when it has been multiplied out, and tidying it back to find the center and radius.

- ① Recognizes which multiplied-out equation is a circle.
- ② Completes the square for both x and y to tidy the equation.
- ③ Reads the center and radius from the tidied equation.
- ④ Finds the circle that passes through three given points.

V.8.17

All circles are similar

All circles are the same shape, just scaled. That is why π , the ratio of distance around to distance across, is the same for every circle.

- ① Finds the scale factor that shrinks or stretches one circle onto another.
- ② Explains why distance around divided by distance across is the same for every circle.
- ③ Traps π between the perimeters of shapes drawn inside and around a circle, as Archimedes did.
- ④ Slices a circle into thin wedges and rearranges them into a near-rectangle to find its area.

Trig functions

This branch turns trigonometry into functions that work for any angle, not just those inside a triangle. Their wave-shaped graphs describe repeating patterns such as tides, Ferris wheels and daylight, and a few skills include short proofs where the learner picks the reason for a step.

V.9.01

Standard position

Drawing an angle on a grid as a turn that starts from the positive x-axis (standard position). This lets angles be any size, even negative or more than a full turn.

- 1 Picks the picture that shows an angle correctly drawn from the positive x-axis.
- 2 Knows positive angles turn counterclockwise and negative ones clockwise, and says where the turn ends.
- 3 Reads angles of more than a full turn from a picture, such as two turns and a bit.
- 4 Names which quarter of the grid (quadrant) an angle ends in, in degrees or radians.

V.9.02

Coterminal angles

Coterminal angles end in the same place, like 30 degrees and 390 degrees, because they differ by whole turns.

- 1 Finds a matching angle by adding or subtracting 360 degrees.
- 2 Does the same in radians (another unit for angles), adding or subtracting 2 times pi.
- 3 Finds the smallest positive angle that ends in the same place, sometimes adding several turns.
- 4 Picks the general rule that lists every angle ending in the same place.

V.9.03

Unit circle definition

Using a circle of radius 1 (the unit circle) to define sine and cosine for any angle. Cosine is a point's left-right position on it and sine is its up-down position.

- 1 Reads cosine as the left-right position and sine as the up-down position of a point.
- 2 Finds tangent by dividing the up-down position by the left-right one.
- 3 Picks the reason for a step in a short proof that this agrees with right-triangle trigonometry.
- 4 Finds the missing coordinate of a point on the circle exactly, using its quadrant.

V.9.04

Special angles

Learning the exact sine, cosine and tangent values for the most common angles all the way around the circle.

- 1 Gives exact values for 30, 45 and 60 degrees.
- 2 Gives the values where the circle crosses the axes, including when tangent is undefined.
- 3 Finds exact values for common angles anywhere around the circle.
- 4 Recalls the common values by heart, judging true-or-false statements in degrees and radians.
- ★ Works out an expression that combines several special angles at once.
- ★★ Adds up a long list of values quickly by pairing terms that combine to make 1.

V.9.05

Reference angles

A reference angle is the sharp angle between an angle's ending side and the x-axis. It lets any angle be worked out from a familiar small one.

- 1 Finds the reference angle for an angle in any quarter of the grid.
- 2 Finds reference angles in radians.
- 3 Uses a reference angle and the quarter to find an exact trig value.
- 4 Handles negative angles and angles of more than a full turn.

V.9.06

Signs by quadrant

Knowing whether sine, cosine and tangent are positive or negative in each quarter of the grid.

- 1 Says whether a given ratio of an angle is positive or negative.
- 2 Uses the memory trick "All Students Take Calculus" to recall the signs.
- 3 Works out which quarter an angle is in from the signs of two ratios.
- 4 Finds the other ratios exactly from just one of them and the quarter.

V.9.07

Trig of any angle

Finding trig values for any angle from a point on its ending side, even if the point is not on the unit circle.

- 1 Finds a ratio from a point's coordinates and its distance from the center.
- 2 Works out that distance first using Pythagoras, then finds a ratio.
- 3 Gives exact answers with square roots rather than decimals.
- 4 Handles points on an axis, where a ratio may be zero or undefined (dividing by zero).

V.9.08

Graph sine & cosine

Drawing the graphs of sine and cosine, which are smooth repeating waves.

- 1 Uses a point moving around the unit circle to tell whether the graph rises or falls.
- 2 Reads key points from the graph: highs, lows and zero crossings.
- 3 Answers questions about one or more full cycles (periods), such as how many peaks.
- 4 Explains that the sine wave is the cosine wave shifted sideways.

V.9.09

Amplitude & period

Changing a wave's height (amplitude) and how long it takes to repeat (period).

- 1 Reads the amplitude from the number in front, knowing a minus sign only flips the wave.
- 2 Finds the period from the number multiplying x inside.
- 3 Reads the amplitude and period from a graph.
- 4 Writes the equation of a wave from its graph.
- ★ Counts how many times a wave reaches a given height in a stretch, including never.
- ★★ Finds how often a sum of two different waves repeats.

V.9.10

Phase & vertical shift

Sliding a wave sideways (phase shift) or up and down (moving the middle line it rises and falls around).

- 1 Reads which way and how far a wave is shifted sideways.
 - 2 Finds the middle line and the highest and lowest values.
 - 3 Handles a wave with all changes combined, tidying the inside to read the shift correctly.
 - 4 Writes the full equation of a wave from its graph.
 - ★ Rewrites a flipped wave as an unflipped one shifted sideways.
 - ★★ Finds every sideways shift that turns one wave exactly into another.
-

V.9.11

Periodic models

Using sine and cosine waves to model real repeating patterns.

- 1 Models the rise and fall of ocean tides from high and low tide times.
 - 2 Models the height of a seat on a Ferris wheel over time.
 - 3 Models hours of daylight through the year.
 - 4 Fits a wave equation to a table of data.
-

V.9.12

Graph tangent

Drawing the graph of tangent, which repeats but shoots off to infinity at regular gaps.

- 1 Lists the asymptotes, the vertical lines the graph approaches but never touches.
 - 2 Finds the period, knowing tangent repeats every π rather than every 2π .
 - 3 Finds key values on the graph, including where it is undefined.
 - 4 Finds the period and asymptotes of a stretched or flipped tangent graph.
-

V.9.13

Reciprocal function graphs

Cosecant, secant and cotangent are 1 divided by sine, cosine and tangent. Their graphs come from the graphs of those three.

- 1 Finds exact cosecant values as 1 divided by sine.
 - 2 Finds exact secant values as 1 divided by cosine.
 - 3 Finds exact cotangent values, even where tangent itself is undefined.
 - 4 Picks the reason for a step in a proof about which values secant can take.
-

V.9.14

Inverse trig functions

Inverse trig functions answer 'which angle has this sine?' To give one answer, each one only returns angles from a set range.

- 1 Explains why the angles must be limited to get a single answer.
- 2 Knows the range of angles each inverse function can give.
- 3 Finds exact values, such as the angle whose sine is one half.
- 4 Picks the correct graph of an inverse function.

Inverse trig compositions

Combining a trig function with its inverse, and seeing when they cancel out and when they do not.

- ① Simplifies cases where the inverse goes first, which simply cancel.
- ② Completes a short proof for the trap where the angle falls outside the allowed range.
- ③ Draws a right triangle to find the exact value of a mixed combination.
- ④ Rewrites such combinations as algebra expressions in x , with no trig left.
- ★ Works out pairs that mix two different functions, such as sine inside an inverse cosine.
- ★★ Finds every x that makes two different inverse functions equal.

Identities & equations

This branch works with identities (trig equations that are true for every angle) and uses them to simplify, to prove other identities, and to solve trig equations. Proofs here mostly mean putting the lines of a short argument in order or picking the reason for a step.

V.10.01

Reciprocal & quotient identities

Cosecant, secant and cotangent are 1 divided by sine, cosine and tangent, and tangent is sine divided by cosine. These rules let any trig expression be rewritten.

- 1 Flips a fraction to get one ratio from its partner, such as cosine from secant.
- 2 Finds tangent or cotangent by dividing sine and cosine.
- 3 Rewrites an expression using only sine and cosine.
- 4 Rewrites and cancels to simplify an expression, often down to a single number.

V.10.02

Pythagorean identities

The Pythagorean identities, such as sine squared plus cosine squared equals 1, come from Pythagoras on the unit circle.

- 1 Finds cosine from sine (or the other way) using sine squared plus cosine squared equals 1, choosing the sign by quarter.
- 2 Uses the version linking tangent and secant to find one from the other.
- 3 Uses the version linking cotangent and cosecant.
- 4 Spots whether a rearranged version of an identity is true or false.

V.10.03

Simplify trig expressions

Making a trig expression shorter using algebra and the basic identities.

- 1 Factors out a shared trig function.
- 2 Combines fractions over a common bottom, then simplifies.
- 3 Replaces part of an expression using a Pythagorean identity, then cancels.
- 4 Simplifies a longer expression down to a single trig function or a number.

V.10.04

Prove identities

Proving that a trig equation is true for every angle. The learner works on one side only, rewriting it step by step until it matches the other side.

- 1 Knows the rules of the game: only one side may be changed.
- 2 Picks the correct first step of rewriting everything in sine and cosine.
- 3 Uses the trick of multiplying by a matching bracket (a conjugate) and simplifies the result.
- 4 Puts the lines of a short proof in the right order.
- ★ Orders the lines of a longer proof that uses several identities.
- ★★ Uses identities to find an exact value, such as sine times cosine from their sum.

V.10.05

Sum & difference formulas

Formulas for the sine, cosine and tangent of two angles added or subtracted. They are not as simple as adding the separate values.

- 1 Picks the correct formula for the cosine of a sum or difference.
 - 2 Picks the correct formula for the sine of a sum or difference.
 - 3 Picks the reason for a step in building the tangent formula from the other two.
 - 4 Finds exact values for angles like 15 or 75 degrees by splitting them into familiar angles.
 - ★ Spots a formula written out in full and runs it backwards to a single value.
 - ★★ Finds the size of a sum of two angles from their tangents alone.
-

V.10.06

Double-angle formulas

Formulas for the sine, cosine and tangent of double an angle.

- 1 Uses the double-angle formula for sine, finding the missing ratio first.
 - 2 Chooses the most convenient of the three forms for the cosine of double an angle.
 - 3 Uses the double-angle formula for tangent.
 - 4 Finds a double-angle value from just one ratio and the quarter.
-

V.10.07

Half-angle formulas

Formulas for the trig values of half an angle, built from the double-angle formulas.

- 1 Shows where a half-angle formula comes from by substituting into a double-angle one.
 - 2 Decides whether a half-angle value is positive or negative from its quarter.
 - 3 Finds exact values, such as for 22.5 or 67.5 degrees.
 - 4 Rewrites a squared sine or cosine without the square, using double the angle.
-

V.10.08

Harmonic form

Any mix of a sine and a cosine wave of the same angle is really one wave, shifted sideways and stretched. Writing it that way makes it easy to study.

- 1 Finds the height and shift that turn a cosine-plus-sine into one cosine wave.
 - 2 Writes the same kind of mix as a single shifted sine wave.
 - 3 Uses the single-wave form to find the largest and smallest values.
 - 4 Uses the single-wave form to solve an equation.
-

V.10.09

Basic trig equations

Solving simple trig equations, such as finding the angles whose sine is one half.

- 1 Rearranges the equation to get the trig function by itself.
- 2 Finds the reference angle (the sharp angle to the x-axis) that fits.
- 3 Works out which quarters of the circle the solutions lie in.
- 4 Finds the second solution in a full turn from the first.

V.10.10

Intervals & general solutions

Finding all solutions to a trig equation, either within a stretch of angles or as a general rule covering every turn.

- 1 Lists every solution within one full turn.
- 2 Finds solutions in other stretches by adding whole turns.
- 3 Solves inequalities, finding the stretches of angles where a trig value is above or below a number.
- 4 Counts how many solutions there are in a longer stretch.

V.10.11

Equations by factoring & identities

Solving harder trig equations by factoring or by swapping in an identity first.

- 1 Treats an equation as a quadratic in sine or cosine to find their possible values.
- 2 Factors out a common function instead of dividing, so no solutions are lost.
- 3 Swaps in an identity so the equation uses just one function.
- 4 Rejects impossible values, such as a cosine bigger than 1.
- ★ Finds a hidden common factor in equations that mix functions.
- ★★ Squares both sides to solve, then checks for false solutions that squaring creates.

V.10.12

Multiple-angle equations

Solving equations with a multiple of the angle, like the sine of $2x$, which have more solutions than usual.

- 1 Solves equations with double the angle within a stretch.
- 2 Widens the range first, listing solutions across several turns.
- 3 Divides back to get the final angles after solving for the multiple.
- 4 Checks the number of solutions, using a graph.

Coordinate geometry

This branch uses a coordinate grid to measure lengths, find midpoints and slopes, and prove facts about shapes using algebra. Several skills include short proofs, where the learner puts the steps in order or picks the reason for a step.

V.11.01

Distance formula

Finding the straight-line distance between two points on a grid. The distance formula is Pythagoras in disguise.

- ① Puts the steps of a proof in order, showing the distance comes from a right triangle.
- ② Finds the distance between two points with the formula.
- ③ Works backwards from a known distance to find a missing coordinate, often getting two answers.
- ④ Gives exact answers as simplified square roots rather than decimals.

V.11.02

Midpoint formula

Finding the point exactly halfway between two points by averaging their coordinates.

- ① Picks the midpoint by averaging the x values and the y values.
- ② Calculates the midpoint of a segment.
- ③ Finds a missing endpoint when the midpoint and the other end are known.
- ④ Finds the center and radius of a circle from the two ends of a diameter.

V.11.03

Partition a segment

Finding a point that splits a segment in a given ratio, such as one third of the way along.

- ① Splits a length in a ratio such as 5 to 7.
- ② Finds the point a given fraction of the way from one end to the other.
- ③ Finds the point that splits a segment in a given ratio.
- ④ Checks a ratio by counting grid steps on a graph.

V.11.04

Slope criteria

Using slope (steepness) to test lines: parallel lines have equal slopes, and perpendicular slopes multiply to negative 1.

- ① Uses slopes to decide which sides of a four-sided shape are parallel.
- ② Finds slopes to show two sides meet at a right angle.
- ③ Finds a missing corner of a parallelogram or square.
- ④ Tests every pair of sides to find which angle of a triangle, if any, is a right angle.

V.11.05

Special lines

Writing equations for special lines in a triangle and finding where they cross.

- 1 Writes the equation of a perpendicular bisector (the line cutting a segment in half at a right angle).
- 2 Writes the equation of an altitude (a line from a corner meeting the opposite side at a right angle).
- 3 Writes the equation of a median (a line from a corner to the middle of the opposite side).
- 4 Finds the point where all three altitudes meet by solving two equations together.
- ★ Finds a point that is the same distance from two or three given points.
- ★★ Spots a shortcut, such as a hidden right angle, that gives a meeting point without long working.

V.11.06

Perimeter & area on the plane

Finding the perimeter and area of shapes drawn on a grid from their corner points.

- 1 Adds up side lengths for the perimeter, keeping exact square roots.
- 2 Finds an area by boxing the shape and subtracting the corner triangles.
- 3 Uses the shoelace formula, a shortcut that gives area from the corner coordinates.
- 4 Spots a mistake from listing the corners out of order, and finds the true area.
- ★ Works backwards from the area to find a missing coordinate.
- ★★ Counts the whole-number grid points inside a shape using area and a clever formula.

V.11.07

Point-to-line distance

Finding the shortest distance from a point to a line, which is measured along a perpendicular.

- 1 Writes the equation of the perpendicular line from the point.
- 2 Finds where the perpendicular meets the line, then measures the distance.
- 3 Uses a formula for the distance directly.
- 4 Finds the distance between two parallel lines.
- ★ Works backwards from a known distance to find a missing coordinate.
- ★★ Uses the gap between parallel lines to find the size of a square or circle that fits between them.

V.11.08

Classify by coordinates

Using lengths and slopes to decide exactly what kind of triangle or four-sided shape a set of points makes.

- 1 Classifies a triangle by its side lengths and whether it has a right angle.
- 2 Tests whether a shape is a parallelogram, for example by checking that its diagonals share a midpoint.
- 3 Tests a parallelogram to see if it is a rectangle, a rhombus, a square or none of these.
- 4 Puts the steps of a proof that a shape is a square in the right order.

V.11.09

Coordinate proofs of triangles

Proving facts about every triangle of a kind by placing it on a grid with letters for its coordinates and using algebra.

- 1 Chooses a smart placement with letters, not numbers, so the proof covers every case.
- 2 Shows that the segment joining two midpoints is parallel to the third side and half as long.
- 3 Picks which fact about an isosceles triangle holds for every possible size.
- 4 Finds the point where the medians meet, two thirds of the way along each.

Conic sections

This branch studies conic sections: the curves made by slicing a cone, namely circles, ellipses (ovals), parabolas and hyperbolas. They describe planet orbits, satellite dishes and headlights, and a few skills ask the learner to put the steps of a short derivation in order.

V.12.01

Slicing a cone

The four conic curves all come from slicing a double cone (two cones tip to tip) with a flat plane at different tilts.

- 1 Sees that a cut straight across the cone makes a circle.
- 2 Sees that a cut less steep than the cone's sides makes an ellipse (an oval).
- 3 Tells a parabola from a hyperbola by comparing the cut's tilt with the cone's sides.
- 4 Names any slice, including special cuts through the tip that give a point, a line or two crossing lines.

V.12.02

Parabola as focus & directrix

A parabola is every point the same distance from a fixed point (the focus) and a fixed line (the directrix).

- 1 Uses the equal-distance rule to find a point on a parabola.
- 2 Puts in order the steps that turn the equal-distance rule into an equation.
- 3 Finds the focus and directrix from an equation.
- 4 Finds the number p , the distance from the turning point to the focus, after rearranging the equation.
- ★ Finds a point's distance to the focus quickly by measuring to the directrix instead.
- ★★ Works with a line through the focus, finding one end's distance from the other's.

V.12.03

Parabola equations

Writing and reading parabola equations that open up, down, left or right.

- 1 Finds the focus of a parabola that opens up or down.
- 2 Finds the focus of a parabola that opens left or right.
- 3 Writes the equation from the turning point and the focus or directrix.
- 4 Picks the graph that matches an equation.

V.12.04

Ellipse definition

An ellipse is every point whose distances to two fixed points (the foci) add up to the same total.

- 1 Uses the constant total to find a missing distance to a focus.
- 2 Connects this to drawing an ellipse with a loop of string around two pins.
- 3 Reads the lengths of the long and short axes from an equation.
- 4 Finds the eccentricity, a number saying how stretched the oval is.

V.12.05

Ellipse equations & graphs

Writing and graphing ellipse equations.

- 1 Writes the standard equation from the center and the ends of the axes.
 - 2 Tells whether the ellipse is wide or tall from its equation.
 - 3 Picks the graph that matches an equation.
 - 4 Writes the equation of an ellipse from its graph.
-

V.12.06

Ellipse foci

Locating the two foci of an ellipse and using them in real settings.

- 1 Finds the distance from the center to each focus with a Pythagoras-like rule.
 - 2 Finds the coordinates of both foci.
 - 3 Uses foci to explain whispering galleries, where a whisper at one focus is heard at the other.
 - 4 Works out how stretched a planet's orbit is from its closest and farthest distances.
 - ★ Works backwards from the foci and one point on the curve to write the equation.
 - ★★ Finds the area of the triangle formed by a point on the ellipse and the two foci.
-

V.12.07

Hyperbola definition

A hyperbola is every point whose distances to two foci differ by the same amount. It has two separate branches.

- 1 Uses the constant difference to find a missing distance to a focus.
 - 2 Finds the vertices (turning points) from an equation.
 - 3 Finds the distance from the center to each focus.
 - 4 Finds the coordinates of both foci.
-

V.12.08

Hyperbola graphs

Writing and graphing hyperbola equations, using a guide box and slanted guide lines (asymptotes) the branches approach.

- 1 Writes the standard equation from the center, a vertex and a focus.
 - 2 Puts in order a short proof that the guide box's corners are as far from the center as the foci.
 - 3 Finds the equations of the asymptotes.
 - 4 Picks the equation that matches a graph.
-

V.12.09

Classify conics

Telling which conic a general equation describes.

- 1 Completes the square to rewrite an equation in standard form and find its center.
- 2 Uses the signs and sizes of the x squared and y squared terms to name the conic.
- 3 Calculates a test number (the discriminant) for equations with an xy term.
- 4 Names the graph, including unusual cases such as two crossing lines or a single point.

V.12.10

Conics in the world

Seeing conics at work in the real world.

- ① Finds the depth of a satellite dish with the receiver at its focus.
- ② Writes the shape of a headlight reflector, with the bulb at the focus.
- ③ Uses eccentricity to find where a star sits in a planet's orbit.
- ④ Uses signal timing from two radio stations (LORAN navigation) to locate a ship.

V.12.11

Systems with conics

Finding where a conic meets a line or another conic by solving their equations together.

- ① Finds where a line crosses a parabola.
- ② Finds where two conics cross.
- ③ Chooses substitution or elimination to solve a system.
- ④ Works out how many crossing points there are.
- ★ Finds the value that makes a line just touch a curve (be tangent to it).
- ★★ Finds which values of a changing number give exactly one meeting point.

Transformations

This branch studies transformations: sliding, flipping, turning and resizing shapes. It gives a precise way to say when two shapes are the same or similar, links geometry to matrices (grids of numbers), and includes a few short proofs where the learner picks the reason for a step.

V.13.01

Rigid motions defined

A rigid motion moves a shape without stretching or bending it, so every length and angle stays the same.

- 1 Matches each point of the original shape with its point in the moved copy (the image).
- 2 Decides whether a rule is an isometry, a move that keeps every distance the same.
- 3 Uses the fact that lengths, angles and perimeters stay the same after a rigid motion.
- 4 Applies a move written as a rule, such as T of (x, y) equals $(x + 4, y + 4)$.

V.13.02

Translation vectors

A translation slides every point the same distance in the same direction, described by a vector (a pair of numbers for across and up).

- 1 Reads a vector as so many units across and so many up or down.
- 2 Slides points of a shape by a given vector.
- 3 Finds the vector from one point and its image, then uses it on another point.
- 4 Combines two slides into one, or finds the missing second slide.

V.13.03

Reflections precisely

A reflection flips a shape over a mirror line. The mirror line cuts each point-to-image segment in half at a right angle.

- 1 Uses the fact that a point and its image are equally far from the mirror.
- 2 Reflects a point over a horizontal or vertical line.
- 3 Finds the mirror line from a point and its image.
- 4 Picks the coordinate rule for a reflection, such as over the line y equals negative x .

V.13.04

Rotations about any point

A rotation turns a shape by an angle around a fixed point, called the center.

- 1 Matches a counterclockwise turn with the same result clockwise.
- 2 Picks the coordinate rule for quarter and half turns about the origin.
- 3 Rotates a point around a center other than the origin.
- 4 Picks the missing reason in an argument that finds the center of a rotation.
- ★ Finds the single turn that a slide followed by a turn adds up to.
- ★★ Finds the center of the single turn made by two turns about different points.

V.13.05

Compositions

Doing one transformation after another (a composition). The order often changes the result.

- 1 Applies two moves in the stated order and sees that order matters.
 - 2 Shows that two reflections add up to a slide or a turn.
 - 3 Names the single move that has the same effect as two moves.
 - 4 Undoes a sequence of moves to find where a point started.
 - ★ Works backwards to find the second mirror line from the resulting slide.
 - ★★ Finds the overall effect of a long chain of reflections without doing each one.
-

V.13.06

Glide reflections

A glide reflection flips a shape over a line and then slides it along that line, like a trail of footprints.

- 1 Picks the correct description of a glide reflection.
 - 2 Applies a glide reflection to a point.
 - 3 Names the move that takes one footprint in a trail to another.
 - 4 Names which single move maps one shape onto another in a picture.
-

V.13.07

Symmetry

A shape has symmetry if some flip or turn lands it exactly on itself.

- 1 Counts the mirror lines (lines of symmetry) of a shape.
 - 2 Counts how many times a shape fits onto itself in one full turn (its order of rotational symmetry).
 - 3 Decides whether a shape has point symmetry, fitting onto itself after a half turn.
 - 4 Works out the symmetries of regular polygons from their number of sides.
-

V.13.08

Dilations about any center

A dilation enlarges or shrinks a shape from a center point by a scale factor.

- 1 Multiplies the distance from the center by the scale factor.
 - 2 Dilates points about the origin by multiplying their coordinates.
 - 3 Uses negative scale factors, which flip the image to the other side of the center.
 - 4 Finds the image of a line, which stays parallel to the original.
 - ★ Finds the center and scale factor from a segment and its image.
 - ★★ Combines two dilations about different centers into one.
-

V.13.09

Congruence by rigid motions

Two shapes are congruent (identical in size and shape) if a sequence of rigid motions moves one exactly onto the other.

- 1 Picks the correct definition of congruence.
- 2 Picks a sequence of moves that maps one triangle onto another.
- 3 Recognizes which congruence shortcut a proof using moves is justifying, such as side-angle-side.
- 4 Finds the exact slide that completes a sequence mapping one triangle onto another.

V.13.10

Similarity by transformations

Two shapes are similar (same shape, possibly different size) if a dilation plus rigid motions maps one onto the other.

- ① Works out lengths after a dilation and a rigid motion.
- ② Picks a sequence of moves that maps one triangle onto a similar one.
- ③ Picks the reason in the angle-angle similarity proof: dilations keep angles the same.
- ④ Finds the scale factor between two similar triangles on a grid.

V.13.11

Transformation matrices

A matrix (a grid of numbers) can describe a transformation. Multiplying a point by the matrix moves it.

- ① Multiplies a matrix by a point to find its image.
- ② Picks the matrix that gives a given reflection or rotation.
- ③ Uses a matrix that enlarges or shrinks points.
- ④ Combines two moves by multiplying their matrices, in the right order.

V.13.12

Tessellations

A tessellation covers a flat surface with repeating tiles, with no gaps or overlaps.

- ① Decides whether one kind of regular polygon can tile on its own.
- ② Finds the missing shape in a pattern mixing regular polygons.
- ③ Checks whether the angles meeting at a corner add up to exactly 360 degrees.
- ④ Explains why an Escher-style tile, made by moving a cut-out piece, keeps its area and still tiles.

Solids

This branch explores three-dimensional solids: what their slices look like, how to find their surface areas and volumes, and why the formulas work. A few skills include short arguments where the learner puts steps in order or picks what they prove.

V.14.01

Cross-sections

A cross-section is the flat shape you see when you slice straight through a solid, like cutting a loaf of bread.

- 1 Names the shape of a slice through a box or other prism.
- 2 Names the shape of a slice through a pyramid.
- 3 Finds the area of a slice through a cylinder or cone, such as a cut through a cone's tip.
- 4 Names the slice through a cube cut at an unusual angle, such as a triangle or hexagon.

V.14.02

Solids of revolution

Spinning a flat shape around a line sweeps out a solid, like a potter's wheel shaping clay.

- 1 Finds the radius and height of the cylinder made by spinning a rectangle.
- 2 Finds the radius and height of the cone made by spinning a right triangle.
- 3 Finds the radius of the sphere made by spinning a half circle.
- 4 Finds the volume of the solid made, taking care which side becomes the radius.

V.14.03

Cavalieri's principle

Cavalieri's principle: if two solids have equal-area slices at every height and the same height, they have equal volume, like a straight or slanted stack of coins.

- 1 Uses equal slices to find the volume of a slanted solid from an upright one.
- 2 Finds the volume or upright height of a slanted box.
- 3 Finds the volume of a slanted cylinder, working out its upright height first.
- 4 Puts in order the steps that find a sphere's volume by comparing it with a cylinder minus a cone.

V.14.04

Surface area of pyramids & cones

Finding the total outside area of pyramids and cones by adding the area of the base and the slanted sides.

- 1 Finds the slant height, measured along the sloping side, using Pythagoras.
- 2 Finds the area of the sloping sides only (the lateral area).
- 3 Adds the base to find the total surface area.
- 4 Works from a flat net (the shape unfolded) to find the size of a cone.

V.14.05

Surface area of spheres

Finding the outside area of a sphere, which is exactly 4 times the area of a circle with the same radius.

- 1 Finds a sphere's surface area from its radius.
 - 2 Finds the area of a solid half-sphere, including its flat face.
 - 3 Works backwards from the area to find the radius.
 - 4 Explores Archimedes' discovery: a sphere fills two thirds of the cylinder that just fits around it.
-

V.14.06

Volume of pyramids & cones

A pyramid or cone holds exactly one third of the prism or cylinder with the same base and height.

- 1 Picks what a short argument proves: by splitting a cube into 6 pyramids, a pyramid is one third of its prism.
 - 2 Finds the volume of a pyramid or cone.
 - 3 Works backwards from the volume to find a missing height or radius.
 - 4 Finds the volume of a frustum, a cone or pyramid with its top sliced off.
 - ★ Finds how full a cone-shaped glass is, knowing volume grows with the cube of the height.
 - ★★ Finds the volume of a pyramid hidden inside a cube by cutting away the corners.
-

V.14.07

Composite solids

Finding the volume and surface area of objects built from several simple solids.

- 1 Adds the volumes of the parts.
 - 2 Subtracts the volume of a hole that has been cut out.
 - 3 Finds the outside area of a combined solid, leaving out the hidden faces.
 - 4 Applies this to real objects, such as a capsule-shaped pill.
 - ★ Works out how many smaller solids can be cast from a melted-down larger one.
 - ★★ Counts every face, including inside tunnels, to find the surface area of a complicated solid.
-

V.14.08

Density & design

Using volume and area in real design problems, such as density and making packages efficient.

- 1 Finds density, how heavy something is for its size, as mass divided by volume.
- 2 Finds population density, people per square kilometer.
- 3 Picks the box shape that holds a set amount using the least material.
- 4 Works out how many items fit in a package, rounding down for partial items.

Constructions

This branch teaches classic constructions: drawing exact figures with only a compass and an unmarked straightedge (ruler with no measurements). Each one is then shown to work, by putting construction or proof steps in order, picking the reason, or working out a length or angle the construction creates.

V.15.01

Copy a segment & an angle

Using a compass to copy a length or an angle exactly, without measuring.

- 1 Understands that a compass carries a length: a copied arc has the same length.
- 2 Puts in order the steps for copying segments, such as building one as long as two others combined.
- 3 Puts in order the steps for copying an angle.
- 4 Picks the reason the copied angle is equal: the triangles match side-side-side (SSS).

V.15.02

Bisect a segment

Constructing the line that cuts a segment exactly in half at a right angle (its perpendicular bisector).

- 1 Knows the compass must open more than half the segment for the arcs to meet.
- 2 Uses the construction to find the midpoint.
- 3 Uses the right angle the construction makes to find a missing angle.
- 4 Puts the lines of a proof that the construction works in order.

V.15.03

Bisect an angle

Constructing the line that cuts an angle exactly in half (its bisector).

- 1 Works out lengths made by the first arc across both sides of the angle.
- 2 Recognizes the kite shape the construction arcs create.
- 3 Uses the bisector to find the two equal half angles.
- 4 Puts the lines of a proof that the construction works in order.

V.15.04

Perpendicular through a point

Constructing a line at a right angle to a given line, through a chosen point.

- 1 Finds lengths made by the first arc, drawn around a point that lies on the line.
- 2 Works out the distance to the line when the point is off the line.
- 3 Puts the construction steps in order.
- 4 Picks the reason the construction gives a right angle.

V.15.05

Parallel through a point

Constructing a line through a point that runs parallel to a given line.

- 1 Knows the method copies an angle to make matching (corresponding) angles equal.
- 2 Puts the construction steps in order.
- 3 Orders the steps of another method: two perpendiculars in a row make a parallel line.
- 4 Picks the reason the new line is parallel.

V.15.06

Triangle & hexagon in a circle

Constructing a regular hexagon and an equilateral triangle inside a circle by stepping the radius around it.

- 1 Finds the angle at the center made by one radius-length step.
 - 2 Uses the fact that the hexagon's sides equal the radius.
 - 3 Finds the exact side of the triangle made by joining every other point.
 - 4 Picks the reason each step makes a 60 degree angle at the center.
 - ★ Uses circle angle facts to find angles inside the hexagon.
 - ★★ Finds the exact area of one point of the six-pointed star made by two triangles.
-

V.15.07

Square in a circle

Constructing a square inside a circle using two diameters that cross at right angles.

- 1 Draws a diameter, a line through the center from edge to edge.
 - 2 Puts in order the steps for a second diameter at a right angle to the first.
 - 3 Joins the four ends and finds the square's exact side.
 - 4 Picks why the shape made must be a square.
-

V.15.08

Tangent from a point

Constructing the lines from an outside point that just touch a circle (tangents).

- 1 Finds the midpoint between the point and the circle's center.
 - 2 Finds the radius of the helper circle drawn on that segment.
 - 3 Finds a tangent's length, using the right angle where it touches.
 - 4 Uses the right angle in a half circle to find missing angles.
 - ★ Finds the area of the kite formed by both tangents and two radii.
 - ★★ Finds the length of a tangent shared by two circles.
-

V.15.09

Inscribed & circumscribed circles

Constructing the circle that passes through a triangle's three corners, and the circle that fits snugly inside it.

- 1 Finds where the perpendicular bisectors meet, the center of the outside circle.
- 2 Finds the radius of the outside circle.
- 3 Uses the fact that the angle bisectors meet at a point equally far from all three sides.
- 4 Finds the radius of the inside circle from the area.
- ★ Uses circle angle facts to find an angle at the outside circle's center.
- ★★ Finds a triangle's area from where its inside circle touches the longest side.

VI

Era VI • Change

About ages 16–19 • final years of high school and first-year college • coming in Mathera 2.0

Era VI is the mathematics of change. It covers calculus from limits to derivatives and integrals, then differential equations, curves described in new ways, and infinite series. It closes with statistical inference (drawing conclusions from data) and a short unit on mechanics, the math of forces and motion.

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Limits & continuity

This branch is about limits: what value a calculation is heading toward as its input gets closer and closer to some number. Limits are the foundation that the rest of calculus is built on.

VI.1.01

Limits informally

A limit is the value something heads toward as you get closer and closer, even if it never quite arrives. Think of walking halfway to a wall, again and again.

- 1 Describes in words what it means for a quantity to approach a value.
 - 2 Explains that where a formula is heading can differ from its actual value there.
 - 3 Reads and writes the standard shorthand for a limit.
 - 4 Uses limits to untangle Zeno's paradox of halving the distance forever.
-

VI.1.02

Limits from tables

Guessing a limit by making a table of values that creep closer and closer to the point of interest.

- 1 Builds a table of values for inputs getting close to a chosen number.
 - 2 Approaches the number from both below and above in the table.
 - 3 Reads the trend in the table to estimate the limit.
 - 4 Recognizes cases where a table gives a misleading answer and explains why.
-

VI.1.03

Limits from graphs

Finding a limit by looking at a graph and following the curve toward a point.

- 1 Traces a curve with a finger to see what height it approaches.
 - 2 Finds limits at a point where the graph has a hole.
 - 3 Reads graphs with jumps, where the two sides head to different heights.
 - 4 Explains that the limit ignores whatever happens exactly at the point itself.
-

VI.1.04

One-sided limits

Looking at the approach from just one side, the left or the right, and comparing the two.

- 1 Finds the value approached when coming from the left only.
 - 2 Finds the value approached when coming from the right only.
 - 3 Explains that a full limit exists only when both sides agree.
 - 4 Finds one-sided limits for functions defined by different rules on different pieces.
-

VI.1.05

When limits fail

Recognizing situations where a limit simply does not exist, and saying why.

- 1 Spots a jump in a graph as a reason a limit fails.
- 2 Spots values shooting off to infinity as a reason a limit fails.
- 3 Spots endless wiggling, faster and faster, as a reason a limit fails.
- 4 Explains in words exactly why a given limit does not exist.

VI.1.06

Limit laws

Rules that let you break a complicated limit into simpler limits and combine the answers.

- 1 Finds the limit of a sum or difference from the separate limits.
 - 2 Finds the limit of a product or quotient from the separate limits.
 - 3 Handles powers and roots of expressions with known limits.
 - 4 Uses the rules one careful step at a time, naming each rule used.
-

VI.1.07

Direct substitution

The simplest way to find a limit: just plug the number in, when that is allowed.

- 1 Recognizes when plugging the number straight in gives the right answer.
 - 2 Plugs in to find limits of polynomials and fractions made of polynomials.
 - 3 Recognizes that getting zero over zero means more work is needed.
 - 4 Recognizes that a nonzero number over zero means the values blow up.
-

VI.1.08

Factor & cancel

When plugging in gives zero over zero, factoring the top and bottom and canceling the troublemaker often reveals the limit.

- 1 Factors the top and bottom of a fraction.
 - 2 Cancels the common factor that caused the zero over zero.
 - 3 Plugs in the number to the simplified expression to find the limit.
 - 4 Explains why canceling is allowed even though that input was a problem.
-

VI.1.09

Rationalize

A trick for limits with square roots: multiply by a matching partner expression to clear the root out of the way.

- 1 Multiplies top and bottom by the partner with the opposite sign (the conjugate).
 - 2 Simplifies the result, which removes the square root from one part.
 - 3 Cancels the factor that was causing zero over zero.
 - 4 Plugs in the number to finish the limit.
-

VI.1.10

Complex-fraction limits

Finding limits of fractions that have smaller fractions stacked inside them.

- 1 Combines the small fractions into one over a common bottom.
- 2 Simplifies the stacked fraction into a single fraction.
- 3 Cancels the factor that was causing zero over zero.
- 4 Plugs in the number to finish the limit.

VI.1.11

Squeeze theorem

The squeeze theorem: if something is trapped between two things heading to the same value, it must head there too.

- 1 Explains the squeeze idea with a picture of one curve trapped between two others.
- 2 Finds two simpler expressions that trap the tricky one.
- 3 Squeezes a wiggling expression, like $x^2 \sin(1/x)$.
- 4 Writes out a complete squeeze argument in clear sentences.

VI.1.12

Special trig limits

Two famous limits involving sine and cosine near zero, which are needed later for the derivatives of trig functions.

- 1 Uses the fact that $\sin x / x$ approaches 1 near zero.
- 2 Uses the fact that $(1 - \cos x) / x$ approaches 0.
- 3 Adjusts for a scaled angle, such as $\sin 3x / x$.
- 4 Follows and explains a proof of the sine limit using a circle diagram.

VI.1.13

Infinite limits

Describing values that grow without end near a point, like dividing by a number shrinking toward zero.

- 1 Recognizes when values grow without bound near a point.
- 2 Decides whether values head to plus or minus infinity from each side.
- 3 Finds the vertical lines (asymptotes) that a graph shoots up or down beside.
- 4 Writes these results using correct limit notation with infinity.

VI.1.14

Limits at infinity

Asking what happens to a formula as the input gets huge, in either the positive or negative direction.

- 1 Describes where a function is heading as x becomes very large or very negative.
- 2 Divides top and bottom by the biggest power of x to find the limit.
- 3 Compares how fast exponentials and logarithms grow for huge inputs.
- 4 Handles limits at infinity that involve square roots.

VI.1.15

Horizontal asymptotes by limits

Using limits at infinity to find the flat lines a graph settles toward far out to the left or right.

- 1 Finds the level a fraction settles to by comparing the highest powers.
- 2 Handles graphs that settle to different levels on the left and right ends.
- 3 Recognizes graphs that wiggle back and forth across their settling level.
- 4 Sketches a graph showing its long-run flat lines.

VI.1.16

Continuity definition

A function is continuous at a point if its graph passes through without a break. Informally, you can draw it without lifting your pencil.

- 1 Checks that the function actually has a value at the point.
 - 2 Checks that the limit at the point exists.
 - 3 Checks that the value and the limit are equal, completing the test.
 - 4 Decides whether a function is continuous over a whole stretch of inputs.
-

VI.1.17

Types of discontinuity

Naming the different ways a graph can break: a single missing point, a jump, or a shoot-off to infinity.

- 1 Identifies a removable break, where just one point is missing or misplaced.
 - 2 Identifies a jump, where the graph leaps to a new height.
 - 3 Identifies an infinite break, where the graph shoots off near a vertical line.
 - 4 Classifies the type of break from the formula alone, without a graph.
-

VI.1.18

Making things continuous

Adjusting a function so its graph has no breaks, by filling a hole or choosing the right numbers.

- 1 Fills a hole by defining the missing value as the limit.
 - 2 Chooses a constant so the two pieces of a split function meet up.
 - 3 Chooses two constants at once so several pieces all join smoothly.
 - 4 Checks carefully that the adjusted function is now continuous.
-

VI.1.19

Intermediate value theorem

The intermediate value theorem: an unbroken curve going from below a height to above it must cross that height somewhere.

- 1 States the theorem in plain words.
 - 2 Uses it to show an equation has a solution between two numbers.
 - 3 Explains why the theorem fails if the function has a break.
 - 4 Uses repeated halving (bisection) to close in on a solution.
-

VI.1.20

Epsilon-delta

The fully precise definition of a limit, using two small tolerances called epsilon and delta. It turns "gets close" into a checkable promise.

- 1 Explains the precise definition in everyday words.
- 2 Finds how close the input must be for a straight-line function.
- 3 Does the same for a squared function, where it takes more care.
- 4 Writes a full formal proof that a limit has a certain value.

Derivatives

This branch introduces the derivative, which measures how fast something is changing at a single moment. It covers what the derivative means and the rules for calculating it quickly.

VI.2.01 Average vs instantaneous rate

The difference between average speed over a trip and your exact speed at one instant. The instant speed is what a speedometer shows.

- 1 Finds the average rate of change over a stretch, like average speed.
 - 2 Shrinks the stretch smaller and smaller to close in on the instant rate.
 - 3 Explains the instant rate as a speedometer-style reading.
 - 4 Estimates an instant rate from a table of measured values.
-

VI.2.02 Tangent as limit of secants

Finding the line that just touches a curve at one point (the tangent) by looking at lines through two nearby points.

- 1 Draws lines through two points on a curve (secant lines).
 - 2 Sees the slopes of these lines settle on one value as the points get closer.
 - 3 Identifies that settled line as the tangent line at the point.
 - 4 Zooms into a smooth curve until it looks straight, matching the tangent.
-

VI.2.03 Limit definition

The official definition of a derivative as a limit: the slope between two points as the gap between them shrinks to nothing.

- 1 Writes the slope formula between x and a nearby point x plus h .
 - 2 Computes the derivative of x squared directly from the definition.
 - 3 Does the same for one over x and the square root of x .
 - 4 Uses a second version of the definition to find the slope at one specific point.
-

VI.2.04 Notation

The different ways derivatives are written, and what each piece of notation means.

- 1 Reads and writes the "f prime" notation for a derivative.
 - 2 Reads and writes the "dy over dx" notation.
 - 3 Writes the derivative at one specific input, a single number.
 - 4 Gives the units of a derivative, such as meters per second.
-

VI.2.05

Differentiability

A function is differentiable where its graph is smooth enough to have a clear slope. Sharp points and vertical spots are where this fails.

- 1 Spots sharp corners, where there is no single slope.
 - 2 Spots cusps, where the graph comes to a pointed tip.
 - 3 Spots places where the tangent line is vertical.
 - 4 Explains why a function with a slope at a point must be unbroken there.
-

VI.2.06

Graph f' from f

Sketching the graph of the derivative by reading slopes off the original graph, and working back the other way.

- 1 Turns steepness on the original graph into height on the new graph.
 - 2 Marks zeros of the derivative where the original graph is flat.
 - 3 Sketches the whole derivative graph from the original.
 - 4 Works backwards, sketching the original graph from the derivative graph.
-

VI.2.07

Power rule

The power rule, a shortcut for the derivative of x raised to any power: bring the power down and lower it by one.

- 1 Uses the power rule for whole-number powers like x cubed.
 - 2 Uses it for negative powers, such as one over x squared.
 - 3 Uses it for fractional powers, such as square roots.
 - 4 Rewrites roots and fractions as powers first, then differentiates.
-

VI.2.08

Sum & constant rules

Basic rules: constants have zero rate of change, multipliers carry through, and sums can be done piece by piece.

- 1 Knows the derivative of a plain constant is zero.
 - 2 Keeps a number multiplier in front while differentiating.
 - 3 Differentiates sums and differences term by term.
 - 4 Differentiates whole polynomials quickly in one line.
-

VI.2.09

Product rule

The product rule, for finding the derivative of two things multiplied together.

- 1 States the product rule formula.
- 2 Applies it to a product of two functions.
- 3 Extends it to a product of three functions.
- 4 Explains why the answer is not simply the two derivatives multiplied.

VI.2.10

Quotient rule

The quotient rule, for finding the derivative of one thing divided by another.

- 1 States the quotient rule formula.
- 2 Applies it to a fraction of two functions.
- 3 Simplifies the often messy result into a neat final answer.
- 4 Recognizes when rewriting the fraction first is easier than the quotient rule.

VI.2.11

Trig derivatives

The derivatives of the trigonometry functions, like sine and cosine, which describe waves and rotation.

- 1 Uses the derivatives of sine and cosine.
- 2 Uses the derivatives of tangent and secant.
- 3 Uses the derivatives of cosecant and cotangent.
- 4 Derives the sine and cosine rules from the limit definition.

VI.2.12

Exponential & log derivatives

The derivatives of exponential functions, which describe growth like compound interest, and of logarithms, which undo them.

- 1 Uses the fact that e to the x is its own derivative.
- 2 Differentiates other bases, like 2 to the x .
- 3 Differentiates the natural logarithm, $\ln x$.
- 4 Differentiates logarithms in other bases.

VI.2.13

Chain rule

The chain rule, for functions nested inside other functions. Multiply the rate of the outer layer by the rate of the inner layer.

- 1 Identifies the inside and outside functions of a nested expression.
- 2 Applies the chain rule once to find a derivative.
- 3 Applies it several times for functions nested more than two deep.
- 4 Combines the chain rule with the product and quotient rules.

VI.2.14

Implicit differentiation

Finding slopes on curves like circles, where y is mixed in with x and cannot easily be separated out.

- 1 Differentiates both sides of an equation involving x and y .
- 2 Recognizes where " dy over dx " appears when differentiating y terms.
- 3 Rearranges to solve for " dy over dx ".
- 4 Finds the slope of the curve at a specific point.

VI.2.15

Inverse function derivatives

Finding the derivative of a function that undoes another, such as arcsine undoing sine.

- 1 Uses the general formula linking the slopes of a function and its inverse.
- 2 Differentiates arcsine and arccosine.
- 3 Differentiates arctangent.
- 4 Finds the slope of an inverse function using only a table of values.

VI.2.16

Higher derivatives

Taking the derivative again, and again. The second derivative measures how the rate itself is changing, like acceleration.

- ① Finds the second derivative by differentiating twice.
- ② Spots patterns to write the derivative after any number of steps.
- ③ Finds second derivatives for curves where x and y are mixed.
- ④ Reads and writes the notation for higher derivatives.

VI.2.17

Logarithmic differentiation

A trick for messy derivatives: take the logarithm of both sides first, which turns products and powers into simpler sums.

- ① Takes the natural log of both sides of an equation.
- ② Differentiates the result and solves for the derivative.
- ③ Handles functions like x to the power x , with the variable in both places.
- ④ Uses the method to tame long products and quotients.

VI.2.18

Hyperbolic functions

Hyperbolic functions are cousins of sine and cosine built from exponentials. One of them describes the shape of a hanging chain.

- ① Defines hyperbolic sine and cosine using e to the x .
- ② Sketches their graphs.
- ③ Checks their key identity: $\cosh^2$ minus $\sinh^2$ equals one.
- ④ Finds their derivatives.

Applications of derivatives

This branch puts derivatives to work: estimating values, describing motion, finding the highest and lowest points of a graph, and solving real "best possible" problems.

VI.3.01 Tangent & normal lines

Writing the equation of the line that just touches a curve at a point, and of the line at right angles to it.

- 1 Finds the slope of the touching line from the derivative at the point.
 - 2 Writes the line's equation using the point and the slope.
 - 3 Writes the equation of the perpendicular line (the normal line).
 - 4 Finds the points where the touching line is flat.
-

VI.3.02 Linear approximation

Using the touching line as a quick estimate for a curve near a known point, like estimating the square root of 4.1.

- 1 Explains why the tangent line stays close to the curve nearby.
 - 2 Uses the tangent line to approximate a hard-to-compute value.
 - 3 Decides whether the estimate is too high or too low from the curve's bend.
 - 4 Uses differentials (tiny changes) to estimate small errors.
-

VI.3.03 Motion

Using derivatives to describe moving objects: position, speed and acceleration.

- 1 Finds velocity by differentiating position.
 - 2 Finds acceleration by differentiating velocity.
 - 3 Decides whether an object is speeding up or slowing down.
 - 4 Finds the total distance traveled, including trips back and forth.
-

VI.3.04 Rates in context

Reading a derivative as a real-world rate, such as dollars per item or people per year.

- 1 Explains the meaning of a derivative at a point, including its units.
 - 2 Interprets marginal cost, the cost of making one more item.
 - 3 Interprets rates of population growth.
 - 4 Estimates a rate from real data in a table.
-

VI.3.05 Related rates setup

Setting up problems where two quantities change together over time, like the radius and area of a spreading puddle.

- 1 Draws a diagram and labels the changing quantities.
- 2 Finds an equation linking the quantities.
- 3 Differentiates that equation with respect to time.
- 4 Plugs in the given numbers only at the end, after differentiating.

VI.3.06

Related rates classics

Classic linked-rates problems, such as a sliding ladder or a filling cone-shaped tank.

- 1 Solves the sliding ladder problem.
 - 2 Solves for how fast water rises in a filling cone.
 - 3 Solves for how fast a shadow grows as someone walks.
 - 4 Solves problems where an angle is changing, using trigonometry.
-

VI.3.07

Critical points

Critical points are where a graph is flat or has a sharp point. They are the candidates for peaks and valleys.

- 1 Finds points where the derivative equals zero.
 - 2 Finds points where the derivative does not exist.
 - 3 Finds all critical points of a function.
 - 4 Explains why not every critical point is a peak or a valley.
-

VI.3.08

Increasing & decreasing

Using the sign of the derivative to tell where a graph is rising and where it is falling.

- 1 Links a positive derivative to rising and a negative one to falling.
 - 2 Makes a sign chart showing where the derivative is positive or negative.
 - 3 Writes the stretches of x where the function rises or falls.
 - 4 Reads rising and falling parts from a graph of the derivative.
-

VI.3.09

First derivative test

Deciding whether a critical point is a peak or valley by checking if the graph switches between rising and falling there.

- 1 Checks whether the derivative changes sign at a point.
 - 2 Identifies a local peak where rising switches to falling.
 - 3 Identifies a local valley where falling switches to rising.
 - 4 Justifies the conclusion in clear sentences.
-

VI.3.10

Concavity

Concavity describes the way a graph bends: cupped upward like a bowl or downward like a dome.

- 1 Links the sign of the second derivative to the bend.
 - 2 Recognizes cupped-up and cupped-down shapes.
 - 3 Finds the stretches of x where the graph bends each way.
 - 4 Explains cupped-up as the slope steadily increasing.
-

VI.3.11

Inflection points

Inflection points are where a graph switches its bend, from bowl-shaped to dome-shaped or back.

- 1 Identifies points where the bend changes direction.
- 2 Explains why a zero second derivative alone does not guarantee a switch.
- 3 Finds all inflection points of a function.
- 4 Finds inflection points from a graph of the derivative.

VI.3.12

Second derivative test

A quicker test for peaks and valleys: at a flat point, a bowl shape means a valley and a dome shape means a peak.

- 1 Identifies a valley where the slope is zero and the graph bends up.
 - 2 Identifies a peak where the slope is zero and the graph bends down.
 - 3 Recognizes when the test gives no answer and another test is needed.
 - 4 Justifies conclusions from the test in words.
-

VI.3.13

Absolute extrema

Finding the very highest and lowest values a function reaches over a whole stretch, not just local peaks.

- 1 States that an unbroken function on a closed stretch has a highest and lowest value.
 - 2 Lists the candidates: critical points and the two endpoints.
 - 3 Compares the candidates' values to pick the highest and lowest.
 - 4 Handles open stretches, where a highest or lowest value may not exist.
-

VI.3.14

Mean value theorem

The mean value theorem: on any smooth trip, at some moment your instant speed equals your average speed.

- 1 Uses Rolle's theorem: if you end where you started, somewhere the slope is zero.
 - 2 States the mean value theorem in plain words.
 - 3 Finds the exact point where the instant rate equals the average rate.
 - 4 Uses it to prove a driver must have broken a speed limit.
-

VI.3.15

Curve sketching

Drawing an accurate graph of a function by gathering all the clues calculus gives about its shape.

- 1 Finds the allowed inputs and where the graph crosses the axes.
 - 2 Finds the lines the graph approaches (asymptotes).
 - 3 Makes charts of rising, falling and bending from the derivatives.
 - 4 Combines everything into a complete, labeled sketch.
-

VI.3.16

Reading f , f' , f'' together

Reading a function, its derivative and its second derivative side by side to see how their shapes connect.

- 1 Matches three graphs to the function and its two derivatives.
 - 2 Describes features of the function from its derivative's graph.
 - 3 Describes features of the function from its second derivative's graph.
 - 4 Justifies each claim with precise language.
-

VI.3.17

Optimization setup

Turning a word problem about the biggest or smallest possible value into a single formula to work with.

- 1 Identifies the quantity to make as large or small as possible.
- 2 Writes the limiting condition, like a fixed amount of fencing, as an equation.
- 3 Combines them into a formula in just one variable.
- 4 States the sensible range of values the variable can take.

VI.3.18

Optimization solve

Solving "best possible" problems, like the largest area a fence can enclose, using derivatives.

- 1 Finds the critical points of the formula.
 - 2 Tests them to confirm which gives the best value.
 - 3 Answers the question actually asked, with units.
 - 4 Solves classic fence, open-box and soup-can problems from start to finish.
-

VI.3.19

L'Hôpital's rule

L'Hôpital's rule: when a limit gives zero over zero, you can often compare the rates of top and bottom instead.

- 1 Recognizes the forms where the rule applies: zero over zero and infinity over infinity.
 - 2 Applies the rule by differentiating top and bottom separately.
 - 3 Applies it repeatedly when the first try still gives zero over zero.
 - 4 Rewrites other tricky forms, like zero times infinity, so the rule applies.
-

VI.3.20

Newton's method

Newton's method: a way to hunt down solutions to hard equations by following tangent lines toward where a graph crosses zero.

- 1 Explains how following a tangent line gives a better guess.
 - 2 Writes the formula that turns one guess into the next.
 - 3 Runs a few steps by hand or calculator to close in on a solution.
 - 4 Recognizes situations where the method fails or wanders off.
-

VI.3.21

Fixed-point iteration

Solving an equation by rewriting it as x equals something, then feeding each answer back in until it settles.

- 1 Rearranges an equation into the form x equals a formula in x .
- 2 Repeats the formula, plugging each answer back in.
- 3 Draws cobweb and staircase diagrams to show the process.
- 4 Decides when the process settles down and when it runs away.

Integrals

This branch introduces the integral, which adds up many tiny pieces to find a total, such as distance from speed or area under a curve. It ends with the key link showing integrals undo derivatives.

VI.4.01

Area as accumulation

Finding a total by measuring area under a graph. For example, the area under a speed graph gives the distance traveled.

- 1 Finds distance traveled from the area under a velocity graph.
 - 2 Explains that rate times time appears as an area on a graph.
 - 3 Treats area below the axis as negative, like driving backwards.
 - 4 Works out the units of area under a curve, such as meters.
-

VI.4.02

Left & right sums

Estimating area under a curve by filling it with rectangles, using the left or right edge to set each height.

- 1 Divides the area into rectangles and adds up their areas.
 - 2 Uses the curve's height at each left edge for the rectangles.
 - 3 Uses the curve's height at each right edge instead.
 - 4 Decides whether the estimate is too high or too low.
-

VI.4.03

Midpoint & trapezoid sums

Better area estimates using the middle of each strip, or using slanted-top strips (trapezoids).

- 1 Estimates area using rectangles set at each strip's midpoint.
 - 2 Estimates area using trapezoids instead of rectangles.
 - 3 Makes these estimates from a table of values.
 - 4 Handles strips of unequal widths.
-

VI.4.04

Limit of Riemann sums

Getting the exact area by using more and more thinner rectangles, and taking the limit as their number grows without end.

- 1 Writes a rectangle sum using sigma (summation) notation.
 - 2 Writes the width of each strip and where each one sits.
 - 3 Takes the limit as the number of strips grows endlessly.
 - 4 Converts such a limit of sums into integral notation.
-

VI.4.05

Definite integral notation

How an integral is written and read: the start and end points, the function, and the "dx" marking thin strips.

- 1 Identifies the start and end points and the function in an integral.
- 2 Explains "dx" as the tiny width of each strip.
- 3 Reads an integral aloud correctly.
- 4 Interprets an integral as area, with below-axis parts counting as negative.

VI.4.06

Integral properties

Rules for working with integrals, such as splitting them into pieces or flipping the start and end.

- 1 Knows that swapping start and end points flips the sign.
- 2 Splits an integral over a stretch into two neighboring stretches.
- 3 Pulls out constants and splits up sums.
- 4 Compares integrals: a bigger function gives a bigger integral.

VI.4.07

Integrals by geometry

Finding integrals exactly using familiar shapes like triangles and semicircles, without any calculus rules.

- 1 Finds integrals from areas of triangles and rectangles.
- 2 Uses semicircle areas for circular graphs.
- 3 Handles V-shaped absolute value graphs.
- 4 Uses symmetry, where equal areas above and below cancel out.

VI.4.08

Antiderivatives

An antiderivative undoes a derivative: given the rate, it recovers the original function.

- 1 Finds a function whose derivative is the given one.
- 2 Adds "plus C", since any constant could have been there.
- 3 Sees the answers as a family of curves stacked vertically.
- 4 Checks an antiderivative by differentiating it.

VI.4.09

Basic rules

The basic rules for finding antiderivatives, starting with running the power rule backwards.

- 1 Uses the power rule in reverse: raise the power by one and divide.
- 2 Handles sums and number multipliers.
- 3 Knows one over x comes from the natural log.
- 4 Rewrites roots and fractions as powers before finding antiderivatives.

VI.4.10

Trig & exponential antiderivatives

Antiderivatives of trigonometry and exponential functions, found by reversing their derivative rules.

- 1 Finds antiderivatives of sine and cosine.
- 2 Finds antiderivatives of secant squared and secant times tangent.
- 3 Finds antiderivatives of e to the x and other exponentials.
- 4 Recognizes forms whose antiderivatives are inverse trig functions.

VI.4.11

Initial value problems

Recovering a specific function from its rate of change plus one known value, like finding position from speed and a starting point.

- 1 Finds the general antiderivative with its "plus C".
- 2 Uses one known point to pin down the value of C.
- 3 Finds position from velocity and a starting position.
- 4 Works back twice, from acceleration to velocity to position.

VI.4.12

FTC part 1

The first part of the Fundamental Theorem of Calculus: the rate at which area builds up equals the height of the curve.

- 1 Differentiates an integral whose end point is x .
- 2 Combines this with the chain rule when the end point is a formula.
- 3 Handles integrals where the start point is variable.
- 4 Handles integrals where both start and end are variable.

VI.4.13

FTC part 2

The second part of the Fundamental Theorem: to find an exact area, take any antiderivative and subtract its values at the two ends.

- 1 Writes the result as the antiderivative at the end minus at the start.
- 2 Evaluates definite integrals using this method.
- 3 Explains why this shortcut gives the area.
- 4 Applies it to functions defined in pieces.

VI.4.14

Accumulation functions

A running total function that tracks how much area has piled up from a starting point to x .

- 1 Defines a running total as an integral with a variable end point.
- 2 Finds values of the running total from a graph.
- 3 Finds where the running total increases.
- 4 Finds its peaks, valleys and bends from the original graph.

VI.4.15

Average value

The average height of a curve over a stretch, like the average temperature across a whole day.

- 1 States the formula: the integral divided by the stretch's length.
- 2 Computes the average value of a function.
- 3 Finds where the function actually hits its average value.
- 4 Interprets the average value in a real-world setting.

Integration techniques

This branch is a toolkit of methods for finding integrals that the basic rules can't handle. Much of the work is recognizing which tool fits a problem.

VI.5.01

u-substitution

Substitution: spotting a function nested inside another and renaming it, so the integral becomes a simple one. It reverses the chain rule.

- 1 Spots the inside function to rename.
 - 2 Finds how the small change in the new variable relates to dx .
 - 3 Rewrites the integral in the new variable and integrates.
 - 4 Switches back to the original variable at the end.
-

VI.5.02

u-substitution with bounds

Using substitution on integrals with start and end points, keeping the endpoints consistent.

- 1 Changes the start and end points to match the new variable.
 - 2 Alternatively, switches back to x and uses the original endpoints.
 - 3 Evaluates the integral to get a number.
 - 4 Uses symmetry shortcuts for balanced (odd and even) functions.
-

VI.5.03

Log & inverse trig results

Recognizing integrals that produce a logarithm or an inverse trigonometry function as the answer.

- 1 Recognizes a derivative over its function, which gives a log.
 - 2 Recognizes one over one plus x squared, which gives arctangent.
 - 3 Recognizes the pattern that gives arcsine.
 - 4 Handles stretched or scaled versions of these patterns.
-

VI.5.04

Divide first

When a fraction has a top as big as its bottom, dividing first splits it into easier pieces.

- 1 Recognizes a fraction whose top has as high a power as its bottom.
 - 2 Uses polynomial long division to split it.
 - 3 Integrates each resulting piece.
 - 4 Splits a fraction into separate simpler fractions.
-

VI.5.05

Complete the square

Rewriting a quadratic in the bottom of a fraction by completing the square, so a known pattern appears.

- 1 Recognizes integrals with a quadratic on the bottom.
- 2 Completes the square to rewrite the quadratic.
- 3 Substitutes to simplify the shifted expression.
- 4 Integrates to an arctangent answer.

VI.5.06

Integration by parts

Integration by parts: a method for integrals of two things multiplied together. It reverses the product rule.

- 1 Derives the formula from the product rule.
- 2 Chooses which part to differentiate using a priority list (LIATE).
- 3 Applies the method to find the integral.
- 4 Applies it to integrals with start and end points.

VI.5.07

Repeated parts

Using integration by parts several times in a row, including a neat trick when the integral loops back to itself.

- 1 Organizes repeated steps with a table (the tabular method).
- 2 Handles a power of x times an exponential.
- 3 Handles an exponential times sine, which loops back on itself.
- 4 Solves for the integral algebraically when it reappears.

VI.5.08

Trig integrals

Integrating powers and products of trigonometry functions using identities to rewrite them.

- 1 Handles an odd power of sine or cosine.
- 2 Handles even powers using the half-angle identities.
- 3 Handles powers of tangent and secant.
- 4 Handles products of sines and cosines of different angles.

VI.5.09

Trig substitution

Replacing x with a trigonometry expression to simplify square roots, guided by a right triangle.

- 1 Uses a sine substitution for roots of a number minus x squared.
- 2 Uses a tangent substitution for roots of a number plus x squared.
- 3 Uses a secant substitution for roots of x squared minus a number.
- 4 Draws a triangle to switch the answer back to x .

VI.5.10

Partial fractions, linear

Partial fractions: breaking one complicated fraction into a sum of simple ones that are easy to integrate.

- 1 Factors the bottom of the fraction into simple pieces.
- 2 Sets up the simple fractions with unknown numbers on top.
- 3 Solves for the unknown numbers.
- 4 Integrates each piece, getting logarithms.

VI.5.11

Partial fractions, harder

Harder partial fractions, where the bottom has repeated pieces or pieces that can't be factored further.

- 1 Sets up fractions for a repeated factor on the bottom.
- 2 Sets up fractions for a quadratic piece that won't factor.
- 3 Solves the resulting system of equations for the unknowns.
- 4 Integrates all the pieces.

VI.5.12

Improper integrals, infinite bounds

Integrals that run out to infinity. Surprisingly, an endless region can still have a finite area.

- 1 Rewrites an integral to infinity as a limit.
 - 2 Evaluates the limit.
 - 3 Decides whether the area is finite (converges) or endless (diverges).
 - 4 Uses the known rule for one over x to a power.
-

VI.5.13

Improper integrals, blow-ups

Integrals where the function shoots off to infinity at some point in the range.

- 1 Finds the point where the function blows up.
 - 2 Splits the integral there and takes a limit on each side.
 - 3 Evaluates the pieces.
 - 4 Decides finite or endless by comparing with a known integral.
-

VI.5.14

Numerical integration

Estimating integrals with numbers when no exact formula can be found, and judging how accurate the estimate is.

- 1 Estimates the error of the trapezoid method.
 - 2 Uses Simpson's rule, which fits curves instead of straight lines.
 - 3 Compares the accuracy of different methods.
 - 4 Recognizes integrals that have no formula and must be estimated.
-

VI.5.15

Choosing a technique

Looking at an unfamiliar integral and deciding which method to try first.

- 1 Simplifies or rewrites the integral before choosing a method.
 - 2 Checks whether substitution will work.
 - 3 Decides between integration by parts and partial fractions.
 - 4 Uses integral tables and technology sensibly.
-

VI.5.16

Inverse hyperbolic functions

The functions that undo the hyperbolic functions. They can be written using logarithms and appear as answers to some integrals.

- 1 Writes inverse hyperbolic functions as logarithms.
- 2 Finds their derivatives.
- 3 Recognizes integrals that give them as answers.
- 4 Compares them with the similar inverse trig patterns.

Applications of integrals

This branch uses integrals to measure real things: distances, areas, volumes of solid shapes, work done by forces, and totals in probability and economics.

VI.6.01

Displacement vs distance

The difference between how far you end up from the start (displacement) and how far you actually traveled (distance).

- 1 Finds displacement by integrating velocity.
 - 2 Finds total distance by integrating speed.
 - 3 Finds where velocity changes sign, meaning the object turns around.
 - 4 Finds final position from a starting position and velocity.
-

VI.6.02

Net change

Integrating a rate of change gives the total change, like water flow giving the total amount added to a tank.

- 1 Finds the total change by integrating a rate.
 - 2 Finds the amount at a later time from a starting amount.
 - 3 Handles an inflow and an outflow at the same time.
 - 4 Estimates net change from a table of rates.
-

VI.6.03

Area between curves in x

Finding the area of the region between two curves by subtracting the lower curve from the upper one.

- 1 Integrates the top curve minus the bottom curve.
 - 2 Finds where the curves cross to get the endpoints.
 - 3 Splits the area where the top and bottom curves switch places.
 - 4 Sets up and evaluates the full area integral.
-

VI.6.04

Area between curves in y

Finding area between curves by slicing sideways instead, which is sometimes much easier.

- 1 Integrates the right curve minus the left curve.
 - 2 Rewrites each curve with x in terms of y.
 - 3 Recognizes when slicing sideways is simpler.
 - 4 Sets up and evaluates the full area integral.
-

VI.6.05

Known cross-sections

Finding the volume of a solid by slicing it like a loaf of bread, where each slice has a known shape.

- 1 Finds volume when each slice is a square.
- 2 Finds volume when each slice is a semicircle.
- 3 Finds volume when each slice is a triangle.
- 4 Sets up the volume integral for any given slice shape.

VI.6.06

Disk method

Finding the volume of a solid made by spinning a region around a line, by stacking thin coin-shaped disks.

- 1 Writes the radius of each disk as a function.
- 2 Uses π times radius squared for each thin slice.
- 3 Finds volumes when spinning around the horizontal axis.
- 4 Finds volumes when spinning around the vertical axis.

VI.6.07

Washer method

Like the disk method, but for spun solids with a hole in the middle, using ring-shaped slices called washers.

- 1 Identifies the outer and inner radius of each ring.
- 2 Sets up the volume integral.
- 3 Handles spinning around lines other than the axes.
- 4 Evaluates the integral to find the volume.

VI.6.08

Shell method

Finding volumes of spun solids by adding up thin nested tubes, like the layers of an onion.

- 1 Identifies the radius and height of each tube.
- 2 Uses the tube's unrolled area, two π times radius times height.
- 3 Recognizes when tubes are easier than disks.
- 4 Handles spinning around lines other than the axes.

VI.6.09

Arc length

Finding the length of a curved path by adding up tiny straight pieces.

- 1 Builds the length formula from Pythagoras' theorem on tiny pieces.
- 2 Sets up the length integral for a curve.
- 3 Evaluates it exactly or estimates it.
- 4 Uses the sideways version with x in terms of y .

VI.6.10

Surface area of revolution

Finding the outer surface area of a shape made by spinning a curve around a line.

- 1 States the surface area formula.
- 2 Sets up the surface area integral.
- 3 Evaluates it.
- 4 Explores Gabriel's horn, which has finite volume but endless surface area.

VI.6.11

Work

Work is force times distance. Integrals handle cases where the force changes, like stretching a spring.

- 1 Finds work as force times distance for a constant force.
- 2 Finds work to stretch a spring, where force grows with stretch.
- 3 Finds work to pump liquid out of a tank.
- 4 Finds work to lift a hanging cable or chain.

VI.6.12

Fluid force

Finding the total push of water on a wall, where the pressure grows deeper down.

- 1 Explains that water pressure increases with depth.
 - 2 Splits the wall into thin horizontal strips.
 - 3 Sets up the force integral.
 - 4 Finds the force on dams and underwater windows.
-

VI.6.13

Center of mass

Finding the balance point of a flat shape or object, where it would sit level on a pin.

- 1 Computes moments, which measure turning effect about a line.
 - 2 Finds the balance point (centroid) of a region.
 - 3 Uses symmetry to find balance points quickly.
 - 4 Uses Pappus's theorem to find volumes from balance points.
-

VI.6.14

Probability densities

Using curves to describe probability, where the area under the curve over a range gives the chance of landing in it.

- 1 Finds a probability as the area under the curve.
 - 2 Checks that the total area under the curve is one.
 - 3 Finds the average outcome using an integral.
 - 4 Works with the exponential distribution, used for waiting times.
-

VI.6.15

Economics

Using integrals in economics, such as totaling costs and measuring how much buyers and sellers gain from a sale.

- 1 Finds a total cost or revenue from its rate per item.
- 2 Finds consumer surplus, the benefit to buyers.
- 3 Finds producer surplus, the benefit to sellers.
- 4 Finds the present value of a stream of future payments.

Differential equations I

This branch introduces differential equations, which describe how something changes and ask what it will look like over time. They model growth, cooling, populations and mixing.

VI.7.01

What a differential equation is

A differential equation links a quantity to its own rate of change. For example, a population grows faster the bigger it already is.

- 1 Recognizes an equation that includes a derivative.
 - 2 Identifies the order, the highest derivative that appears.
 - 3 Explains that the solution is a whole function, not a single number.
 - 4 Translates a differential equation into an everyday sentence.
-

VI.7.02

Verify solutions

Checking whether a suggested function really solves a differential equation by plugging it in.

- 1 Differentiates the suggested function.
 - 2 Substitutes it and its derivative into the equation.
 - 3 Checks that both sides of the equation match.
 - 4 Finds a constant that makes a suggested solution work.
-

VI.7.03

Slope fields

A slope field is a picture made of tiny line segments showing which way solutions flow at every point, like wind arrows on a weather map.

- 1 Draws short slope marks at points using the equation.
 - 2 Matches slope field pictures to their equations.
 - 3 Sketches a solution curve by following the slopes.
 - 4 Uses lines of equal slope to predict long-run behavior.
-

VI.7.04

Euler's method

Euler's method: estimating a solution by taking small straight steps in the direction the equation points.

- 1 Explains stepping along the tangent line to a new point.
 - 2 Runs several steps by hand.
 - 3 Sees how smaller steps give a better estimate.
 - 4 Decides whether the estimate is too high or too low.
-

VI.7.05

Separable equations

Solving equations where all the y terms can be moved to one side and all the x terms to the other.

- 1 Moves y terms and x terms to opposite sides.
- 2 Integrates both sides.
- 3 Rearranges to get y on its own.
- 4 Uses a starting value, and watches for solutions lost when dividing.

VI.7.06

Growth & decay

Modeling things that grow or shrink at a rate proportional to their size, like savings earning interest or radioactive decay.

- 1 Recognizes the equation where the rate is a constant times the amount.
- 2 Writes its solution as a starting amount times an exponential.
- 3 Finds the growth constant from data, like a half-life.
- 4 Uses the model to make predictions.

VI.7.07

Newton's law of cooling

Newton's law of cooling: a hot drink cools faster when it is much hotter than the room, and slower as it nears room temperature.

- 1 Sets up the cooling equation.
- 2 Solves it.
- 3 Finds the cooling constant from measurements.
- 4 Works backwards to find when cooling began, as in detective problems.

VI.7.08

Logistic growth

Logistic growth models a population that grows quickly at first, then levels off as it nears the most its environment can support.

- 1 Explains the carrying capacity, the population ceiling.
- 2 Writes the logistic equation.
- 3 Shows growth is fastest at half the carrying capacity.
- 4 Solves it, reads off its features, and adds removal like fishing.

VI.7.09

General vs particular

A differential equation has a whole family of solutions. One known starting value picks out the specific one.

- 1 Describes the family of all solutions.
- 2 Uses a starting value to pick one solution from the family.
- 3 Spots unusual solutions that the general formula misses.
- 4 Graphs several members of the family together.

VI.7.10

Mixing problems

Modeling a tank where salty water flows in and mixed water flows out, to track how much salt is inside.

- 1 Writes the change as rate in minus rate out.
- 2 Sets up the equation for the tank.
- 3 Solves it.
- 4 Finds the amount the tank settles toward in the long run.

VI.7.11

Equilibrium & stability

Finding steady states where nothing changes, and whether things drift back to them or run away if nudged.

- ① Finds equilibrium values, where the rate of change is zero.
- ② Draws a phase line showing which way values move.
- ③ Classifies each equilibrium as stable or unstable.
- ④ Studies how equilibria appear or vanish as a setting changes.

VI.7.12

Modeling from words

Turning a description in words into a differential equation, and checking it makes sense.

- ① Translates "proportional to" into a constant times a quantity.
- ② Writes the full equation from a word description.
- ③ States the assumptions the model makes.
- ④ Checks that the units on both sides match.

Parametric & polar

This branch describes curves in two new ways: by tracking a moving point over time (parametric), or by angle and distance from a center (polar). Calculus then finds slopes, lengths and areas for them.

VI.8.01 Parametric curves

A parametric curve gives x and y each as a formula in time, like tracing the path of a moving bug.

- 1 Reads a curve given as separate formulas for x and y in terms of time.
 - 2 Plots the curve from a table of time values.
 - 3 Marks the direction the point moves along the curve.
 - 4 Writes parametric formulas for circles and ellipses.
-

VI.8.02 Eliminate the parameter

Removing the time variable to get an ordinary equation linking x and y directly.

- 1 Solves one formula for time and substitutes into the other.
 - 2 Uses a trig identity to remove time from sine and cosine formulas.
 - 3 Notices when only part of the ordinary graph is actually traced.
 - 4 Compares the parametric graph with the ordinary one.
-

VI.8.03 Parametric slope

Finding the slope of a parametric curve by dividing how fast y changes by how fast x changes.

- 1 Uses the formula: rate of y over rate of x .
 - 2 Writes tangent lines to parametric curves.
 - 3 Finds points where the curve has a flat tangent.
 - 4 Finds points where the curve has a vertical tangent.
-

VI.8.04 Parametric second derivative

Finding how the slope of a parametric curve changes, which tells which way it bends.

- 1 States the formula for the second derivative.
 - 2 Computes it for a given curve.
 - 3 Uses it to decide which way the curve bends.
 - 4 Avoids the common mistake of just differentiating the slope in time.
-

VI.8.05 Parametric arc length

Finding the length of a parametric curve, like the distance a moving point travels.

- 1 States the arc length formula for parametric curves.
- 2 Sets up the length integral.
- 3 Evaluates it.
- 4 Finds the total distance traveled by a moving point.

VI.8.06

Motion in the plane

Describing motion in two dimensions, like a thrown ball, using arrows (vectors) for position, velocity and acceleration.

- 1 Writes position as an arrow from the origin (a position vector).
- 2 Finds velocity and speed.
- 3 Finds acceleration.
- 4 Models the curved paths of thrown objects.

VI.8.07

Polar coordinates

Polar coordinates locate a point by its distance from the center and its angle, like giving directions by compass bearing and distance.

- 1 Plots points given as a distance and an angle.
- 2 Converts polar points to ordinary x and y coordinates.
- 3 Converts ordinary coordinates to polar.
- 4 Recognizes that one point has many different polar names.

VI.8.08

Polar graphs

Sketching curves described in polar form, which produces shapes like flowers, hearts and spirals.

- 1 Graphs polar circles and lines.
- 2 Graphs rose curves with petals.
- 3 Graphs heart-shaped cardioids and looped limaçons.
- 4 Graphs spirals.

VI.8.09

Polar slope

Finding the slope of a polar curve at a point.

- 1 Rewrites a polar curve as a parametric curve.
- 2 Uses the slope formula for polar curves.
- 3 Finds tangent lines where the curve passes through the center.
- 4 Finds flat and vertical tangent points.

VI.8.10

Polar area

Finding the area inside a polar curve by adding up thin pizza-slice wedges.

- 1 Uses the wedge formula: half the integral of r squared.
- 2 Finds the area of one petal of a rose.
- 3 Finds the correct angles where the curve passes through the center.
- 4 Uses symmetry to simplify the calculation.

VI.8.11

Area between polar curves

Finding the area between two polar curves.

- 1 Finds where the two curves meet.
- 2 Subtracts the inner curve's area from the outer one's.
- 3 Sets up the area integral.
- 4 Evaluates it.

Polar arc length

Finding the length of a polar curve.

- 1 States the polar arc length formula.
- 2 Sets up the length integral.
- 3 Evaluates it exactly.
- 4 Estimates it with technology when no exact answer is possible.

Sequences & series

This branch asks whether adding up infinitely many numbers can give a finite total. It builds a set of tests for deciding when an endless sum settles down.

VI.9.01

Limits of sequences

A sequence is an endless list of numbers. The question is whether the list settles toward one value.

- 1 Decides whether a sequence settles (converges) or not (diverges).
 - 2 Uses limits of functions to find where a sequence heads.
 - 3 Uses the squeeze idea to trap a sequence.
 - 4 Uses the fact that a list always rising but capped must settle.
-

VI.9.02

Series & partial sums

A series is the sum of an endless list. It has a total only if the running totals (partial sums) settle down.

- 1 Computes running totals of a series.
 - 2 Explains that a series has a total when its running totals settle.
 - 3 Sums telescoping series, where most terms cancel out.
 - 4 Writes a series using sigma (summation) notation.
-

VI.9.03

n th term test

A first check: if the terms don't shrink to zero, the endless sum cannot have a finite total.

- 1 Explains that terms must shrink to zero for a total to exist.
 - 2 Applies the test to show a series has no total.
 - 3 Explains that shrinking terms alone do not guarantee a total.
 - 4 Uses one plus a half plus a third and so on, which grows forever.
-

VI.9.04

Integral test

Comparing an endless sum to the area under a curve to decide whether it has a finite total.

- 1 Checks the conditions needed to use the test.
 - 2 Applies the test to a series.
 - 3 Estimates how much is left over after adding many terms.
 - 4 Links the test to integrals that run out to infinity.
-

VI.9.05

p-series

A family of benchmark sums, one over n to a power, whose behavior depends only on that power.

- 1 Recognizes series of the form one over n to a power.
 - 2 Knows they have a total when the power is bigger than one.
 - 3 Knows the power-one case (the harmonic series) grows forever.
 - 4 Uses these as benchmarks to judge other series.
-

VI.9.06

Direct comparison

Deciding about a series by comparing it term by term with one whose behavior is already known.

- 1 Picks a known series to compare with.
- 2 Concludes no total when the series is bigger than one that grows forever.
- 3 Concludes a total exists when it is smaller than one that settles.
- 4 Writes the full comparison argument clearly.

VI.9.07

Limit comparison

A more flexible comparison: if two series' terms are roughly in proportion in the long run, they behave the same way.

- 1 Divides the terms of one series by the other's.
- 2 Checks that the ratio settles to a positive, finite number.
- 3 Chooses a good partner series to compare with.
- 4 States the conclusion correctly.

VI.9.08

Alternating series test

A test for series whose signs flip back and forth, plus and minus.

- 1 Recognizes a series with alternating signs.
- 2 Checks that the terms shrink steadily to zero.
- 3 Concludes the series has a total.
- 4 Bounds the error when stopping after some terms.

VI.9.09

Ratio test

The ratio test compares each term to the one before. If terms shrink by a steady factor below one, the sum is finite.

- 1 Finds the long-run ratio of each term to the one before.
- 2 Concludes a total exists when the ratio is less than one.
- 3 Knows the test gives no answer when the ratio is exactly one.
- 4 Applies it to series with factorials (products like 5 times 4 times 3 times 2 times 1).

VI.9.10

Root test

The root test, a cousin of the ratio test, which looks at the n th root of the n th term.

- 1 Takes the n th root of each term.
- 2 Applies the test to a series.
- 3 Recognizes when it is easier than the ratio test.
- 4 Compares the root and ratio tests.

Absolute vs conditional

Some alternating sums settle only because positives and negatives cancel. These behave very differently from sums that settle anyway.

- ① Identifies series that settle even with all signs made positive.
- ② Identifies series that settle only thanks to the alternating signs.
- ③ Classifies a series into one of these types.
- ④ Explores how reordering the second kind can change its total.

Power & Taylor series

This branch shows how functions like sine or e to the x can be written as endless polynomials. That is how calculators actually compute them.

VI.10.01 Power series

A power series is a polynomial that never ends, with powers of x going on forever.

- 1 Reads a power series written in sigma notation.
 - 2 Describes it as a polynomial that never ends.
 - 3 Adds up the first few terms to estimate a value.
 - 4 Identifies the center and the coefficients (the multiplying numbers).
-

VI.10.02 Radius of convergence

A power series only works for x values within a certain distance of its center, called the radius of convergence.

- 1 Applies the ratio test to a power series.
 - 2 Finds the radius of convergence.
 - 3 Handles series that work nowhere except the center, or everywhere.
 - 4 Handles series centered away from zero.
-

VI.10.03 Interval & endpoints

Finding exactly which x values a power series works for, including checking the two edges.

- 1 Tests each edge point separately.
 - 2 Decides whether each edge is included or not.
 - 3 Writes the full range of x values that work.
 - 4 Checks how the edges can change after differentiating the series.
-

VI.10.04 Taylor polynomials

A Taylor polynomial is a simple polynomial built to imitate a complicated function near one point.

- 1 Matches the function's value and derivatives at a point.
 - 2 Builds a Taylor polynomial of a given degree.
 - 3 Centers the polynomial at any chosen point.
 - 4 Graphs the polynomial against the function to see the fit.
-

VI.10.05 Series for e^x

Writing e to the x as an endless polynomial, one of the most useful series in math.

- 1 Derives the series for e to the x.
- 2 Uses it to approximate the number e.
- 3 Substitutes to get new series, like e to the minus x squared.
- 4 Shows the series works for every value of x.

VI.10.06

Series for sin & cos

Writing sine and cosine as endless polynomials.

- 1 Derives the series for sine and cosine.
- 2 Notices sine uses only odd powers and cosine only even powers.
- 3 Uses the series to approximate a value.
- 4 Uses small-angle shortcuts, like sine of x being nearly x .

VI.10.07

Series for $1/(1 - x)$ & $\ln(1 + x)$

Series for one over one minus x , built from repeated multiplication, and for related functions like the logarithm.

- 1 Writes the repeated-ratio (geometric) series as a power series.
- 2 Integrates it to get the series for the logarithm.
- 3 Substitutes to get the series for arctangent.
- 4 Shows these series work only for x between minus one and one.

VI.10.08

New series from old

Making new series quickly from known ones instead of starting from scratch.

- 1 Substitutes an expression for x in a known series.
- 2 Multiplies a series by a power of x .
- 3 Adds two series together.
- 4 Finds a particular coefficient quickly using these tricks.

VI.10.09

Differentiate & integrate series

Differentiating and integrating series one term at a time, like ordinary polynomials.

- 1 Differentiates or integrates a series term by term.
- 2 Knows the radius of convergence stays the same.
- 3 Finds the total of a series by differentiating a known one.
- 4 Integrates a series to build a new one.

VI.10.10

Multiply & divide series

Multiplying and dividing series to find the first few terms of a new series.

- 1 Finds the first few terms of a product of two series.
- 2 Divides the sine series by the cosine series to get tangent.
- 3 Keeps track of which powers are needed.
- 4 Checks results against known series.

VI.10.11

Lagrange error bound

A guarantee on how far off a Taylor polynomial's estimate can be.

- 1 States the error bound formula.
- 2 Finds the largest size of the next derivative on the range.
- 3 Uses the formula to bound the error.
- 4 Chooses how many terms are needed for a desired accuracy.

VI.10.12

Alternating series error

For alternating series, the error from stopping early is no bigger than the first term left out.

- 1 Uses the next term as a bound on the error.
 - 2 Applies this to an estimate.
 - 3 Decides whether the estimate is too high or too low.
 - 4 Compares this with the general error bound.
-

VI.10.13

Binomial series

Expanding one plus x raised to any power, even a fraction or negative number, as an endless series.

- 1 Writes the series for one plus x to any power.
 - 2 Computes the coefficients.
 - 3 Approximates square roots of numbers close to one.
 - 4 Shows the series works only for x between minus one and one.
-

VI.10.14

Euler's formula

Euler's formula links exponentials to sine and cosine using imaginary numbers. It leads to a famous equation joining five key numbers.

- 1 Derives $e^{i\theta}$ by substituting into the exponential series.
 - 2 Shows it equals $\cos \theta + i \sin \theta$.
 - 3 Derives $e^{i\pi} + 1 = 0$.
 - 4 Links the formula to the polar form of imaginary numbers.
-

VI.10.15

Integrals by series

Using series to handle integrals that have no ordinary formula, such as the one behind the bell curve.

- 1 Recognizes integrals with no standard formula answer.
 - 2 Integrates the function's series term by term.
 - 3 Approximates the integral's value from a few terms.
 - 4 Bounds the error of the approximation.
-

VI.10.16

Limits by series

Finding tricky limits by replacing functions with their series and seeing which terms matter.

- 1 Replaces each function in the limit with its series.
- 2 Cancels the leading terms that cause trouble.
- 3 Finds the limit from what remains.
- 4 Compares this method with L'Hôpital's rule.

Statistical inference

This branch is about drawing trustworthy conclusions from data: how to collect a fair sample, how much results can wobble by chance, and how to test a claim.

VI.11.01 Sampling methods

How to choose a smaller group (a sample) to learn about a larger one (the population) without skewing the results.

- ① Distinguishes the whole population from the sample taken.
 - ② Describes random, stratified and systematic ways to choose a sample.
 - ③ Identifies ways a sample can be biased, or unfair.
 - ④ Chooses a suitable sampling method for a situation.
-

VI.11.02 Studies & experiments

The difference between watching what happens and running a controlled experiment, and what each can prove.

- ① Describes observational studies, which watch without interfering.
 - ② Describes experiments with a comparison (control) group.
 - ③ Explains random assignment and blinding.
 - ④ Decides whether a study can show cause and effect.
-

VI.11.03 Sampling distributions

Different samples give slightly different results. The pattern of those results is called a sampling distribution.

- ① Explains that results vary from sample to sample.
 - ② Simulates many samples to see the spread of results.
 - ③ Finds the standard error, the typical size of that wobble.
 - ④ Explains why bigger samples wobble less.
-

VI.11.04 Central limit theorem

The central limit theorem: averages of large samples follow a bell-shaped curve, whatever the original data looked like.

- ① Explains that sample averages are roughly bell-shaped.
- ② Finds the center and spread of the sample averages.
- ③ Finds probabilities about a sample average.
- ④ Judges whether a sample is large enough for this to hold.

VI.11.05

Confidence intervals for a proportion

A confidence interval gives a range likely to contain a true percentage, such as a poll saying 52% plus or minus 3%.

- ① Writes an interval as an estimate plus or minus a margin of error.
 - ② Computes an interval for a percentage.
 - ③ Explains correctly what 95% confidence means.
 - ④ Chooses a sample size to get a desired margin of error.
-

VI.11.06

Confidence intervals for a mean

A range likely to contain a true average, like the average height of all adults in a country.

- ① Builds an interval when the population's spread is known.
 - ② Explains the adjustment (the t idea) when the spread must be estimated.
 - ③ Computes and interprets an interval for an average.
 - ④ Explains how sample size affects the interval's width.
-

VI.11.07

Hypothesis testing ideas

Hypothesis testing asks whether data give strong evidence against a default assumption, or could just be chance.

- ① States the default claim (null) and the alternative claim.
 - ② Explains the significance level, the cutoff for strong evidence.
 - ③ Explains what a p-value means, and what it does not mean.
 - ④ Decides whether to test in one direction or both.
-

VI.11.08

Tests with the binomial

Testing a claim about a proportion, like whether a coin is fair, by counting successes.

- ① Sets up a test about a proportion.
 - ② Finds the critical region, the results that would reject the claim.
 - ③ Computes the p-value for the observed count.
 - ④ States the conclusion in the context of the problem.
-

VI.11.09

Tests for a mean

Testing a claim about an average, and understanding the two ways a test can get it wrong.

- ① Sets up a test about an average using the bell curve.
- ② Computes the test statistic, a measure of how unusual the data are.
- ③ Draws a conclusion from the test.
- ④ Explains false alarms (Type I) and missed effects (Type II).

Mechanics

This branch applies math to physical motion and forces: falling objects, things sliding on slopes, pulleys and balancing beams.

VI.12.01 Kinematics graphs

Reading motion from graphs, where the steepness and the area under the line each tell part of the story.

- 1 Finds velocity from the steepness (gradient) of a position–time graph.
 - 2 Finds acceleration from the steepness of a velocity–time graph.
 - 3 Finds displacement from the area under a velocity–time graph.
 - 4 Distinguishes total distance traveled from change in position.
-

VI.12.02 Constant acceleration

A set of five equations (often called "suvat") for objects speeding up or slowing down at a steady rate.

- 1 Knows the constant-acceleration equations and what each letter means.
 - 2 Chooses the right equation from the information given.
 - 3 Solves for the unknown quantity.
 - 4 Handles journeys with two stages, like speeding up then braking.
-

VI.12.03 Vertical motion

Modeling objects thrown straight up or dropped, with gravity pulling them down.

- 1 Uses gravity as a steady downward acceleration.
 - 2 Models an object thrown up that falls back down.
 - 3 Finds the maximum height reached.
 - 4 Finds the total time spent in the air.
-

VI.12.04 Forces & Newton's laws

Newton's laws describe how forces change motion. A bigger push gives a bigger acceleration.

- 1 Identifies weight, the surface's push back (normal reaction) and string tension.
 - 2 Combines forces into one overall (resultant) force.
 - 3 Uses force equals mass times acceleration.
 - 4 Applies the third law: every push has an equal push back.
-

VI.12.05 Resolving forces

Splitting a slanted force into sideways and up-down parts, which makes problems much easier.

- 1 Splits a force into two parts using sine and cosine.
- 2 Uses balanced forces (equilibrium) to find unknowns.
- 3 Handles objects resting or moving on a slope.
- 4 Finds an unknown force from the other forces present.

VI.12.06

Connected particles

Objects linked by strings or rods, like two weights over a pulley, that move together.

- ① Finds the tension (pulling force) in a connecting string.
 - ② Handles strings running over pulleys.
 - ③ Solves for the shared acceleration of the linked objects.
 - ④ Models lifts and vehicles towing trailers.
-

VI.12.07

Friction

Friction is the grip between surfaces that resists sliding. It has a maximum strength that depends on how hard the surfaces press together.

- ① Uses the rule that friction is at most a constant times the normal reaction.
 - ② Identifies limiting friction, when an object is just about to slip.
 - ③ Includes friction for objects on a slope.
 - ④ Decides whether an object will slide or stay put.
-

VI.12.08

Projectiles

Modeling the curved path of a thrown or kicked object by treating sideways and up-down motion separately.

- ① Splits the motion into horizontal and vertical parts.
 - ② Finds how long the object stays in the air.
 - ③ Finds how far it travels and how high it goes.
 - ④ Chooses the launch speed or angle to hit a target.
-

VI.12.09

Moments

A moment is the turning effect of a force, like pushing a door far from its hinge versus close to it.

- ① Finds a moment as force times the perpendicular distance to the pivot.
- ② Balances turning effects on a level rod.
- ③ Handles a rod just on the point of tipping over.
- ④ Finds the force needed from a support.

VII

Era VII • Space

About ages 18–20 • second-year college mathematics • coming in Mathera 2.0

Era VII takes math into many dimensions. Linear algebra studies vectors and matrices, the language behind computer graphics, data science and engineering. Multivariable calculus extends calculus to surfaces and 3-D space. Differential equations model how real systems change over time.

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Vectors in $\mathbb{R}^n$

Vectors are quantities with both a size and a direction, like a push, a velocity, or a step on a map. This branch builds the basic toolkit for working with them, the foundation for everything else in this era.

VII.1.01 What a vector is

A vector is an arrow with a length and a direction, or equally a short list of numbers. A wind of 10 mph blowing northeast is a vector.

- 1 Draws and reads vectors as arrows with a length and a direction.
- 2 Writes a vector as a list of numbers, such as 3 steps right and 2 up.
- 3 Tells apart an arrow pinned to a spot from one that can slide anywhere.
- 4 Works with vectors that have any number of entries, not just two or three.

VII.1.02 Vector addition

Adding vectors means combining two moves into one, like walking 3 blocks east and then 4 blocks north.

- 1 Adds two arrows by placing the tail of the second at the tip of the first.
- 2 Adds two arrows from the same starting point by completing a parallelogram.
- 3 Adds vectors as lists by adding matching entries.
- 4 Subtracts vectors and sees the result as the arrow from one tip to the other.

VII.1.03 Scalar multiplication

Multiplying a vector by an ordinary number (a scalar) stretches or shrinks it, like doubling a speed while keeping the direction.

- 1 Stretches or shrinks an arrow by multiplying it by a number.
- 2 Sees that multiplying by a negative number flips the arrow around.
- 3 Multiplies each entry of a vector list by the number.
- 4 Recognizes when two vectors point along the same line (are parallel).

VII.1.04 Magnitude

The magnitude of a vector is its length, found with the Pythagorean theorem. It tells how big the push or movement is.

- 1 Finds the length of a flat, two-entry vector.
- 2 Finds lengths of vectors in 3D and with more entries.
- 3 Finds the distance between two points as the length of the arrow joining them.
- 4 Uses basic facts about length, such as a detour never being shorter.

VII.1.05

Unit vectors

A unit vector has length exactly 1 and just records a direction, like a compass heading with no distance attached.

- 1 Shrinks or stretches a vector to length 1 while keeping its direction.
- 2 Uses the three standard unit arrows that point along the x, y and z axes.
- 3 Builds a direction vector from a given angle.
- 4 Writes any vector as its length times its direction.

VII.1.06

Linear combinations

A linear combination mixes vectors using chosen amounts of each, like a recipe: 2 parts of one move plus 3 parts of another.

- 1 Combines vectors using given amounts (weights) of each.
- 2 Finds the amounts needed to reach a given target vector.
- 3 Sets up that search as a set of equations to solve together.
- 4 Pictures combinations as a slanted grid covering the plane.

VII.1.07

Span, first look

The span of some vectors is every point you can reach by combining them. One arrow gives a line; two different arrows usually give a flat plane.

- 1 Sees that one vector's span is a line through the starting point.
- 2 Decides whether two flat vectors reach the whole plane or just a line.
- 3 Describes what two vectors in 3D reach: a plane or a line.
- 4 Checks whether a given vector can be reached from a set of vectors.

VII.1.08

Dot product

The dot product multiplies two vectors to get a single number that measures how much they point the same way.

- 1 Computes a dot product by multiplying matching entries and adding them up.
- 2 Uses its basic rules, such as order not mattering.
- 3 Finds a vector's length from its dot product with itself.
- 4 Computes work done as force times distance moved, using the dot product.

VII.1.09

Angle between vectors

The dot product can reveal the angle between two arrows. When it is zero, the arrows are at right angles.

- 1 Learns the formula linking the dot product to the angle's cosine.
- 2 Computes the angle between two given vectors.
- 3 Recognizes perpendicular vectors because their dot product is zero.
- 4 Uses the fact that a dot product can never exceed the product of the lengths.

Projections

Projecting one vector onto another finds its shadow along that direction, like the part of a slanted push that moves a cart forward.

- ① Finds how long the shadow of one vector on another is.
 - ② Finds the shadow itself as a vector.
 - ③ Finds the leftover part of the vector that sits at right angles.
 - ④ Finds the point on a line closest to a given point.
-

Cross product

The cross product takes two 3D arrows and gives a third arrow at right angles to both. Its length equals the area they outline.

- ① Computes a cross product with a set recipe of multiplications.
 - ② Uses the right-hand rule to tell which way the result points.
 - ③ Finds the area of a parallelogram from its two side vectors.
 - ④ Finds the volume of a slanted box from its three edge vectors.
-

Lines & planes in 3D

Vectors describe straight lines and flat planes in 3D space, the way a pilot might describe a flight path or a runway surface.

- ① Writes a line as a starting point plus any amount of a direction vector.
- ② Writes a plane's equation from an arrow sticking straight out of it.
- ③ Finds where lines and planes meet each other.
- ④ Finds shortest distances between points, lines and planes.

Linear systems

A linear system is a set of equations that must all be true at once. This branch teaches a reliable, step-by-step method for solving any such system, however many unknowns it has.

VII.2.01 Systems as matrices

Writing a set of equations as a matrix, a rectangular table of numbers, so only the numbers that matter are kept.

- 1 Writes the numbers in front of the unknowns as a table (the coefficient matrix).
 - 2 Adds the right-hand-side numbers as an extra column (the augmented matrix).
 - 3 Writes the whole system in the short form: matrix times unknowns equals answers.
 - 4 Writes the system as a combination of column vectors that must hit a target.
-

VII.2.02 Row operations

Row operations are three safe moves on the rows of a matrix that simplify a system without changing its solutions.

- 1 Swaps two rows.
 - 2 Multiplies a row by a nonzero number.
 - 3 Adds a multiple of one row to another row.
 - 4 Explains why these moves never gain or lose any solutions.
-

VII.2.03 Row echelon form

Row echelon form is a staircase shape the matrix is tidied into, which makes the system easy to solve from the bottom up.

- 1 Recognizes the staircase shape, with zeros below each step.
 - 2 Uses row operations to clear out numbers below each step, working downward.
 - 3 Identifies the pivots, the first nonzero number on each step.
 - 4 Solves from the bottom row up, substituting as it goes.
-

VII.2.04 Reduced row echelon

Reduced row echelon form is the tidiest possible version of the matrix, where the answers can be read off directly.

- 1 Scales rows so each pivot becomes a 1.
- 2 Clears the numbers above each pivot to zero as well.
- 3 Runs the full process from start to finish (Gauss–Jordan elimination).
- 4 Understands that every matrix tidies to exactly one such final form.

VII.2.05

Pivots & free variables

Unknowns that land on a pivot are fixed by the others; the rest (free variables) can be anything, which is how systems get infinitely many answers.

- 1 Picks out the unknowns tied to pivots (the basic variables).
- 2 Picks out the free variables, which can take any value.
- 3 Decides whether a system has no solution, one, or infinitely many.
- 4 Predicts the possible outcomes just from the matrix's shape.

VII.2.06

Parametric solutions

When there are infinitely many solutions, they can be described neatly by letting the free variables vary, like tracing all points along a line.

- 1 Writes each basic variable in terms of the free ones.
- 2 Writes the full solution as a fixed vector plus multiples of other vectors.
- 3 Pictures the solution set as a point, line or plane.
- 4 Checks that a proposed solution really satisfies every equation.

VII.2.07

Consistency

A system is consistent if it has at least one solution. Some systems contradict themselves and have none.

- 1 Spots a row that says 0 equals a nonzero number, meaning no solution.
- 2 Decides whether a given system has any solution at all.
- 3 Finds which right-hand-side values make a system solvable.
- 4 Separates two questions: is there a solution, and is it the only one.

VII.2.08

Homogeneous systems

A homogeneous system has every equation equal to zero. It always has the all-zeros solution, and sometimes many more.

- 1 Recognizes and sets up a system where every right-hand side is zero.
- 2 Notes that setting every unknown to zero always works.
- 3 Decides when other, nonzero solutions exist too.
- 4 Describes all the solutions as everything reachable from a few vectors.

VII.2.09

Applications

Linear systems model real problems with many interlocking quantities, from traffic on streets to chemistry to electronics.

- 1 Sets up and solves traffic or pipe networks where flow in equals flow out.
- 2 Balances chemical equations so atoms on both sides match.
- 3 Finds currents in simple electrical circuits.
- 4 Finds a smooth curve that passes exactly through given points.

Matrix algebra

Matrices (rectangular tables of numbers) can be added and multiplied much like ordinary numbers, but with some surprising differences. This branch learns that arithmetic and how to undo a matrix with its inverse.

VII.3.01 Matrix notation

The basic vocabulary of matrices: rectangular tables of numbers arranged in rows and columns.

- 1 Identifies the rows (across) and columns (down) of a matrix.
- 2 Locates an entry by its row number and column number.
- 3 States a matrix's size as rows by columns.
- 4 Recognizes special shapes such as square, diagonal and triangular matrices.

VII.3.02 Add & scale matrices

Adding matrices and multiplying them by a single number, done one entry at a time.

- 1 Knows only matrices of the same size can be added.
- 2 Adds two matrices by adding matching entries.
- 3 Multiplies every entry of a matrix by the same number.
- 4 Uses the familiar rules, such as order not mattering when adding.

VII.3.03 Matrix times vector

Multiplying a matrix by a vector produces a new vector. It is the basic way a matrix acts on things.

- 1 Multiplies by pairing each row with the vector and adding products.
- 2 Sees the result as a mix of the matrix's columns.
- 3 Checks the sizes match before multiplying.
- 4 Uses the fact that this multiplication respects adding and scaling.

VII.3.04 Matrix multiplication

Multiplying two matrices together, which combines two actions into one.

- 1 Checks which sizes can be multiplied and the size of the result.
- 2 Computes each entry of the product from a row and a column.
- 3 Computes the product one column at a time.
- 4 Sees the product as doing one action and then the other.

VII.3.05 Algebra rules

The rules of matrix arithmetic, including the surprise that order matters: A times B is often different from B times A.

- 1 Finds examples where swapping the order changes the product.
- 2 Uses the rules that do still hold, such as regrouping and expanding brackets.
- 3 Sees that you cannot simply cancel a matrix from both sides.
- 4 Computes powers of a matrix by repeated multiplication.

VII.3.06

Identity & zero

The identity matrix acts like the number 1 and the zero matrix like 0. Some odd matrices become zero when multiplied by themselves.

- 1 Recognizes the identity matrix: 1s down the diagonal, 0s elsewhere.
- 2 Recognizes the zero matrix and its role.
- 3 Shows multiplying by the identity leaves any matrix unchanged.
- 4 Finds nonzero matrices whose powers eventually become zero.

VII.3.07

Transpose

The transpose flips a matrix over its diagonal, so rows become columns.

- 1 Writes the transpose by turning rows into columns.
- 2 Uses the rule that transposing a product reverses the order.
- 3 Recognizes symmetric matrices, which equal their own transpose.
- 4 Writes a dot product as a row times a column.

VII.3.08

2×2 inverse

The inverse of a matrix undoes it, like dividing undoes multiplying. For 2-by-2 matrices there is a short formula.

- 1 Uses the formula to find the inverse of a 2-by-2 matrix.
- 2 Recognizes when the formula breaks down and no inverse exists.
- 3 Solves a system of equations by multiplying by the inverse.
- 4 Checks an inverse by multiplying back to get the identity.

VII.3.09

Inverse by row reduction

Finding the inverse of a larger matrix using the same row operations used to solve equations.

- 1 Places the identity matrix beside the matrix to start.
- 2 Row-reduces the combined table.
- 3 Reads the inverse from the right-hand half.
- 4 Recognizes during the process when no inverse exists.

VII.3.10

Invertible matrix theorem

A single big theorem listing many facts about a square matrix that are all true together or all false together, such as having an inverse.

- 1 Lists the equivalent facts, such as having an inverse and a pivot in every column.
- 2 Uses one known fact to conclude the others.
- 3 Knows the theorem applies only to square matrices.
- 4 Uses quick checks to decide whether a matrix is invertible.

Elementary matrices

Each row operation can be carried out by multiplying by a simple matrix, which explains why row reduction works.

- ① Writes a single row operation as a matrix.
- ② Finds the matrix that undoes each such operation.
- ③ Writes an inverse as a product of these simple matrices.
- ④ Explains why row reduction finds inverses and solutions.

LU factorization

LU factorization splits a matrix into two triangular pieces, a shortcut computers use to solve many systems quickly.

- ① Recognizes a matrix written as lower-triangular times upper-triangular.
- ② Builds the two pieces from the steps of elimination.
- ③ Solves a system in two easy triangular stages.
- ④ Explains why this saves work when solving many similar systems.

Vector spaces

A vector space is any collection of things that can be added and scaled the way arrows can, including lists of numbers, polynomials and even functions. This branch learns the shared language for measuring the size and shape of such collections.

VII.4.01 Vector space axioms

A vector space is a collection where adding and scaling always work and follow the usual rules. Arrows in the plane are the standard example.

- 1 Checks that adding or scaling never takes you outside the collection.
- 2 Checks there is a zero element and every element has a negative.
- 3 Checks the remaining rules, such as brackets and order not mattering.
- 4 Tests whether a proposed collection really is a vector space.

VII.4.02 Subspaces

A subspace is a smaller vector space sitting inside a bigger one, like a flat plane through the center of 3D space.

- 1 Checks that the collection contains the zero vector.
- 2 Checks that adding two members always gives another member.
- 3 Checks that scaling a member always gives another member.
- 4 Sorts examples into subspaces and non-subspaces, with reasons.

VII.4.03 Span

The span of a set is everything you can build by mixing its members in any amounts.

- 1 Describes a span as all possible combinations of the vectors.
- 2 Shows a span is always a subspace.
- 3 Finds sets of vectors that span a given space.
- 4 Decides whether a set reaches every vector in the whole space.

VII.4.04 Linear independence

Vectors are linearly independent if none of them can be built from the others, so none is redundant.

- 1 Uses the definition: only the all-zero mix gives the zero vector.
- 2 Tests independence by row reduction.
- 3 Finds the exact relation when vectors do depend on each other.
- 4 Uses quick rules, such as too many vectors always being dependent.

VII.4.05

Basis

A basis is a set of building blocks with no redundancy that can build everything in the space, like the directions east, north and up.

- ① Checks a set is both independent and spanning.
- ② Finds a basis for a given space.
- ③ Uses the standard basis for common spaces.
- ④ Removes extra vectors from a set, or adds missing ones, to make a basis.

VII.4.06

Dimension

The dimension of a space is the number of vectors in any basis, a precise version of how many directions it has.

- ① Understands why every basis of a space has the same size.
- ② Finds the dimension of a given space.
- ③ Relates the dimension of a subspace to the space containing it.
- ④ Meets spaces that no finite basis can cover, such as all polynomials.

VII.4.07

Null space

The null space of a matrix is every input the matrix sends to zero.

- ① Describes it as all solutions of the system that equals zero.
- ② Finds a basis for it.
- ③ Shows it is a subspace.
- ④ Connects its size to the number of free variables.

VII.4.08

Column space

The column space of a matrix is every output it can produce, built from its columns.

- ① Describes it as the span of the matrix's columns.
- ② Finds a basis using the pivot columns of the original matrix.
- ③ Decides whether a given vector is in it.
- ④ Links it to when a system has a solution.

VII.4.09

Row space

The row space of a matrix is everything built from its rows.

- ① Describes it as the span of the rows.
- ② Finds a basis from the nonzero rows of the tidied matrix.
- ③ Sees it as the column space of the transpose.
- ④ Compares it with the column space and their sizes.

VII.4.10

Rank

The rank of a matrix is how many truly independent rows or columns it has, a measure of how much information it carries.

- 1 Finds the rank by counting pivots.
- 2 Learns that counting by rows or by columns gives the same rank.
- 3 Recognizes full-rank matrices.
- 4 Uses rank to decide whether a matrix has an inverse.

VII.4.11

Rank-nullity theorem

The rank-nullity theorem says the directions a matrix keeps plus the directions it crushes to zero add up to the number of inputs.

- 1 States the theorem: rank plus nullity equals the number of columns.
- 2 Uses it to find one number from the other.
- 3 Predicts how big a solution set will be.
- 4 Explains why the theorem is true.

VII.4.12

Coordinates

Coordinates describe a vector by how much of each basis vector it takes, like giving an address in a chosen grid.

- 1 Understands coordinates relative to a chosen basis.
- 2 Finds a vector's coordinates in a given basis.
- 3 Treats turning vectors into coordinates as a map.
- 4 Sees that any space of dimension n behaves just like lists of n numbers.

VII.4.13

Change of basis

Changing basis means translating coordinates from one grid to another, like converting between two map systems.

- 1 Builds the matrix that converts between bases.
- 2 Uses it to convert coordinates.
- 3 Converts between two unusual bases.
- 4 Explains why a well-chosen basis can make a problem much simpler.

VII.4.14

Polynomial & function spaces

Polynomials and functions can also be added and scaled, so they form vector spaces too.

- 1 Treats polynomials up to a certain degree as a vector space.
- 2 Finds and uses bases made of polynomials.
- 3 Works with spaces made of functions.
- 4 Tests whether a set of functions is independent.

VII.4.15

Solution sets as shifted subspaces

Every solution set of a system is one particular solution plus all solutions of the matching zero system, like a plane shifted off the center.

- ① Writes solutions as one particular solution plus the zero-system solutions.
- ② Pictures the solution set as a shifted line or plane.
- ③ Writes any solution set in this form.
- ④ Sees the same pattern in differential equations later on.

VII.4.16

The four fundamental subspaces

Every matrix comes with four key subspaces that together give a complete picture of what the matrix does.

- ① Names the four: column space, row space, null space, and the null space of the transpose.
- ② Finds the dimension of each.
- ③ Identifies which pairs sit at right angles to each other.
- ④ Draws the big-picture diagram connecting all four.

Linear transformations

A linear transformation is a rule that moves vectors around while keeping straight lines straight, such as a rotation or a stretch. This branch shows that every such rule can be captured by a matrix.

VII.5.01 Definition

A linear transformation is a function on vectors that respects adding and scaling. Rotating a picture is an example.

- ① Checks that transforming a sum equals the sum of the transformed parts.
 - ② Checks that scaling before or after transforming gives the same result.
 - ③ Tests whether a given rule is linear.
 - ④ Uses the fact that a linear map always sends zero to zero.
-

VII.5.02 Standard matrix

Every linear transformation of ordinary space can be written as a matrix, found by seeing where the basic directions go.

- ① Finds where the basic unit arrows are sent.
 - ② Builds the matrix from those images.
 - ③ Applies the matrix to transform any vector.
 - ④ Understands that every such linear map comes from a matrix.
-

VII.5.03 Geometric maps

Familiar geometric moves such as turning, mirroring and slanting are all linear transformations with simple matrices.

- ① Writes and uses matrices for rotations.
 - ② Writes and uses matrices for reflections in a line.
 - ③ Writes matrices for shears (slants) and stretches.
 - ④ Writes matrices that project vectors flat onto a line.
-

VII.5.04 Kernel & range

The kernel is everything a transformation squashes to zero; the range is everything it can reach.

- ① Identifies the kernel with the matrix's null space.
- ② Identifies the range with the matrix's column space.
- ③ Finds bases for both.
- ④ Finds their dimensions and relates them.

VII.5.05

One-to-one & onto

A map is one-to-one if different inputs never land in the same place, and onto if every possible output is reached.

- 1 Tests one-to-one by checking only zero goes to zero.
 - 2 Tests onto by checking the range fills the whole space.
 - 3 Uses pivot positions to test both quickly.
 - 4 Sees that a map with both properties can be undone.
-

VII.5.06

Composition

Composition means doing one transformation and then another, which matches multiplying their matrices.

- 1 Applies one map and then a second.
 - 2 Finds the combined matrix by multiplying.
 - 3 Gets the order of multiplication right, since it matters.
 - 4 Combines geometric moves, such as a rotation then a reflection.
-

VII.5.07

Inverse transformations

An inverse transformation undoes the original, like rotating back the other way.

- 1 Describes the map that undoes a given one.
 - 2 Finds it using the inverse matrix.
 - 3 Decides when an inverse exists.
 - 4 Describes inverses of geometric moves, such as undoing a shear.
-

VII.5.08

Maps between abstract spaces

Linear maps also act on polynomials and functions. Taking a derivative is itself a linear transformation.

- 1 Sees taking the derivative as a linear map.
 - 2 Represents such a map by a matrix once bases are chosen.
 - 3 Computes that matrix for a given map.
 - 4 Sees integration as a linear map too.
-

VII.5.09

Similar matrices

Similar matrices describe the same transformation seen through different coordinate systems.

- 1 Recognizes the change-of-basis sandwich that makes matrices similar.
- 2 Explains it as one map described in two bases.
- 3 Lists properties that similar matrices share.
- 4 Chooses a basis that makes the matrix especially simple.

Determinants

The determinant is a single number computed from a square matrix. It tells how much the matrix stretches areas or volumes, and whether the matrix can be undone.

VII.6.01 **2 × 2 determinant**

The determinant of a 2-by-2 matrix is a quick cross-multiplication that measures how the matrix scales area.

- 1 Learns the formula: top-left times bottom-right minus the other diagonal.
 - 2 Computes 2-by-2 determinants.
 - 3 Knows a zero determinant means the matrix has no inverse.
 - 4 Reads it as the signed area of the parallelogram from the columns.
-

VII.6.02 **Cofactor expansion**

Cofactor expansion computes a larger determinant by breaking it into smaller ones.

- 1 Finds minors, the smaller determinants left after crossing out a row and column.
 - 2 Uses the checkerboard pattern of plus and minus signs.
 - 3 Expands a determinant along a chosen row.
 - 4 Picks the row or column with the most zeros to save work.
-

VII.6.03 **By row reduction**

Computing a determinant efficiently by row-reducing the matrix while tracking how each step changes it.

- 1 Learns how each row operation changes the determinant.
 - 2 Reduces the matrix to triangular form.
 - 3 Multiplies the diagonal entries of a triangular matrix.
 - 4 Keeps track of all changes to get the original determinant.
-

VII.6.04 **Properties**

Useful rules that make determinants quicker to find without full calculation.

- 1 Finds a triangular matrix's determinant from its diagonal.
 - 2 Uses the fact that transposing does not change the determinant.
 - 3 Knows two equal rows or a zero row gives determinant zero.
 - 4 Works out what happens when the whole matrix is multiplied by a number.
-

VII.6.05 **Determinant & invertibility**

A matrix has an inverse exactly when its determinant is not zero, a single-number test.

- 1 Uses a nonzero determinant to confirm an inverse exists.
- 2 Finds the values that make a matrix lose its inverse.
- 3 Adds this test to the invertible matrix theorem's list.
- 4 Finds the determinant of an inverse from the original's.

VII.6.06

Area & volume

The determinant measures how a matrix scales area in 2D or volume in 3D, and whether it flips things over.

- 1 Finds a parallelogram's area from its side vectors.
- 2 Finds the volume of a slanted box from its edge vectors.
- 3 Uses the determinant as the area or volume scaling factor of a map.
- 4 Reads a negative sign as a mirror-image flip.

VII.6.07

Product rule

The determinant of a product equals the product of the determinants, which makes many calculations quick.

- 1 States the product rule.
- 2 Uses it to find determinants of products.
- 3 Finds determinants of powers of a matrix.
- 4 Explains with an example why sums do not work the same way.

VII.6.08

Cramer's rule

Cramer's rule solves a system of equations using only determinants.

- 1 Learns the formula: each unknown is a ratio of two determinants.
- 2 Solves a system of two equations with it.
- 3 Solves a system of three equations with it.
- 4 Explains why it is too slow for large systems.

VII.6.09

Adjugate & inverse

A formula that gives a matrix's inverse directly from determinants of its pieces.

- 1 Builds the matrix of cofactors.
- 2 Transposes it to get the adjugate.
- 3 Divides the adjugate by the determinant to get the inverse.
- 4 Checks the result by multiplying back to the identity.

Eigenvalues & eigenvectors

An eigenvector is a special direction that a matrix only stretches or shrinks, never turns, and the eigenvalue is the stretch amount. These ideas reveal a matrix's long-run behavior, from population models to web search rankings.

VII.7.01 Definition

An eigenvector is a direction a matrix only stretches, without turning it. The eigenvalue is how much it stretches.

- 1 Learns the defining equation: the matrix times the vector equals a number times the vector.
- 2 Pictures eigenvectors as directions that only get stretched or shrunk.
- 3 Checks whether a given vector is an eigenvector.
- 4 Spots eigenvector directions in a 2D picture of a transformation.

VII.7.02 Characteristic equation

The characteristic equation is a polynomial equation whose solutions are exactly the eigenvalues.

- 1 Sets up the equation using a determinant.
- 2 Expands the determinant into a polynomial.
- 3 Knows an n -by- n matrix gives a polynomial of degree n .
- 4 Relates the eigenvalues to the diagonal sum and the determinant.

VII.7.03 Find eigenvalues

Finding the eigenvalues of a matrix by solving its characteristic equation.

- 1 Finds eigenvalues of 2-by-2 matrices.
- 2 Reads eigenvalues straight off the diagonal of a triangular matrix.
- 3 Finds eigenvalues of 3-by-3 matrices.
- 4 Counts how many times an eigenvalue repeats (its multiplicity).

VII.7.04 Eigenspaces

The eigenspace for an eigenvalue is the set of all its eigenvectors, plus zero.

- 1 Finds it as the null space of a related matrix.
- 2 Finds a basis for the eigenspace.
- 3 Finds its dimension (the geometric multiplicity).
- 4 Compares that with how often the eigenvalue repeats as a root.

VII.7.05 Diagonalization

Diagonalizing a matrix rewrites it using its eigenvectors so it acts as simple stretches along each direction.

- 1 Recognizes a matrix written as a change of basis around a diagonal matrix.
- 2 Decides when diagonalizing is possible.
- 3 Builds the eigenvector matrix and the diagonal eigenvalue matrix.
- 4 Checks the result by multiplying out.

VII.7.06

Matrix powers

Diagonalizing makes it easy to raise a matrix to a high power, useful for predicting where a process ends up.

- 1 Uses diagonalization to write a power of a matrix simply.
- 2 Computes a high power, such as the 50th.
- 3 Predicts long-run behavior from the eigenvalues.
- 4 Finds a formula for the Fibonacci numbers using matrices.

VII.7.07

Complex eigenvalues

Some matrices have eigenvalues that are complex numbers. These signal rotation mixed with stretching.

- 1 Recognizes when complex eigenvalues appear.
- 2 Knows they come in matching conjugate pairs.
- 3 Interprets them as a rotation combined with a scaling.
- 4 Pictures the resulting spiral motion.

VII.7.08

Markov chains

Markov chains model things that move between states with fixed chances, like customers switching brands each month.

- 1 Recognizes matrices of probabilities whose columns add to 1.
- 2 Tracks the chance of being in each state over time.
- 3 Finds the steady state as an eigenvector.
- 4 Predicts the long-run share in each state.

VII.7.09

Discrete dynamical systems

A discrete dynamical system updates a state step by step using a matrix, like a population counted each year.

- 1 Computes the next state from the current one with a matrix.
- 2 Splits the starting state into eigenvector parts to see the future.
- 3 Classifies whether states are pulled in or pushed away over time.
- 4 Models predators and prey, such as owls and mice.

Orthogonality

Orthogonal means at right angles. This branch uses right angles to find closest points and best approximations, which underlies fitting a line to messy data.

VII.8.01 Inner products

An inner product is a way of multiplying vectors to measure length and angle. The dot product is the standard one.

- ① Uses the dot product on lists of any length.
 - ② Finds lengths and distances from it.
 - ③ Works with other inner products, such as ones for functions.
 - ④ Uses the rules every inner product must follow.
-

VII.8.02 Orthogonal vectors

Two vectors are orthogonal when they meet at a right angle, which happens exactly when their dot product is zero.

- ① Tests for a right angle by checking the dot product is zero.
 - ② Uses the Pythagorean theorem for perpendicular vectors in any dimension.
 - ③ Tests whether every pair in a set is perpendicular.
 - ④ Finds a vector perpendicular to given ones.
-

VII.8.03 Orthogonal complements

The orthogonal complement of a subspace is every vector at right angles to all of it, like the vertical line beneath a flat floor.

- ① Understands the definition.
 - ② Finds a basis for it.
 - ③ Shows the row space and null space are perpendicular to each other.
 - ④ Uses the fact that the two dimensions add up to the whole.
-

VII.8.04 Orthogonal bases

An orthogonal basis is a set of building-block vectors that are all at right angles to each other, which makes calculations much easier.

- ① Shows perpendicular nonzero vectors are automatically independent.
- ② Finds coordinates quickly using dot products.
- ③ Checks whether a set is an orthogonal basis.
- ④ Explains why such bases save so much work.

VII.8.05

Orthonormal bases

An orthonormal basis has perpendicular vectors that are also each of length 1, the tidiest possible basis.

- 1 Scales perpendicular vectors to length 1.
 - 2 Checks a matrix's columns are orthonormal by one multiplication.
 - 3 Finds coordinates instantly using dot products.
 - 4 Shows these coordinates keep lengths unchanged.
-

VII.8.06

Projection onto a subspace

Projecting onto a subspace finds the nearest point in it, like dropping a straight shadow onto a floor.

- 1 Uses the projection formula with a perpendicular basis.
 - 2 Computes a projection.
 - 3 Splits a vector into a part in the subspace and a perpendicular part.
 - 4 Builds a matrix that performs the projection.
-

VII.8.07

Best approximation

The best approximation theorem says the projection is the closest point in the subspace to a given vector.

- 1 States the closest-point theorem.
 - 2 Finds the distance from a vector to a subspace.
 - 3 Computes the closest point in examples.
 - 4 Explains why the projection beats every other point.
-

VII.8.08

Gram-Schmidt

Gram-Schmidt is a step-by-step recipe for turning any basis into one where every vector is at right angles to the rest.

- 1 Learns the key move: subtract the shadow on earlier vectors.
 - 2 Runs the process on two vectors.
 - 3 Runs it on three vectors.
 - 4 Scales the results to length 1 at the end.
-

VII.8.09

QR factorization

QR factorization splits a matrix into a part with perpendicular columns and a triangular part, a standard tool in computing.

- 1 Builds the perpendicular-column part using Gram-Schmidt.
- 2 Finds the triangular part by a single multiplication.
- 3 Checks the factors multiply back to the original.
- 4 Describes how computers use it.

VII.8.10

Least squares

Least squares finds the best approximate solution when a system has no exact one, keeping the total error as small as possible.

- ① Recognizes systems with no exact solution.
- ② Sets up the normal equations, a solvable related system.
- ③ Solves them for the best approximate answer.
- ④ Measures how large the leftover error is.

VII.8.11

Least-squares lines

Fitting the best straight line or curve through scattered data points, the method behind a trend line on a chart.

- ① Sets up the design matrix from the data.
- ② Finds the best-fit straight line.
- ③ Fits curves, such as a parabola, to data.
- ④ Connects this to regression in statistics.

VII.8.12

Orthogonal matrices

An orthogonal matrix moves things without distorting them, keeping every length and angle the same, like a rotation or mirror.

- ① Recognizes matrices whose columns are perpendicular and of length 1.
- ② Finds the inverse simply by transposing.
- ③ Shows these matrices keep lengths and angles unchanged.
- ④ Identifies them as rotations and reflections.

Symmetric matrices & SVD

Symmetric matrices, which mirror across their diagonal, have especially clean eigenvectors. This branch builds up to the singular value decomposition, a tool used in image compression and data science.

VII.9.01 Symmetric matrices

A symmetric matrix equals its own mirror image across the diagonal. Such matrices have unusually well-behaved eigenvalues and eigenvectors.

- 1 Recognizes a matrix that equals its own transpose.
 - 2 Learns that its eigenvalues are always real numbers.
 - 3 Learns its eigenvectors for different eigenvalues are perpendicular.
 - 4 Works through examples to see these facts in action.
-

VII.9.02 Orthogonal diagonalization

Orthogonal diagonalization rewrites a symmetric matrix as pure stretches along perpendicular directions.

- 1 Recognizes the form using a rotation-like matrix and its transpose.
 - 2 Builds that matrix from length-1 eigenvectors.
 - 3 Handles repeated eigenvalues by making their eigenvectors perpendicular.
 - 4 Checks the result by multiplying out.
-

VII.9.03 Spectral theorem

The spectral theorem guarantees every symmetric matrix can be broken into simple stretches along perpendicular directions.

- 1 States the theorem.
 - 2 Writes the matrix as a sum of simple pieces, one per eigenvalue.
 - 3 Sees each piece as a scaled projection onto one direction.
 - 4 Explains why this result is so widely useful.
-

VII.9.04 Quadratic forms

A quadratic form is an expression where every term has two variables multiplied, like x^2 plus $3xy$. A matrix records it compactly.

- 1 Writes a quadratic form using a symmetric matrix.
- 2 Builds the matrix from a given expression.
- 3 Changes variables to remove the mixed xy terms.
- 4 Finds the main axes of the ellipse or other shape it describes.

VII.9.05

Definiteness

Definiteness says whether a quadratic form is always positive, always negative, or mixed, like a bowl, a dome or a saddle.

- ① Recognizes forms that are always positive (bowl-shaped).
- ② Recognizes always-negative and mixed forms.
- ③ Decides which by checking the signs of the eigenvalues.
- ④ Links this to finding maximums and minimums of surfaces.

VII.9.06

Constrained optimization

Finding the largest and smallest value of a quadratic form when the input must have length 1.

- ① Sets up the problem on a circle or sphere of radius 1.
- ② Shows the maximum equals the largest eigenvalue.
- ③ Shows the minimum equals the smallest eigenvalue.
- ④ Finds the directions where these extremes occur.

VII.9.07

Singular values

Singular values measure how much any matrix, even a non-square one, stretches things in its main directions.

- ① Finds eigenvalues of the matrix's transpose times the matrix.
- ② Takes square roots of those to get the singular values.
- ③ Interprets them as the stretch amounts in key directions.
- ④ Finds the rank by counting nonzero singular values.

VII.9.08

Computing the SVD

The singular value decomposition (SVD) breaks any matrix into a rotation, a stretch and another rotation, used in data compression.

- ① Finds the first rotation from eigenvectors.
- ② Builds the middle stretch matrix from the singular values.
- ③ Finds the final rotation from the other two.
- ④ Checks that the three pieces multiply back to the original.

Functions of several variables

Many real quantities depend on more than one input, like temperature depending on both location and time. This branch learns to picture and handle functions of two or three variables.

VII.10.01 Domains

A function of several variables takes two or more inputs and gives one output. The domain is the set of inputs that make sense.

- 1 Reads and evaluates functions written with two inputs, x and y .
 - 2 Finds which input pairs are allowed (the domain).
 - 3 Sketches the domain as a region in the flat plane.
 - 4 Handles functions of three inputs.
-

VII.10.02 Surfaces

A function of two inputs can be drawn as a surface, like a landscape where the height depends on where you stand.

- 1 Pictures the graph as a landscape over a flat map.
 - 2 Slices the surface to see cross-section curves (traces).
 - 3 Recognizes common surfaces such as bowls, saddles and cones.
 - 4 Matches surface pictures to their formulas.
-

VII.10.03 Level curves

Level curves join all points at the same height, exactly like the contour lines on a hiking map.

- 1 Reads a contour map of a function.
 - 2 Draws level curves for a given function.
 - 3 Reads steepness from how close the contour lines are.
 - 4 Interprets weather maps and topographic maps this way.
-

VII.10.04 Level surfaces

For functions of three inputs, points with the same output form surfaces, like shells of equal temperature around a heater.

- 1 Understands a level surface as all points giving one output.
- 2 Recognizes spheres and ellipsoids (stretched spheres) as level surfaces.
- 3 Recognizes other curved surfaces from their equations.
- 4 Matches level surfaces to their equations.

VII.10.05 **Limits in 2D**

A limit asks what value a function approaches near a point. With two inputs, you can approach from infinitely many directions.

- ① Sees that a point can be approached along many different paths.
- ② Shows a limit does not exist by finding two paths that disagree.
- ③ Switches to distance-and-angle coordinates to find a limit.
- ④ Traps a function between two simpler ones to find its limit.

VII.10.06 **Continuity in 2D**

A function is continuous where it has no sudden jumps or holes, so small moves in the inputs give small changes in the output.

- ① Understands the definition of continuity with two inputs.
- ② Knows sums, products and compositions of continuous functions stay continuous.
- ③ Finds where a given function is continuous.
- ④ Fills in a single missing point to remove a break.

Partial derivatives

A partial derivative measures how fast a quantity changes when just one input changes. This branch extends the idea of a derivative (rate of change) to functions of several inputs, including the gradient, which points straight uphill.

VII.11.01 Partial derivative idea

A partial derivative measures how the output changes when one input moves and the others stay fixed, like walking due east on a hillside.

- 1 Holds all other inputs fixed and varies just one.
- 2 Interprets it as the slope in one direction.
- 3 Reads and writes the notation for partial derivatives.
- 4 Uses the limit definition to compute one from scratch.

VII.11.02 Compute partials

Computing partial derivatives using the usual derivative rules, treating the other variables as fixed numbers.

- 1 Finds the partial derivative with respect to x .
- 2 Finds the partial derivative with respect to y .
- 3 Finds partial derivatives of functions with three inputs.
- 4 Estimates partial derivatives from a contour map.

VII.11.03 Higher partials

Taking partial derivatives again to measure how rates of change themselves change.

- 1 Finds second partial derivatives.
- 2 Finds mixed partials, taken first in one variable then another.
- 3 Uses the fact that the order of mixed partials usually does not matter.
- 4 Checks whether a function satisfies the Laplace equation from physics.

VII.11.04 Tangent planes

A tangent plane is the flat surface that just touches a curved surface at one point, like a sheet of paper resting on a ball.

- 1 Learns the formula for a tangent plane.
- 2 Finds the tangent plane at a given point.
- 3 Finds tangent planes to level surfaces.
- 4 Finds the line sticking straight out of the surface (the normal line).

VII.11.05 **Linear approximation**

Using the tangent plane to estimate values of a function near a known point.

- 1 Uses the tangent plane as an estimate of the surface.
 - 2 Approximates a function value near a known point.
 - 3 Estimates small changes in output from small changes in inputs.
 - 4 Estimates how measurement errors grow in a calculation.
-

VII.11.06 **Differentiability**

A function is differentiable at a point if it is smooth enough there to have a good tangent plane.

- 1 Sees that having partial derivatives is not quite enough.
 - 2 Uses the rule that continuous partial derivatives guarantee it.
 - 3 Studies a counterexample where partials exist but smoothness fails.
 - 4 Connects differentiability to the tangent plane being a good estimate.
-

VII.11.07 **Multivariable chain rule**

The chain rule tracks how changes pass through a chain of dependencies, such as temperature depending on position, which depends on time.

- 1 Draws tree diagrams showing what depends on what.
 - 2 Applies the chain rule when everything depends on one variable.
 - 3 Applies it when everything depends on two variables.
 - 4 Solves related-rates problems in three dimensions.
-

VII.11.08 **Implicit differentiation**

Finding slopes of curves or surfaces defined by an equation, without solving for one variable first.

- 1 Uses the formula for curves given by an equation in x and y .
 - 2 Extends it to surfaces given by an equation in x , y and z .
 - 3 Applies it to specific examples.
 - 4 Compares it with the single-variable method learned earlier.
-

VII.11.09 **Gradient**

The gradient collects all the partial derivatives into one arrow that points in the steepest uphill direction.

- 1 Understands the gradient as the vector of partial derivatives.
 - 2 Computes gradients.
 - 3 Shows the gradient points straight uphill.
 - 4 Shows its length equals the steepest slope.
-

VII.11.10 **Directional derivatives**

A directional derivative measures the rate of change in any chosen direction, like the slope felt walking northeast on a hill.

- 1 Describes a direction with an arrow of length 1.
- 2 Computes the rate as the gradient dotted with that direction.
- 3 Finds the directions of fastest increase and decrease.
- 4 Interprets directional slopes on a contour map.

VII.11.11 **Gradient and level sets**

The gradient always points at right angles to level curves and level surfaces, straight across the contour lines.

- ① Shows the gradient is perpendicular to level curves.
- ② Shows it is perpendicular to level surfaces.
- ③ Finds tangent planes using the gradient.
- ④ Traces the path of steepest climb up a hill.

VII.11.12 **Space curves**

A space curve is a path through 3D, like the flight of a drone, described by position over time.

- ① Describes a path by position over time and finds its velocity.
- ② Finds the length of a curved path.
- ③ Finds the direction of travel and the direction of turning.
- ④ Measures how sharply a path bends (its curvature).

Optimization

Optimization means finding the best possible value: the highest profit, the least material, the shortest distance. This branch finds peaks, valleys and saddle points of functions with several inputs, with and without restrictions.

VII.12.01 Critical points

Critical points are the spots where a surface is flat in every direction, where peaks, valleys and saddles can occur.

- 1 Knows critical points are where the gradient is zero.
- 2 Solves the resulting set of equations to find them.
- 3 Includes points where the partial derivatives do not exist.
- 4 Lists all candidate points for a maximum or minimum.

VII.12.02 Second derivative test

The second derivative test uses second partial derivatives to decide if a flat spot is a peak, a valley or a saddle.

- 1 Computes the test number from the second partial derivatives.
- 2 Identifies maximums and minimums with it.
- 3 Identifies saddle points with it.
- 4 Recognizes when the test gives no answer.

VII.12.03 Saddle points

A saddle point is flat but is neither a peak nor a valley, like the middle of a horse saddle or a mountain pass.

- 1 Recognizes the saddle shape.
- 2 Sees the surface rises one way and falls another way.
- 3 Finds saddle points of a function.
- 4 Spots saddles on a contour map.

VII.12.04 Closed regions

Finding the absolute highest and lowest values of a function over a fenced-in region, checking the inside and the edges.

- 1 Finds critical points inside the region.
- 2 Checks each piece of the boundary.
- 3 Checks the corners.
- 4 Compares all candidates to pick the overall largest and smallest.

VII.12.05 **Lagrange multipliers**

Lagrange multipliers find the best value when the inputs must obey a rule, like the largest area for a fixed length of fence.

- 1 Learns the key condition: the two gradients point along the same line.
- 2 Sets up the system of equations.
- 3 Solves the system to find the best points.
- 4 Explains the picture: the best point is where a level curve touches the constraint.

VII.12.06 **Two constraints**

Optimizing when the inputs must obey two rules at once.

- 1 Uses two multipliers, one for each rule.
- 2 Sets up the larger system of equations.
- 3 Solves it.
- 4 Pictures the allowed points as the curve where two surfaces meet.

VII.12.07 **Applied optimization**

Applying these methods to real problems in design, business and geometry.

- 1 Designs boxes that use the least material for a given volume.
- 2 Balances cost and production to find the best plan.
- 3 Finds the closest point on a surface to a given point.
- 4 Checks that an answer makes sense in context.

VII.12.08 **Gradient descent**

Gradient descent finds a low point by repeatedly taking small steps downhill, like walking down a foggy hill.

- 1 Takes one step in the steepest downhill direction.
- 2 Chooses a sensible step size.
- 3 Runs several steps by hand.
- 4 Explains how machine learning uses this method.

Multiple integrals

A multiple integral adds up a quantity spread over an area or a volume, such as the total rain falling on a field or the mass of an object. This branch extends integration (adding up many tiny pieces) to two and three dimensions.

VII.13.01 Double integrals over rectangles

A double integral adds up values over a rectangle, and can be pictured as the volume under a surface, like soil piled on a garden bed.

- 1 Interprets a double integral as the volume under a surface.
- 2 Estimates it by adding up many thin boxes.
- 3 Reads and writes the notation.
- 4 Estimates a double integral from a table of values.

VII.13.02 Iterated integrals

Iterated integrals compute a double integral as two ordinary integrals, one inside the other.

- 1 Does the inner integral first, then the outer one.
- 2 Uses the fact that the order can be swapped over a rectangle.
- 3 Evaluates iterated integrals.
- 4 Splits products of separate x and y parts into two simple integrals.

VII.13.03 General regions

Integrating over regions with curved edges, not just rectangles.

- 1 Handles regions between two curves stacked top and bottom.
- 2 Handles regions between two curves side by side.
- 3 Finds the correct limits from a sketch of the region.
- 4 Evaluates the integral over such regions.

VII.13.04 Reverse the order

Swapping the order of integration can turn an impossible integral into an easy one.

- 1 Sketches the region from the given limits.
- 2 Rewrites the limits for the other order.
- 3 Explains why switching order can help.
- 4 Solves integrals that look impossible in their original order.

VII.13.05 Polar double integrals

Using distance and angle (polar coordinates) instead of x and y, which suits circles and round regions.

- 1 Learns how a small area piece looks in polar coordinates.
- 2 Converts both the region and the function to polar form.
- 3 Evaluates polar double integrals.
- 4 Evaluates the famous bell-curve integral using polar coordinates.

VII.13.06 **Applications**

Using double integrals to find real quantities such as area, mass, balance points and probabilities.

- 1 Finds areas and masses of flat plates.
 - 2 Finds the balance point (center of mass).
 - 3 Finds how hard an object is to spin (moment of inertia).
 - 4 Finds probabilities involving two random quantities.
-

VII.13.07 **Triple integrals**

Triple integrals add up a quantity throughout a solid, such as the total mass of an object with uneven density.

- 1 Uses triple integrals to find volume and mass.
 - 2 Sets up limits for a solid region.
 - 3 Knows the six possible orders of integrating and when to switch.
 - 4 Evaluates triple integrals.
-

VII.13.08 **Cylindrical & spherical**

Using coordinate systems that suit round shapes: cylindrical for cans and pipes, spherical for balls and cones.

- 1 Uses cylindrical coordinates: a polar angle and distance plus height.
 - 2 Uses spherical coordinates: distance from center plus two angles.
 - 3 Uses the correct small-volume piece in each system.
 - 4 Chooses the best coordinate system for a given problem.
-

VII.13.09 **Change of variables**

Changing variables reshapes an awkward region into a simple one, with a correction factor (the Jacobian) for how areas stretch.

- 1 Learns what the Jacobian measures: how much the change stretches area.
- 2 Computes the Jacobian as a determinant.
- 3 Transforms a region into a simpler shape.
- 4 Evaluates an integral after changing variables.

Vector calculus

Vector calculus studies quantities that have a direction at every point, like wind, water flow or gravity. It culminates in theorems linking what happens inside a region to what happens on its edge, key tools in physics and engineering.

VII.14.01 Vector fields

A vector field attaches an arrow to every point, like a weather map showing wind speed and direction everywhere.

- 1 Pictures a field as an arrow at every point.
- 2 Sketches a field from its formula.
- 3 Draws flow lines that follow the arrows.
- 4 Describes wind, water currents and gravity as vector fields.

VII.14.02 Gradient fields

A gradient field comes from a single height-like function, with every arrow pointing uphill on it.

- 1 Recognizes fields that are the gradient of some function.
- 2 Understands potential functions, the height functions behind such fields.
- 3 Finds a potential function for a given field.
- 4 Describes gravity as the gradient of a potential.

VII.14.03 Scalar line integrals

A scalar line integral adds up a quantity along a curve, like the total mass of a bent wire.

- 1 Understands adding up a quantity along a curve.
- 2 Describes the curve with a parameter, like time along a path.
- 3 Uses the small piece of arc length correctly.
- 4 Finds the mass of a wire with uneven density.

VII.14.04 Vector line integrals

A vector line integral adds up how much a field pushes along a path, such as the work done by wind on a moving boat.

- 1 Understands the integral of the field along the direction of travel.
- 2 Interprets it as work done along a path.
- 3 Evaluates vector line integrals.
- 4 Sees that reversing the path's direction flips the sign.

VII.14.05 **Fundamental theorem for line integrals**

For gradient fields, a line integral depends only on the start and end points, like height gained on a hike regardless of the route.

- 1 Computes the integral as potential at the end minus potential at the start.
- 2 Applies this shortcut to examples.
- 3 Shows any closed loop gives zero.
- 4 Compares it with the fundamental theorem of ordinary calculus.

VII.14.06 **Conservative field tests**

Tests that check whether a field is conservative, meaning it comes from a potential function.

- 1 Uses a mixed-partials test for flat 2D fields.
- 2 Uses the curl test for 3D fields.
- 3 Knows the tests need a region without holes.
- 4 Studies the swirling vortex field where the test misleads.

VII.14.07 **Path independence**

Path independence means the result of a line integral does not depend on the route taken between two points.

- 1 Lists statements that all mean the same thing as path independence.
- 2 Picks an easy path to compute the integral.
- 3 Connects this to conservation of energy.
- 4 Checks whether a claim of path independence is correct.

VII.14.08 **Green's theorem, circulation**

Green's theorem links the swirl of a field around a closed loop to the total swirl inside it.

- 1 States Green's theorem in its circulation form.
- 2 Goes around the boundary in the correct direction.
- 3 Evaluates both sides and checks they agree.
- 4 Finds an area using only a trip around its edge.

VII.14.09 **Green's theorem, flux**

The flux form of Green's theorem measures net flow out across a loop by adding up sources inside.

- 1 States the flux form of the theorem.
- 2 Computes the outward flow across a boundary.
- 3 Evaluates both sides.
- 4 Applies it to regions with holes.

VII.14.10 **Curl**

Curl measures how much a field makes things spin, like a tiny paddle wheel placed in flowing water.

- 1 Learns the formula for curl.
- 2 Pictures curl with a spinning paddle wheel.
- 3 Computes the curl of a field.
- 4 Recognizes fields with zero curl (no spin).

VII.14.11 Divergence

Divergence measures how much a field spreads out from a point, like water from a sprinkler or draining into a sink.

- 1 Learns the formula for divergence.
 - 2 Pictures sources (spreading out) and sinks (drawing in).
 - 3 Computes the divergence of a field.
 - 4 Recognizes fields with zero divergence, like water that cannot be squeezed.
-

VII.14.12 Parametric surfaces

Describing a curved surface using two parameters, like latitude and longitude on a globe.

- 1 Describes a surface with two parameters.
 - 2 Writes descriptions of spheres, cylinders and cones.
 - 3 Writes graphs of functions as parametric surfaces.
 - 4 Finds tangent planes to parametric surfaces.
-

VII.14.13 Surface area

Finding the area of a curved surface, like the amount of fabric needed to cover a dome.

- 1 Uses the cross product of two tangent arrows for a tiny area piece.
 - 2 Sets up the surface area integral.
 - 3 Evaluates it.
 - 4 Uses the shortcut formula for surfaces that are graphs.
-

VII.14.14 Scalar surface integrals

A surface integral adds up a quantity over a curved surface, like the total mass of a thin shell.

- 1 Understands adding a quantity over a curved surface.
 - 2 Sets up the integral.
 - 3 Evaluates it.
 - 4 Finds the mass of a thin shell.
-

VII.14.15 Flux integrals

Flux measures how much of a flow passes through a surface, like water through a fishing net.

- 1 Understands flux as flow across the surface in the normal direction.
 - 2 Sets up flux integrals.
 - 3 Evaluates them.
 - 4 Computes flow through a net-like surface.
-

VII.14.16 Orientation

Orientation means choosing which side of a surface counts as outside, which fixes the sign of flux.

- 1 Chooses a normal direction for a surface.
 - 2 Uses outward normals for closed surfaces.
 - 3 Matches the boundary's direction to the surface's side.
 - 4 Studies the Möbius strip, a surface with only one side.
-

VII.14.17 **Stokes' theorem**

Stokes' theorem links the swirl of a field over a surface to the circulation around its edge.

- ① States Stokes' theorem.
 - ② Matches the surface and edge directions correctly.
 - ③ Evaluates whichever side is easier.
 - ④ Shows any surface with the same edge gives the same answer.
-

VII.14.18 **Divergence theorem**

The divergence theorem links flow out through a closed surface to the total spreading-out inside it.

- ① States the divergence theorem.
- ② Evaluates whichever side is easier.
- ③ Computes flux out of closed surfaces such as boxes and spheres.
- ④ Connects it to Gauss's law for electric fields.

First-order equations

A differential equation connects a quantity to how fast it changes, like a cooling cup of coffee losing heat faster when it is hotter. This branch solves the simplest kind, which involve only the first rate of change.

VII.15.01 Classify ODEs

Sorting differential equations by type, since each type has its own solving method.

- 1 Finds the order: the highest derivative (rate of change) involved.
- 2 Tells linear equations from nonlinear ones.
- 3 Recognizes autonomous equations, where time does not appear directly.
- 4 Tells equations with one input variable from those with several.

VII.15.02 Linear first-order

Solving linear first-order equations using an integrating factor, a clever multiplier that makes the equation easy to integrate.

- 1 Rewrites the equation in standard form.
- 2 Finds the integrating factor.
- 3 Solves the equation.
- 4 Finds the one solution that matches a given starting value.

VII.15.03 Exact equations

Exact equations come from a hidden function whose level curves are the solutions.

- 1 Tests whether an equation is exact.
- 2 Finds the hidden function.
- 3 Solves the equation.
- 4 Multiplies by a factor to make a non-exact equation exact.

VII.15.04 Substitutions

Using a clever change of variable to turn a hard equation into one already known how to solve.

- 1 Solves homogeneous equations, where the slope depends only on y divided by x .
- 2 Solves Bernoulli equations with a standard substitution.
- 3 Uses the substitution y equals v times x .
- 4 Chooses the right substitution for a given equation.

VII.15.05 Existence & uniqueness

A theorem that says when an equation is guaranteed to have exactly one solution through a starting point.

- 1 States the existence and uniqueness theorem.
- 2 Checks whether its conditions hold.
- 3 Studies an example with more than one solution.
- 4 Finds how far a solution is guaranteed to last.

Circuits & mixing

Applying first-order equations to electrical circuits and to tanks where liquids mix.

- ① Solves circuits with a resistor and a capacitor.
 - ② Solves circuits with a resistor and an inductor (coil).
 - ③ Tracks salt concentration in a tank with liquid flowing in and out.
 - ④ Interprets what happens in the long run.
-

Numerical methods

Numerical methods approximate a solution step by step when no formula can be found, as computers do.

- ① Uses Euler's method: repeated small steps along the current slope.
- ② Uses an improved version that averages two slopes.
- ③ Uses the Runge-Kutta method, a more accurate standard.
- ④ Compares how accurate the methods are.

Second-order linear

Second-order equations involve acceleration as well as speed, so they describe things that vibrate and swing, like springs, pendulums and electrical circuits. This branch solves them and uses them to understand bouncing, damping and resonance.

VII.16.01 Structure of solutions

The solutions of these equations have a neat structure: any solution is a mix of two basic ones.

- 1 Adds and scales known solutions to make new ones (superposition).
- 2 Knows two genuinely different solutions are needed.
- 3 Uses the Wronskian, a determinant, to check they are different enough.
- 4 Writes the general solution covering every case.

VII.16.02 Distinct real roots

Solving constant-coefficient equations whose helper polynomial has two different real roots, giving two exponential solutions.

- 1 Writes the characteristic equation, a helper polynomial.
- 2 Builds the solution from two exponential growth or decay terms.
- 3 Finds the one solution matching given starting values.
- 4 Describes what the solution does in the long run.

VII.16.03 Repeated roots

Solving when the helper polynomial has one root repeated, which needs a second, slightly different solution.

- 1 Recognizes a repeated root.
- 2 Uses t times the exponential as the second solution.
- 3 Finds the solution matching given starting values.
- 4 Explains why the extra factor of t works.

VII.16.04 Complex roots

Solving when the helper polynomial has complex roots, which produces wavy, oscillating solutions.

- 1 Recognizes complex roots and their real and imaginary parts.
- 2 Writes solutions using exponentials times sines and cosines.
- 3 Finds the solution matching given starting values.
- 4 Rewrites the answer as a single wave with a size and a shift.

VII.16.05 Reduction of order

Reduction of order finds a second solution when one solution is already known.

- 1 Starts from one known solution.
- 2 Guesses the second as an unknown function times the first.
- 3 Solves for that unknown function.
- 4 Applies the method to equations with changing coefficients.

VII.16.06 **Undetermined coefficients**

Undetermined coefficients is an educated-guess method for equations with an outside push, such as a steady or wavy force.

- 1 Guesses the form of the solution from the push.
- 2 Solves for the unknown numbers in the guess.
- 3 Adjusts the guess when it overlaps a basic solution (resonance).
- 4 Handles pushes made of several parts added together.

VII.16.07 **Variation of parameters**

Variation of parameters is a general formula for equations with an outside push, working even when guessing fails.

- 1 Learns the formula.
- 2 Sets it up for a given equation.
- 3 Evaluates the integrals to find the solution.
- 4 Uses it for pushes where guessing does not work.

VII.16.08 **Free spring motion**

Modeling a weight bouncing on a spring with no friction, which moves back and forth forever in a smooth wave.

- 1 Sets up the spring equation from mass and stiffness.
- 2 Finds how long one bounce takes and how often it repeats.
- 3 Finds how far the weight swings (the amplitude).
- 4 Approximates a swinging pendulum the same way.

VII.16.09 **Damped motion**

Adding friction to the spring model, so the motion dies away over time.

- 1 Analyzes light damping, where it bounces while slowly settling.
- 2 Analyzes critical damping, the fastest settling with no bounce.
- 3 Analyzes heavy damping, a slow creep back with no bounce.
- 4 Explains how car shock absorbers are tuned.

VII.16.10 **Forced motion & resonance**

Pushing a spring repeatedly, which can build huge swings when the push matches its natural rhythm (resonance).

- 1 Finds the steady motion that lasts once things settle.
- 2 Finds the short-lived motion that fades away.
- 3 Explains resonance, when swings grow larger and larger.
- 4 Explains beats, a slow pulsing when two rhythms nearly match.

VII.16.11 **RLC circuits**

Circuits with a resistor, coil and capacitor obey the same equation as a spring, so the same math applies.

- 1 Sets up the circuit equation.
- 2 Matches circuit parts to spring parts.
- 3 Solves the circuit equation.
- 4 Explains how tuning a radio uses resonance.

VII.16.12 **Series solutions**

Series solutions build an answer as an endless sum of powers of x , useful when no simple formula exists.

- 1 Assumes the solution is an endless polynomial with unknown coefficients.
- 2 Substitutes it in and lines up matching powers.
- 3 Finds a rule giving each coefficient from earlier ones.
- 4 Solves a simple spring equation this way and recognizes sine and cosine.

VII.16.13 **Cauchy–Euler equations**

Cauchy–Euler equations are a special type with changing coefficients that powers of x solve neatly.

- 1 Recognizes the special pattern of these equations.
- 2 Tries a power of x as a solution.
- 3 Handles the three cases for the resulting roots.
- 4 Finds the solution matching given starting values.

Systems of ODEs

Many real situations involve several quantities that change together and affect each other, like foxes and rabbits. This branch solves such linked equations using matrices and eigenvalues, and pictures how they behave over time.

VII.17.01 Writing a system

A system of differential equations describes several quantities that change together, each affecting the others.

- 1 Writes two linked equations for two changing quantities.
- 2 Chooses the quantities to track (the state variables).
- 3 Includes the starting values.
- 4 Builds systems from real contexts such as linked tanks.

VII.17.02 Higher order to first order

Any higher-order equation can be rewritten as a system of first-order ones by giving each rate of change its own name.

- 1 Introduces new variables for the derivatives (rates of change).
- 2 Converts a higher-order equation into a system.
- 3 Writes the system in matrix form.
- 4 Explains why this makes solving and computing easier.

VII.17.03 Matrix form

Writing a linear system compactly as a vector of unknowns whose rate of change equals a matrix times itself.

- 1 Writes the system in matrix form.
- 2 Checks that a proposed solution works.
- 3 Combines solutions to make new ones.
- 4 Builds a fundamental matrix that holds a full set of solutions.

VII.17.04 Real distinct eigenvalues

When the matrix has distinct real eigenvalues, each eigenvector gives a simple solution that grows or shrinks along a straight line.

- 1 Builds one solution from each eigenvalue and eigenvector.
- 2 Writes the general solution.
- 3 Finds the solution that fits given starting values.
- 4 Identifies solutions that travel along straight lines.

VII.17.05 **Phase portraits**

A phase portrait is a picture of how every possible solution moves, like a map of currents.

- 1 Draws paths traced out by solutions (trajectories).
- 2 Recognizes nodes, where all paths flow in or out.
- 3 Recognizes saddles, where paths come in one way and leave another.
- 4 Sketches a portrait from the eigenvectors.

VII.17.06 **Complex eigenvalues**

Complex eigenvalues produce solutions that circle or spiral around a center point.

- 1 Builds real solutions from the complex ones.
- 2 Recognizes spirals moving inward or outward.
- 3 Recognizes centers, where paths loop around forever.
- 4 Works out whether the rotation is clockwise or counterclockwise.

VII.17.07 **Repeated eigenvalues**

Handling the tricky case where an eigenvalue repeats but there are not enough eigenvectors.

- 1 Recognizes when only one eigenvector exists.
- 2 Finds a generalized eigenvector to fill the gap.
- 3 Writes the general solution.
- 4 Sketches the resulting lopsided node.

VII.17.08 **Stability map**

A single chart, based on two numbers from the matrix, that classifies how every linear system behaves.

- 1 Places a system on the chart using the diagonal sum and determinant.
- 2 Classifies an equilibrium, a resting point, as node, saddle, spiral or center.
- 3 Decides whether nearby paths settle in or run away.
- 4 Handles borderline cases on the dividing lines.

VII.17.09 **Nonhomogeneous systems**

Solving systems that also have an outside push or input.

- 1 Recognizes systems with an added outside term.
- 2 Solves them by educated guessing.
- 3 Solves them with the general variation of parameters formula.
- 4 Finds the steady state the system settles into.

VII.17.10 **Nonlinear systems**

Nonlinear systems cannot usually be solved exactly, but they can be understood near their resting points by zooming in.

- 1 Finds the equilibrium (resting) points.
- 2 Computes the Jacobian, the matrix of partial derivatives.
- 3 Replaces the system near each resting point with a simpler linear one.
- 4 Classifies the behavior near each resting point.

VII.17.11 **Applications**

Using systems to model real situations in ecology, physics and public health.

- ① Models predators and prey, like foxes and rabbits.
- ② Models two species competing for the same food.
- ③ Models two springs connected together.
- ④ Models how a disease spreads through a population.

Laplace transform

The Laplace transform turns a differential equation into an algebra problem, solves it there, and translates the answer back. It is especially handy for systems that are switched on and off or struck suddenly.

VII.18.01 Definition

The Laplace transform converts a function of time into a new function using an integral, turning calculus problems into algebra problems.

- 1 Understands the definition as an integral running to infinity.
- 2 Finds the transforms of a constant and of an exponential.
- 3 Checks when the transform exists.
- 4 Explains the big idea: calculus becomes algebra.

VII.18.02 Transform table

A table of common transforms that saves recomputing them each time.

- 1 Uses table entries for powers of t .
- 2 Uses table entries for sine and cosine.
- 3 Reads the table to transform a given function.
- 4 Derives one table entry from the definition.

VII.18.03 Properties

Rules that let complicated transforms be built from simple ones.

- 1 Uses linearity: sums and multiples transform piece by piece.
- 2 Uses the shift rule for functions multiplied by an exponential.
- 3 Transforms derivatives, which is what makes solving equations possible.
- 4 Transforms functions multiplied by t .

VII.18.04 Inverse transforms

Inverse transforms translate an answer back from the algebra side to an ordinary function of time.

- 1 Reads the table backwards.
- 2 Splits sums and pulls out constants.
- 3 Completes the square to match a table entry.
- 4 Undoes shifts to recover the original function.

VII.18.05 Partial fractions for inverses

Breaking a complicated fraction into simple pieces (partial fractions) so each piece can be transformed back.

- 1 Splits fractions with different simple factors below.
- 2 Handles repeated factors.
- 3 Handles squared factors that do not break down further.
- 4 Transforms each simple piece back.

VII.18.06 **Solve IVPs**

Solving differential equations with given starting values using the Laplace transform from start to finish.

- 1 Transforms every term of the equation.
 - 2 Solves the resulting algebra for the unknown transform.
 - 3 Transforms back to find the solution.
 - 4 Checks the solution matches the starting values.
-

VII.18.07 **Step functions**

Step functions model something switched on at a certain moment, like a motor turned on at 5 seconds.

- 1 Uses the unit step, which jumps from 0 to 1 at a set time.
 - 2 Writes pushes that switch on and off.
 - 3 Uses the delay rule for shifted functions.
 - 4 Solves an equation with a switch in it.
-

VII.18.08 **Impulses**

An impulse is a sudden, instant kick, like a hammer strike, modeled by the Dirac delta.

- 1 Understands the Dirac delta as an idealized instant kick.
 - 2 Finds its transform.
 - 3 Solves problems where a system is struck suddenly.
 - 4 Finds a system's impulse response, its reaction to one kick.
-

VII.18.09 **Convolution**

Convolution is a special way of blending two functions, and the transform turns it into ordinary multiplication.

- 1 Understands the definition of convolution as an integral.
- 2 Uses the convolution theorem: blending becomes multiplying.
- 3 Transforms back products of two known transforms.
- 4 Uses transfer functions to describe how a system responds to any input.

Fourier series, a first look

A Fourier series builds any repeating pattern out of simple sine and cosine waves, the way a musical chord is built from pure notes. It is the math behind sound, signals and heat flow.

VII.19.01 Periodic functions & orthogonality

Periodic functions repeat over and over. Sine and cosine waves of different frequencies are orthogonal, meaning they do not overlap in a precise sense.

- 1 Works with functions that repeat over a fixed length.
 - 2 Shows sines and cosines of different frequencies are orthogonal.
 - 3 Computes integrals of products of sines and cosines.
 - 4 Explains how this lets each wave be picked out from a mix.
-

VII.19.02 Fourier coefficients

Fourier coefficients are the amounts of each wave needed to build a given pattern, like a recipe for a sound.

- 1 Finds the average level, the constant term.
 - 2 Finds the amount of each cosine wave.
 - 3 Finds the amount of each sine wave.
 - 4 Writes out the full Fourier series.
-

VII.19.03 Square wave

Building a square wave, which jumps between two levels, from smooth sine waves.

- 1 Computes the coefficients for a square wave.
 - 2 Sees that only the odd-numbered waves appear.
 - 3 Graphs partial sums getting closer to the square shape.
 - 4 Listens to how adding waves changes the sound.
-

VII.19.04 Sawtooth & triangle waves

Building sawtooth and triangle waves from sine and cosine waves, and comparing them.

- 1 Computes the coefficients for a sawtooth wave.
 - 2 Computes the coefficients for a triangle wave.
 - 3 Compares how fast the amounts of higher waves shrink.
 - 4 Compares how the two waves sound.
-

VII.19.05 Even & odd extensions

A pattern given on half an interval can be extended to use only cosines or only sines, whichever suits the problem.

- 1 Builds a cosine-only series by mirroring the pattern.
- 2 Builds a sine-only series by flipping the pattern.
- 3 Finds half-range series for a pattern given on part of an interval.
- 4 Chooses which series suits a given problem.

VII.19.06 **Convergence & Gibbs**

How well a Fourier series matches the original pattern, including a persistent overshoot at sudden jumps.

- 1 Knows at a jump the series lands halfway between the two levels.
- 2 Observes the Gibbs overshoot near jumps that never fully goes away.
- 3 Links smoother patterns to faster-shrinking coefficients.
- 4 Uses Parseval's identity to add up the energy in all the waves.

VII.19.07 **Complex form**

Writing Fourier series with complex exponentials, a compact form that combines sines and cosines.

- 1 Uses complex exponential waves as building blocks.
- 2 Computes the complex coefficients.
- 3 Converts between the complex and sine-cosine forms.
- 4 Draws a spectrum showing how much of each frequency is present.

VII.19.08 **Heat equation**

The heat equation describes how temperature spreads along a rod. Fourier series solve it by splitting the starting temperature into waves.

- 1 Understands the heat equation and what it describes.
- 2 Splits the problem into a time part and a position part.
- 3 Matches the starting temperature with a sine series.
- 4 Explains how heat smooths out sharp differences over time.